

**INFORMATION AND COMMUNICATION
TECHNOLOGY**

S3

STUDENT BOOK

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FOREWORD

Dear Student,

Rwanda Basic Education Board is honored to present to you this ICT Student's Book for Senior Three which serves as a guide to competence-based teaching and learning to ensure consistency and coherence in the learning of ICT. The Rwandan educational philosophy is to ensure that you achieve full potential at every level of education which will prepare you to be well integrated in society and exploit employment opportunities.

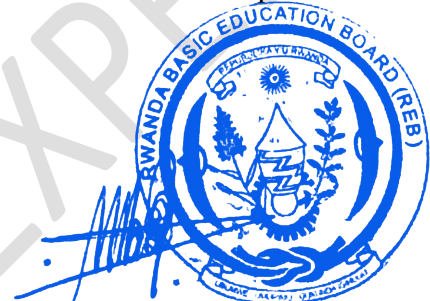
The government of Rwanda emphasizes the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn and the competences you acquire. Those factors include the instructional materials available among others. In this book, special attention was paid to the activities that facilitate the learning process in which you can develop your ideas and make new discoveries during concrete activities carried out individually or with peers.

In competence-based curriculum, learning is considered as a process of active building by the learner where concepts are mainly introduced by an activity, a situation or a scenario that helps the learner to construct knowledge, develop skills and acquire positive attitudes and values.

For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of knowledge and skills;
- Share relevant information with other learners through presentations, discussions, group work and other active learning techniques such as field visit, laboratory work, investigation, role play, case studies and research in the library, from the internet or from your community;
- Participate and take responsibility for your own learning;
- Draw conclusions based on the findings from the learning activities.

I wish to sincerely extend my appreciation to REB staff who organized the editing process of this book. Special gratitude goes to the lecturers, teachers, illustrators and designers who diligently worked to successful completion of this book. Any comment or contribution would be welcome for the improvement of this textbook for the next edition.



Dr. MBARUSHIMANA Nelson
Director General, REB

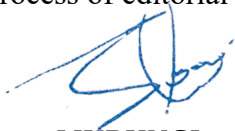
EXPERIMENTAL VERSION

ACKNOWLEDGEMENT

I wish to express my appreciation to all the people who played a major role in editing process of ICT Student's Book for Senior Three. It would not have been successful without their active participation.

Special thanks are given to those who gave their time to read and refine this textbook to meet the needs of competence-based curriculum. I owe gratitude to different Universities and secondary schools in Rwanda that allowed their staff to work with REB to edit this book. I therefore, wish to extend my sincere gratitude to lecturers, teachers, illustrators, designers and all other individuals whose efforts in one way or the other contributed to the completion of this edition.

Finally, my word of gratitude goes to the Rwanda Basic Education Board staff particularly those from Curriculum, Teaching and Learning Resources Department who were involved in the whole process of editorial work.



Joan MURUNGI,

Head of Department of Curriculum, Teaching and Learning Resources

Contents

FOREWORD	3
ACKNOWLEDGEMENT	5
UNIT 1: OPERATING SYSTEM, APPLICATION AND UTILITY SOFTWARE	12
1.1. Operating system	13
1.1.1. Types of operating systems	13
1.1.2. Functions of OS	14
1.1.3. OS issues and their solution	15
1.2. Application software	16
1.2.1. Categories of application software	16
1.2.2. Irembo application	17
1.2.3. Software add ons	18
1.3. OS interaction with hardware	18
1.4. Installing application software	20
1.5. Utility software	24
1.5.1. Role of utility software	24
1.5.2. Types of utility software	24
1.5.3. Installation of an antivirus	24
1.5.4. Role of an antivirus	25
1.6. Uninstalling a program	25
UNIT 2: WORD AUTOMATIC TABLES	29
2.0. Introduction.....	29
2.1. Table of content.....	30
2.1.1. Inserting page numbers	30
2.1.2. Heading styles	31
2.1.3. Creation of table of content	33
2.1.4. Updating the table of content	34
2.2. List of figures	35
2.2.1. Figure captions	35
2.2.2. List of figure creation	36
2.2.3. List of figure updating	37

2.3. List of tables.....	38
2.3.1. Table captions	38
2.3.2. List of table creation	39
2.3.3. List of table updation	40
2.4. Mail merge	41
2.4.1. Using a new list of recipients	41
2.4.2. Using existing list	44
UNIT 3. CHARTS AND OBJECTS IN SPREADSHEET	49
3.0. Introduction.....	49
3.1. Creating charts	50
3.1.1. Role of charts	50
3.1.2. Types of charts:	50
3.1.3. Charts insertion:	53
3.2. Charts formatting	56
3.2.1. Charts elements	56
3.2.2. Format changing	57
3.3. Format cell data as a table.....	62
3.4. Printable datasheet	64
3.4.1. Margin settings	64
3.4.2. Headers & footers	66
3.4.3. Page numbers	70
3.4.4. Printing an Excel worksheet	71
3.6. End unit assessment	74
3.6.1. Project activity	74
3.6.2. Assessment questions.....	74
UNIT 4: PRESENTATION	76
4.0. Introduction.....	76
4.1. Introduction on presentation	77
4.1.1. Role of presentation applications	77
4.1.2. Creating, saving and opening a presentation	78
4.2. Slides.....	84

4.2.1. Insert slide	84
4.2.2. Slide navigation pane	86
4.2.3. Slide manipulation	86
4.3. Objects	88
4.3.1. Textbox	89
4.3.2. Images	90
4.3.3. Table	93
4.3.4. WordArt	95
4.3.5. SmartArt	97
4.3.6. Shapes	99
4.3.7. Chart	101
4.3.8. Media clips	102
4.4. Formatting a slide	106
4.4.1. Changing the font	106
4.4.2. Slide design	107
4.4.3. Slide layout	108
4.5. Presentation views	110
4.5.1. Normal view	111
4.5.2 Slide sorter	111
4.5.3. Reading view	111
4.5.4. Slide show	112
4.6. Master views	113
4.6.1. Slide master	113
4.6.2. Handout master	114
4.7. Animation of a presentation	116
4.7.1. Slide transition	116
4.7.2. Transition sound and speed	117
4.7.3. Preview presentation	118
4.8. End unit assessment	118
4.8.1. Project activity	118
4.8.2. Assessment questions	119

UNIT 5: PLATFORM GAME PROGRAMMING WITH AI	120
5.0. Introduction.....	120
5.1. Platform games	121
5.1.2 Scratch interface	121
5.1.3 Engaging games through interactivity	122
5.2: AI ethics in game programming.....	124
5.2.1 Fairness and non discrimination	124
5.2.2 Data privacy and security	124
5.2.3 Copyright and intellectual property.....	124
5.2.4 Inclusive and ethical storytelling	125
5.3 Artificial Intelligence in games.....	125
5.3.1. AI behaviors.....	126
5.3.2. Algorithms in AI.....	126
5.4. Building the game environment.....	127
5.4.1 Platform and background creation	127
5.4.2 Sprites (Player, Obstacles, Goals).....	128
5.4.3 Plan basic game layout	129
5.4.4 Example of AI chase challenge.....	130
5.5 Player movement and controls.....	132
5.5.1. Player movement with arrows	132
5.5.2 Jump, fall, and collision control.....	133
5.5.3 Coordinate movement in Scratch	136
5.6 Adding obstacles and scoring.....	136
5.6.1 Interactive elements	136
5.6.2 Variable based scoring.....	138
5.6.3 Testing and feedback.....	140
5.7 AI behaviors in games.....	141
5.7.1 Program AI movement.	142
5.7.2 AI decision processes.....	143
5.7.3 Programmed and adaptive behavior.....	145
5.8. Teachable Machine for AI.....	147

5.8.1 AI prediction with Scratch	148
5.8.2 Training data and AI performance	149
5.9. End unit assessment	152
5.9.1. Project activity	152
5.9.2. Assessment questions.....	153
UNIT 6. ROBOT LOGIC IN SCRATCH	155
6.0. Introduction.....	155
6.1 Introduction to robotics.....	156
6.1.1. Robot parts	156
6.1.2. Types of robots	157
6.1.3. Importance of robots	158
6.1.4. Robot ethics	158
6.1.5 Hardware required for a robotics programming	158
6.2 Getting started with Scratch.....	160
6.2.1 Categories of blocks for a robot	160
6.2.2 Robot simulation	162
6.3. Scratch variables	165
6.3.1 Introduction to Scratch variables	165
6.3.2 Data types stored in Scratch variables	166
6.3.3 Examples of variable use	166
6.4. Simulating one sensor.....	166
6.4.1 Touch sensor	167
6.4.2 Color sensor	167
6.4.3 Distance sensor	168
6.4.4 Temperature sensor	168
6.4.5 Sound sensor	169
6.4.6 Other sensors	169
6.5. Simulating many sensors	170
6.5.1. Combined conditions	170
6.5.2 Sensor fusion	171
6.6. End unit assessment	173

6.6.1. Project activity	173
6.6.2. Assessment questions.....	174

EXPERIMENTAL VERSION

UNIT 1: OPERATING SYSTEM, APPLICATION AND UTILITY SOFTWARE



Key Unit Competency: Manipulate application software, utility software and add-ons through installation and practical use

Learning Objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Explain the functions of an operating system and how it interacts with application software to manage hardware and tasks
- Differentiate various types of software and explain the purpose and use of each.
- Describe computer user privileges while installing a program.

Skills

- Resolve OS related issues
- Download software from the internet.
- Use administrative rights to install different software application Software

Attitudes and values

- Appreciate the role of software namely operating system, application software and utility software in a computer.
- Show concern over virus infection from the internet through different downloaded files

Introduction

Computers cannot function without software that manages their hardware and enables users to perform tasks. At the heart of this software is the Operating System (OS), which acts as the foundation for all other programs. This unit introduces the concept of operating systems, their types, functions, and common issues, while also exploring application software, utility software, and the interaction between software and hardware. This unit introduces both theoretical knowledge and practical skills in installing, managing, and uninstalling software responsibly.

Introductory activity

The school has received a set of new computers as a gift for the students. In the smart classroom, the Senior Two class sits a round table, and each student has a computer in front of them.

The teacher says:

"These computers are ready to use, but first we need to install the software that will make them work properly. Some software are more important because they provide the working environment to others.

- 1) State the software that you will install on the computer
- 2) For each software you give, explain why you chose it.

1.1. Operating system

Activity 1.1

- 1) What do you think makes a computer or phone able to work?
- 2) Can a computer run without any software? Explain.
- 3) If your computer doesn't start what would you do?
- 4) What happens when you switch on a computer?

Operating system is a large and powerful program that controls and manages computer hardware and application software. It is the most important computer program.

Operating system software runs and manages all the operations of the computer. All electronic devices use operating systems to work, for example; laptop computers, tablet, desktop computers, smartphone, router, etc.

The operating system that runs a laptop or desktop computer is commonly called **Desktop Operating System (DOS)**. This includes; **Microsoft Windows**, **MacOS**, Chrome OS (designed by Google) and **Linux (e.g. Ubuntu, Fedora)**.

Mobile operating system runs on smartphones, PDAs, Tablet PCs and other mobile devices. Examples include **Android** (by Google Inc.) and **iOS** (by Apple) etc.

Server operating system or server OS is installed and runs on servers. Examples of server OS are: Windows Server, Linux.

1.1.1. Types of operating systems

There are different types of operating systems helping electronic devices to function and provide working environment for other applications.

- **Single user, single task OS:** This type of OS allows only one user and one task at a time. It is simple and used on small devices. An example is DOS (Disk operating system).
- **Single user, multi-tasking OS:** This OS allows one user to use many applications at the same time. It is common on personal computers. Examples include Windows, macOS, and Linux.
- **Multi user OS:** This kind of OS is found in server computers. It allows many people to use the computer at the same time. Examples include Windows Server.

- **Distributed OS:** It manages a group of computers so they work together like one single system.
- **Embedded OS:** It is used inside small devices like ATMs, smart TVs, microwaves, and traffic lights. It is designed to do one main job. An example is Windows IoT.
- **Mobile OS:** It is made for smartphones and tablets. It manages touch screens, apps, and mobile networks. Examples include Android and iOS.

1.1.2. Functions of OS

The operating system is the main program that helps the user, the applications, and the computer hardware work together. It takes commands from the user through applications, sends them to the hardware, and brings the results back. It also controls how the computer's memory, files, and devices are used.



Picture 1. 1: Functions of an Operating System

a. Managing the booting process

The operating system is responsible for completing booting process of the computer. Booting is starting or restarting the computer. **Cold** boot starts the computer that has been turned off completely whereas **warm** boot restarts the computer's operating system.

b. Providing an appropriate User interface

The *interface* enables the user to interact with the computer software and hardware. The user interface can be **Command-line interface (CLI)** or **Graphical user interface (GUI)**. With CLI, the user interacts with OS by typing in commands at the command prompt to perform specific tasks. Examples include Disk operating system (DOS) such as MS.DOS (Microsoft Disk Operating System).

c. Managing applications

The OS helps applications work well on the computer by controlling how they start, run, and use the computer's resources. It starts an application when the user opens it and stops it when the user closes it. It gives each application the memory, files, and processing power it needs. The OS also makes sure that many applications can run at the same time without problems. If an application stops working, the OS can close it safely. In this way, the OS keeps all applications running well and protects the whole system.

d. Managing Input/Output peripherals

The operating system manages reading from the input peripheral devices and writing to all peripheral devices. It controls access to all peripherals through small programs called **device drivers**. A device driver is a program that controls how a specific device works with specific operating system. Examples are: Printer driver, network driver and graphics driver.

e. Memory and CPU management

Memory management: The operating system controls how the computer's memory is used. It gives each application the memory it needs and makes sure applications do not use each other's memory. When an application is closed, the OS frees that memory so another application can use it. This explains the reason why when many applications are opened, the computer becomes slowly.

CPU management: The OS decides which application the CPU should work on at each moment. It gives each application a small amount of time to use the CPU. This allows many applications to run at the same time without slowing the computer too much.

1.1.3. OS issues and their solution

- **The computer is too slow**

A computer may become too slow when many applications are running or when the memory is almost full. The operating system tries to manage all these tasks, but sometimes it becomes overloaded. To fix this problem, the user can close unused applications, restart the computer, or delete old files to free space.

- **Applications freezing**

Sometimes an application stops working and does not respond. This happens when the application uses too much memory or when there is a small software error. For this case, close the applications by using the task manager. The shortcut keys to use are CTRL+ALT+DELETE.

- **System crashes**

A system crash happens when the computer suddenly stops working or shows an error screen. This can be caused by hardware problems, old drivers, or viruses. Restarting the computer, updating drivers, and scanning for viruses can help solve this issue and keep the system safe.

- **Bootting problems**

This is when the computer does not start correctly. This may happen because some system files are damaged or because the settings are wrong. The operating system may offer tools like Startup Repair to fix the problem. The user can also check if the hard drive is connected properly or restore the system to an earlier working state.

- **File access problems**

A user may not be able to open or save a file if the permissions are wrong or if the file is damaged. The operating system controls who can open or change files, so it may block access for safety. The user can check the file permissions, try opening the file on another device, or restore it from a backup if it is corrupted.

- **Update errors**

Sometimes the operating system cannot install updates. This may happen because there is not enough storage space or because the internet connection is weak. The user can free up space, restart the computer, and try to update again. Keeping the OS updated helps the computer stay secure and work better.

Application activity 1.1

- 1) Explain briefly the meaning of 'operating system' in the computer system.
- 2) Identify the operating system(s) that is installed in your school computers.
- 3) Discuss any five functions performed by the operating system in a computer.
- 4) Give an example of operating system that can be installed on each of the following devices:
 - a) Smart phone
 - b) Laptop computer
 - c) Microwave

1.2. Application software

Activity 1.2

The school has provided students with new computers, each installed with several application software. The teacher asks the students to explore different types of software on their computers and discuss how they are useful for everyday tasks.

- 1) Identify and classify application software into general application software, utility software, and software add-ons.
- 2) Explain the functions and uses of application software.
- 3) Demonstrate the use of one installed application software and one online application

1.2.1. Categories of application software

a. General application software

These are the software used in a computer in order to perform tasks and functions for the benefits to the user. Application software available in a specific computer depends on the needs of the computer user. For example one may have VLC to play videos while another may not have it.

Examples: Microsoft Office Word (for writing text), Adobe Photoshop (for editing images)

Application software can be used without needing having access to the internet because they are installed to the local computer. However, some others applications software such as **irembo.rw**

requires access to the internet because these applications are stored to a remote computer known as server.

b. Utility software

It is a type of software used for maintaining, managing and optimizing a computer's performance. It is a kind of toolbox that helps a computer stay healthy, secure, and efficient.

They help with file management, like copying, organizing, and compressing files. They also improve security by using antivirus tools and firewalls. These programs clean and organize the computer's storage, and they help with backup and recovery when data is lost. Utility programs also monitor the system, checking things like CPU, memory, and temperature.

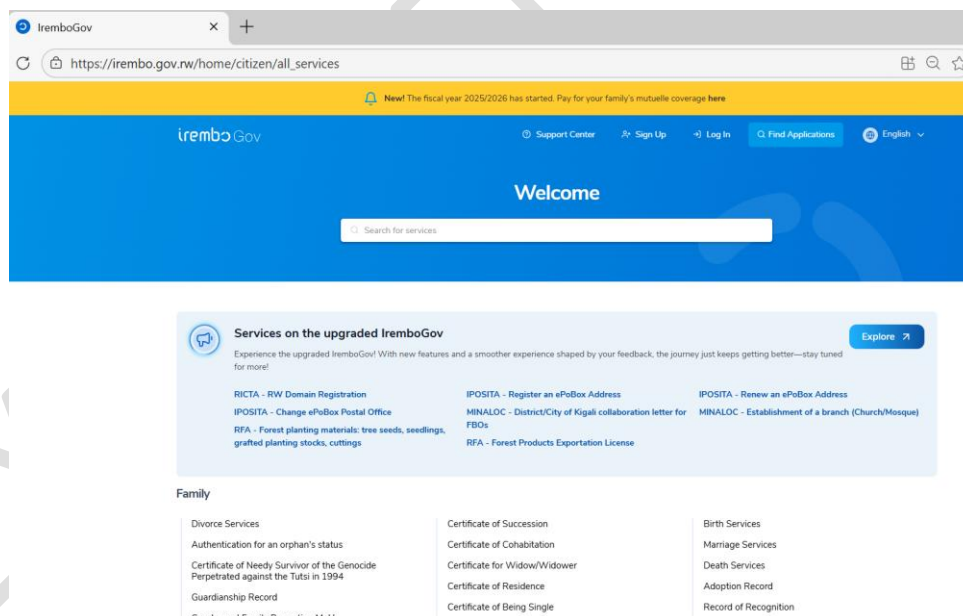
Examples: antiviruses, Winzip,

1.2.2. Irembo application

Irembo.rw is an online application software which is an initiative by the Government of Rwanda aiming at improving service delivery to the citizens. Irembo is the one-stop portal for e-Government services.

To use irembo.rw first write this URL in the browsers address bar and see the services to access then click on anyone you want and follow the guidance given by that application.

As it is common for many online applications, they send verification codes to the user's email or phone number. Make sure you fill those codes in the designated fields. Once the service you wanted is finished, you will download a paper (for example a birth certificate)



Picture 1. 2: Home page for the irembo online service platform

Note that for many services or for better service experience, it may be required that you log in to the irembo.rw platform.

1.2.3. Software add ons

This is a piece of software that adds a specific feature or capability to an existing software application. A software add-on cannot be run independently, it is just a software extension.

Examples of common add-ons are **Adobe Flash**, **Ad blocker** to stops ads, **Password manager** for saving passwords and **Download helper**.

Role of add-ons

- **Interface change:** The application can be dynamically extended to include new features which have friendly interface suitable for different users.
- **Adding features:** More features are added to the browser to increase its usability.

Application activity 1.2

- 1) Define application software and explain its purpose in a computer system.
- 2) Differentiate between general application software and utility software, giving one example for each.
- 3) Classify the following software into their correct categories:
Microsoft Word, Antivirus, Ad blocker, WinZip.
- 4) Explain why some application software, such as Microsoft Word, can be used without internet access, while others like irembo.rw require internet.
- 5) Compress a folder using WinZip or any file compression utility and explain why compression is useful.
- 6) Access irembo.rw using a web browser and identify one service you can apply for on the platform
- 7) Discuss the importance of keeping utility software such as antivirus updated on school computers.
- 8) Describe responsible behavior you should show when using software add-ons in a web browser.

1.3. OS interaction with hardware

Activity 1.3

The school ICT laboratory has a set of new computers. Senior three are using different applications, such as a music player, a word processor, and a web browser. During the lesson, students observe how their computer saves files, plays sound, and displays images.

Discuss how applications can perform tasks such as saving files, playing sound, or displaying images on their own?

Applications cannot talk to the hardware directly. They need the operating system (OS) to help them. The OS uses device drivers to change instructions into commands containing 0 and 1 that

the hardware can understand. After the hardware does the work requested and sends the results back to the OS and then the OS sends results to the applications which then show the the result in the language that can be understood by humans.

The main components of this OS interaction with the hardware include user applications, the Operating System, the driver commands, the hardware abstraction and hardware.

a. User applications

User applications such as a browser, a music player request the operating system for help when they need to do something, like save a file, show a picture, or play sound. These requests are written in a language such as C programming language which is understandable by both the application software and the OS.

Application software send their requests through system calls (or API) which tells the OS exactly what the application wants.

b. Operating System

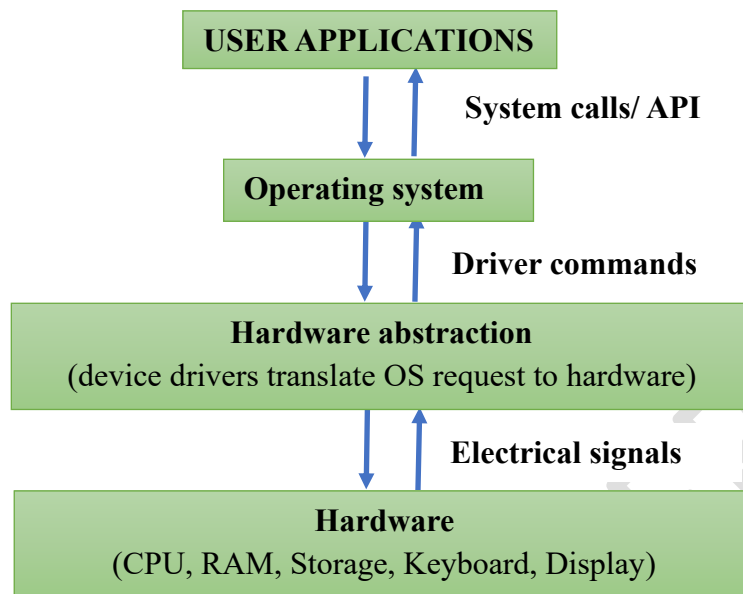
Through system calls or APIs, the OS receives the application requests such as “Save this file” and uses the device driver to send the request to the hardware. To communicate with the hardware, the OS uses a kind of language consisting of 0s and 1s called **machine language**.

c. Hardware abstraction

This is when device drivers translate OS requests to hardware. The Hardware abstraction layer helps the operating system work with different hardware devices without needing to know how each device is built inside.

d. Hardware

The hardware is the bottom layer in interaction between the OS and the hardware. It receives electrical signals from the upper layer (hardware abstraction). There are two main electrical signals for a computer hardware one representing 0 and another representing 1. When these signals are received, they are transmitted back to the hardware abstraction, then to the OS and the application software.



Picture 1. 3: OS interaction with hardware

Application activity 1.3

- 1) Which role does the operating system play between applications and hardware?
- 2) What are device drivers used for?
- 3) Give two examples of user applications.
- 4) Why can't applications talk directly to hardware?
- 5) What is hardware abstraction and why is it important?
- 6) If a music player wants to play sound, explain the steps it takes through the OS to reach the hardware.

1.4. Installing application software

Activity 1.4

- 1) Explain why it is necessary to install application software in a computer.
- 2) Explain what happens if you install too many applications in a computer
- 3) Identify the role of the Operating System in helping applications perform tasks such as saving files and playing sound.
- 4) Demonstrate saving a file in a Word processor and explain the OS's role in the process.
- 5) Discuss why you should follow proper procedures when using applications to avoid damaging hardware.

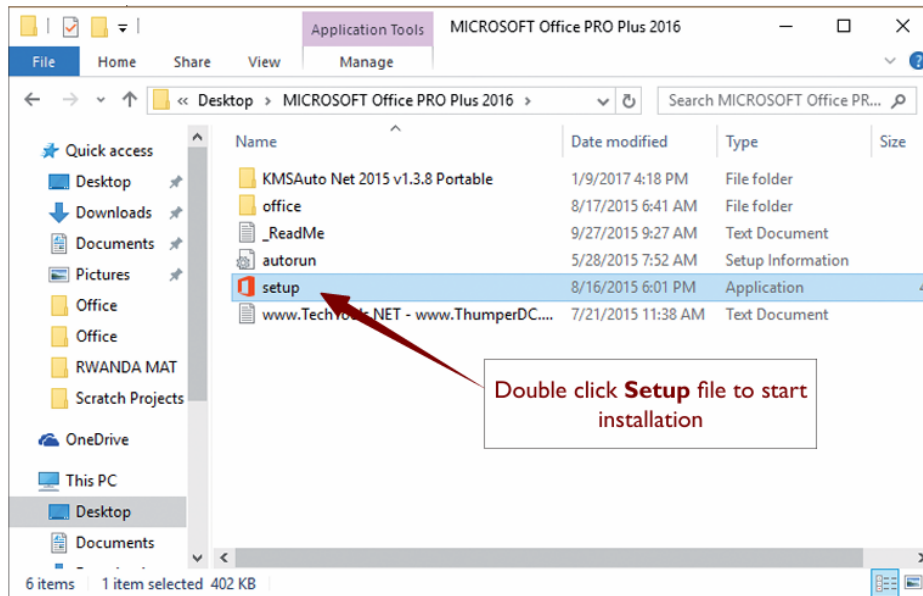
You are provided with Office 2016 Installation DVD. Install this application on your computer. Follow the steps below.

Step 1: Obtain MS Office installation on a DVD, flash disk, external hard disk (or

download Installation file from the Internet.)

Step 2: Insert the disc in the optical drive (e.g. DVD Drive)

The installation disc automatically launches *setup* wizard. (If the setup wizard doesn't start automatically, open the disk drive or folder for Office 2016 and double-click **Setup** file. See the picture below.



Picture 1. 4: Installing application software

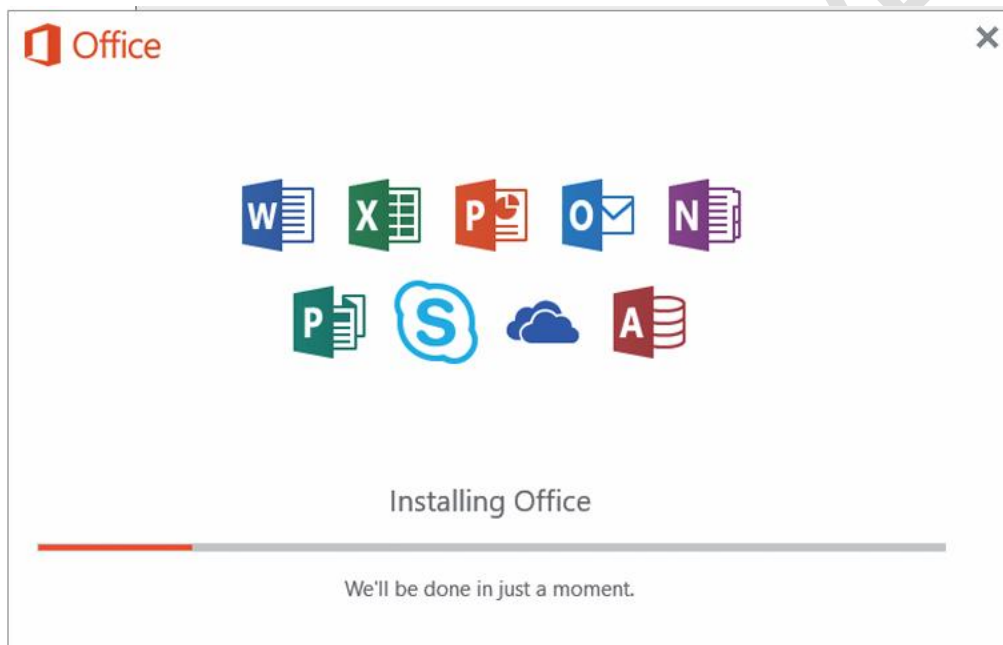
Remember that:

To install application software successfully you must use an Administrator account, without which installation cannot take place. Other types of accounts such as standard or limited accounts and Guest accounts do not have the privilege to install or uninstall software.

If you normally use limited accounts for doing other practical activities, this time you should use an administrator account to install Microsoft Office 2016 and other applications.

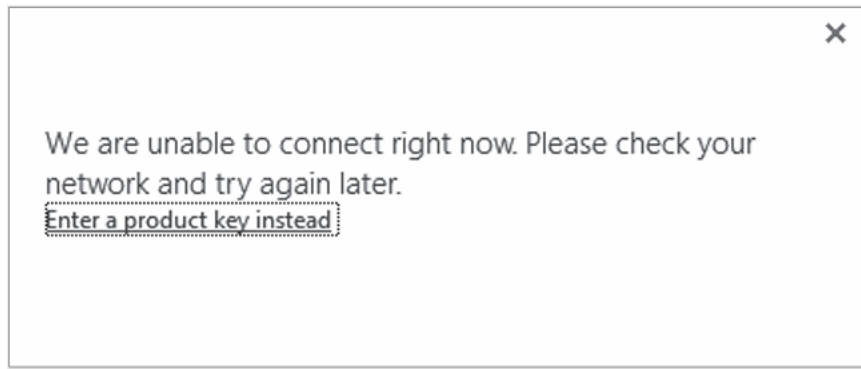


After a short time, the Office installation screen appears to show the status of installation process.



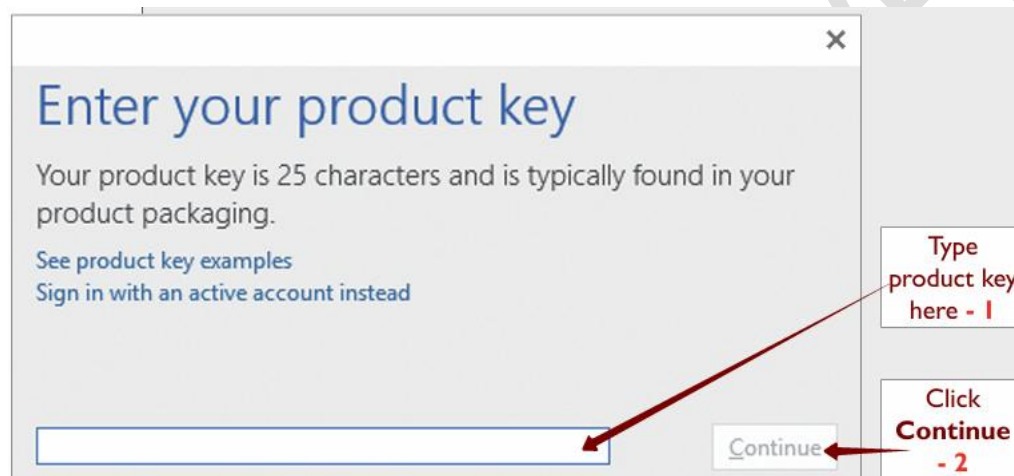
Step 3: Wait as Office installs to complete the installation process. Click **Close** after the installation is complete.

Step 4: Open an office application such as Word 2016. The message requesting for product key may display on top of Office 2016. In case you purchased a licensed product, close it and will not show again, start using office normally. If you purchased a non-licensed product, and you have the product key, click on "Enter a product key instead"



Picture 1. 5. Office connection Error screen

The *Product key* screen displays as shown below. Type in your product key which must be 25 characters and then click on **continue**.



Picture 1. 6. The product key screen

MS Office 2016 may not require a product key once you purchase a licensed copy.

Note: Product key is a specific software-based key for a computer program and certifies that the copy of the program is original. If you do not have product key, you may fail to use the software installed normally or if the product key is required before installation you may fail to install the software. The location of your product key depends on how you got a copy of the software.

Step 5: Afterwards Office installs; click **close** after installation is complete.

Step 6: If the activation wizard displays, click "*I want to activate the software over the Internet*" and then follow prompts.

Application activity 1.4

- 1) Which storage devices can be used to obtain the Office installation file?
- 2) Which type of account is required to install application software?
- 3) What is the purpose of a product key?

- 4) Why can't a guest or limited account install application software?
- 5) Why do you think Microsoft uses activation over the internet after installation?

1.5. Utility software

Activity 1.5

- 1) Why are utility software called so?
- 2) Give 3 examples of utility software
- 3) Why do you think computers need protection from viruses or malware?
- 4) Why computers need special programs to manage files and storage?
- 5) What do you do when you see a message "This file may be harmful"?
- 6) What are the use of the Disk cleanup tool?

1.5.1. Role of utility software

Utility software plays a big role in ensuring that a computer system functions well. It works behind the scenes to support the operating system by performing tasks that maintain and optimize system performance. Below are its main roles:

- **System maintenance:** It keeps the computer running smoothly by cleaning unnecessary files, fixing disk errors, defragmenting the storage
- **Security and protection:** It protects the system from threats such as viruses and malware
- **File and data management:** It helps in organizing files and folders, compressing and decompressing files, backing up data and recovering lost files.
- **Hardware and device management:** Some utility software help in managing and troubleshooting hardware components by updating drivers, diagnosing device problems, and managing printers and storage devices.

1.5.2. Types of utility software

- **Security software:** They protect the computers from threats. They include antivirus software, anti-malware tools, firewall tools and encryption tools.
- **File management software:** They help organize, manage, and manipulate files. Examples are file explorers, file search tools, file compression tools (ZIP, RAR) and file recovery tools.
- **Storage and disk management software:** They are used in managing disk space and storage devices.

Example: Disk defragmenter, disk partitioning tools.

- **Device management software:** They are used in managing hardware devices. They include driver managers and device diagnostics tools.

1.5.3. Installation of an antivirus

Before installing an antivirus first have in mind the trusted antivirus to use which will protect

more the computer. Secondly, consider the properties of your computer such as the processor speed and storage size because when the computer speed and memory size are low, it may become slow depending on the antivirus installed.

Now, install the antivirus by plugging to the computer the media on which it is kept. The installation start by opening the antivirus and opening the .exe file or clicking on the appropriate menu. Follow the instructions given as you install. There are different antiviruses such as Norton, Kaspersky, AVG, McAfee, Avast, etc.

Note that it is better to restart the computer after the installation.

1.5.4. Role of an antivirus

Detecting threats: Antivirus software helps a computer stay safe by checking all files, programs, and websites for dangers. It looks for many types of harmful software, such as viruses, worms, trojans, ransomware, spyware, and other malware. By finding these threats early, the antivirus protects the computer from damage and keeps it safe.

Blocking harmful activities: An antivirus detects dangerous activities and prevents them. Such activities are unauthorized access, dangerous downloads and malicious scripts.

Removing malware: If a threat is found, the antivirus deletes it or isolates it in a safe area. It stops the malware from spreading or damaging the system.

Protecting personal data: An antivirus protects users' personal information when working online. That information can be passwords, banking information, personal files.

Improving system stability: Antivirus software helps improve system stability by preventing infections. When the computer stays free from viruses and malware, it works more smoothly, with fewer crashes, faster performance, and less risk of losing or damaging important files.

Application activity 1.5

- 1) Define utility software and explain its role in a computer.
- 2) What should be considered before installing an antivirus on a computer?
- 3) Name three examples of antivirus software.
- 4) Which utility software helps in managing disk space and storage devices?
 - a. Antivirus software
 - b. Disk defragmenter
 - c. File explorer
 - d. Firewall
- 5) Antivirus software improves system stability by:
 - a. Increasing processor speed
 - b. Preventing infections and crashes
 - c. Compressing files
 - d. Updating drivers
- 6) Explain how utility software contributes to system maintenance with examples.
- 7) Describe the steps involved in installing an antivirus program.

1.6. Uninstalling a program

Activity 1.6

- 1) Why is it necessary to remove a program from a computer?

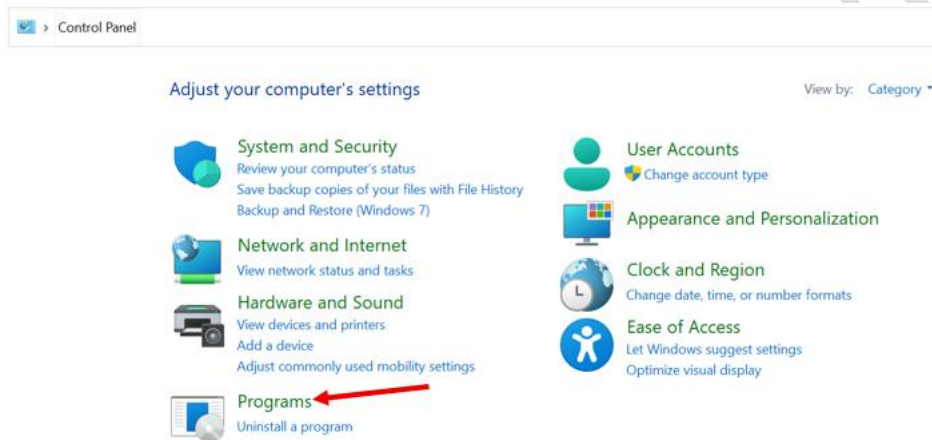
- 2) Look at your computer and identify some programs that are not used and can be uninstalled
- 3) What happens in a computer when a program is uninstalled?
- 4) What is the difference between deleting a program and uninstalling it?

Sometimes, it may become necessary to remove an application software because it is no longer needed. On the other hand, if applications are installed without being removed, the computer memory will get full.

Take the following steps to uninstall an application software from a computer:

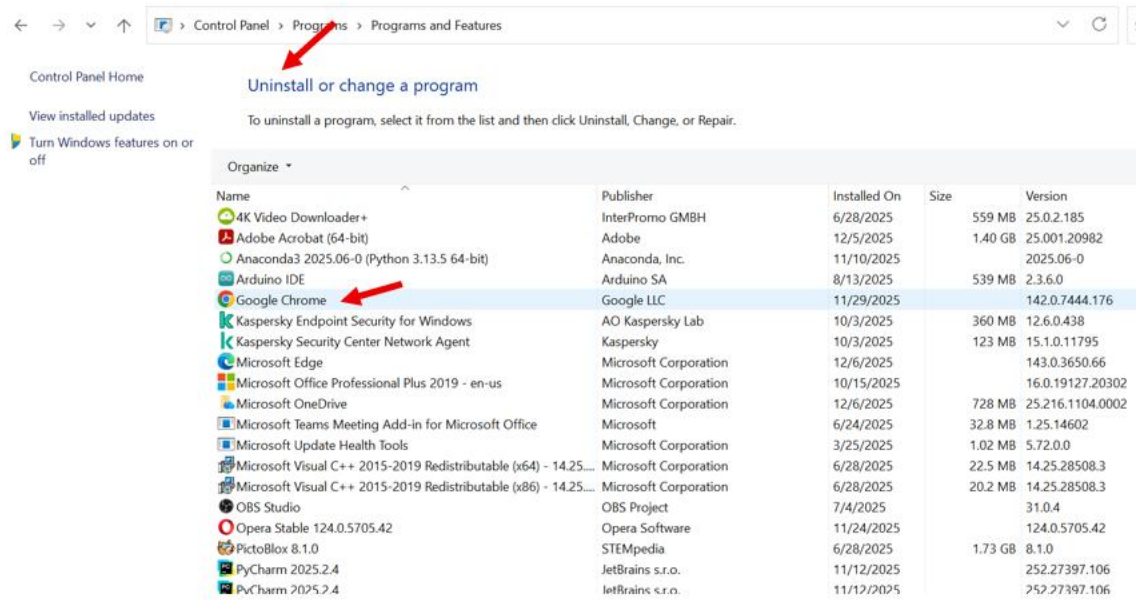
Step 1: Click the Start button and select Control Panel. (Or type “Control panel” in the Search box at the start and press enter).

Step 2: In the control panel window that displays select Uninstall a program under Programs option



Picture 1. 7. Control Panel window

Step 3: In the **Programs and Features** screen, select an application (**Google Chrome**) Then click on **Uninstall** tab above programs list. You can select another program you are sure you do not need and uninstall it.



Picture 1. 8. Window for uninstalling a program

Step 4: Wait for the **Windows Installer** to gather all the features of (Google Chrome) program and remove them. Lastly close **Control Panel**.

Application activity 1.6

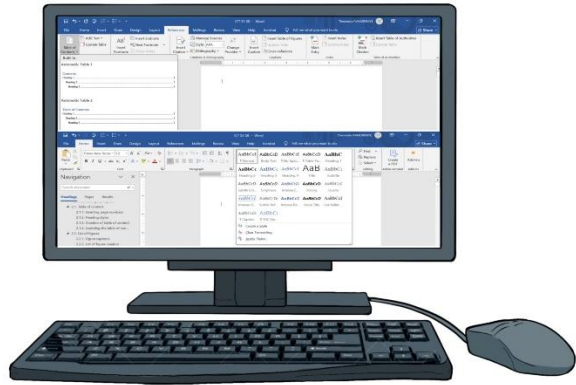
- 1) What is the first step when uninstalling a program in Windows?
- 2) What happens after you click the Uninstall tab for a selected program?
- 3) Why is it recommended to restart the computer after uninstalling software?
- 4) If applications are installed without being removed, what is most likely to happen?
 - a) The computer will run faster
 - b) The computer memory will get full
 - c) The antivirus will stop working
 - d) The operating system will uninstall itself
- 5) In the Control Panel, which option do you select to uninstall a program?
 - a) Devices and Printers
 - b) Programs and Features
 - c) Network and Sharing Center
 - d) Appearance and Personalization
- 6) Explain step by step how you would uninstall Google Chrome from a Windows computer.
- 7) A computer is running slowly because it has too many unused applications. How would uninstalling help solve this problem?

End of unit assessment 1

- 1) Distinguish between system software and application software, giving one example of each.
- 2) Explain why an operating system is considered the bridge between hardware and application software.
- 3) Describe the role of utility software in supporting the operating system.

- 4) Evaluate the importance of device drivers in ensuring compatibility between hardware and software.
- 5) Explain the consequences of installing too many applications on a computer system.
- 6) Propose a strategy for maintaining a school computer laboratory that ensures operating systems, application software, and utility software remain effective and secure.
- 7) Explain how hardware abstraction simplifies the interaction between applications and hardware.
- 8) Without an operating system, a computer is just a collection of hardware. Discuss this statement with reference to OS functions.
- 9) Explain why uninstalling unused programs reflects responsible digital behavior.
- 10) Discuss why you should be responsible when using software add-ons in a web browser

UNIT 2: WORD AUTOMATIC TABLES



Key Unit Competency: Generate structured Word documents using tables of contents, lists of figures and tables, and mail merge tools

Learning Objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Explain the role of a table of content, list of tables, list of figures in a document.
- Explain how to insert automatic tables in a document
- Discuss the need to write and print a letter to multiple recipients.

Skills

- Generate a table of contents, list of figures and list of tables for a document.
- Apply and use mail merge to produce a personalized document for multiple recipients.
- Use a table of contents to indicate document's content

Attitudes and values

- Demonstrates care in applying heading styles and in adding captions
- Appreciate the importance of having captions on tables and figures
- Appreciate the ease with which mail merge can create personalised documents to various recipients.

2.0. Introduction

Microsoft Word is a software used in writing text. However, for better organization of documents, its content should include **Table of Contents, List of Figures, and List of Tables**, which make navigation easier, improve clarity, and enhance professionalism in big documents.

Also, it sometimes becomes necessary to write many a letter to many people while its content is the same like for example in the case of a school inviting parents for a school meeting. For this purpose Mail Merge option gives is used to write such letter

This unit teaches about those important tools and shows how they are used in order to well arranged documents and facilitate the task in the case of one letter being mailed to many people

Introductory Activity

Imagine you are preparing a research document having 100 pages containing many titles tables and figure. You are often scrolling through that document to look at a specific title and its content or to see if the table and image captions are well written or are in correct order. You are finding this difficult to scroll through your document every time you want to look at a specific content

- 1) Explain why having a Table of Contents, List of Figures, and List of Tables is necessary in this situation.
- 2) What challenges would you face if these elements were missing,
- 3) Explain how they improve professionalism in long documents?

2.1. Table of content

Activity 2.1

- 1) Why are table of contents in books?
- 2) What are the different locations in a document where page numbers can be inserted?
- 3) You have already a book with a table of content then you add new content in the middle of that document. Is your table of content still valid? Explain.
- 4) Explain why it is necessary to insert page numbers before creating a table of content.

A Table of Contents is a piece of information used by readers to quickly find or locate what they are looking for in a book, magazine or any piece of literature. The creation of a table of content can be said to be done in 4 steps namely inserting page numbers, formatting titles into appropriate heading styles, inserting the table of content in the appropriate place and updating the table of content if necessary.

2.1.1. Inserting page numbers

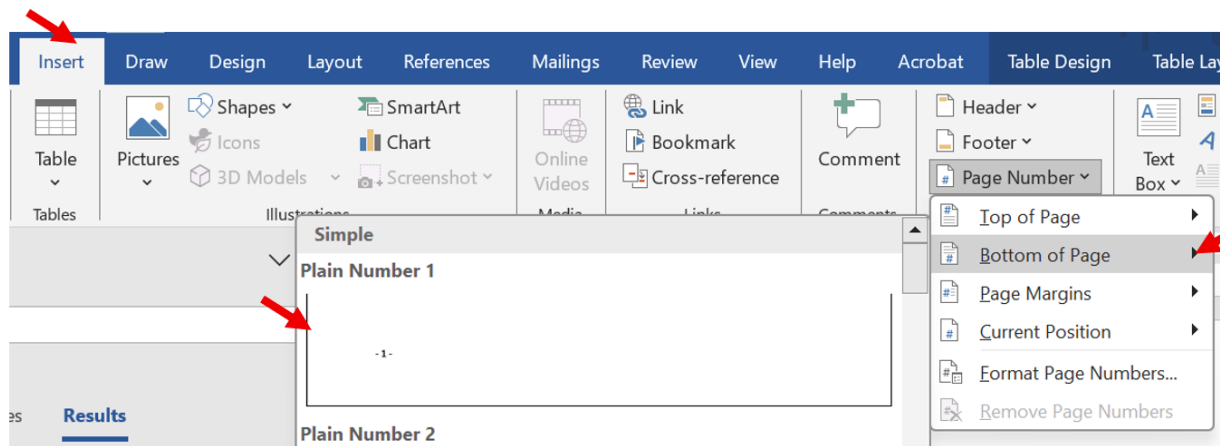
Inserting page numbers is very important because they will appear in the table of content to inform about where to find information on a given title.

Here is how to insert them in a Word document:

Step 1: Go to the **Insert** tab at the top of the screen.

Step 2: Click on **Page Number** then click on **Format Page Number** to change the kind of page numbers to insert (numbers or letters). Under this tab you also choose the position where to insert the page number such as **Top of Page**, **Bottom of Page**, **Page Margins** and **Current Position**.

Step 3: After choosing the position, choose the type of page number. In the image below the page number chosen is **Bottom of Page** while the type of page number is **Plain Number 1**.



Picture 2. 1. Steps for inserting page numbers

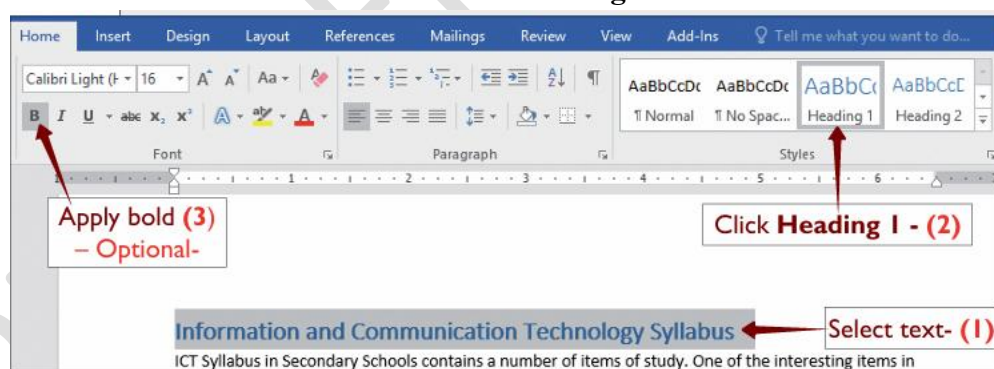
2.1.2. Heading styles

A heading style is text format used in word-processing software to organize a document into clear sections. It controls the appearance of headings such as font size, color, spacing, and numbering so that a document looks neat, consistent, and easy to navigate.

a. Applying heading styles

Follow this procedure to make text heading styles:

- First have a Word document having at least 5 pages and titles at different levels.
- Select text you want to be included in Table of Contents as the first heading
- Click on the **Home** menu then click on **Heading 1**.



Picture 2. 2: Options for applying heading styles

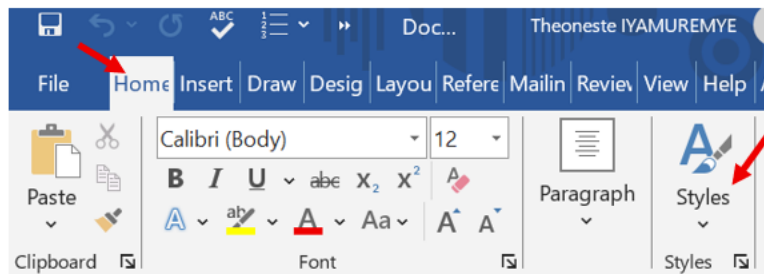
Continue this process for other heading styles until all titles have been formatted. It is advised to apply the heading styles up to Heading 3 because beyond this, the table of content has many titles.

b. Modifying a heading styles

This is changing the appearance of an existing heading style so that all text using that style automatically update to the new format. This allows you to control the font, size, color, spacing, alignment, and other formatting features of headings throughout the entire document without needing to go to every single title (heading).

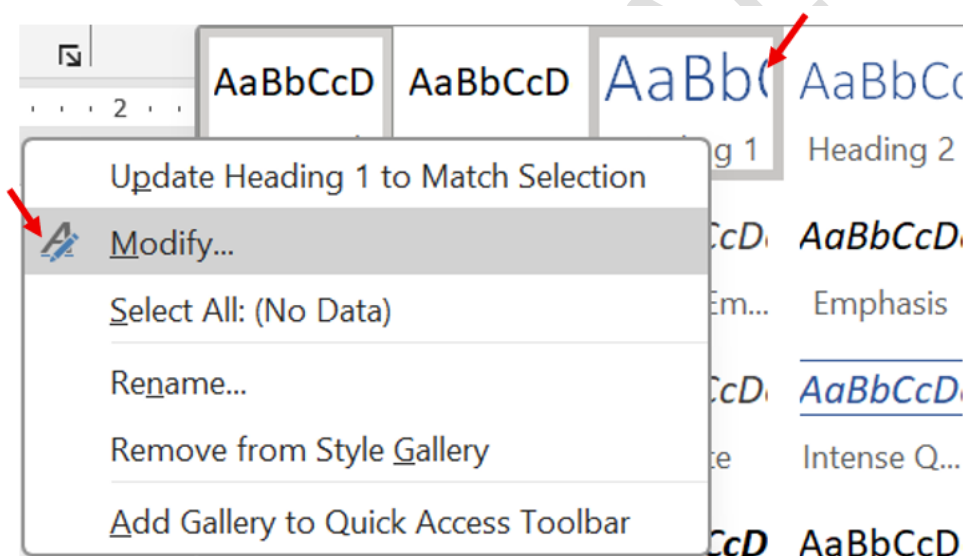
Follow this process to modify a heading style:

- Click on the Home menu then on one heading style



Picture 2. 3. The home menu and heading styles

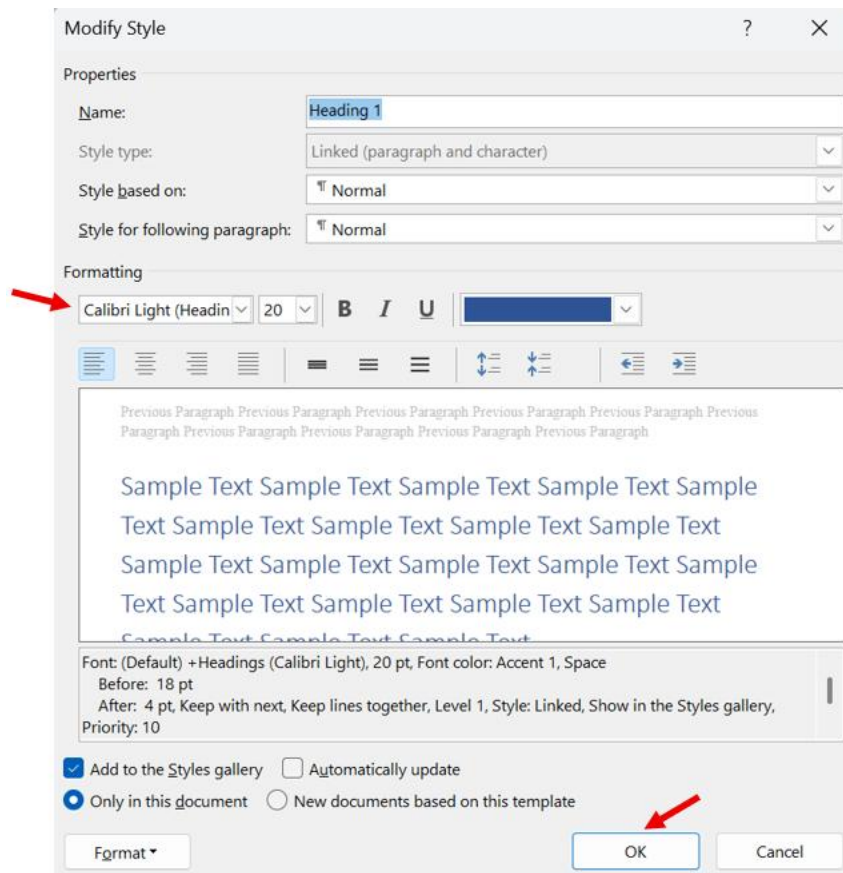
- Right click one of the headings such as Heading 1 and click on **Modify**



Picture 2. 4. Different heading styles

In the new dialog box, choose the kind of formatting you want such as the font size, color, bold, italic or underlined.

- Click on **OK** to apply the new changes.



Picture 2. 5: Format options of a heading style

2.1.3. Creation of table of content

Follow the following steps carefully to create a Table of Contents:

Step 1: Start MS Word and then create a new document in which you need to add Table of Contents. You will save the document as **My-TOC**.

Now use Microsoft Word and type text that will be used to create the table of content.

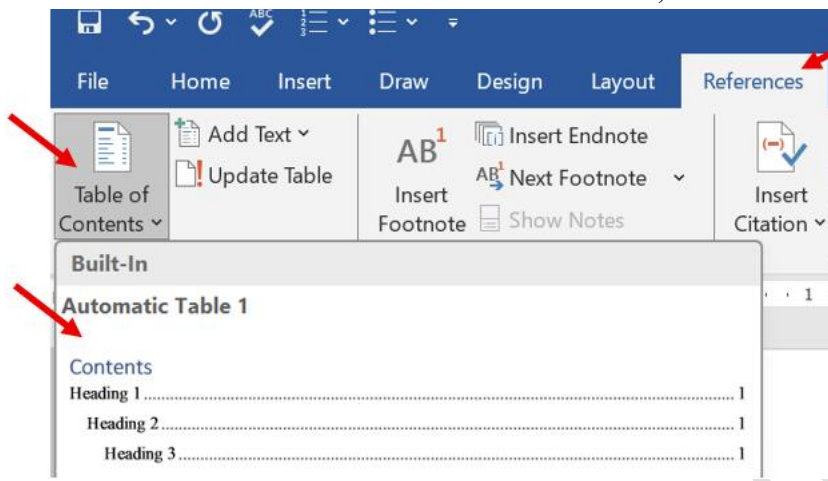
Make sure you insert page numbers in the document by clicking on **Insert tab** then on **Page Number** then choose the position of the page numbers (top, bottom, ...) and choose the page number format such as **Plain number 1**.

Step 2: Apply Heading Styles to the text that you want to include in the Table of Contents by using the procedure learned under the title “Applying heading styles”. You select the title, click on **Home**, click on **Styles** then click on **Heading 1** (or **Heading 2**, or **Heading 3**).

Step 3: Generate an automatic **Table of Contents** for the document at the beginning of your document.

To create Table of contents, do the following below:

- Click where you want to insert the Table of contents (usually at the beginning of the document).
- Click **References** tab then on **Table of Contents**, and then choose **Automatic Table 1**.



Picture 2. 6. Menu for ToC generation

Note that there are different Table of Content styles apart from Automatic Table 1. The kind of style you choose will determine the look of the inserted table of content.

Here is an example of a table of content inserted after going through the steps above:

UNIT 2: WORD AUTOMATIC TABLES.....	17
2.1. Table of content.....	17
2.1.1. Heading styles.....	17
2.1.2. Creation of table of content	19

Picture 2. 7. A table of content

2.1.4. Updating the table of content

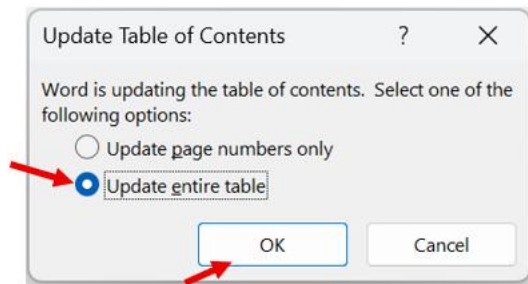
If some titles or information are removed from a document, the table of content is no longer valid because it contains headings which are not in the document. On the other side, new titles and text may be added to the document but are not yet in the table of content. Therefore, there is a need to update the table of content so that it has all the titles that are in the document.

You can update quickly using the following procedure:

Step 1: On the **References** tab, in the *Table of Contents* group, click **Update Table**. You can also click “Update Table” button in the selected Table of Contents

Step 2: In the Update Table of Contents dialog box that displays, select “**Update entire table**” and click **OK**. The new table of contents displays.

You can also click in the table of content, then do a right click, click on **Update Field** then choose **Update entire table**.



Picture 2. 8. Dialog box for updating Table of Content

Application activity 2.1

Create a table of content for a document having at least 20 pages. To create that document, choose any topic in any subject and generate that document using generative AI such as ChatGPT. Insert a table of content by:

- | | |
|----------------------------------|--|
| a) Adding titles and number them | b) Applying heading styles to the titles |
| c) Inserting page numbers | d) Inserting the table of content |

2.2. List of figures

Activity 2.2

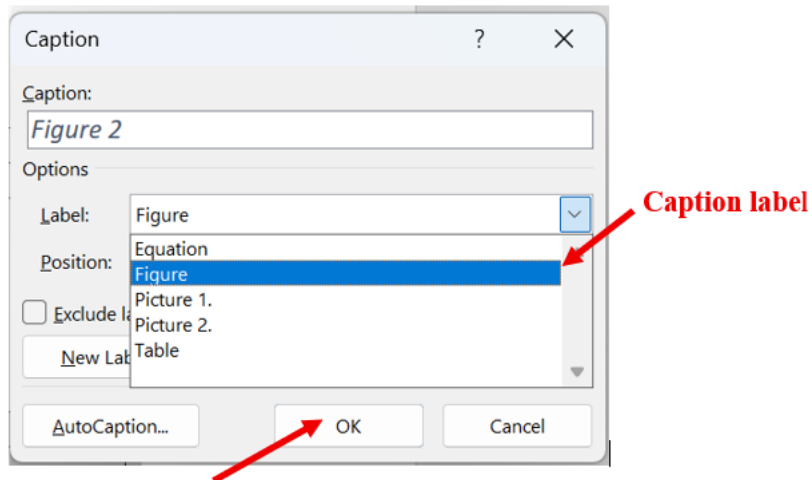
- 1) Why do people add pictures to a document?
- 2) If there are many pictures in a Word document how can one access them very quickly?
- 3) Why is it helpful to have a list of all pictures in a document?
- 4) What is the difference between a list of figures and a table of content
- 5) What happens to the list of figures if you add a new picture to a document

2.2.1. Figure captions

A figure caption is a short description or title that appears below a figure in a document. It tells the reader what the figure shows and helps Word automatically create a list of figures.

To add captions to an image (figure) do the following:

- Select the image (or figure) for which to add a caption then do a right click and choose **Insert caption**.
- In the caption dialog box change the caption label to **Figure** and click **OK**.

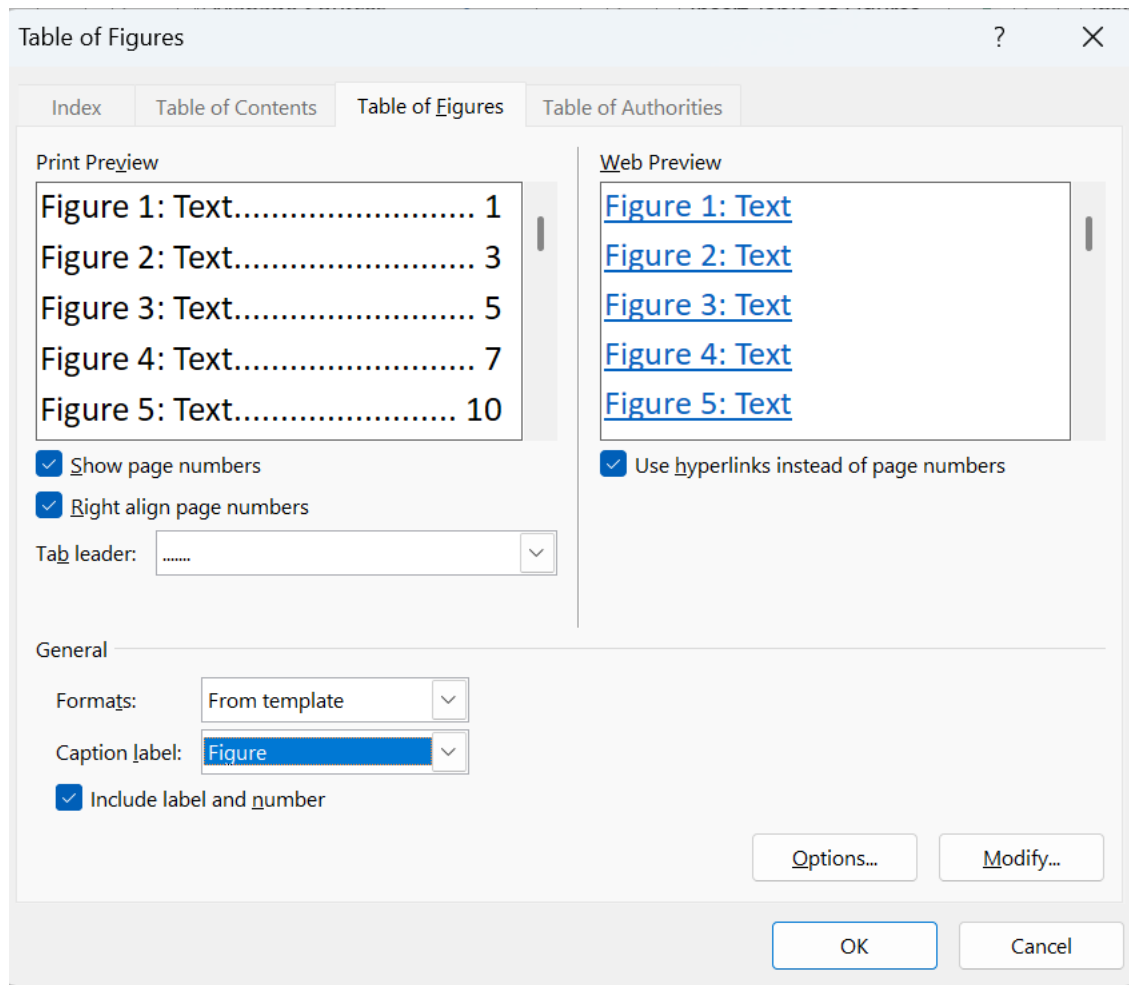


Picture 2. 9: Dialog box for changing the caption label

Afterwards write the caption in the dialog box or edit the caption which is inserted below the image after you have clicked on **OK**.

2.2.2. List of figure creation

- Click on **References** then on **Insert Table of Figures**
- In the dialog box that appears, change the caption label to make it become “Figure”



Picture 2. 10: Dialog box for inserting a table of figures

Note that like other lists, the created list of figure can be updated by selecting it then doing a right click and choosing **Update Field** then choose **Entire table**.

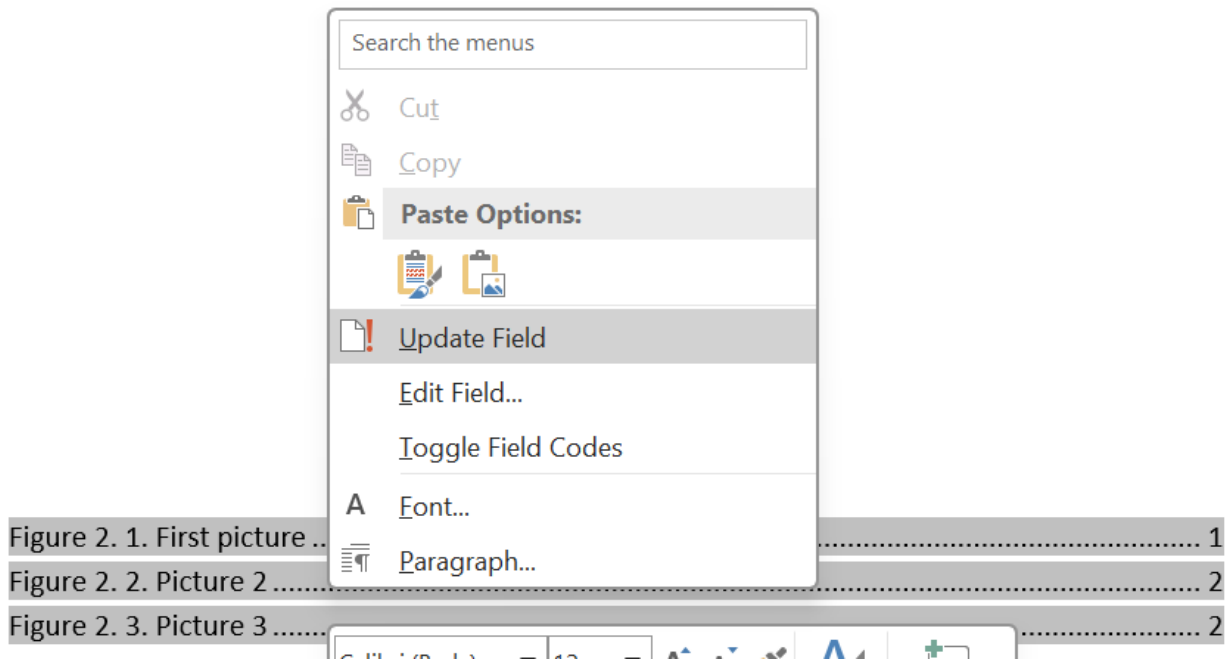
2.2.3. List of figure updating

A List of Figures is like a map of all the figures in a document. This map must stay accurate by updating it if there is a change.

Follow these steps to update the list of figures:

Step 1. Click anywhere inside the **List of Figures** and do a right click

Step 2. In the dialog box that appears, choose Update Window Click **Update Field**.



Step 3. Click on **Update Entire Table** and click **OK**.

Application activity 2.2

- 1) Which problems may happen to a list of figures if images in a document do not have captions?
- 2) Describe the steps you would follow to insert a figure caption in Word document.
- 3) What is the difference between typing text under an image manually and using the Insert Caption feature?
- 4) What steps would you follow to insert a list of figures in a document?
- 5) What happens to the List of Figures if a new figure is added but the list is not updated?

2.3. List of tables

Activity 2.3

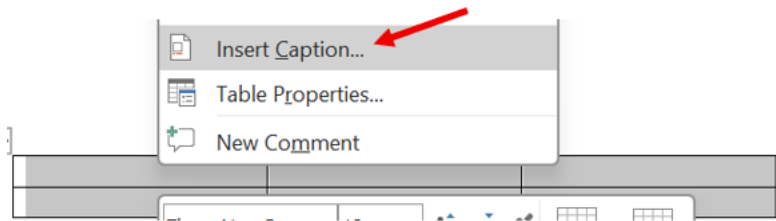
- 1) Why is it necessary to have a list of tables in a document?
- 2) a) what is a table caption?
b) How does a table caption help in creating a list of tables?
- 3) What are the similarities and differences between a table of contents and a list of tables?
- 4) What is the importance of a list of tables in a document?

List of tables is used to keep information organised and provide easy access to a specific element needed in the document. The creation of a list of table involves 3 activities namely adding captions, creating the list and updating the list.

2.3.1. Table captions

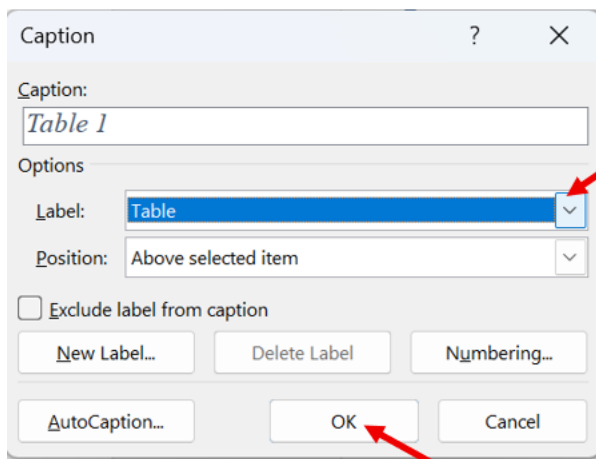
For creating a list of tables, first add captions to the different tables in the document by:

- Select the table then do a right click and choose **Insert Caption**



Picture 2. 11. Insert caption menu

- In the caption dialog box, choose **Table** from the label type and click on **OK**.



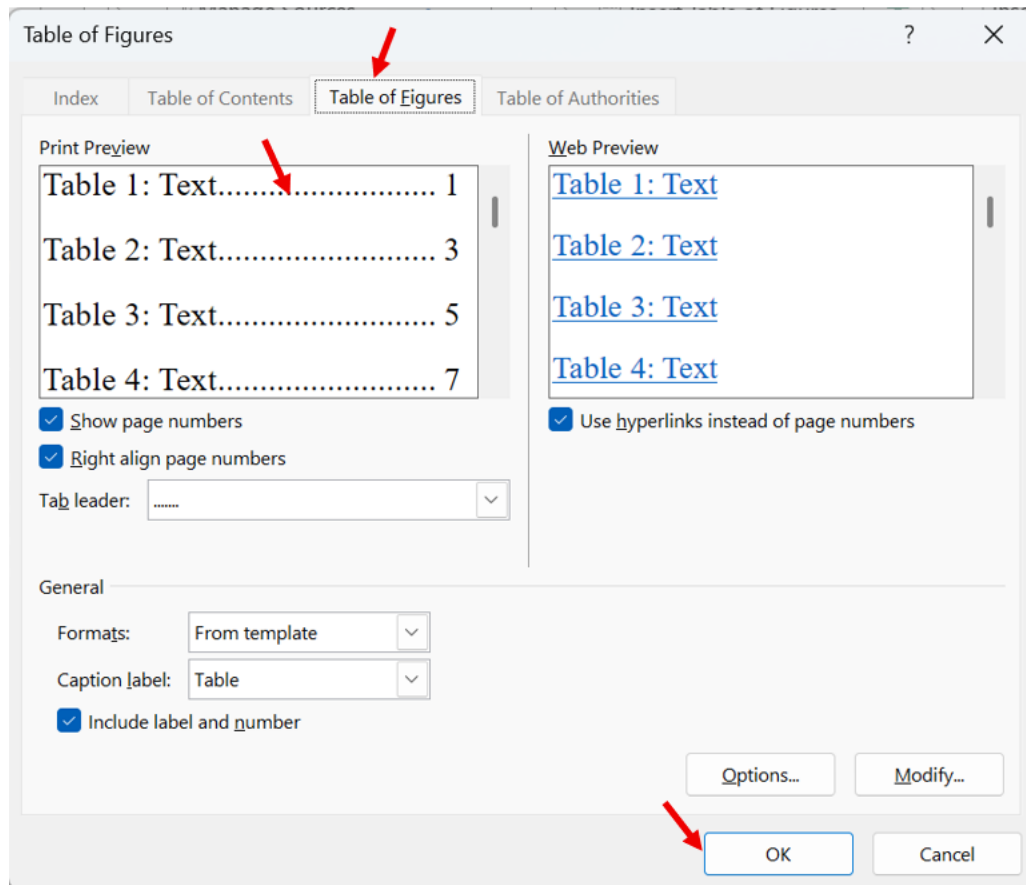
- Under the table that you had selected it is now written “Table 1”. Now add some text for the caption (example: The population of Rwanda from 2000)
- Add table captions to every table by following the same procedure as above and do this for all the tables.

2.3.2. List of table creation

Now that all tables have captions, insert the list of table by following these steps:

Click on **References** then on **Insert Table of Figures**

In the new dialog box, click on **Table of Figure** then click on one template under **Print Preview** then click **OK**.



Picture 2. 12. Dialog box for creating a list of tables

Automatically a list of table like the one below will be inserted.

Table 1. Population of Rwanda from 2000	1
Table 2. Surface area of the 416 sectors in Rwanda	1
Table 3. Projection of economic growth of Rwanda up to 2050.....	2
Table 4. Cash crops and areas they occupy	3

Page where the table is

Picture 2. 13. Created list of table

Note that the names of the tables as displayed in the image above are the captions which are under the tables in the document for which that list of table was created.

2.3.3. List of table updation

As you work on the document, many tables may be added while it may be necessary to delete some tables and this requires updating the list of tables.

To update the list of tables: Select the list, then do a right click and choose **Update Field**. Choose **Update entire table** and click on **OK**.

Application activity 2.3

- 1) What is the purpose of a list of tables in a document?
- 2) What is a table caption, and why is it important?
- 3) Explain how to insert a caption for a table in Word?
- 4) What steps do you follow to update the list of tables?
- 5) Which three activities are involved in creating a list of tables?
 - a) Adding captions, creating the list, updating the list
 - b) Formatting tables, inserting images, printing the document
 - c) Writing captions, numbering pages, saving the file
 - d) Adding charts, inserting references, updating figures
- 6) When inserting a caption, which label type should be chosen in the dialog box?
 - a) Figure
 - b) Chart
 - c) Table
 - d) Diagram
- 7) Which menu option is used to insert the list of tables after captions are added?
 - a) Layout → Insert Caption
 - b) References → Insert Table of Figures
 - b) Insert → Table
 - d) View → Navigation Pane

2.4. Mail merge

Activity 2.4

Create a letter in Microsoft Word, that will invite parents to their meeting at GS Bumbogo School. Save the letter as “Parents Meeting”.

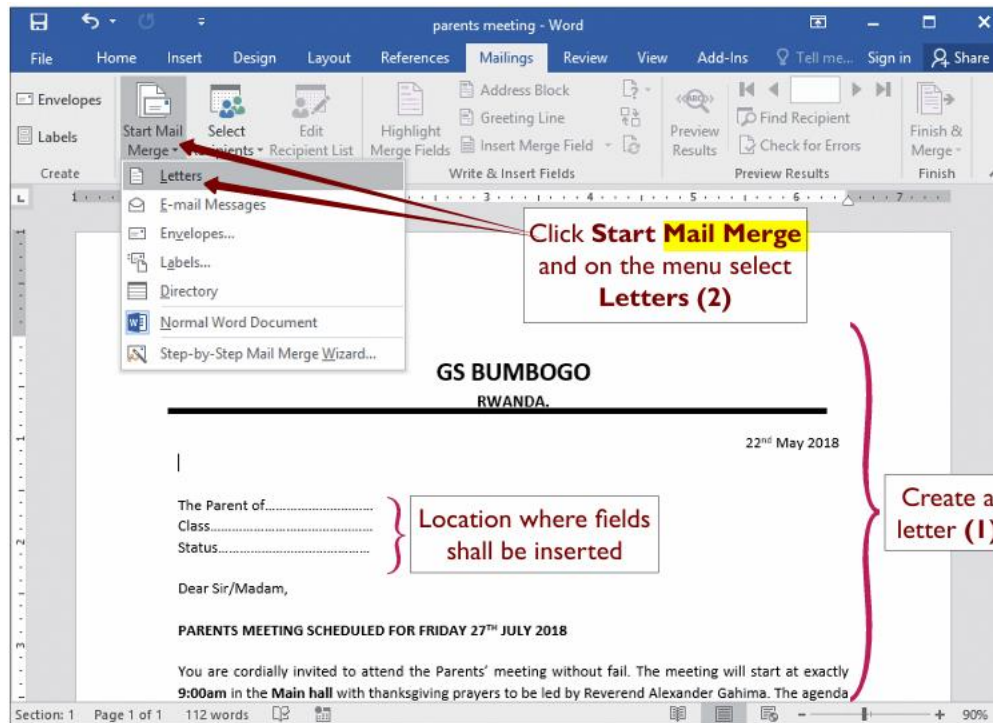
Now you have one letter and you want that letter to each parents (the school has 600 students). You are now going to print 600 letters and fill the parents’ names on each letter and this is not easy.

- a) Explain how Mail merge option can be used to make the task easy.
- b) Explain how Mail merge works
- c) Open a Word document and locate the tool for Mail Merge.

A mail merge is the process that produces personalized letters by combining a database of names and addresses with a formal letter created in a word processor.

2.4.1. Using a new list of recipients

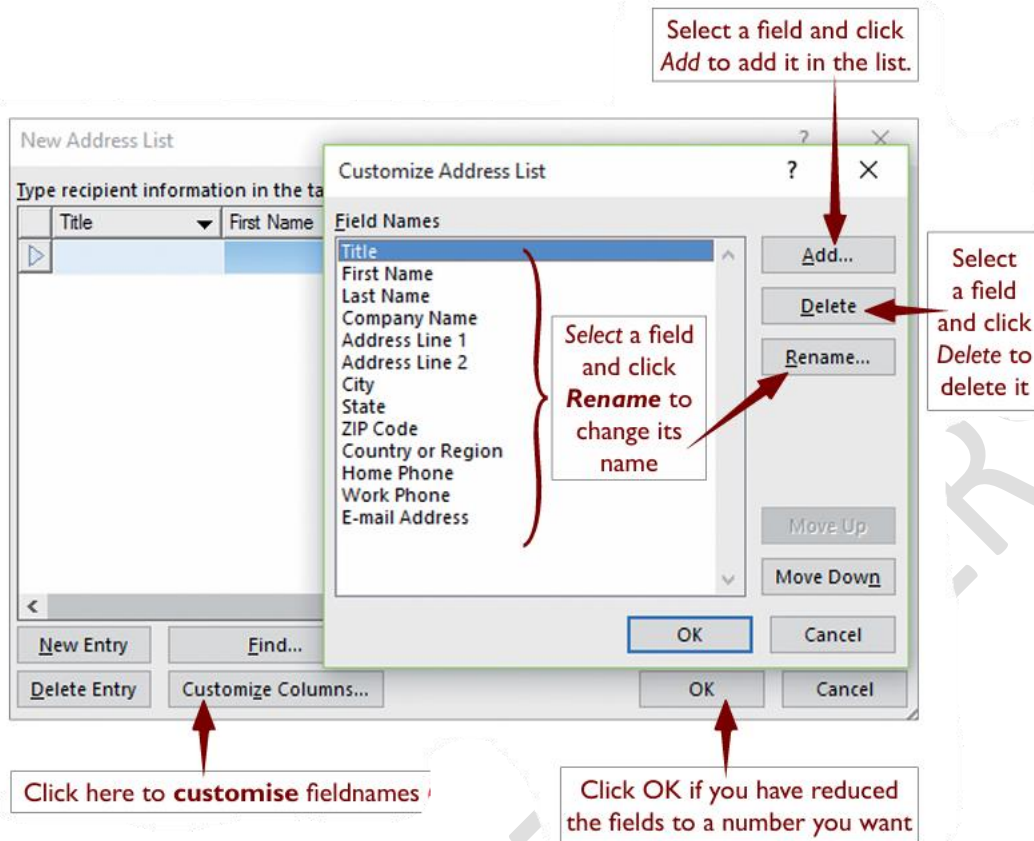
Step 1: Create a letter which you will send to various persons save it as ‘parents meeting’.



Picture 2. 14. Word document showing an invitation letter to parents

Step 2: On the Mailings Tab, click the **Start Mail Merge** button and then select Letters in the menu.

Step 3: Click on the Select Recipients button which is next and in the list that appears select **Type a New List**. The New Address List box appears

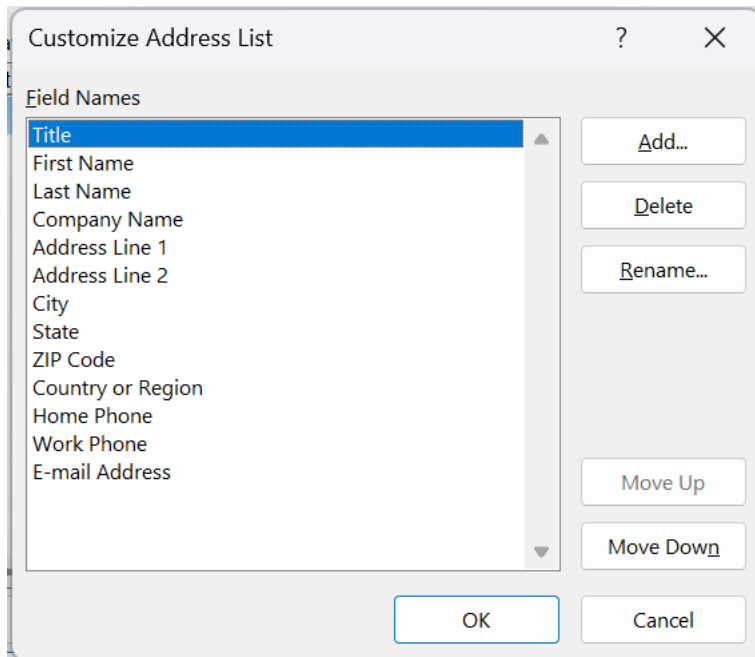


Picture 2. 15. Window for adding recipients

Step 4. Write the details for one recipient in the window that appears. The details above include Title, First Name, last Name, etc.

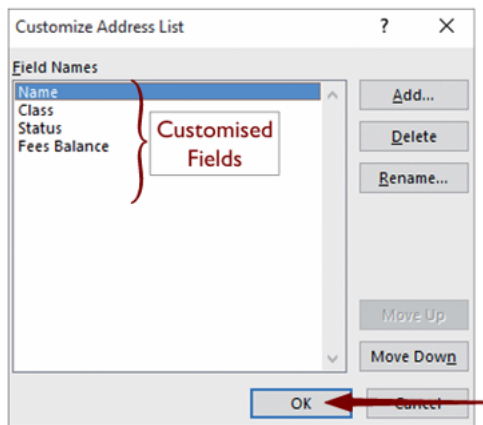
Add other recipients by clicking on **New Entry** and write details in the new row and click **OK**.

Note that you can delete a recipient by clicking on **Delete Entry**. You can also change the details for each recipient by clicking on **Customize column**. This option is necessary because it allows you to have only the columns you want.



Picture 2. 16. Window for customizing the mailing list

In the window above, a column can be removed by clicking on **Delete**, a new column can be added by clicking on **Add** while a column can be renamed by clicking on **Rename**. After the customization click on **OK**.



Picture 2. 17. New field names after customization

2.4.2. Using existing list

Making a list of recipient in Word may be difficult while you may have a list already available. Word provides an option to use the available list written in Excel. Follow the steps below:

Step 1: Prepare a list in Excel by making sure it is in a clear Excel table. The table should have columns such as FirstName, LastName, Address, Phone, and Email and the first row must show the names of the columns. Also make sure there are no empty columns.

	A	B	C	D
1	SN	Name of student	Class	Parent email
2	1	TUYISHIME Onesphore	S1	tuyishime@gmail.com
3	2	HABAGUSENGA Theogene	S3	habagusenga@gmail.com
4	3	ISIMBI Alphonsine	S1	Isimbi@gmail.com
5	4	HAKORIMANA J. Pierre	S2	hakorimana@gmail.com
6	5	NIRAGIRE Onesphore	S1	niragire@gmail.com
7	6	MUSHIMIYIMANA Daniel	S3	mushimiyimana@gmail.com
8	7	NSHIMIYIMANA J. de Dieu	S3	nshimiyimana@gmail.com
9	8	SIBOMANA Monique	S2	sibomana@gmail.com
10	9	NIYOMUGABO Protogene	S2	niyomugabo@gmail.com
11	10	ABIMANA Cyplier	S1	abimana@gmail.com
12	11	NGIZWENIMANA J. Claude	S2	ngizwenimana@gmail.com

Picture 2. 18. List of recipients created in Excel

Step 2: Prepare the letter in Word. In the case of the letter below, the name of the parents, the student's class and parent's email are empty because the recipient of this letter are many.

The Parent of:

Class:

Email:

Dear Sir/Madam,

RE: PARENTS' MEETING ON SATURDAY THE 24TH JULY 2026

You are cordially invited to attend the Parents' meeting without fail. The meeting will start at exactly 9:00am in the Main Hall of the school. The agenda for the meeting shall be given to you on arrival.

HEAD TEACHER

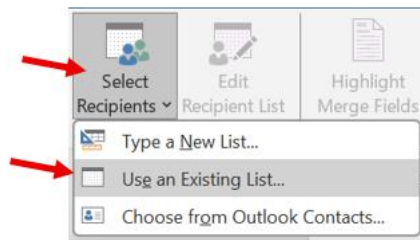
Cc:

- Parents' Teacher Committee

Picture 2. 19. A letter to be filled automatically by mail merge option

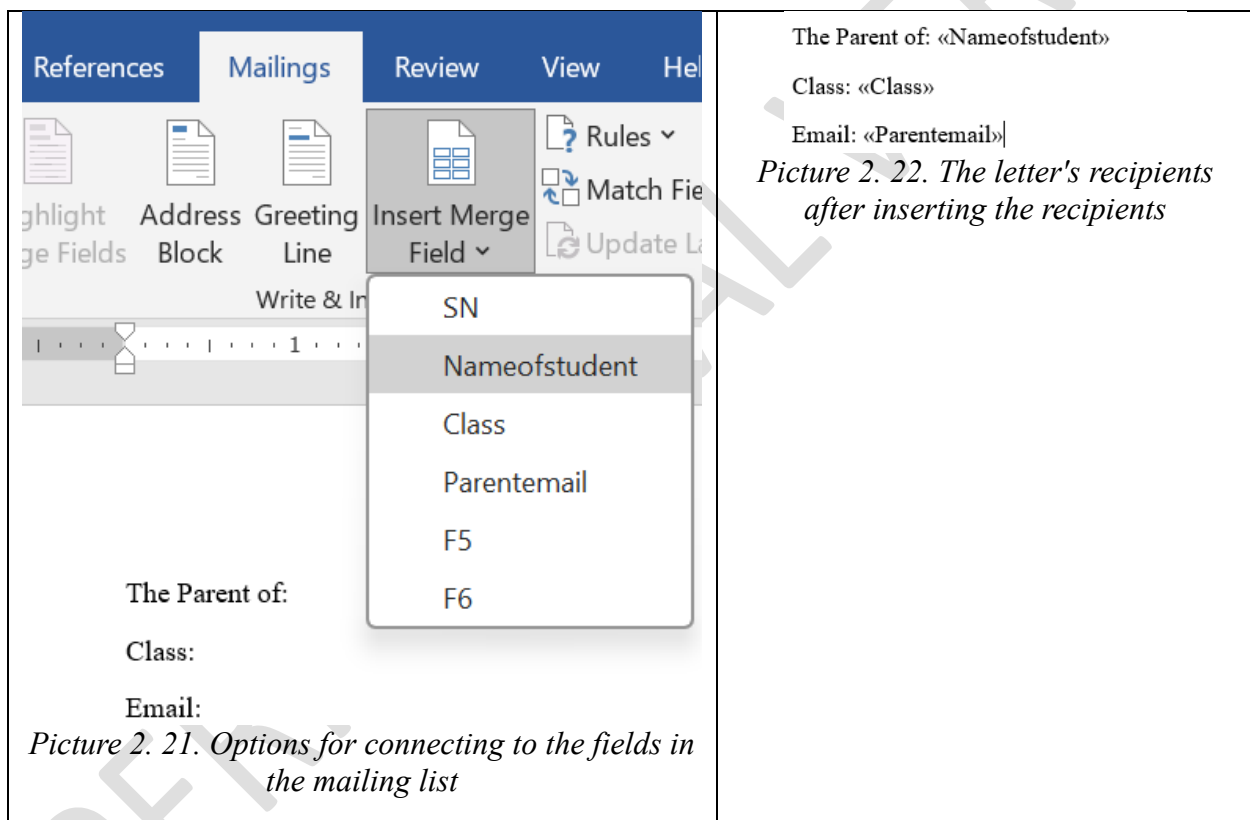
Step 3: Start Mail Merge by clicking the Mailings tab then click on Start Mail Merge and then choose the type of document you want to create, such as Letters.

Step 4: Insert the existing list by opening the letter then clicking Select Recipients then choose Use an Existing List then browse and open the Excel file. After that, select the sheet that has your data and click OK. This will connect the document to the list.



Picture 2. 20. Inserting an existing list in Mail merge

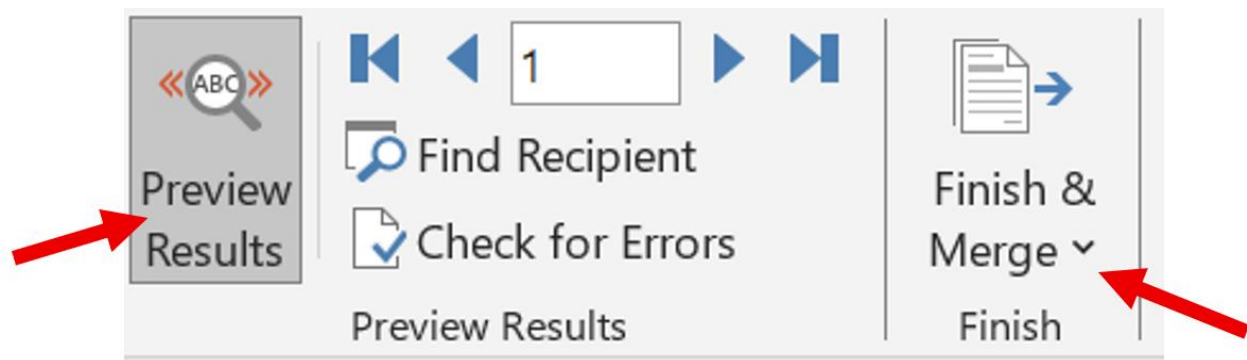
Step 5: Insert Merge fields by putting the cursor where the personalized information will appear in the document. Then click **Insert Merge Field** and choose the fields to use. After inserting the Merge fields, the columns from which to fetch information will appear in quotation marks as «Nameofstudent», «Class» and «Parentemail».



Picture 2. 21. Options for connecting to the fields in the mailing list

Picture 2. 22. The letter's recipients after inserting the recipients

Step 6: Preview the letters and print them by clicking on **Preview Results** to see how the letters will look then clicking on **Finish Merge** and then clicking on **Print Documents**.



Picture 2. 23. Tabs for previewing results and finishing the merge

When you print, you will get 11 letters while you wrote only one letter. In the case you need to send the letter to 600 parents, what you need is only a list of the 600 parents with all the needed details. You will not need to write all the 600 letters.

Application activity 2.4

- 1) Explain how the Mail Merge option can make the task of sending letters to 600 parents easier.
- 2) What is the difference between creating a new recipient list in Word and using an existing list from Excel?
- 3) Why is it important to customize columns when creating a new recipient list?
- 4) Which tab in Microsoft Word contains the Mail Merge tools?
 - a) Home b) Insert c) Mailings d) Review
- 5) What happens when you click “Preview Results” during Mail Merge?
 - a) The document is automatically printed
 - b) You see how the personalized letters will look with recipient data
 - c) The recipient list is deleted
 - d) The merge fields are hidden

End of unit assessment 2

- 1) Is updating a Table of Contents or List of Figures important after editing a document?
- 2) Give and explain two real-life cases where Mail Merge can be applied.
- 3) In your own words, explain why Mail Merge is considered more efficient than manually editing each letter.
- 4) Which menu in Word is used to insert a Table of Contents?
 - a) Home b) References c) Insert d) Layout
- 5) Which of the following is not a step in creating a Table of Contents?
 - a) Applying heading styles b) Inserting page numbers
 - c) Inserting captions under tables d) Choosing an automatic table style
- 6) Answer by True or False
 - a) A list of figures updates itself automatically whenever new tables are added.

- b) Captions are required for Word to generate a List of Figures or List of Tables.
- c) Updating “page numbers only” in a table of contents will also add new headings.
- 7) Create a document about the development of industries in Rwanda using Google (or generative AI tools). Make sure the document contains titles and picture
 - a) Create a table of content for that document
 - b) Create a list of figures for that document

UNIT 3. CHARTS AND OBJECTS IN SPREADSHEET



Key Unit Competency: Design Spreadsheet documents containing charts and objects

Learning Objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Distinguish different types of charts by specifying a scenario where each type can be used.
- Identify Spreadsheet elements used to format charts.
- Explain how to format a chart and its impact on the clarity of data.
- Describe how margins, page numbers, header and footers affect the appearance of a document

Skills

- Use charts and graphics to present data graphically
- Format a range of data cells as a table to improve readability.
- Format page numbers, headers, and footers in a datasheet.

Attitudes and values

- Value clarity and simplicity when choosing chart types to communicate information
- Show aesthetic sensitivity by choosing colours and patterns that enhance readability
- Appreciate the importance of margins, page numbers, headers&footers in Spreadsheets

3.0. Introduction

Spreadsheets are powerful tools not only for storing and calculating data but also for presenting information in clear, visual formats. Charts and objects transform raw numbers into meaningful insights, making it easier to detect patterns, compare values, and communicate findings effectively.

By the end of this unit, you will not only understand how to design charts and objects in spreadsheets but also appreciate the importance of clarity, simplicity, and aesthetics in presenting data.

Introductory activity

The teacher of ICT is preparing an end-of-term report to present the performance of students from different classes to the parents. He has marks for ICT and English from Senior One, Senior Two, and Senior Three.

- 1) If the teacher only shows parents a table of marks, what difficulties will they face in understanding the performance?
- 2) How could charts make it easier to compare results between Senior One, Senior Two, and Senior Three?
- 3) Why should the teacher set margins, headers, and footers before printing the report?

A chart is a graphical representation of numerical data. It is a sheet of information in form of a table, graph, diagram or object. Charts enable users to see results of data in detail, interpret and predict current and future data in a much easier way.

3.1. Creating charts

Activity 3.1

- 1) Why is it preferable to use charts instead of only numbers in tables?
- 2) How would a line chart differ from a bar chart in showing student performance over time?
- 3) What steps are necessary before creating a chart in Excel?
- 4) Why are charts commonly used in presenting data?
 - a) To make numbers look more colorful
 - b) To summarize and visualize numerical information
 - c) To replace tables completely
 - d) To avoid using mathematics

3.1.1. Role of charts

Charts are used for this purpose:

- To summarize numerical information.
- View relationships between different variables e.g. Price against Sales volume.
- Detect trends overtime and make forecasts.
- Search for patterns among a large amount of figures.

3.1.2. Types of charts:

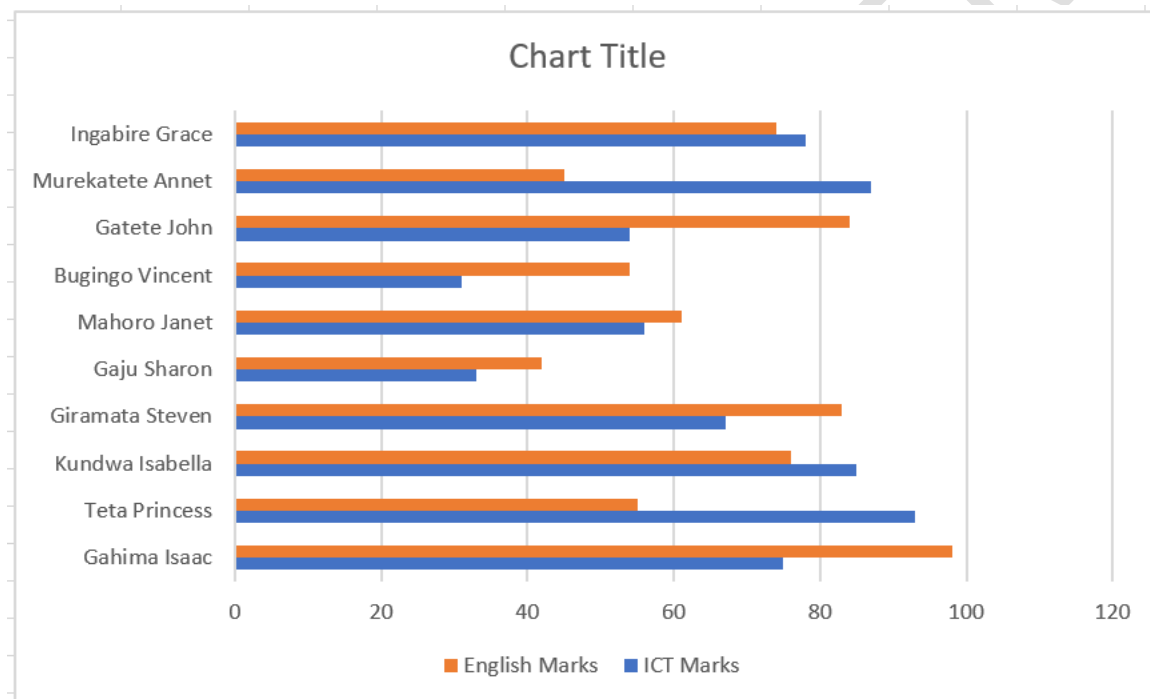
There are several types of charts used to present data pictorially. The charts mainly include; column, bar pie, line and scatter graphs. The information in the table below is going to be used to represent those different charts

Name	ICT Marks	English Marks
Gahima Isaac	75	98
Teta Princess	93	55
Kundwa Isabella	85	76
Giramata Steven	67	83

Gaju Sharon	33	42
Mahoro Janet	56	61
Bugingo Vincent	31	54
Gatete John	54	84
Murekatete Annet	87	45
Ingabire Grace	78	74

a. Bar Chart

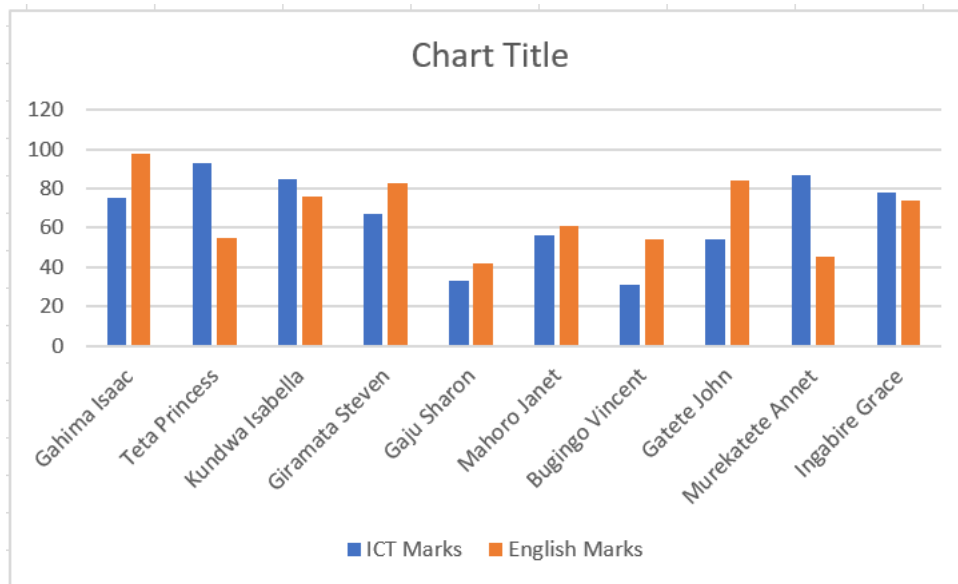
A bar chart is a graphic representation of data with horizontal bars or objects. Bar charts usually display horizontal bars or objects going across the chart vertically, with the values displayed at the bottom of the chart.



Picture 3. 1. Bar Chart

b. Column Chart

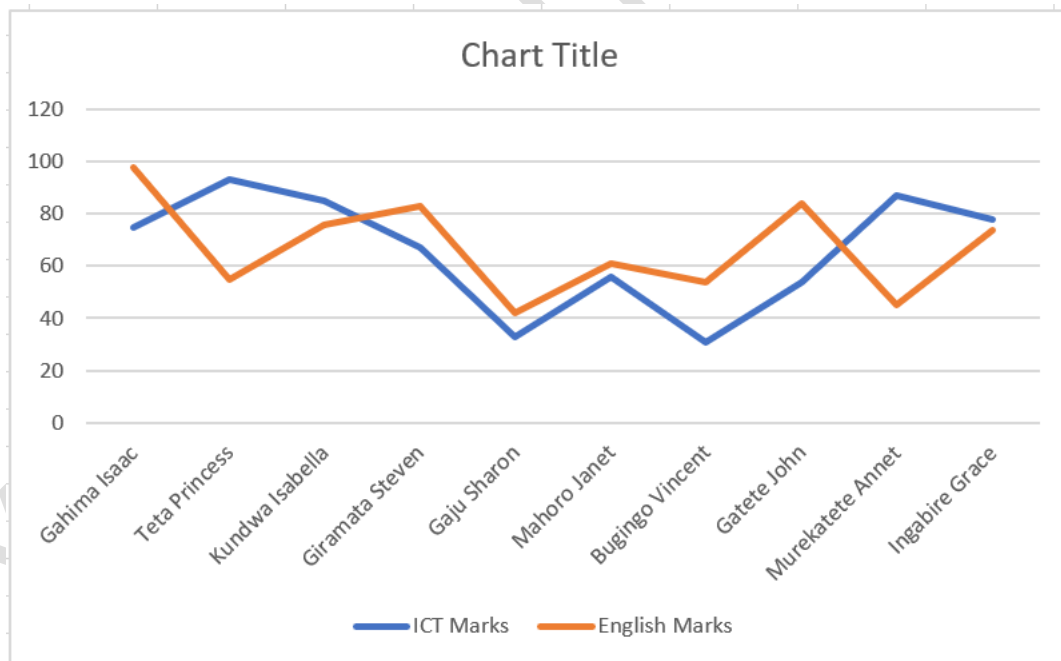
A column chart is a graphic representation of data with vertical bars or objects. Column charts display vertical bars (categories) going across the chart horizontally (along the horizontal axis), with the values organised along the vertical axis (on the left side of the chart).



Picture 3. 2. Column chart

c. Line chart

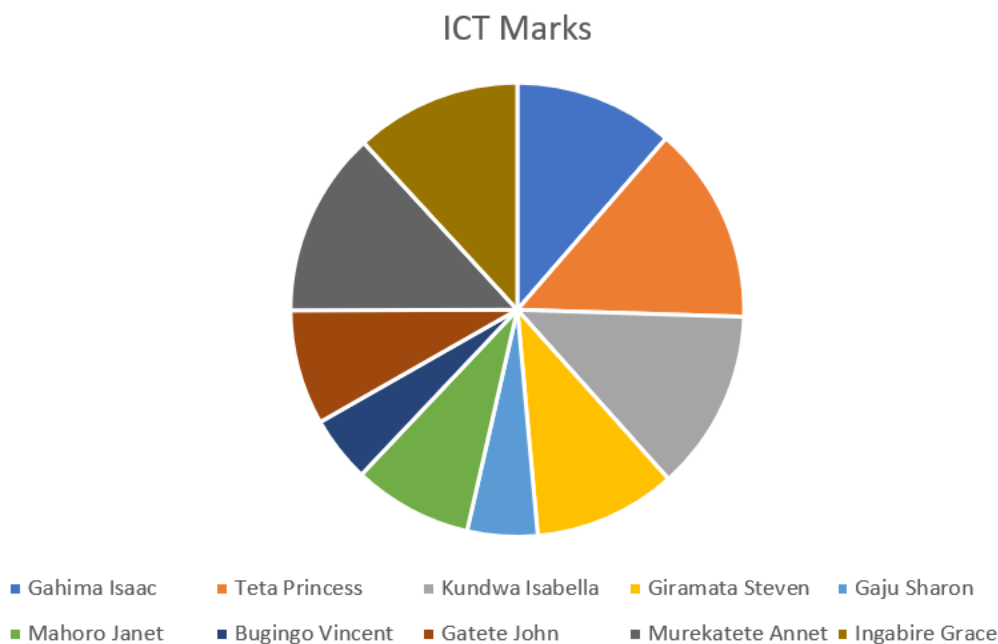
A line chart is a graphic representation of data plotted using a series of lines. Line charts display lines going across the chart horizontally, with the values axis being displayed on the vertical axis (left side of the chart).



Picture 3. 3. Line chart

d. Pie Chart

Pie chart is a circular chart sliced into sections; each section represents a percentage of the whole. Pie charts do not display horizontal and vertical axes as other charts.



Picture 3. 4. Pie chart

3.1.3. Charts insertion:

a. Table creation

Before creating charts, you must first have information which is arranged in a table for which a table will be created. For this, open Excel by clicking on **Start** then search for Excel using the search box. Once the Excel icon appears, click on it.

Now create a data in a tabular format with table headings and data in columns and rows.

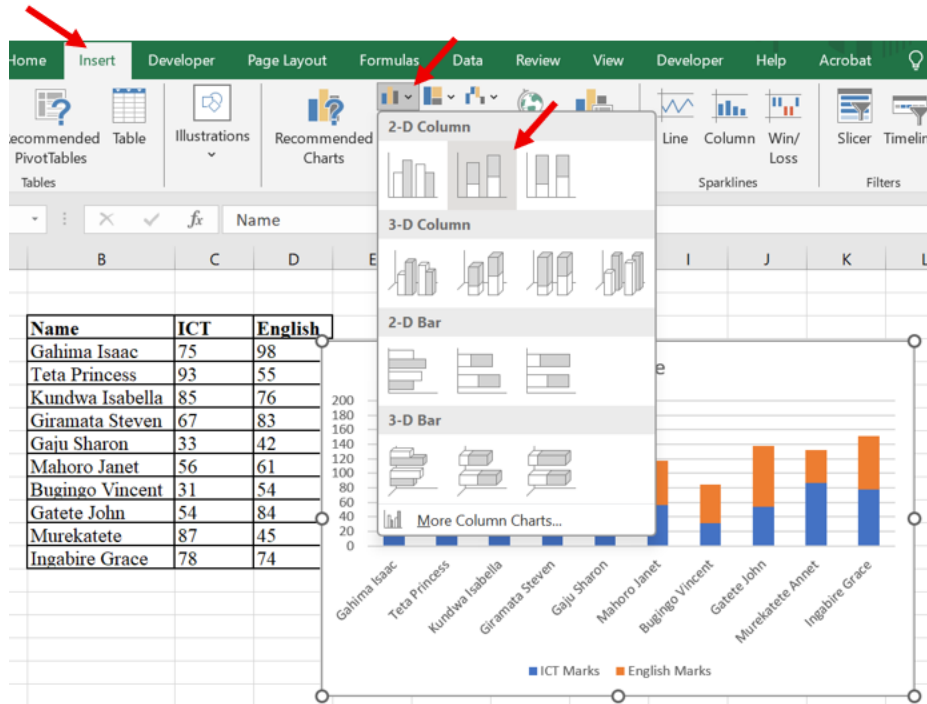
b. Generating charts from data

To create a chart follow the steps given below:

Step 1: Create or open an existing workbook with well-prepared numerical data. In this case write in Excel the information in the table under the title 3.1.2. and save it as **ICT marks**.

Step 2: Select all the cell data you need to be represented in the chart (including all the labels and titles). If the data is not nearby to each other, press Ctrl as you select other pieces of data.

Step 3: On the **Insert tab**, in the **Charts group**, click on the chart type you want.



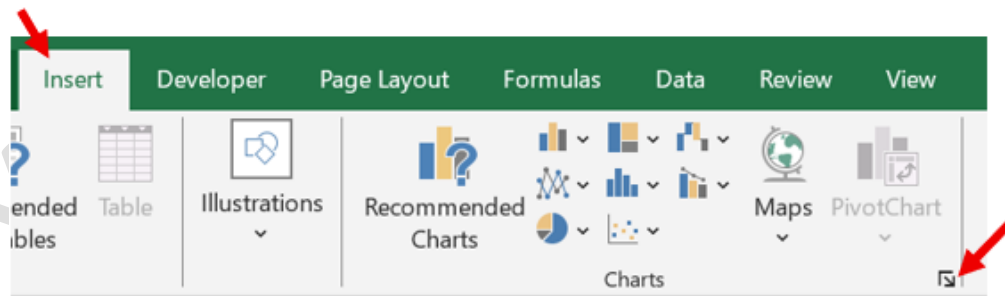
Picture 3. 5. Steps to go through in inserting charts

Step 4: The new chart displays automatically when put the cursor on the chart type and when you click on it the chart get immediately inserted.

c. Accessing different types of charts

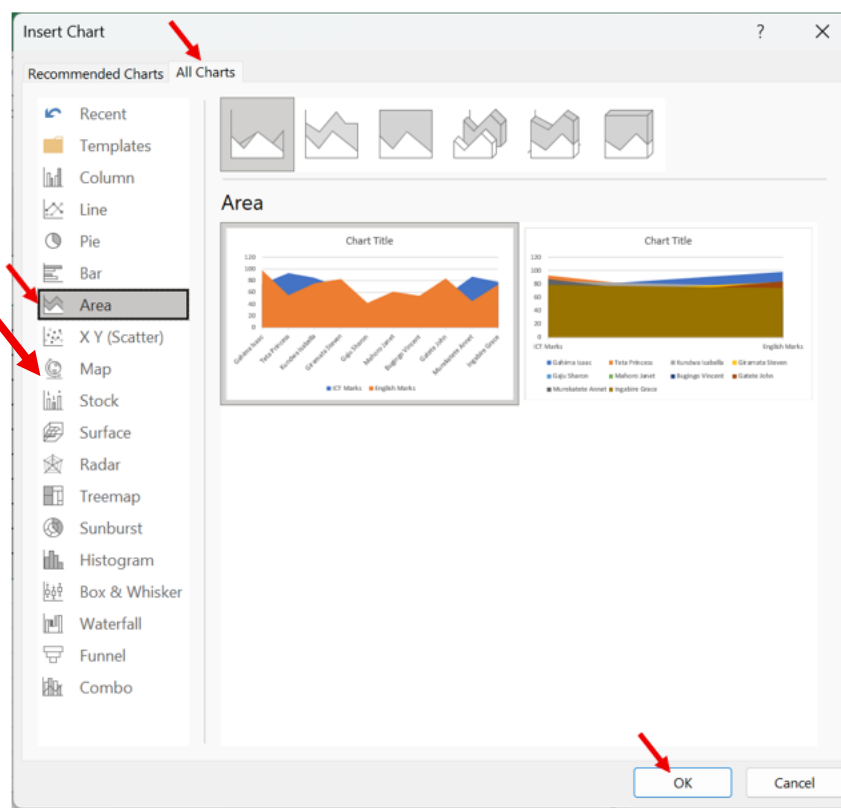
There are many other types of charts apart from those shown above such as **Surface, Radar, Waterfall**, etc. The process to insert those charts is the same as for other charts as illustrated above.

- Select the table or the data for which you want to create the chart then click on **Insert tab** and then on the small icon found on the toolbar (Dialog Box Launcher)



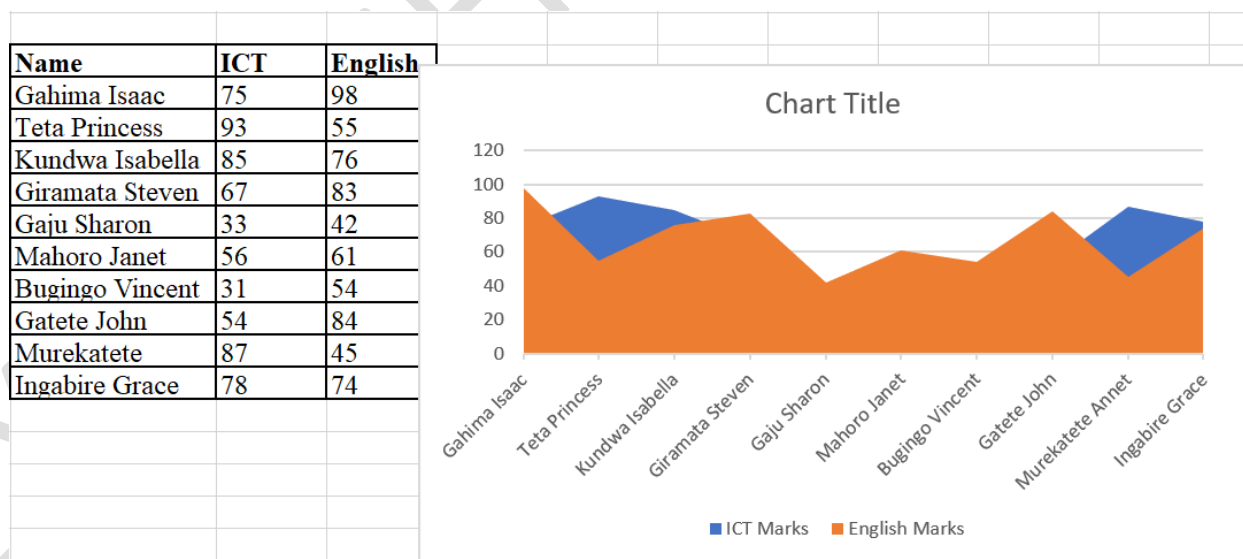
Picture 3. 6. Accessing other types of charts

In the new window, click on **All Charts** then click on one type of chart such as **Area** and click **OK**.



Picture 3. 7. Options for inserting other types of charts

Immediately the new Area chart will be inserted and it will look like this:



Picture 3. 8. Area chart

Application activity 3.1

1) Create a column chart in Excel using the data in the table below:

First Name	ICT (20)	Mathematics (60)	English (50)
MUKAMANA Alice	16	48	40
NIYONZIMA Brian	12	35	30
UWIMANA Chantal	18	55	45
HABIMANA David	14	42	33
NKURUNZIZA Emmanuel	19	52	44

2) Change the chart you have created into a line chart.

3) Which of the following correctly describes a bar chart?

- a) Vertical bars with values on the left axis
- b) Horizontal bars with values displayed at the bottom
- c) Circular slices showing percentages
- d) Lines connecting points across the chart

3.2. Charts formatting

Activity 3.2

1) Charts have title, axes, labels, legend and data series. How does each of these parts make the chart more understandable?

2) What risks might there be if a chart is beautiful but is not formatted correctly?

3) Are accessibility features such as “Alt Text” important in a chart? Explain.

4) What happens when you apply a chart style in Excel?

- a) The data changes
- b) The chart’s appearance changes instantly
- c) The chart type changes
- d) The chart is deleted

5) Which of the following can be modified in chart layout?

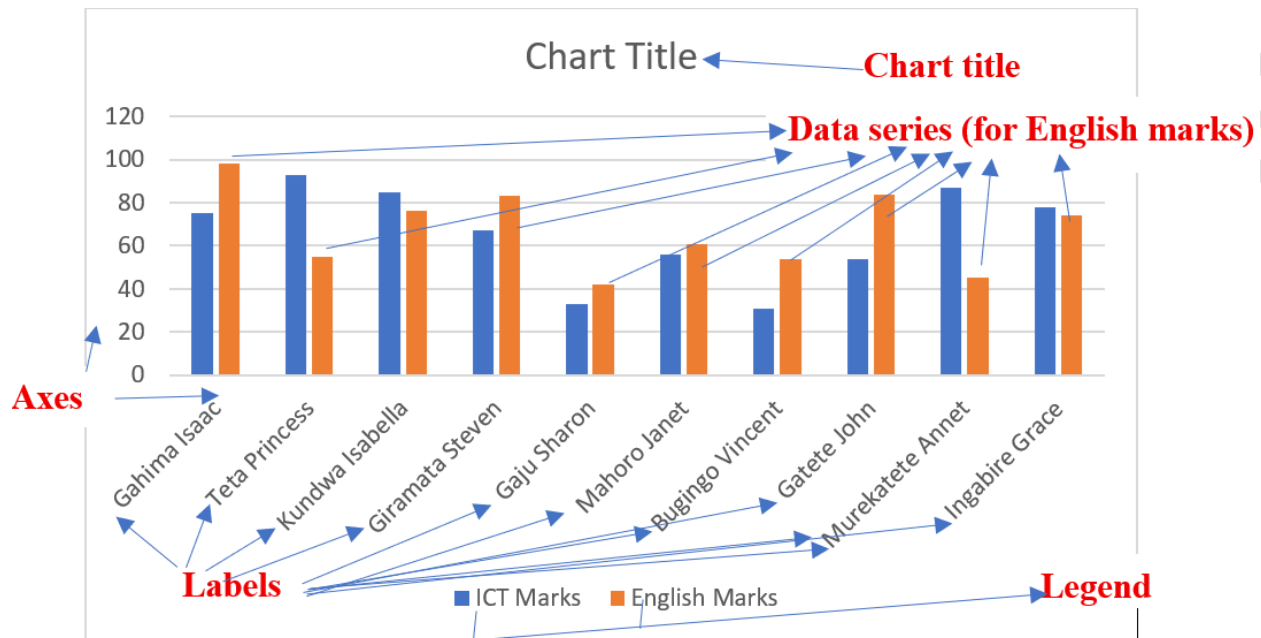
- a) Chart title position
- b) Legend position
- c) Gridlines
- d) All of these options

3.2.1. Charts elements

A chart has main elements that help readers to easily understand its meaning. Those are title, axes, labels, legend and data series.

- **Title:** It is the main heading that tells the reader what the chart is about. It summarizes the information being presented so the audience immediately understands the purpose of the chart.
- **Axes:** Charts have two axes. They are the horizontal axis (X axis) and the vertical axis (Y-axis) which shows numerical values such as scores.
- **Labels:** Labels are the text that identifies the categories or values on the chart. They appear along the axes or directly on the data points. Labels make the chart readable by showing what each bar, line, or point represents.
- **Legend:** The legend explains the meaning of the colors, patterns, or symbols used in the chart.

- **Data series:** A data series is the actual set of numbers being plotted on the chart. Each series represents one group of related data.



Picture 3. 9. Parts of a chart

3.2.2. Format changing

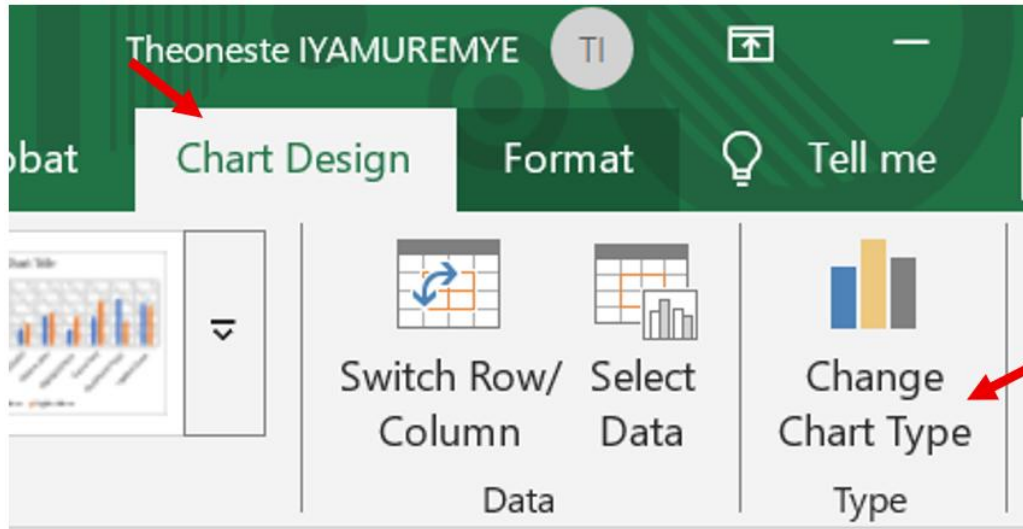
To change the chart format, first select the chart element to change then click on one of the tool to use. When the tool can not be used on a special chart element, that tool becomes grey. For example when you have selected the line graph the WordArt Styles tool become grey because it can not be used in this case.

a. Changing chart type

We are going to use ICT marks file to change chart type. Follow the steps given below

Step 1: Select the chart you want to change.

Step 2: On the **Chart Design** tab in the **Type** group, click **Change Chart Type**. You can also right click the chart and select **Change Chart Type**.



Picture 3. 10. Options for changing chart type in Excel

Step 3: In the **Change Chart Type** dialog box that displays, select the chart type and sub type you want.

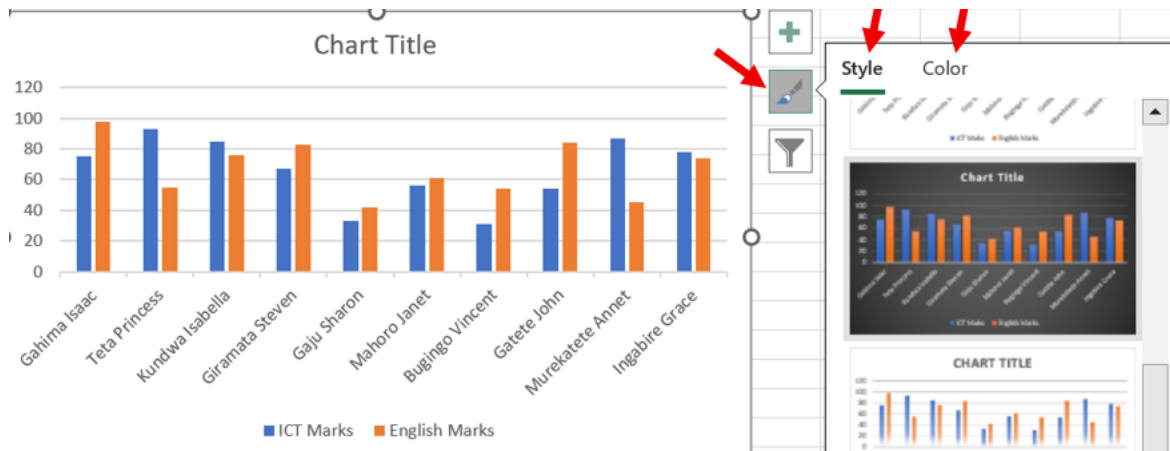
b. Changing chart style

A Chart style in Microsoft Excel is a pre-designed combination of formatting options that you can apply to a chart with one click. It instantly changes the chart's colors, background, borders, effects, and overall visual appearance without changing the actual data.

Follow the steps below, using the chart already created in previous activities

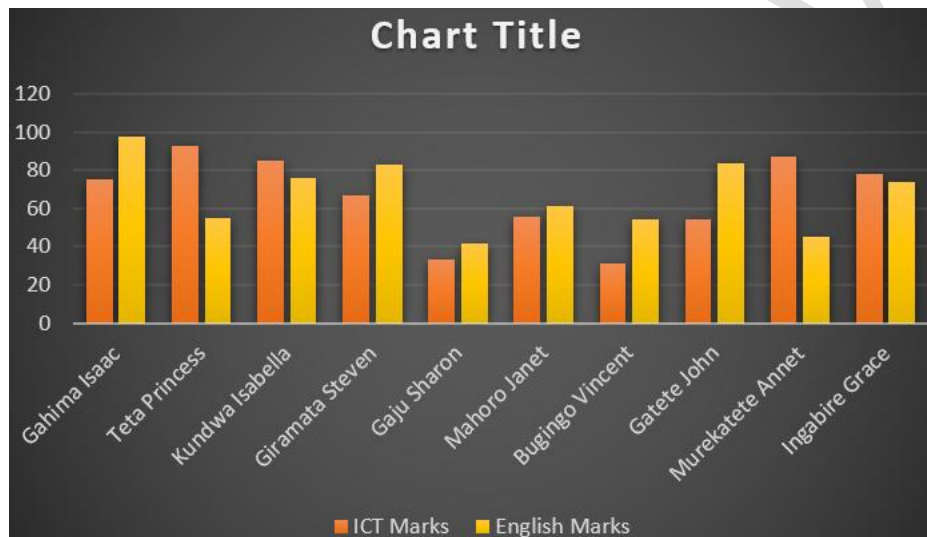
Step 1: Click in the chart on which you want to apply a style. The **Chart Style** tool will display on the right of the chart

Step 2: Click on **Chart Style** tool  and click on one of the style you want to apply. After, select the color you want the chart to have.



Picture 3. 11. Steps for changing the chart style

After changing the style and its color, the chart looks like the one below:



Picture 3. 12. The new chart after changing the style (and its color)

c. Changing chart layout

This is the overall arrangement of all the parts of a chart including how the title, legend, axes, labels, and data are positioned and displayed. It controls what appears on the chart and where it appears.

Here is what it can help to change:

SN	Chart element	Possible modification
1	Chart title	On top or removed
2	Legend	Right, left, top, bottom, or hidden
3	Axis titles	Shown or not shown
4	Data labels	Numbers on the bars/points

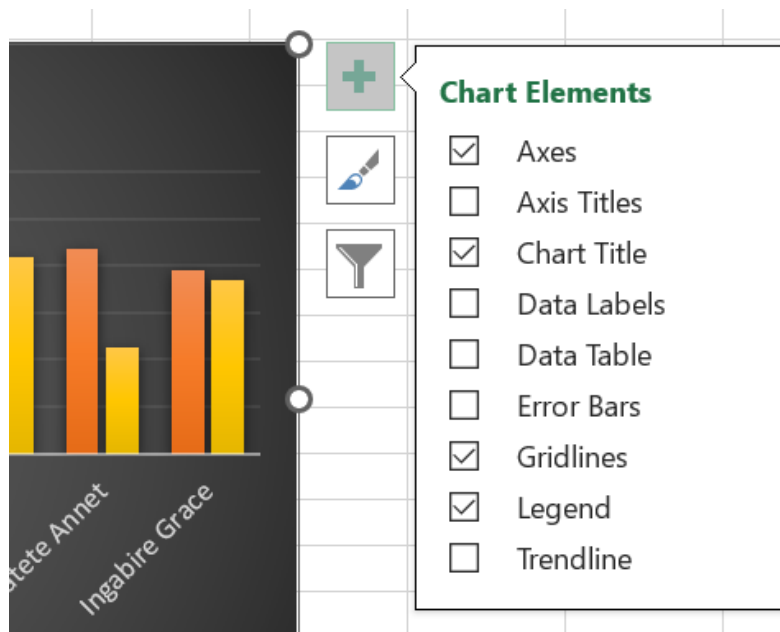
5	Gridlines	Horizontal or vertical
6	Chart background	Color or no color

Do the following to change the chart layout:

Step 1: Click the chart for which you want to change the layout. Immediately the **Chart**

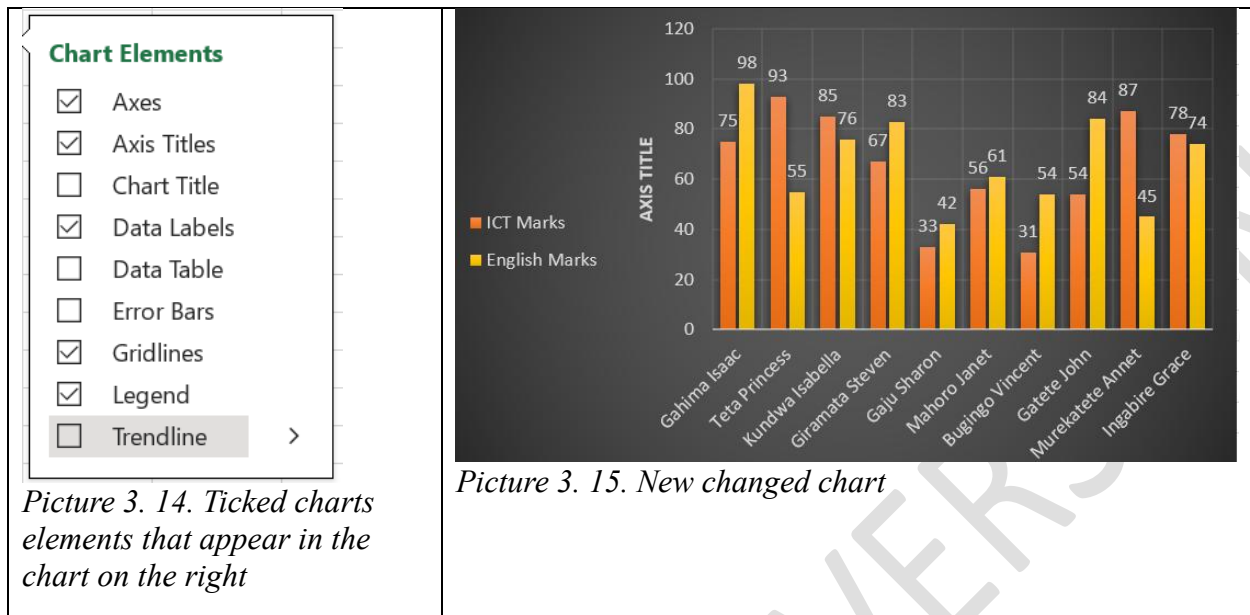
Elements tool  will appear.

Step 2: Click on the Chart element tool then tick the chart element you want to have in the chart.



Picture 3. 13. Changing chart elements

Here is the chart below after having removed the Chart title, adding labels, moving legends to the left and adding gridlines.



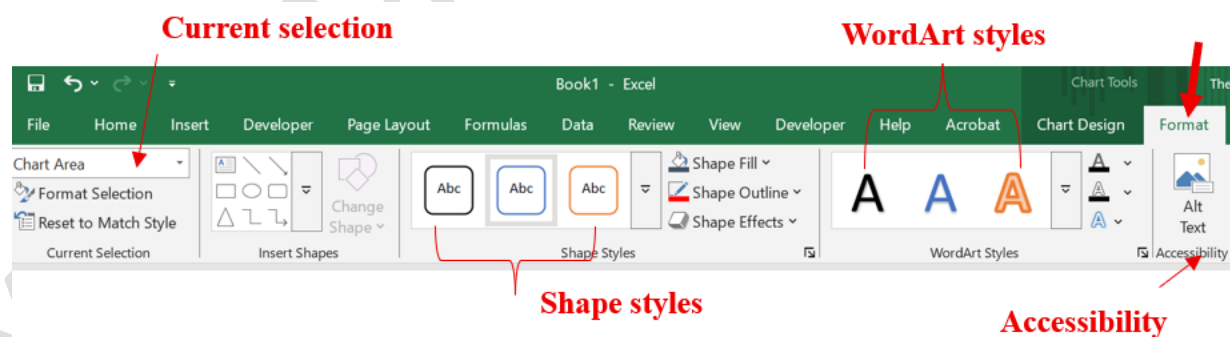
d. Chart format tab

When you click a chart in Excel, two special tabs appear: Chart Design and Format. The Format tab is used to customize the appearance of every part of a chart. It includes shapes, text, colors, borders, alignment, and accessibility.

To use the chart format tool, do the following:

Step 1: Select the chart for which to change the format.

Step 2: Under the **Chart Tools**, click on **Format**. Immediately the formatting tools below are displayed on the toolbar.



Picture 3. 16. Different tools for chart formatting

- **Current selection:** Lets you choose specific chart elements (like axis, legend, or title) to format individually using the “Format Selection” button.
- **Shape Styles:** Provides quick style including “Shape Fill,” “Shape Outline,” and “Shape Effects,” to change the color, border, and visual effects of chart elements.

- **WordArt Styles:** Allows you to stylize text in the chart using effects like shadows, outlines, and color gradients.
- **Accessibility:** Includes the “Alt Text” tool, which lets you add descriptive text for screen readers, improving accessibility for visually impaired users.

Application activity 3.2

1) Write the following table in Excel:

Last Name	Theory (/10)	Practical (/10)	Total (/20)
NDAYISHIMIYE Jean	7	8	15
MUKAMANA Aline	8	7	15
HABIMANA Eric	6	6	12
UWIMANA Diane	9	9	18
NIYONZIMA Samuel	7	7	14

- Using the data in that table create a Column Chart
- Change the chart style to one with a dark background.
- Change the colors so that bars become blue
- After formatting, explain how the changes improved the clarity of the chart compared to the unformatted version created at the beginning.

3.3. Format cell data as a table

Activity 3.3

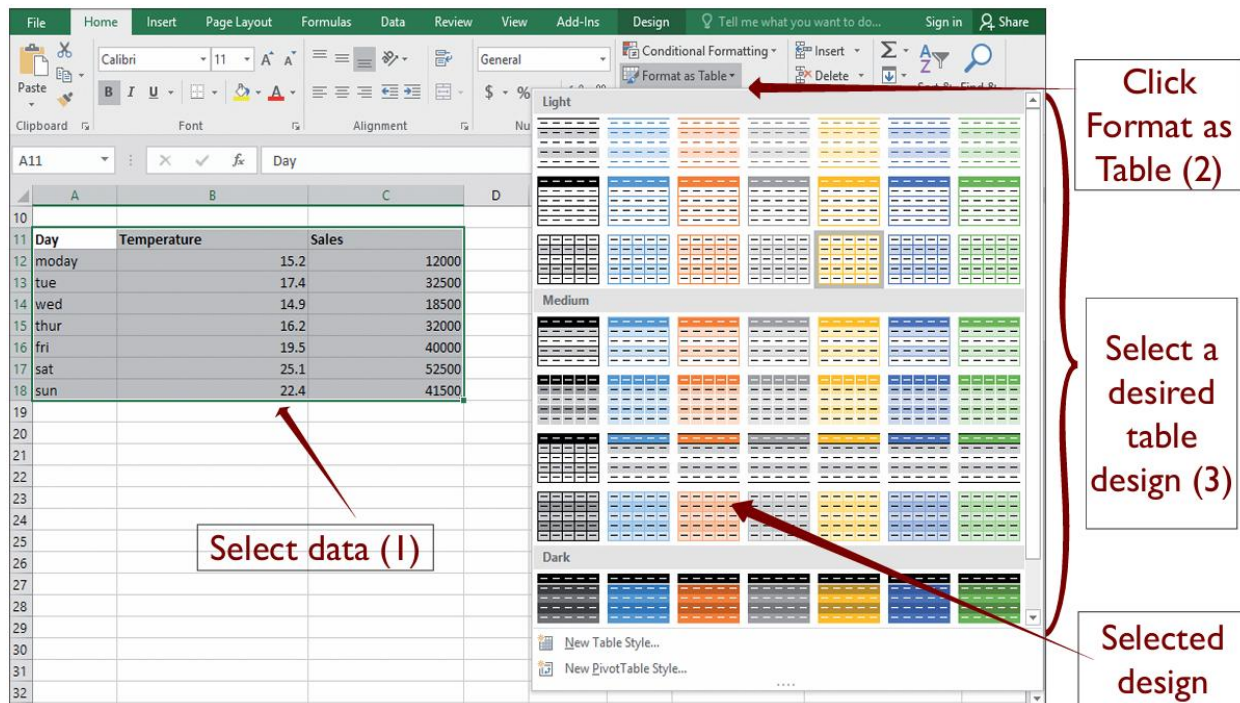
- When you enter data into Excel, how does it usually look by default?
- You are preparing student marks for a school report. Why is it important to format the data as a table?
- What is the first step when formatting cell data as a table in Excel?
 - Click Insert then Table
 - Select the spreadsheet data you want to format
 - Open the Design tab directly
 - Apply a chart style

Cell data is any piece of data you want to format as a table contained within spreadsheet cells.

Steps to format cell data as table

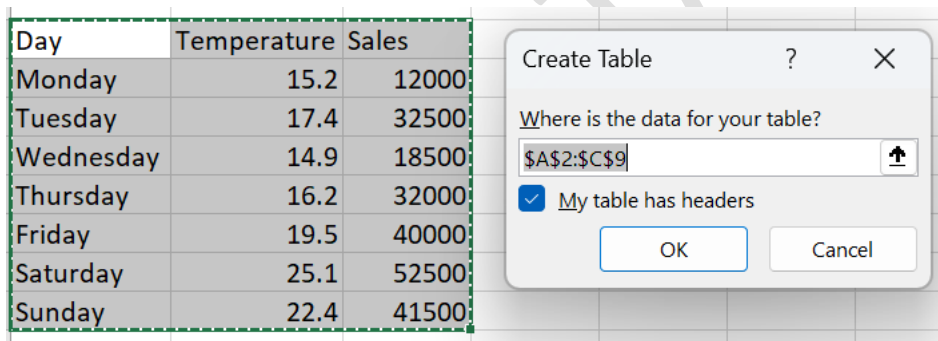
Step 1: Select the spreadsheet data you want to format as table or click in it (must be nearby each other). In this case, open **Comb** file.

Step 2: On the **Home tab**, in the **Styles group**, click **Format as Table** drop down list and select a desired table style from the list.



Picture 3. 17. Selected data to be formatted as a table using inbuilt table designs

Step 3: The **Create table** dialog box appear, click **OK**.



Picture 3. 18. Create Table dialog box

Step 4: Table Tools display on the Design tab. Use Table tools to modify your table as desired.

2	Day	Temperature	Sales
3	Monday	15.2	12000
4	Tuesday	17.4	32500
5	Wednesday	14.9	18500
6	Thursday	16.2	32000
7	Friday	19.5	40000
8	Saturday	25.1	52500
9	Sunday	22.4	41500

Picture 3. 19. Data transformed into a table

Application activity 3.3

1) Using the data in the table below, format it as a table in Excel

Student	Mathematics	English	ICT
KARIZA Alice	55	48	16
MUGABO Ben	36	30	14
INEMA Claire	40	30	11
KARISA David	38	25	18
SHYAKA Emma	30	40	16

2) Under which tab you get the table tools to format data as a table.

- a) Home b) Design c) Data d) View

3.4. Printable datasheet

Activity 3.4

- 1) Explain why it may be necessary to print an Excel worksheet
- 2) Why is it important to adjust the page setup before printing a worksheet?
- 3) In Excel normal view, margins are not visible. What happens if it is printed?
- 4) Explain why it is necessary to set the print area in Excel?
- 5) What steps do you follow to print a document in Microsoft Office applications?

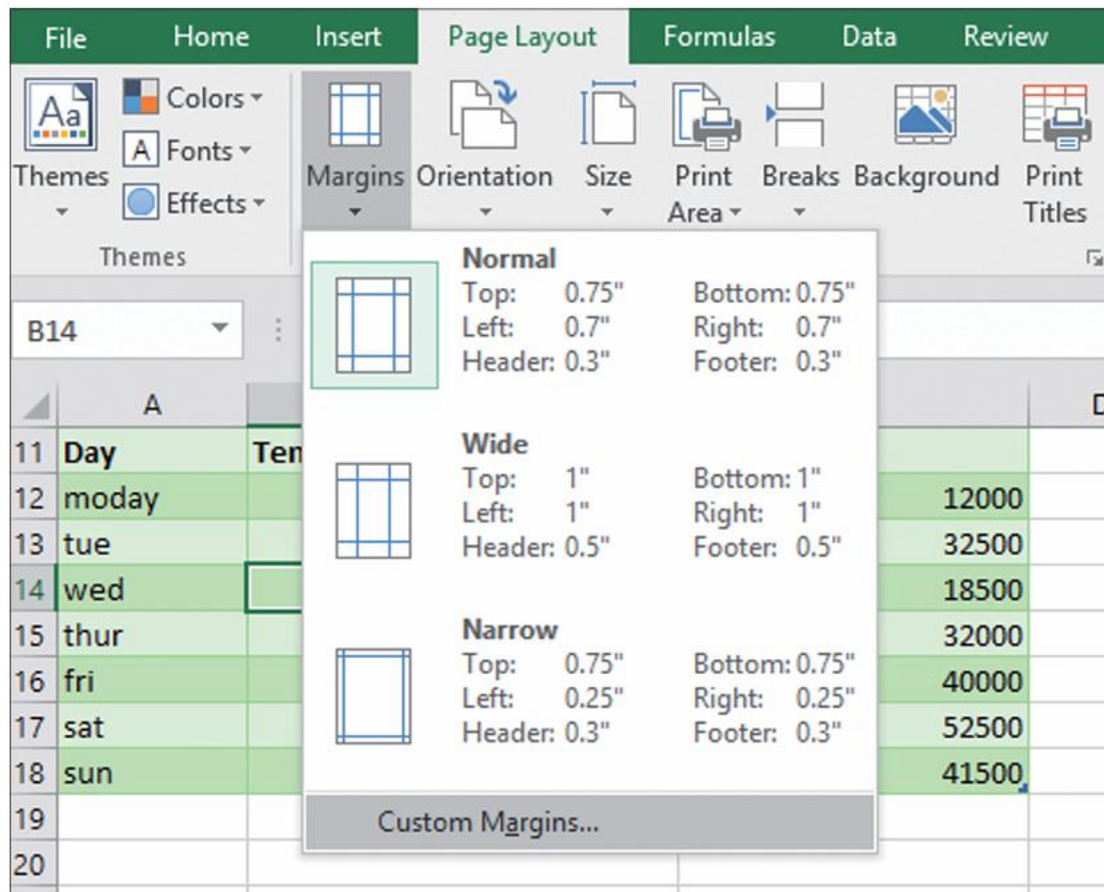
Before printing a worksheet, it is important to set a print area. Printing is the action of producing a hardcopy information using a printer. Page setup options control the layout of a printed sheet and therefore must be used before a worksheet is printed. Page setup options include; margins, headers and footers.

3.4.1. Margin settings

In Microsoft Excel, document margins are the blank spaces around the edges of the printed page on the top, bottom, left, and right. Margins do not show in an Excel document when seen in normal view but they control how a worksheet will look when printed.

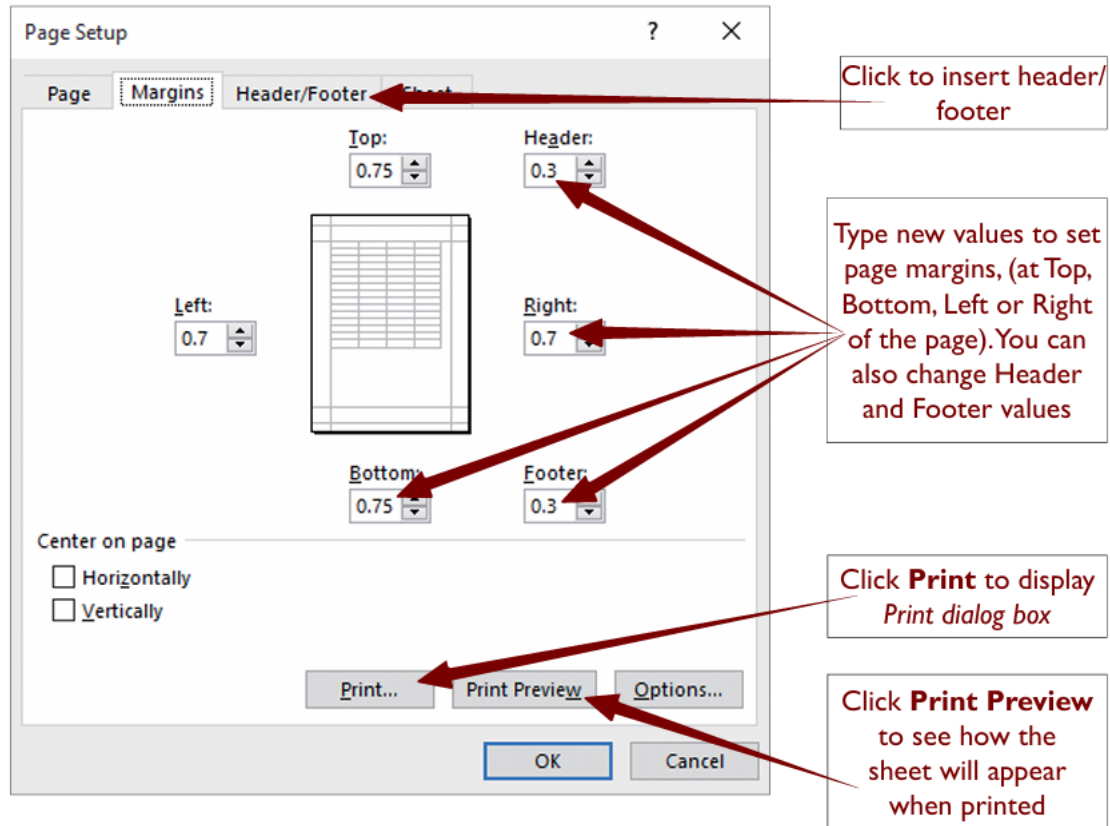
Follow these steps to set page margins in Excel:

Step 1: On the **Page Layout** tab, in the **Page Setup** group, click **Margins** and in the menu select “**Custom Margins**”. Depending on your choice you can also click on Normal, Wide or Narrow which are the inbuilt margin settings.



Picture 3. 20. Accessing Custom Margins in Excel

Step 2: In the **Page Setup** dialog box for **Margins**, increase or reduce the margins and click **OK**.



Picture 3. 21. Page Setup dialog box displaying features for changing page margins

3.4.2. Headers & footers

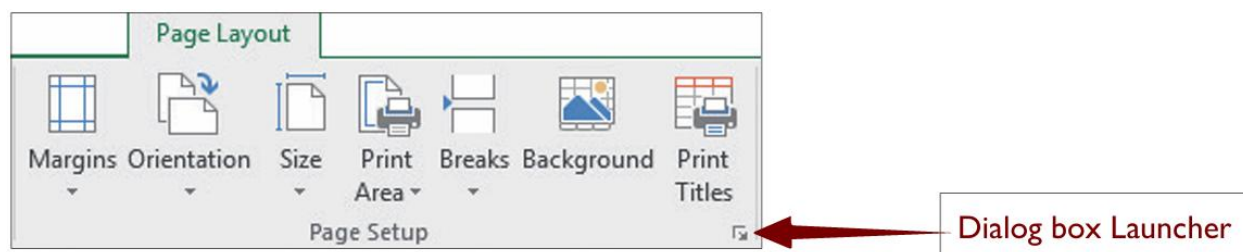
A header is information printed at the top of every page and footer is information printed at the bottom of every page. Headers are suitable for Company names, page titles; footers are good for page numbers and printout dates or times.

Note: Headers and footers are not displayed on the worksheet in normal view.

On ordinary worksheets you can insert header/footer by clicking on **Insert tab**, in the **Text group**, click **Header/Footer**. Automatically the cursor goes in the centre of the header.

Follow these steps given below to insert header and footer:

Step 1: On the **Page Layout tab**, in the **Page Setup group**, click the **Dialog box Launcher**.



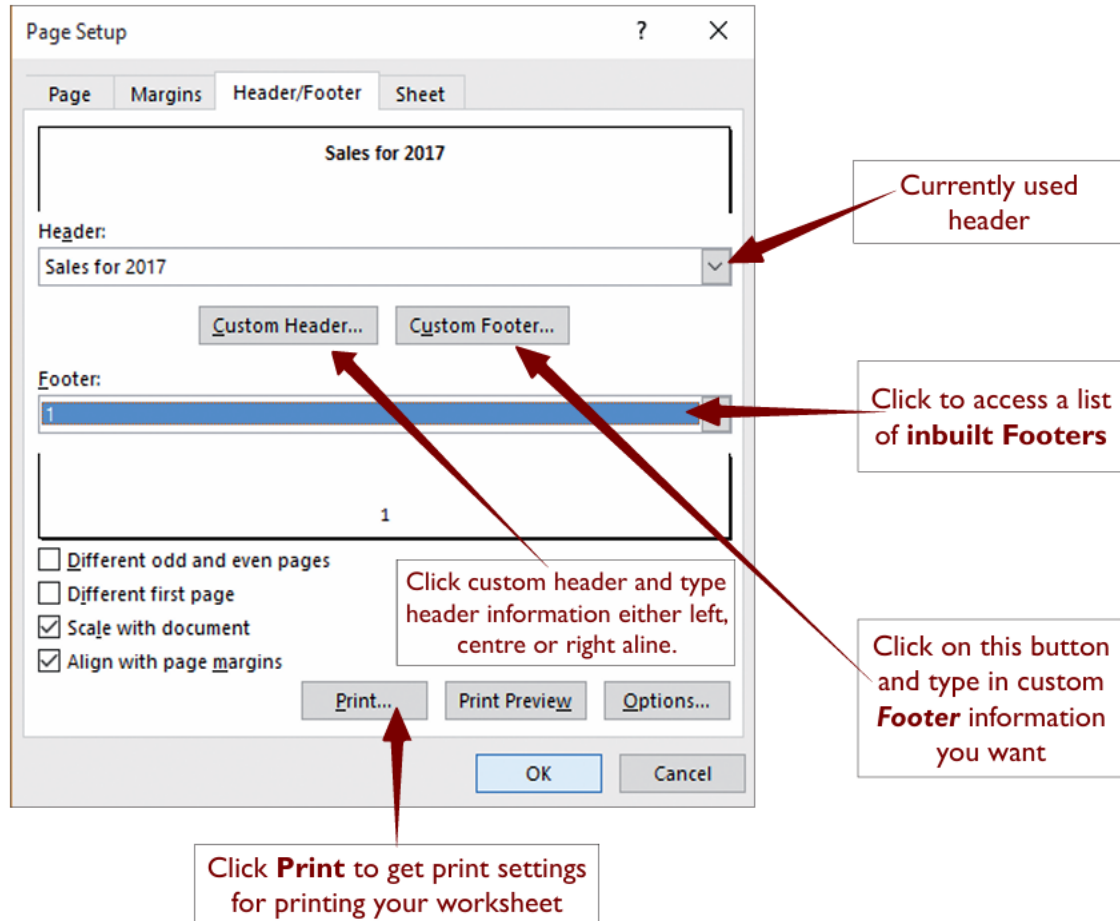
Picture 3. 22. Page Layout tab displaying Page setup group of commands

Step 2: In the **Page Setup** dialog box that displays, click on **Header/ Footer tab**.

Step 3: Click on **Custom Header** to insert your header or **Custom Footer** to insert footer.

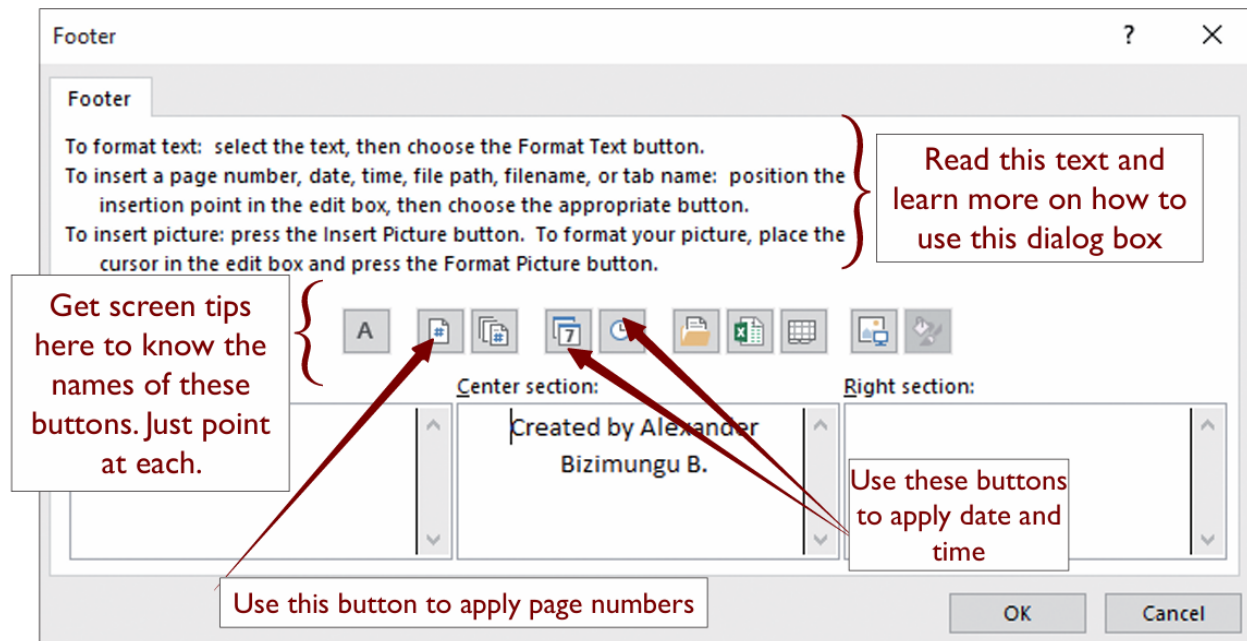
Step 4: Click in the Left section, Centre section, or Right section box, and then click the buttons to insert the header or footer information that you want in that section.

The page setup dialog box is shown in picture above, if you click **either Custom Header or Custom Footer**, the Header or Footer dialog box displays. To add or change the header or footer text, type additional text or edit the existing text in the **Left section, Centre section, or Right section** box.



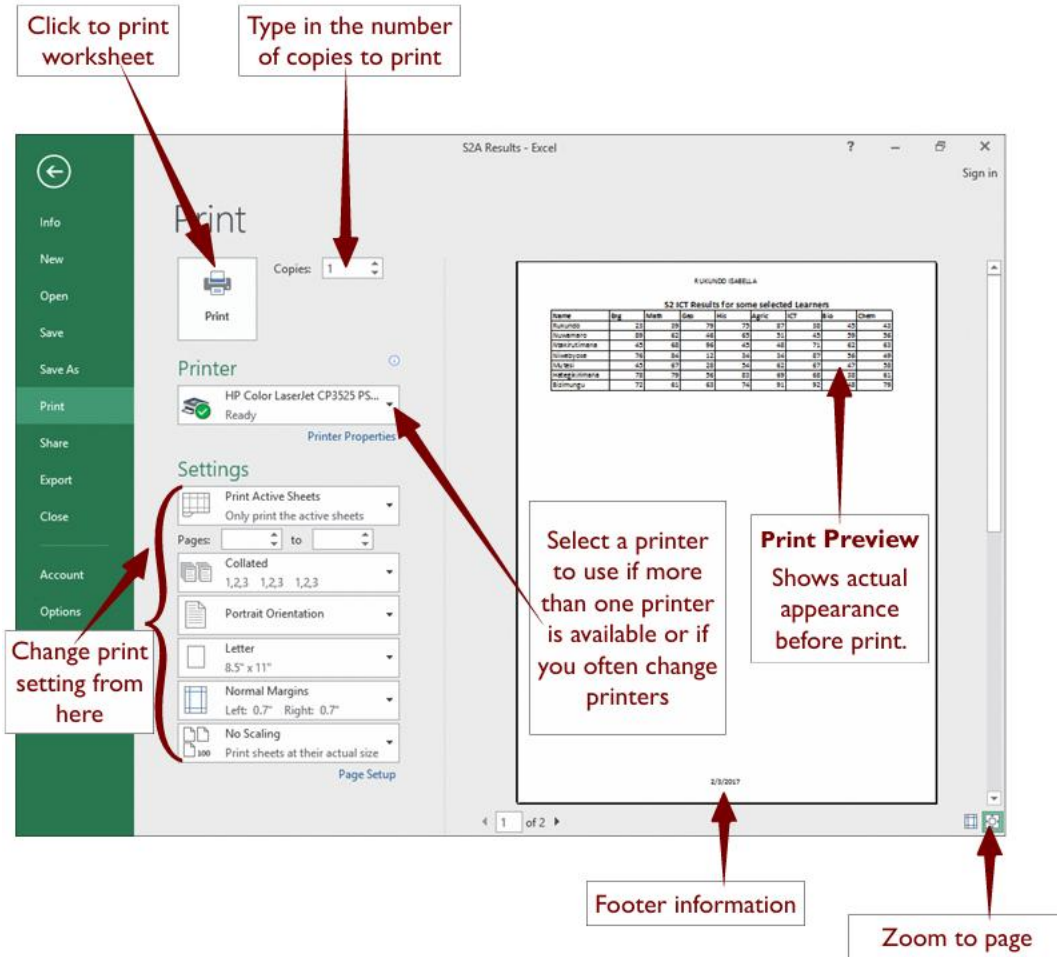
Picture 3. 23. Page Setup dialog box displaying Header/Footer options

When you click on **Custom Header** or **Custom Footer** you get a Header and Footer dialog box respectively. When you use these dialog boxes, you can position your text or object in the left, centre or right section of the page in the header and footer.



Picture 3. 24. Footer dialog box with footer text in centre section

Header/Footers are displayed only in **Page Layout** view and on the printed pages. You can insert headers or footers in **Page Layout view** where you can see them, or you can use the **Page Setup dialog box** if you want to insert headers or footers for more than one worksheet at the same time. For other sheet types, such as chart sheets, you can insert headers and footers only by using the **Page Setup dialog box**.

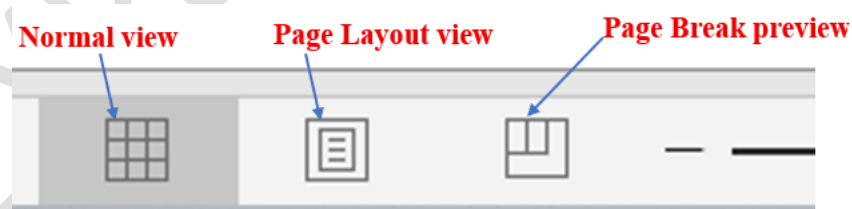


Picture 3. 25. The Back Stage View showing options for printing a worksheet

3.4.3. Page numbers

Page numbers in Excel help you **organize and present printed worksheets** in a clear and professional way. They don't appear in the worksheet itself except when a worksheet is seen in **Page Layout view** or when it is printed. They are inserted in the page header and footer.

Step 1: In the bottom left corner, click on the **Page Layout view** tab . Immediately the document will show headers and footers

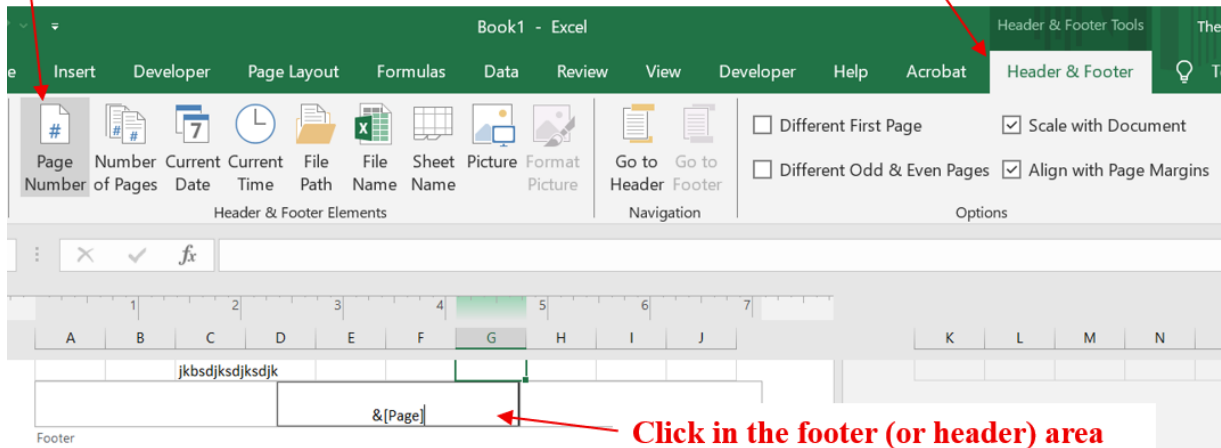


Picture 3. 26. Different page view in Excel

Step 2: Click in the header or footer area where the page number will be inserted and on the menu ribbon, under the **Header & Footer Tools**, click on **Header & Footer**.

Click on Page Number

Click on Header & Footer



Step 3: Click on **Page Number** tool so that the page number gets immediately inserted.

Note that if you want to have page numbers in the format of “1 of 6”, you will click on the tools **Page Number** then write “of” and click on **Number of Pages**.

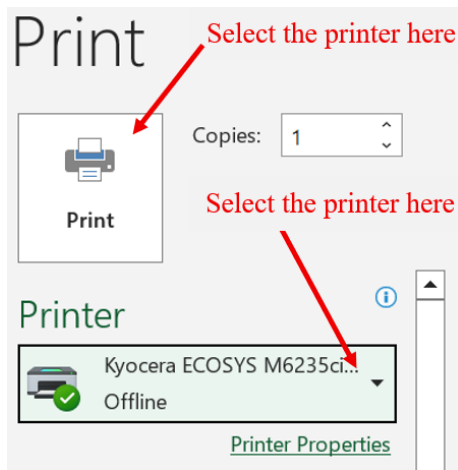
Note that when you switch back to **Normal view** the page numbers will be hidden but when the document is printed or when you see it again in **Page Layout view** the page numbers will reappear.

3.4.4. Printing an Excel worksheet

a. To print the whole content of a worksheet follow these steps:

Step 1: In the opened Excel worksheet, click on **File** then click on **Print**.

Step 2: Choose the printer to use in printing. You may also change the print settings such as printing on both sides, etc.



Picture 3. 27. Options for printing a worksheet

b. Print area

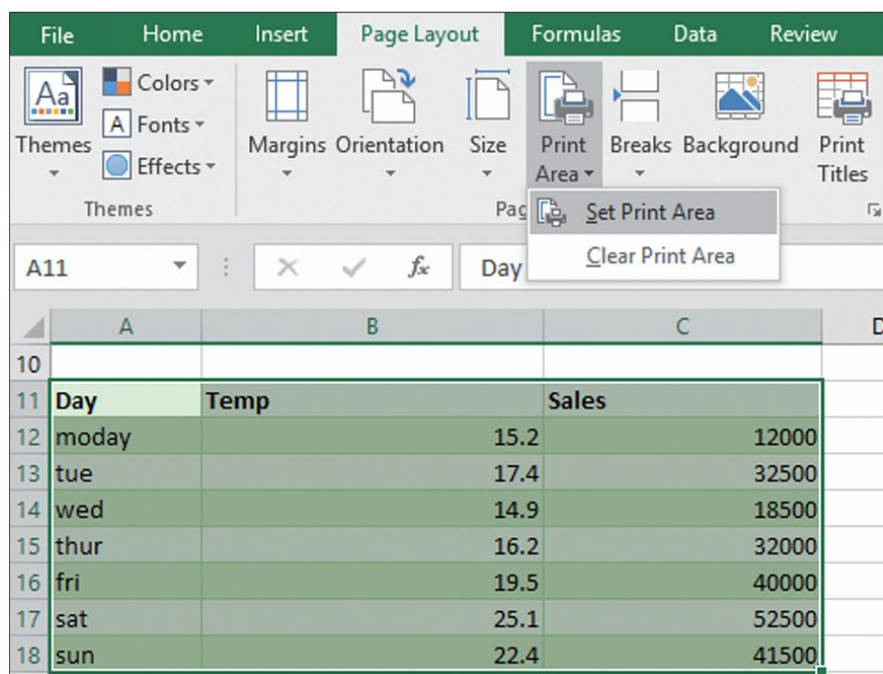
This is one or more ranges of cells that you designate to print when you don't want to print the entire worksheet. This means only the print area set is printed. A worksheet can have multiple print areas. Each print area will print as a separate page.

Use the steps given below to specify print area

Step 1: Create or open an existing worksheet.

Step 2: Select the cells you want, define as print area (data range).

Step 3: On the **Page Layout** tab, in the **Page Setup** group, click **Print Area**, and then click **Set Print Area**.



Picture 3. 28. Excel window showing how to set Print Area from Page

Note: A line around the data marks the print area. When you save a worksheet the print area is also saved. Use the same steps given above to **add** and **clear** the print area.

Application activity 3.4

1) Open an Excel worksheet and make the following table which you save as StudentMarks.

Student Name	History/20	Geography/20	French/20
Rukundo Jean	18	12	9
Uwimana Diane	20	18	20
Habimana Samuel	9	17	18
Mukamana Claudine	17	21	11
Nkurunziza Patrick	14	11	9
Nyirahabimana Alice	14	14	16
Nsengiyumva Eric	5	15	14
Uwamahoro Chantal	20	19	11
Bizimana Emmanuel	11	10	12
Mukeshimana Beatrice	18	15	12
Nishimwe Jean Claude	17	9	8
Muryango Aline	19	13	13
Ndayishimiye Theogene	8	16	16

a) Set custom margins to 1 inch on all sides. Print preview the document and describe how the layout changes.

b) Insert a header with your school name and a footer with today's date.

c) Change the view to Page Layout view. What differences do you see?

3.6. End unit assessment

3.6.1. Project activity

Project Title: Class Performance Analysis

Guidance: Your task is to help your class teacher analyze the performance of students in all the subjects you learn in S3. Instead of just looking at a list of numbers, you will create a charts that makes the results easy to compare and present to the school administration. You will use your skills in data entry, chart creation, and page layout to produce a professional report. Your work will use the classroom's report that you will enter in Excel sheets.

- Do a data entry of all the S3 students' reports. To make the work quick and easy, every student can do data entry of his/her report then they will be pasted in one worksheet.
- Create charts to help in the analysis of the students' performance. The charts will cover:
 - Analysis of marks in 3 groups such as best performer, middle performers and last performers
 - Comparison of subjects depending on how students performed (in those subjects).
 - Comparison of how students performed in Science, mathematics and Humanities subjects
 - Comparison of classes performance using the class's mean

Make a word document and put your analysis charts in it. Write some text to accompany your charts in the document you have created.

Questions on the Project:

- 1) Why are charts more effective than a table when comparing the performance of many students?
- 2) Explain how you created the word document

3.6.2. Assessment questions

- 1) State two reasons why charts are preferable to tables of numbers.
- 2) Name any two elements of a chart.
- 3) What is a print area in Excel?
- 4) True or False: Headers and footers are visible in Normal view.
- 5) Open the StudentMarks file created in the previous activity (application activity 3.3) and do the following:
 - a) Create a column chart for the StudentMarks data.
 - b) Change the column chart created above into a line chart.

- c) Format StudentMarks data as a table and add a total row.
- 6) What risks might there be if a chart is beautiful but not formatted correctly?
- 7) Why is it important to adjust page setup before printing a worksheet?
- 8) Why is it important to include headers and footers in printed worksheet

EXPERIMENTAL VERSION

UNIT 4: PRESENTATION



Key unit competency:

Create a presentation using different PowerPoint tools

Learning objectives

At the end of this unit, you will be able to:

Knowledge and understanding:

- Explain how to insert objects into slides.
- Distinguish different presentation views and their purpose.
- Describe how slide formatting enhances the readability of a presentation.
- Explain the steps to create different presentation effects

Skills:

- Organize inserted objects to enhance message readability and maintain slide aesthetics.
- Arrange and format presentation slides.
- Customize the Slide Master and Handout Master to apply consistent formatting.
- Use various slide presentation modes to display the presentation.

Attitudes and values:

- Appreciate the importance of a good design and proper sequencing of slides in conveying the intended message

4.0. Introduction

This unit introduces presentation software with special emphasis Microsoft PowerPoint. It explains what presentations are, why they are important, and how they help in communicating ideas clearly to an audience. The unit covers working with slides, inserting and managing different objects such as textboxes, images, tables, charts, SmartArt, shapes, and media clips. In addition, it guides learners on formatting slides, applying designs and layouts, using different presentation views, adding animations and slide transitions, and finally presenting effectively using a computer and projector.

Introductory activity

- 1) Why do people use presentations when speaking to an audience?
- 2) Which computer programs can be used to present information?
- 3) Why do presentations use many slides instead of one long page?
- 4) Why are pictures and charts important in a presentation?
- 5) How can sound or video help people understand a message better in a presentation?
- 6) Why is it important to organize slides in the correct order?

A presentation is an organised report or message prepared as a talk before audience, with the help of a computer program. Examples of programs/software which you can use to create presentations include among others the following: Microsoft PowerPoint, Harvard Graphics, Corel Presentations, Open Office .org Impress, Lotus Freelance Graphics, Kingsoft Presentations, etc.

Presentation software is a program used to create slide shows for presentation on screen to an audience. Presentation program/software can supplement or at times can replace use of visual aids like handouts, chalkboards, posters, etc.

PowerPoint presentations can be done on computer screen if the audience is very small. If the audience is large the computer can be connected to a projector that projects the image onto a large screen/wall. When using Microsoft PowerPoint, the data file you create is known as a Presentation.

4.1. Introduction on presentation

Activity 4.1

- 1) Give examples of situations where a presentation is more effective than a simple explanation?
- 2) What steps do you usually follow when opening a program on a computer?
- 3) Why is it important to save before closing a presentation?
- 4) What could happen if you use very small text in a presentation slide?
- 5) What is the first step to launch Microsoft PowerPoint?
 - a) Double-click on the desktop background
 - b) Click Start button and search for PowerPoint
 - c) Press Ctrl + O on the keyboard
 - d) Select File → New
- 6) After clicking the PowerPoint icon, what should you select to start a new presentation?
 - a) Save As
 - b) Blank Presentation
 - c) Open Presentation
 - d) Slide Show

4.1.1. Role of presentation applications

In many professional and educational settings, presenters use techniques to communicate ideas clearly for better understanding. The following points highlight the role of using presentations software:

- Conveying information about an important issue such as disease control measures.

- Introducing a new idea for business.
- Reporting progress using charts or graphs.
- Training by demonstrating how it is done by using screen recorder or animation programs.

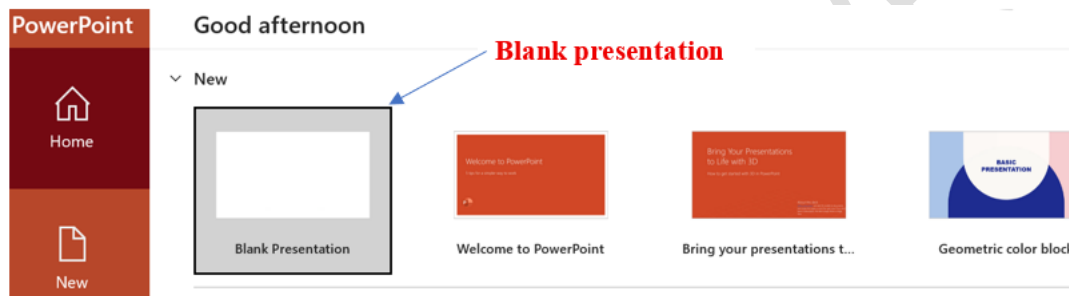
4.1.2. Creating, saving and opening a presentation

a. Launching PowerPoint

Follow these steps to launch Microsoft PowerPoint presentation:

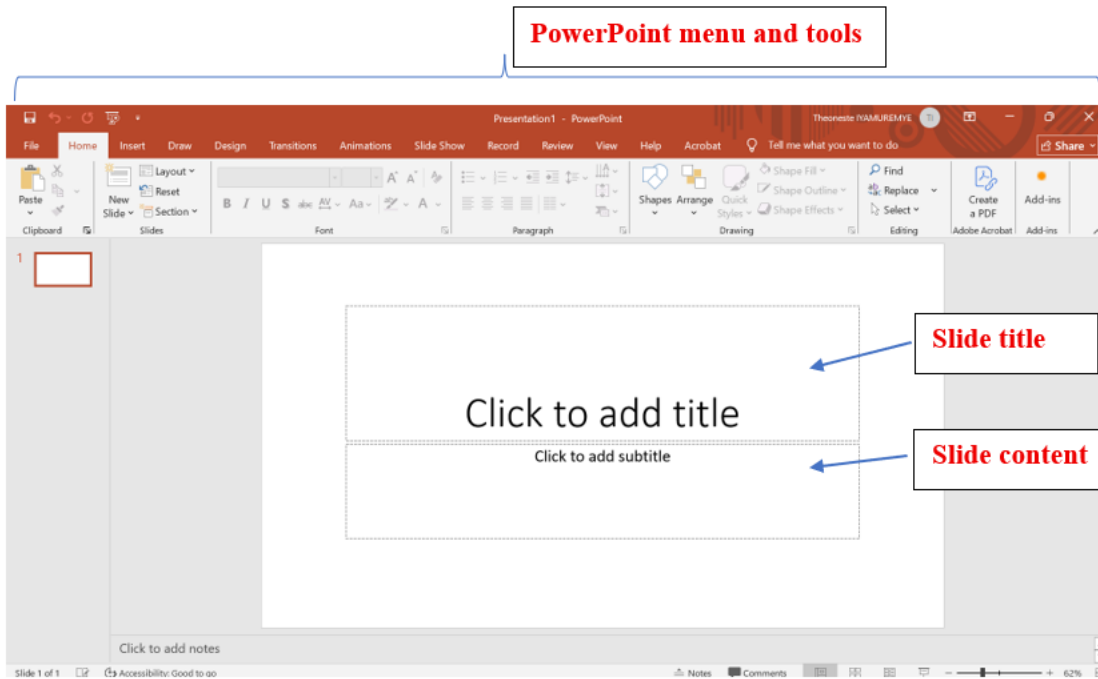
Step 1: Click Start button and search for PowerPoint by writing it in the search box of the new window that appears.

Step 2: Click on the PowerPoint icon  and click on **Blank Presentation**.



Picture 4. 1. Different PowerPoint templates

Immediately the new PowerPoint presentation window gets opened and it looks like the one below:



Picture 4. 2. PowerPoint main window

Step 4: You can now add a title in the Slide title area and add text in the Slide content area

a. Writing in PowerPoint

As shown in the previous image, there are two main areas in PowerPoint where to write:

- a) In the **Slide title** area write the title of the slide. The title must be short and clear so that it makes the reader or the audience have an image of what is going to be presented. Unlike Word, PowerPoint requires a big size of characters for better readability. The title size should not be below 32. It is good to make it bold and colored to attract attention.
- b) In the **Slide content** area write the content to be presented. Here are general rules to observe when making a PowerPoint presentation:
 - Make sure there is only one idea per slide. If one idea can not fit, have many slides with that one idea but with the same title. Add some text to the other slides such as “**Continued**” so that the audience can differentiate them with the first slide.
 - **Write short text:** A PowerPoint presentation only shows the main ideas not all the explanations. The best way is to write ideas in bulleted list. Have in mind the explanation to give for each idea or write them in the **Note** area.
 - **Use big fonts:** Use a font that will be readable from a distance.
 - **Use animations and slide transitions:** These will help attract the audience attention.

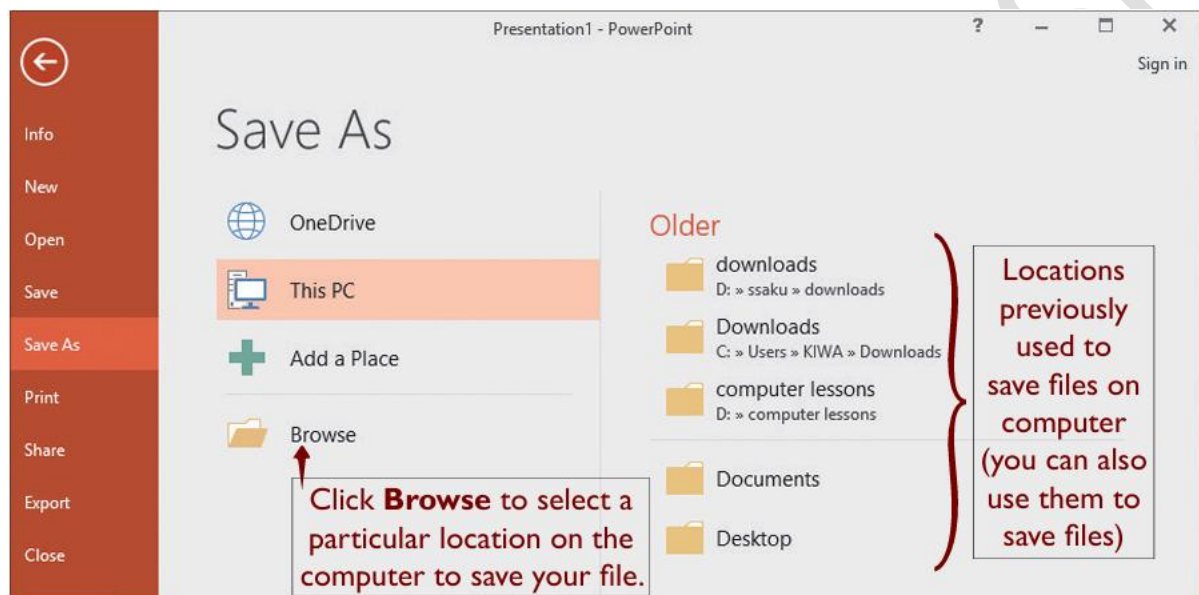
b. Saving a PowerPoint presentation

When you open MS PowerPoint by default you have created a presentation. Now add some content to create a presentation on any topic of a subject you choose.

Now follow the steps below to save that presentation:

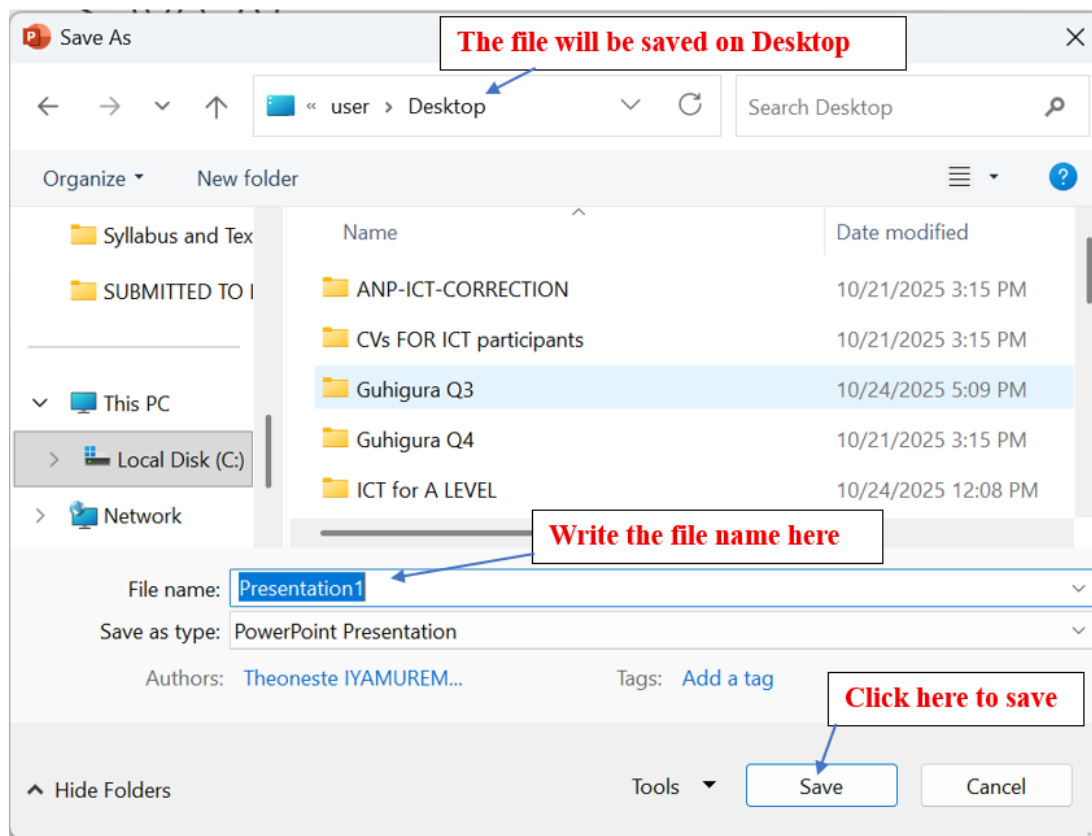
Step 1: On the **File** tab, click **Save As**.

Step 2: Select a location or browse a location where to save your file. The **Save As** dialog box displays.



Picture 4. 3. Window for browsing save locations

Step 3: Type in the desired file name and then click **Save**.



Picture 4. 4. The Save As dialog box

Step 4: Close the application window or file created.

Note:

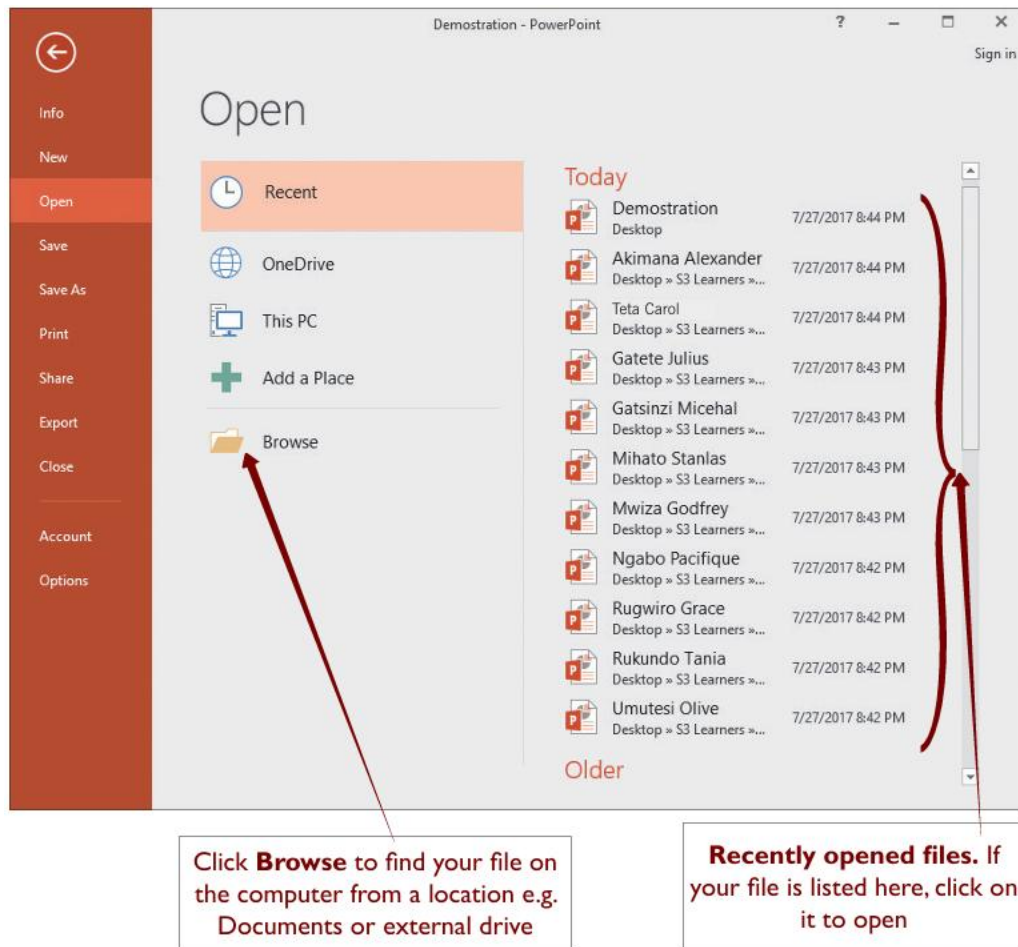
- You can also save by pressing **Ctrl + S** on keyboard or clicking Save button  on the **Quick Access Toolbar**.
- By default, PowerPoint saves files in the PowerPoint Presentation (.pptx) file format. To save in another file format, click on the Save As type list and select the file format you want.

c. Opening an existing presentation

Follow the steps given below to open an existing presentation:

Step 1: Start Microsoft PowerPoint program.

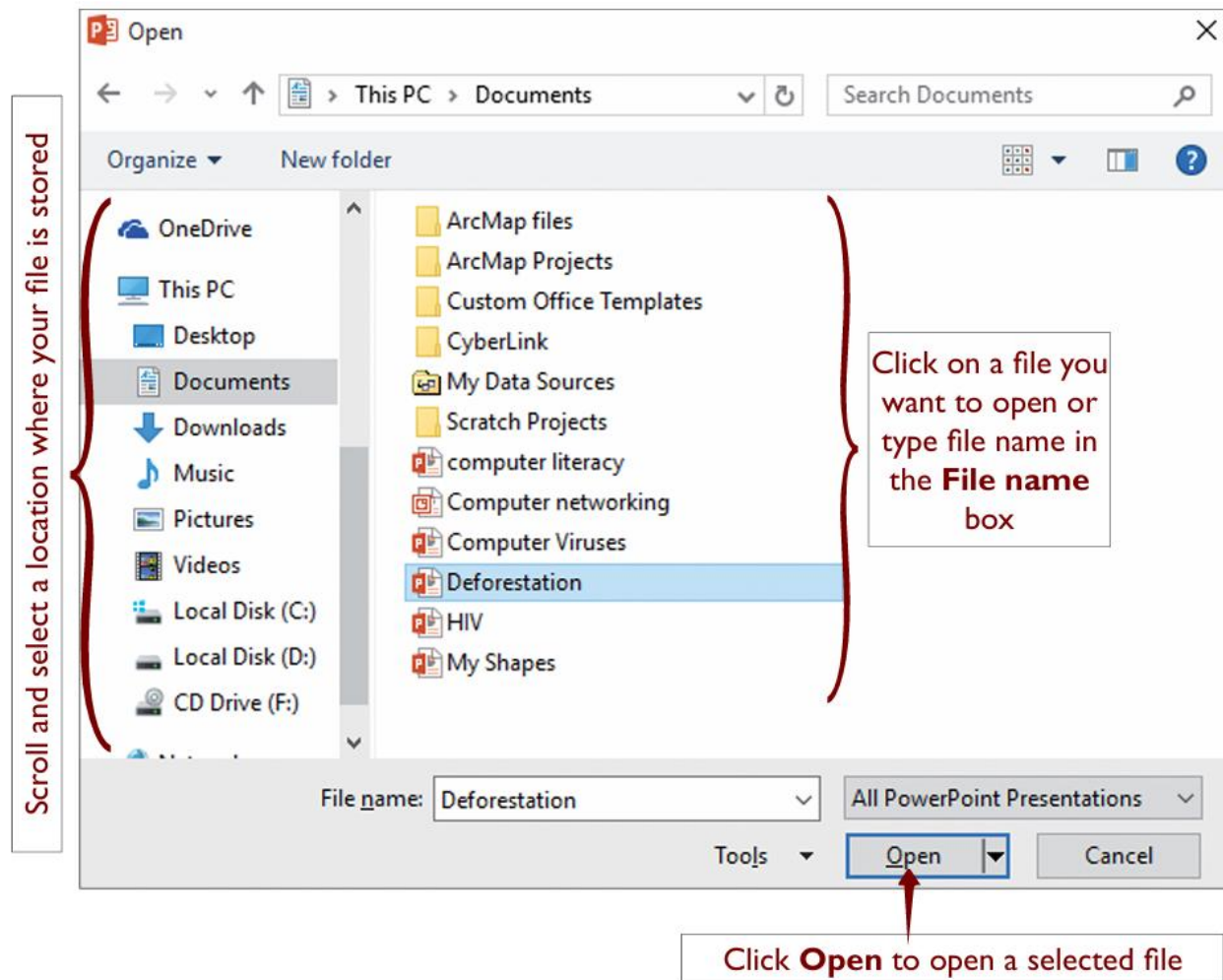
Step 2: On the **File tab**, click **Open**. The Open Backstage view opens. See picture 6.8



Picture 4. 5. The Open Backstage view, showing recently opened files and locations

Step 3: Select a location where your file is located or **browse** in folder to find the file. In the **Open dialog box**, in the **File name** box, type the name of the file you want to open (or scroll, locate and double click on the file).

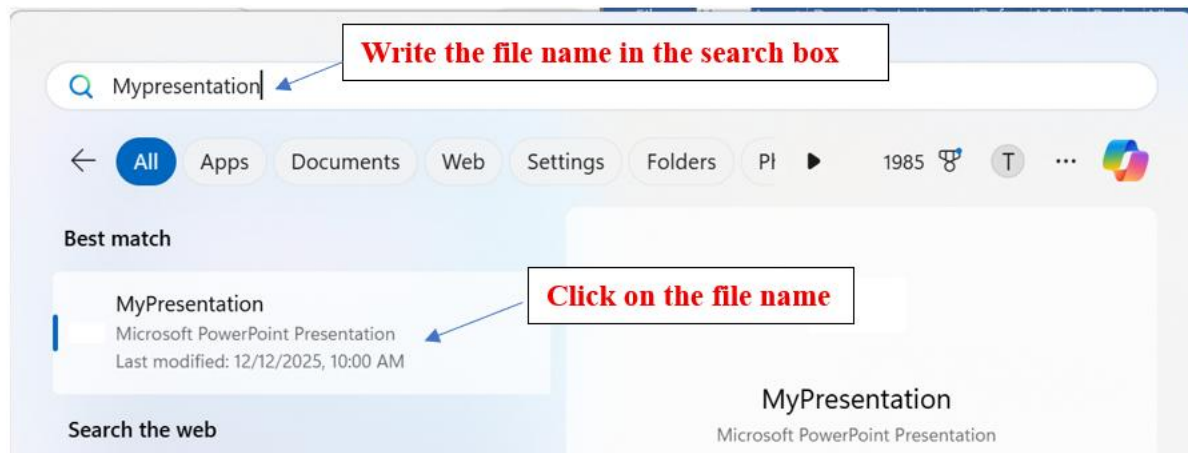
Step 4: Click **Open button** and the file will open.



Picture 4. 6. The Open file dialog box

Note:

- You can quickly access the window for browsing where files to open are located by pressing **Ctrl + O** on the keyboard.
- You can also open a file by clicking on **Start** then write the file name in the **Search** box and pressing **Enter** or double clicking on the file name.



Picture 4. 7. Opening a file using search box

Application activity 4.1

- 1) What is the difference between the **Slide title area** and the **Slide content area** in PowerPoint?
- 2) State the steps to launch Microsoft PowerPoint from the Start menu.
- 3) Which of the following is **NOT** a role of presentation software?
 - a) Conveying information about disease control
 - b) Introducing a new business idea
 - c) Cooking food for an audience
 - d) Reporting progress using charts
- 4) To open an existing presentation, which tab should you click first?
 - a) Home
 - b) File
 - c) Insert
 - d) Design
- 5) a) Create a new PowerPoint presentation and add a slide with the title “**MY LIFE TARGET**”. Write three bullet points in the content area.
 b) Save your presentation as “Mytarget”.
 c) Close PowerPoint, then reopen the file.

4.2. Slides

Activity 4.2

- 1) What is the difference between a slide and a presentation?
- 2) What are the steps the steps to follow to insert a new slide in PowerPoint?
- 3) What is the purpose of the Slide Navigation Pane?
- 4) Create a presentation about Environmental Protection in Rwanda and save the presentation as “Nature protect”.

A slide is a single page of information in a presentation created with programs such as PowerPoint or Corel Presentation. A presentation is made up of one or several slides.

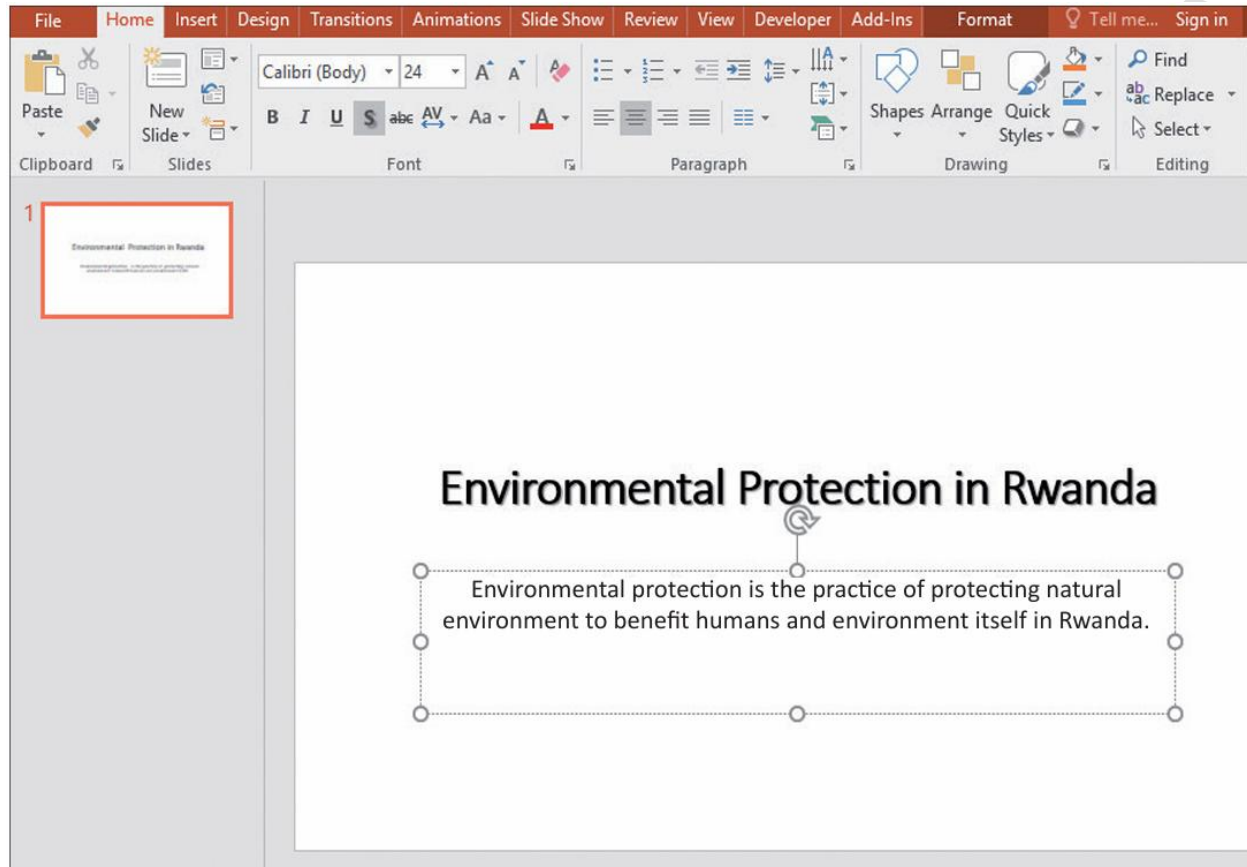
4.2.1. Insert slide

- a) Use the following steps to create the presentation

Step 1: Open a blank presentation. The presentation opens with only one slide.

Step 2: Type the title as Environmental protection in Rwanda.

Step 3: Type in the **Slide content** area the content about Environment Protection in Rwanda



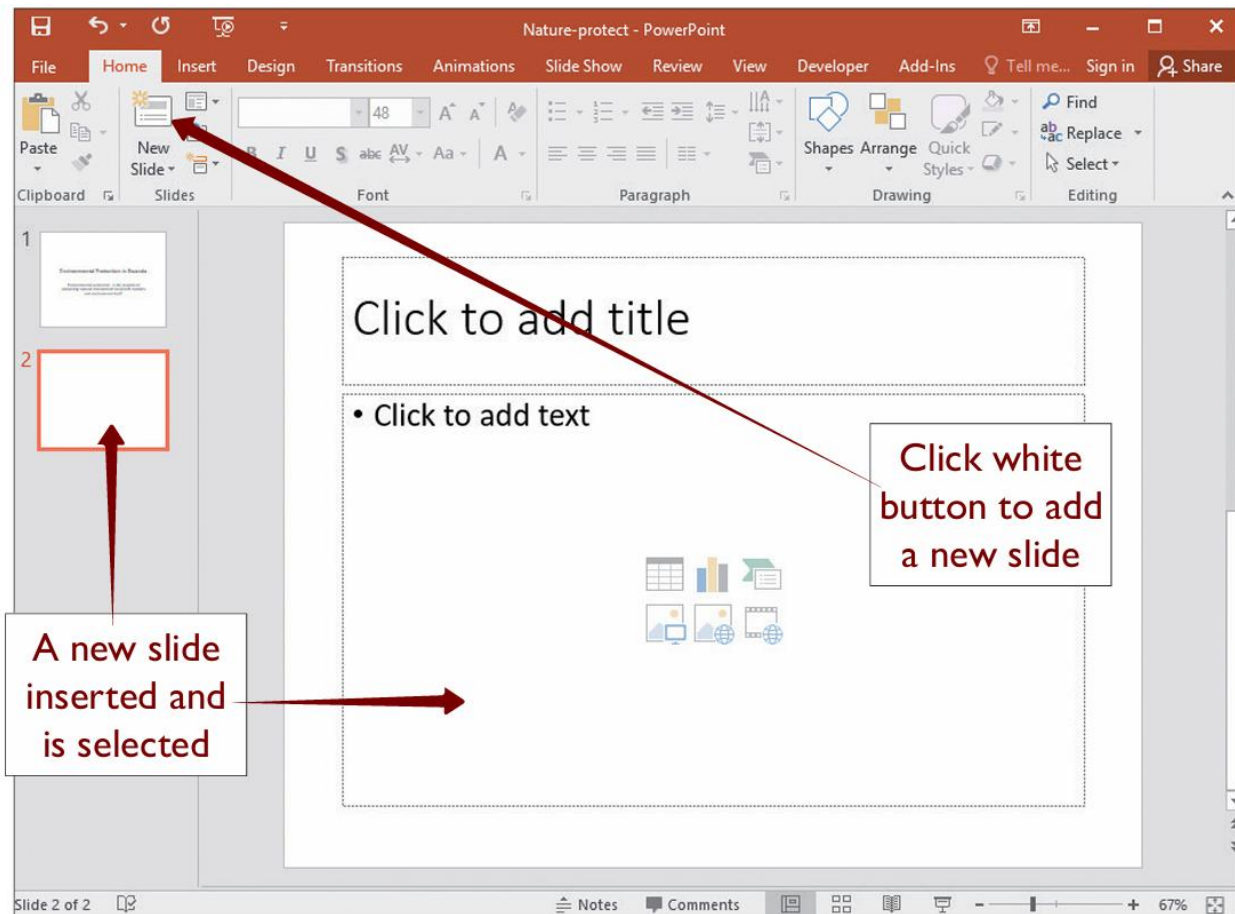
Picture 4. 8. A PowerPoint presentation with one slide

b) Insert new slides by following these steps:

After working on the first slide, you need more slides to complete your presentation. To add new slides, do the following:

Step 1: On **Home** tab, click **New Slide** button (or just press **Ctrl + M** on keyboard).

Step 2: Add some text in the second slide and the other slides that you will add



Picture 4. 9. The presentation after adding a new slide

4.2.2. Slide navigation pane

The Slide Navigation Pane is the vertical panel on the left side of the PowerPoint window that shows thumbnail previews of all your slides in order. It is used for organizing and managing a presentation.

Uses of the navigation pane:

- **View all slides at once:** You see small versions (thumbnails) of every slide, making it easy to understand the flow of a presentation.
- **Selecting one, many or all slides:** To select one slide just click on a slide in the navigation pane. To select many slides, click on one slide then press the **Control key** and click another slide.

4.2.3. Slide manipulation

Slide manipulation refers to all the actions you perform to manage, organize, and modify slides in a PowerPoint presentation.

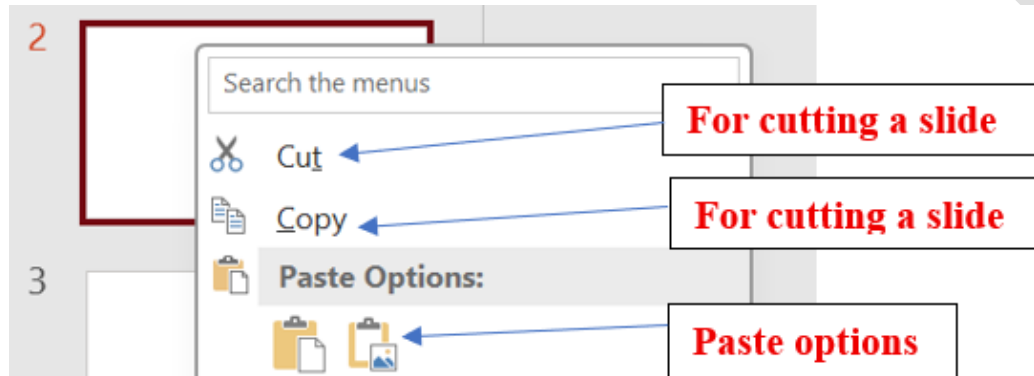
a. Cutting and copying slides

Copying is taking a copy of text while that text remains in its original place while pasting is putting that text somewhere else.

Follow these steps to cut or copy slides:

Step 1: In the **Slide Navigation Pane**, click (select) on the slide you want to cut or copy.

Step 2: Do a right click and choose **Cut** (for cutting a slide) or on **Copy** (for copying a slide). For the Cut option, the slide gets immediately removed from the other slides.



Picture 4. 10. Cut, Copy and Paste options

b. **Pasting slides:** This is an action of inserting slides copied or cut from one place and pasting them to another place within the same presentation or in a different presentation. Pasting a slide must be always preceded by cutting or copying it.

To paste a slide, click where to put it, do a right click and click on **Paste**.

c. **Rearrange slides:** This is changing the order of the slides. You can click and drag slides up or down to change their order. For example the second slide can be moved from the second position to the fourth.

To drag a slide just click it, press and hold down the left button and move the mouse up or down depending on where you want to move the slide (up or down). You can also use the option of cutting a slide and pasting it where you want it.

d. **Add or delete slides:** To delete a slide right-click it then click on **Delete Slide**. You can also just select it in the navigation pane and press the **Delete key** on the keyboard.

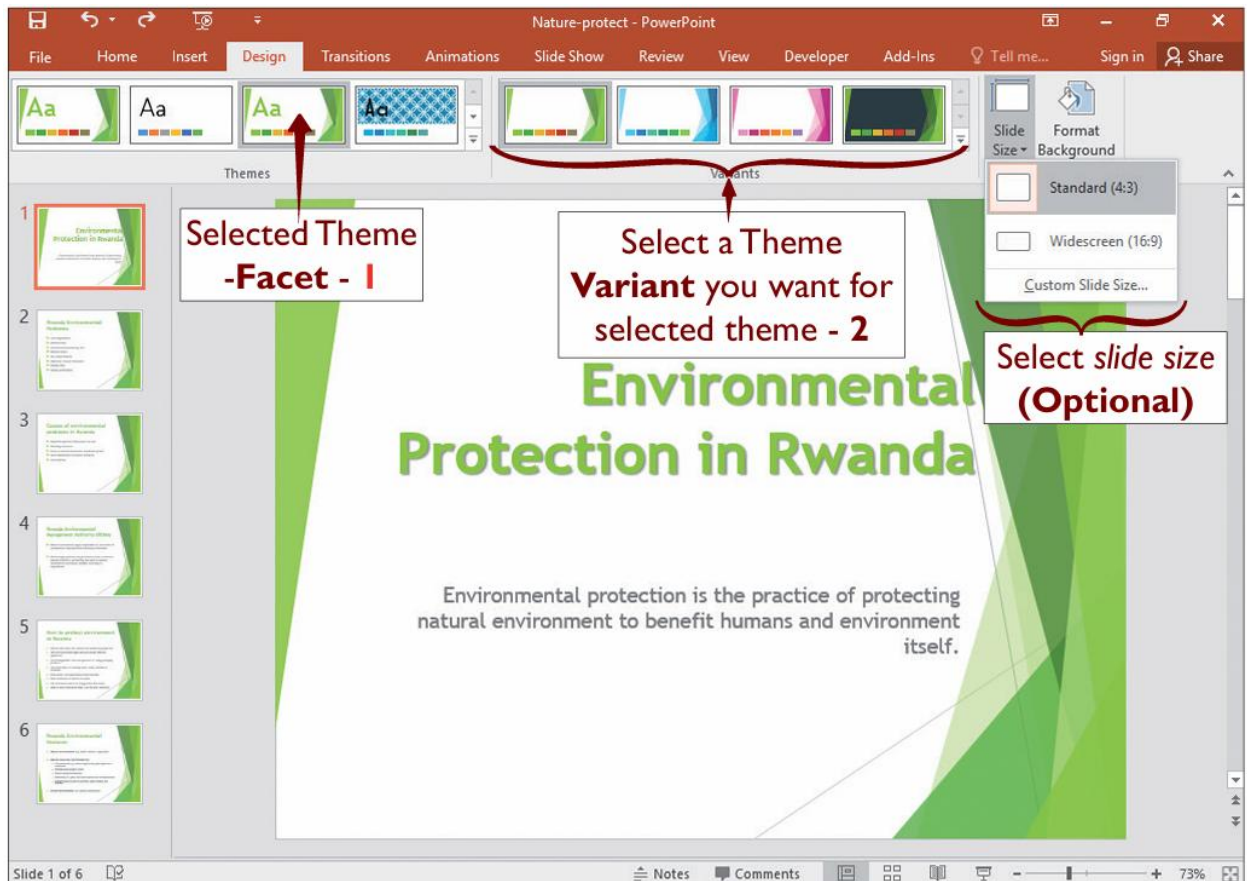
e. **Duplicate slides:** You can have a copy of the same slide by doing a right click and clicking on **Duplicate Slide**.

f. **Apply new settings to one or many slides:** By selecting one or all the slides then selecting the functionality or appearance to apply, the slides immediately change. For example you can change the design of all slides by following these steps:

Step 1: Click on the Design tab.

Step 2: From the Themes group select a theme you want.

Step 3: Select a Theme variant in case you need it and then save changes to your presentation.



Picture 4. 11. A presentation where a (Design) Theme has been applied

Application activity 4.2

- 1) Create a new presentation on “The importance of AI”
- a) In the content area of the first slide write the introduction on that topic. Write the title as “Introduction on AI”
- b) Insert other slides for the development of the topic. For each slide write the title and the content. The content must be written in a bulleted format
- c) Duplicate the second slide
- d) Cut the second slide and paste it in a new PowerPoint presentation you open.

4.3. Objects

Activity 4.3

- 1) In PowerPoint, an object is:
 - a) Only text typed on a slide
 - b) Anything you can insert, move, resize, or edit
 - c) Only images and videos
 - c) The background design of a slide

2) Which tab is commonly used to insert objects such as images and shapes in a PowerPoint presentation?

- a) Home b) Insert c) Design d) View

3) Which one is not an object in PowerPoint?

- a) WordArt b) SmartArt c) Transition d) Chart

4) Why is it good to use pictures, tables, or charts in a presentation instead of only text?

5) Explain how audio or video clips can make a presentation more understandable?

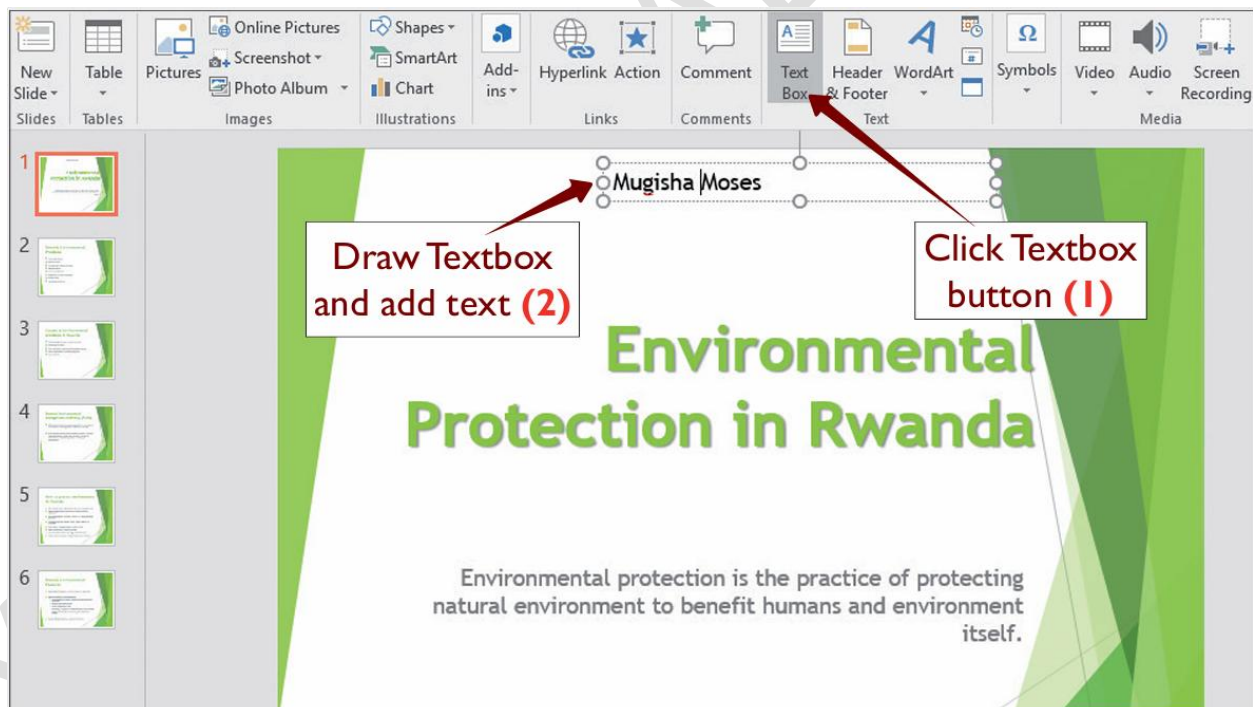
In PowerPoint, an object is anything you can insert, move, resize, or edit on a slide. They are like the building blocks that can be put in a slide.

There are several objects that can be inserted in presentation software. These include: Textbox, Images, Table, Clip Art, Word Art, Smart Art, Shapes, Charts and Media Clips.

4.3.1. Textbox

This is a box in the program from where the user can type or edit text, dates or numbers. A textbox can be drawn and used to place text in a particular location in a presentation.

Step 1: On the **Insert** tab, click **Textbox** command from the **Text** group. The pointer changes to an upside down cross.



Picture 4. 12. A presentation where a textbox is added in the upper middle of the slide

Step 2: Move the pointer to the location where you need to draw the textbox and then draw it. Automatically the insertion point appears in the textbox to indicate where text will appear when you type it.

Step 3: Type in your information in the textbox. If you move the textbox in another place, the text will move together with the textbox.

4.3.2. Images

An image is the actual picture or mental picture of something. Adding pictures can make your presentation more interesting and attractive. You can insert in a slide a picture from a file on your computer on or a picture from the internet.

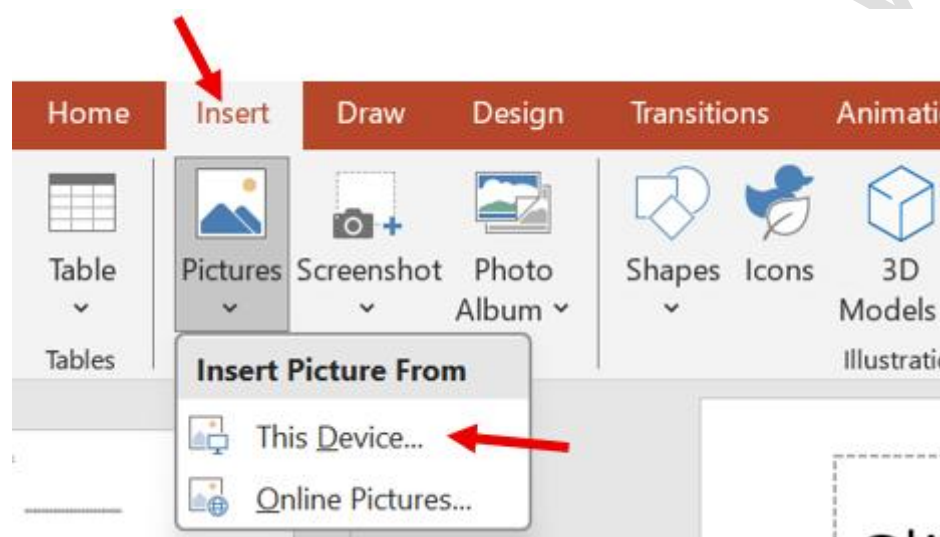
a. Inserting a picture from the computer storage

To insert a picture in a presentation do the following:

Step 1. Select a slide where you need a picture.

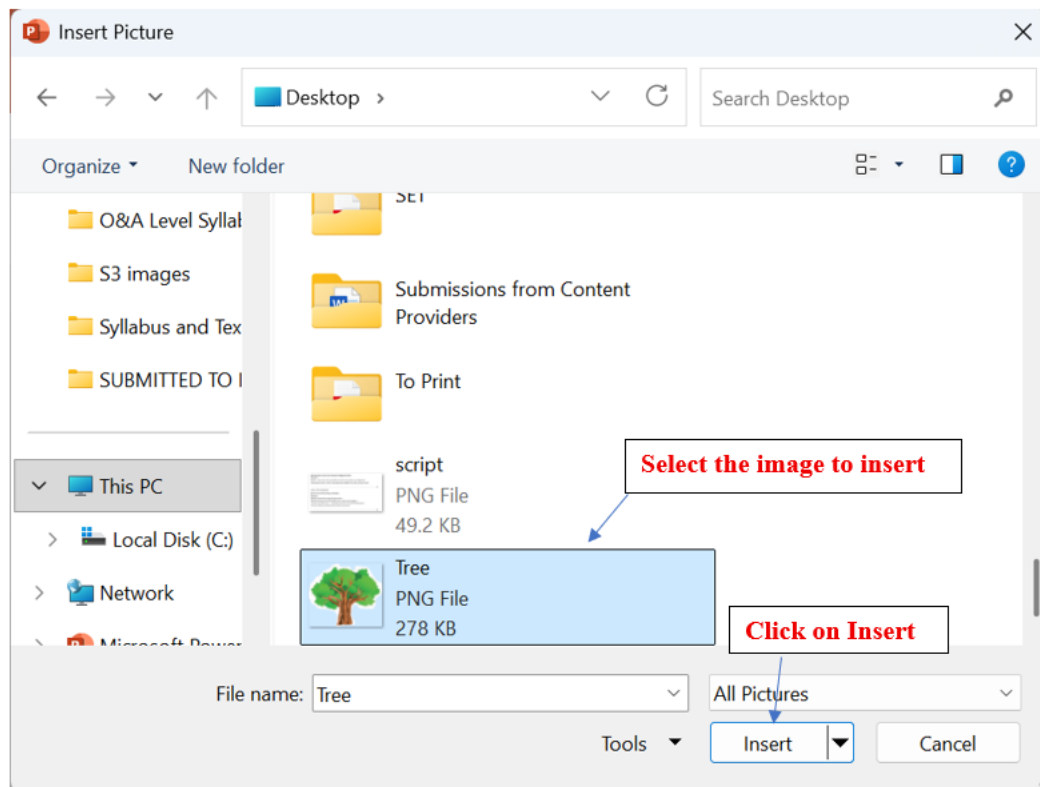
Step 2: Select the **Insert** tab, then click the **Pictures** command in the **Images** group.

Step 3: Click on **This Device** if the image to insert is on the computer memory. If the image to insert is online (from the internet), click on **Online Pictures**.



Picture 4. 13. Inserting pictures from the computer

Step 4: Browse the image from the computer and when found select it and click on **Insert**.



Picture 4. 14. Browsing and inserting selected image

The picture will display on the currently selected slide. Repeat these steps if you want to insert other images. You can also use another process by which you can just copy a picture and then click in the slide where to put it then do a right click and click **Paste**.



Picture 4. 15. The slide after inserting an image

b. Inserting online pictures

In case you don't have a desired picture on your computer, you can find a suitable picture online and then add it to your presentation. In PowerPoint, Online pictures are used instead of ClipArt that exist in earlier versions of the program.

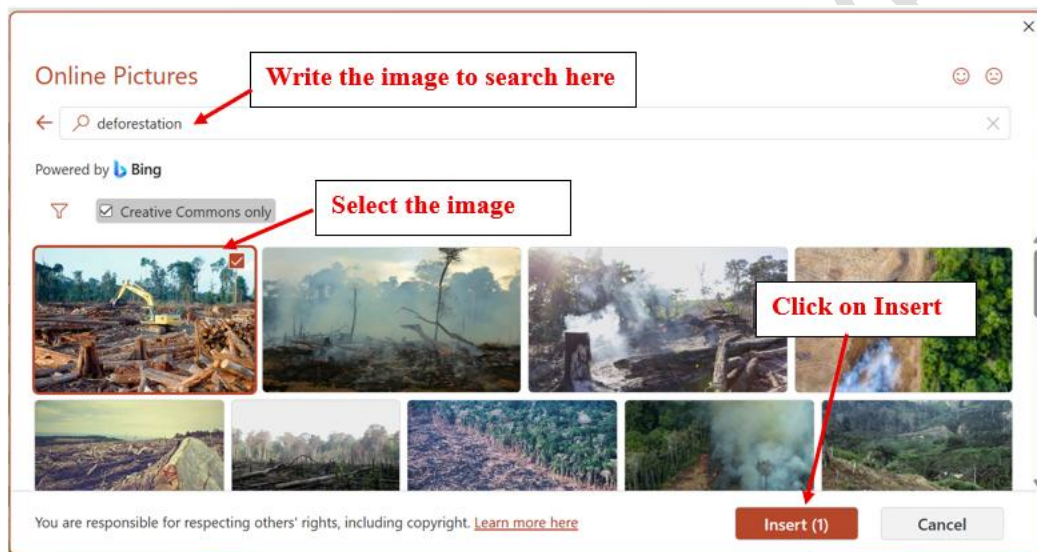
Use the following steps to add online pictures:

Step 1: Click in the slide where the online picture will be inserted.

Step 2: On the **Insert** tab, in the **Images** group, click **Online Pictures** command.

Step 3: In the Search box, search for the image to insert and hit the Enter key. In this case the image searched is "apple".

Step 4: Once the image is found select it and click on **Insert**.



Picture 4. 16. Inserting online picture

Once the image is inserted resize it so that it fits in the slide.

Rwanda Environmental Problems

- Land degradation
- Deforestation
- Wetland and biodiversity loss
- Natural causes
- Man-made disaster
- Reduction of water resources
- Human settlements



Picture 4. 17. Slide after inserting an online picture

How to resize a picture:

Follow these steps to resize an image:

Step 1: Click on the picture. The picture will display selection circles **on its borders**

Step 2: Press and hold down the left button then move the cursor depending on how you want the image to be resized.

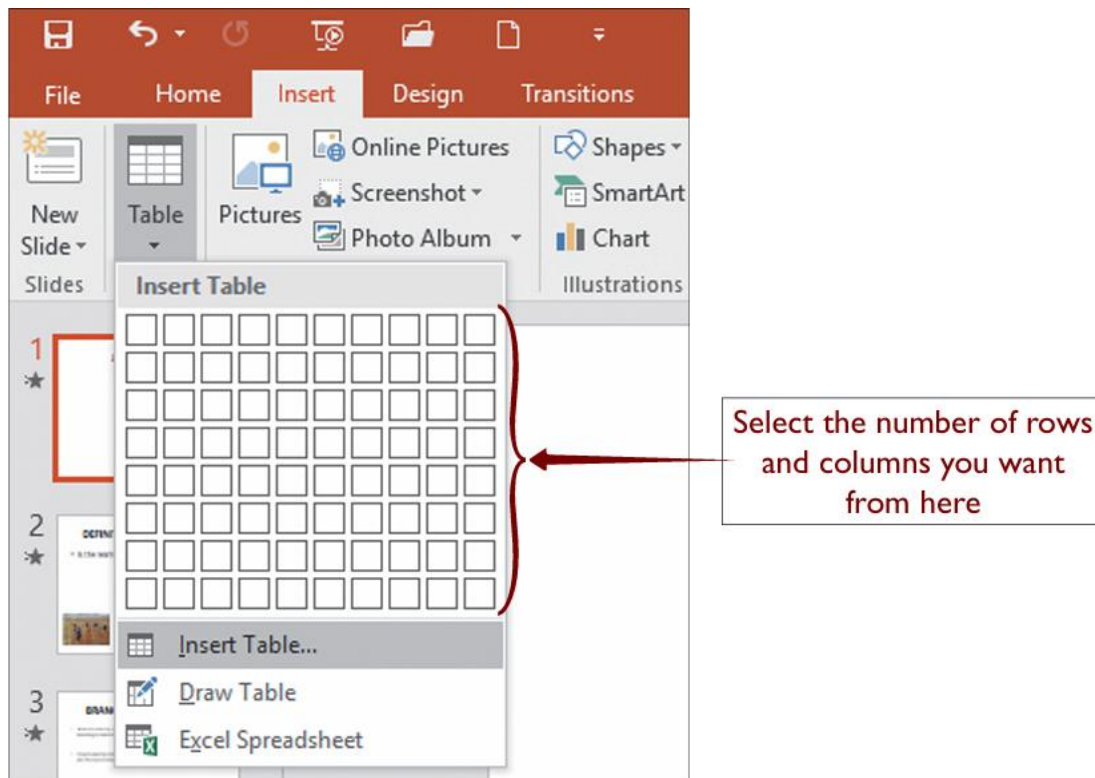
4.3.3. Table

A table is made of rows and columns. A row is a space between two horizontal lines in a table while a column is a space between vertical line. Data or information can be arranged using these rows and columns. A table can be inserted in PowerPoint by using one of the two methods.

a. Method 1

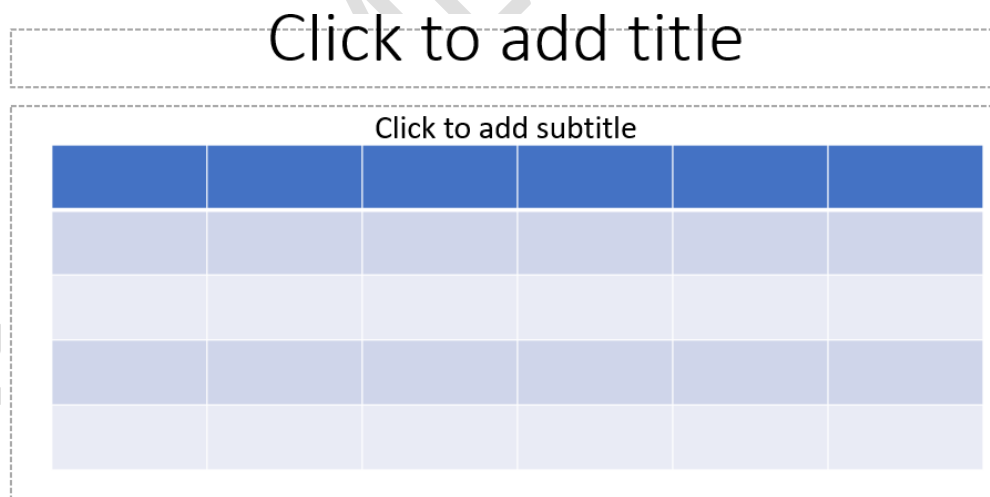
Step 1: Select the slide that you want to add a table to and click where the table will be inserted

Step 2: On the **Insert** tab, select **Table** then select the rows and columns that will make the table.



Picture 4. 18. The options for inserting a table

Step 4. Now the table is inserted but needs resizing and moving it to the right place. To resize it select it then press and hold down the left button then move the cursor depending on how you want to resize it. To move the table, select it so that the cursor becomes a cross with arrows, press and hold down the left button then move the mouse.



Picture 4. 19. A table with 6 columns and 5 rows inserted in PowerPoint

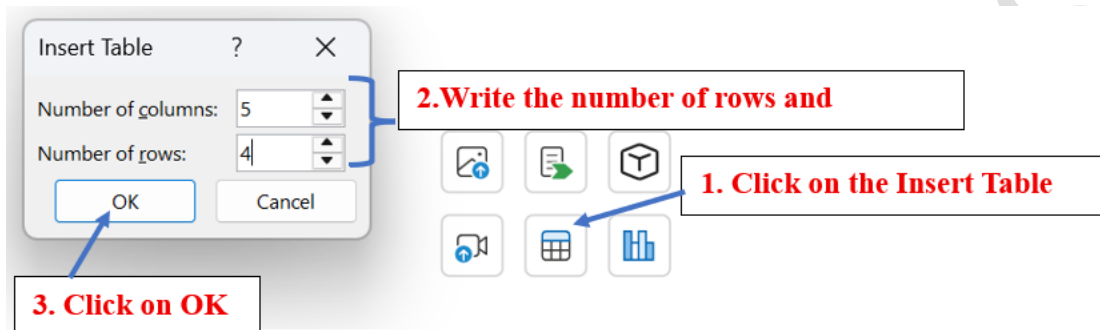
Step 5: To add text to the table cells, click a cell and then enter your text. Afterwards click outside the table.

b. Method 2

Step 1. Click where the table will be inserted

Step 2. In the Slide content area, click on the **Insert table** tool.

Step 3. Write the number of rows and columns and click on **OK**. In this case the number of columns written is 5 while rows are 4. Immediately the new table is inserted.



Picture 4. 20. Options to insert a table using the Insert Table tool

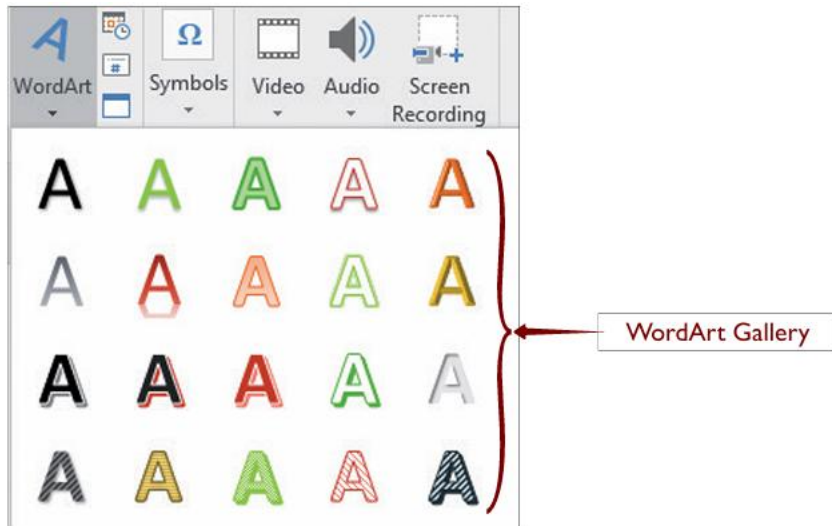
4.3.4. WordArt

Word art is a text with a special design or look. It is an object that you can move, resize or rotate within the document. Word art is applied to text to appear in a unique form such as a circle, curve or oval.

Follow the steps given below for adding WordArt:

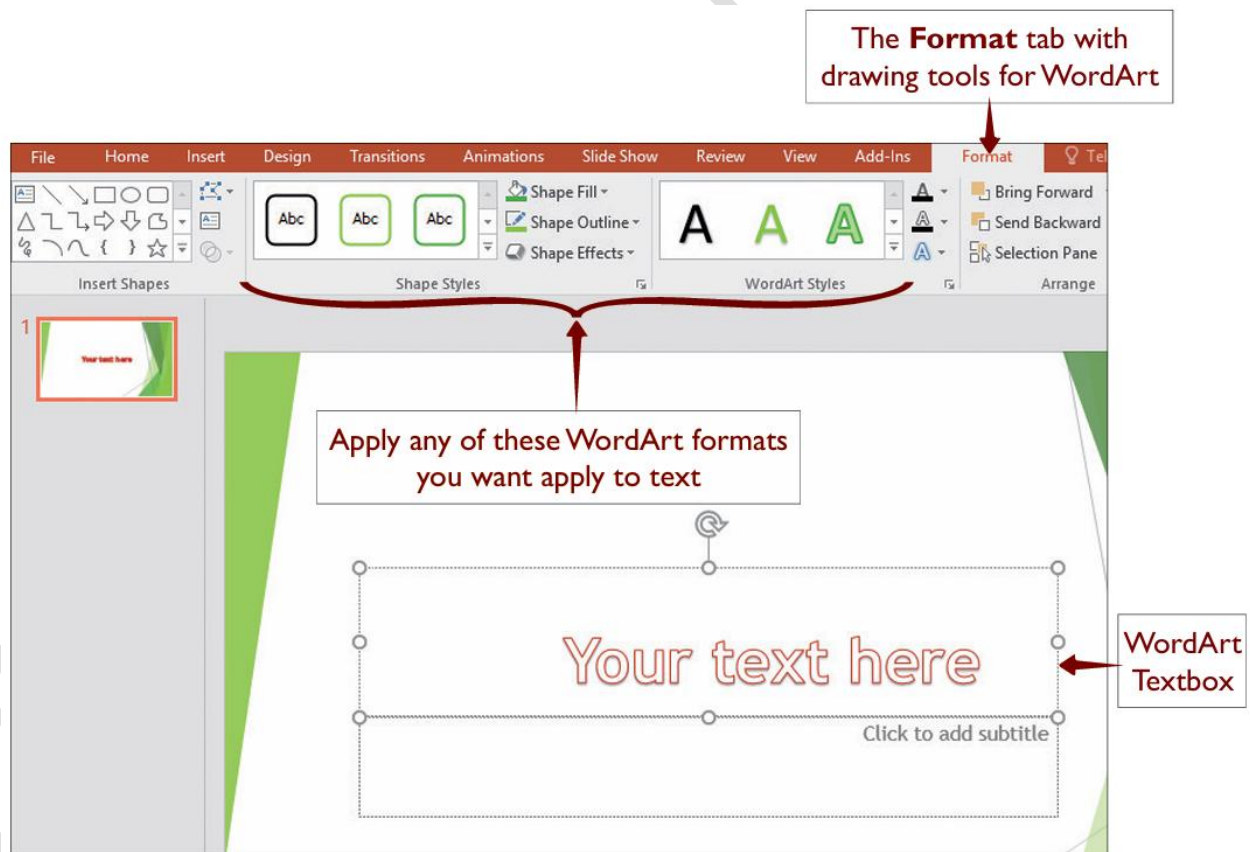
Step 1: Select the slide where you want to insert a WordArt.

Step 2: Click on the **Insert tab** and then click on the **WordArt button** in the **Text group**. A gallery of WordArt styles is displayed as shown below.



Picture 4. 21. WordArt gallery

Step 3: Select a WordArt style you want to use. The WordArt textbox will immediately be inserted in the slide. You may then start writing the text.



Picture 4. 22. A slide with WordArt textbox added

Step 4: Move the WordArt in suitable location. And then delete the original text and textbox.

Step 5: Repeat steps 1-4 and apply a desired WordArt to all slide titles.

Step 6: Click on the **WordArt textbox** and then type the desired text.

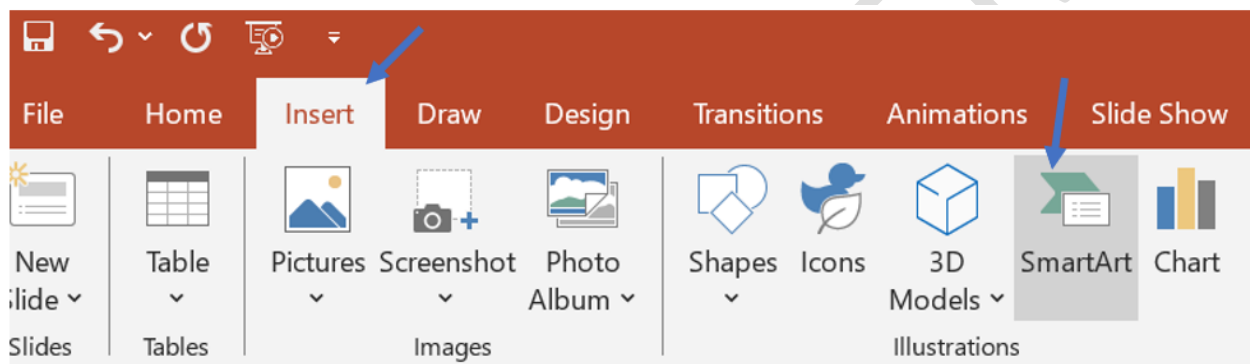
4.3.5. SmartArt

SmartArt is a picture used to communicate information in many different ways. It is an option that allows the user to create diagrams easily.

Use the steps given below to insert a SmartArt graphic to a presentation:

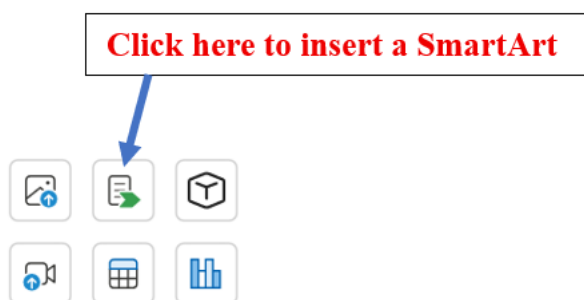
Step 1: Open PowerPoint presentation and click where the SmartArt will be inserted.

Step 2: On the **Insert tab**, in the **Illustrations group**, click **SmartArt**.



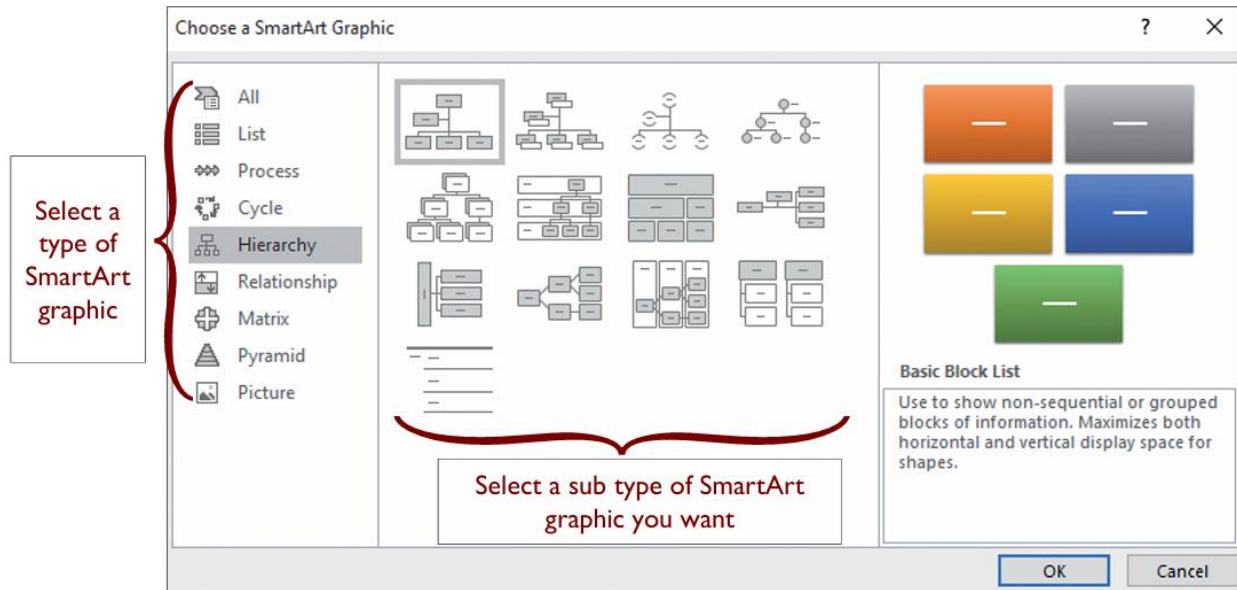
Picture 4. 23. The SmartArt command in the Illustrations group

Note: Alternatively insert a SmartArt by clicking where it will be inserted then click on SmartArt Graphic button which is shown in the image below.



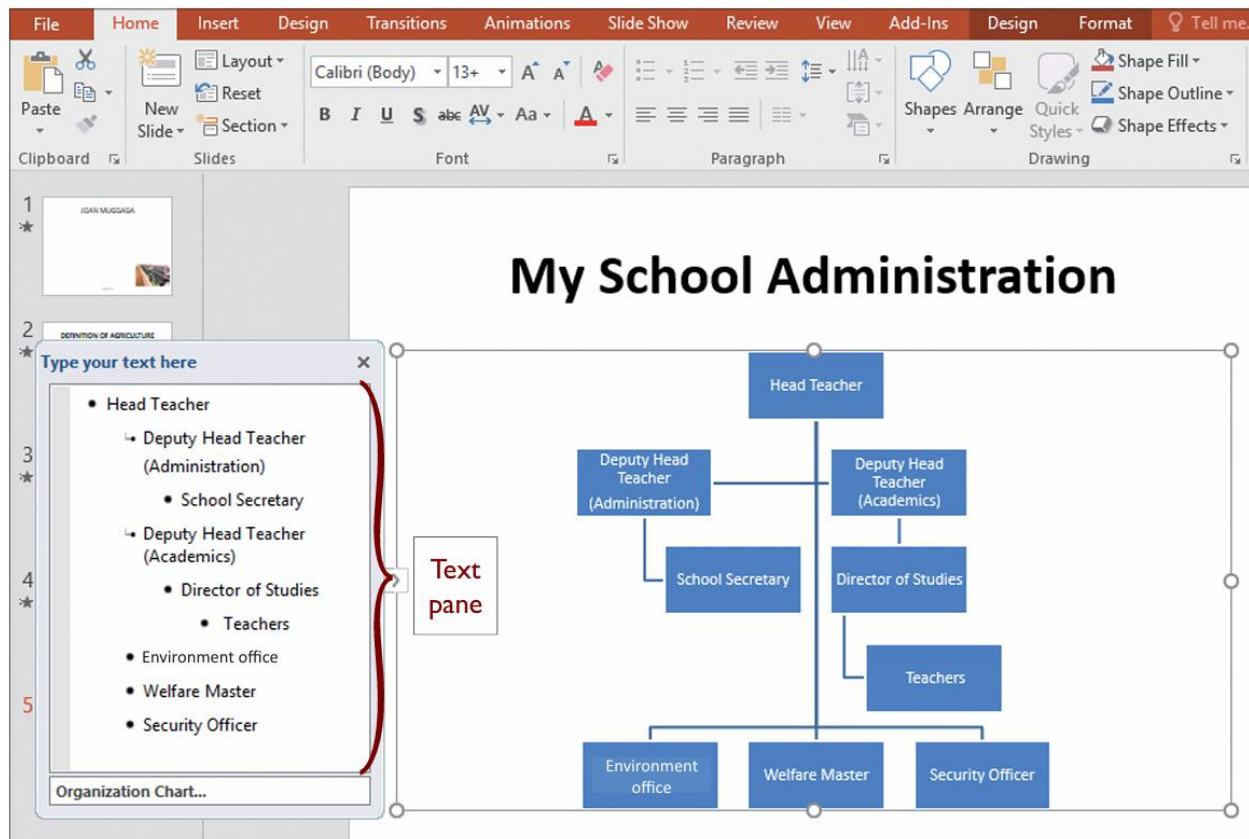
Picture 4. 24. Button for inserting a SmartArt

Step 3: In the Choose a Smart Graphic dialog box that displays, click the type and layout that you want and click on **OK**. In our case, we are going to create a hierarchy of officers in our school.



Picture 4. 25. The dialog box for choosing a SmartArt Graphic

Step 4: Enter the text by either: Clicking in the text pane (left) and then type a text or clicking directly in the Smart Graphic and type the desired text. Afterwards save the file.



Picture 4. 26. A SmartArt graphic organisation chart

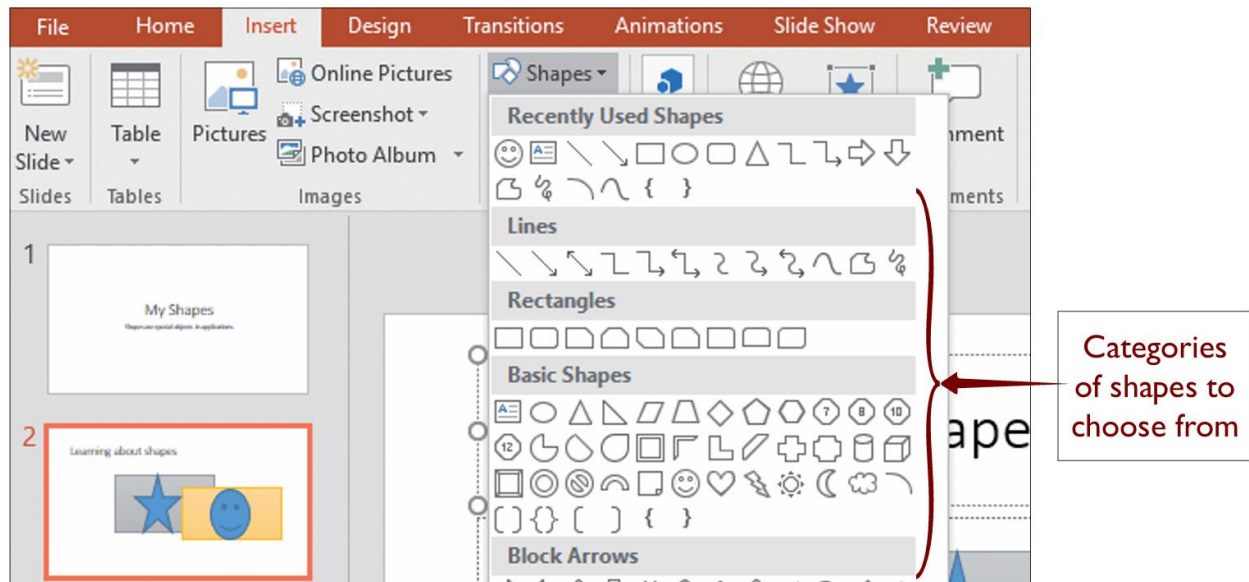
4.3.6. Shapes

A shape is an outline form of an object. In office applications such as Microsoft PowerPoint, you can add shapes such as boxes, circles and arrows to your slide.

Use the steps given below to insert any type of shape in a PowerPoint presentation you have opened.

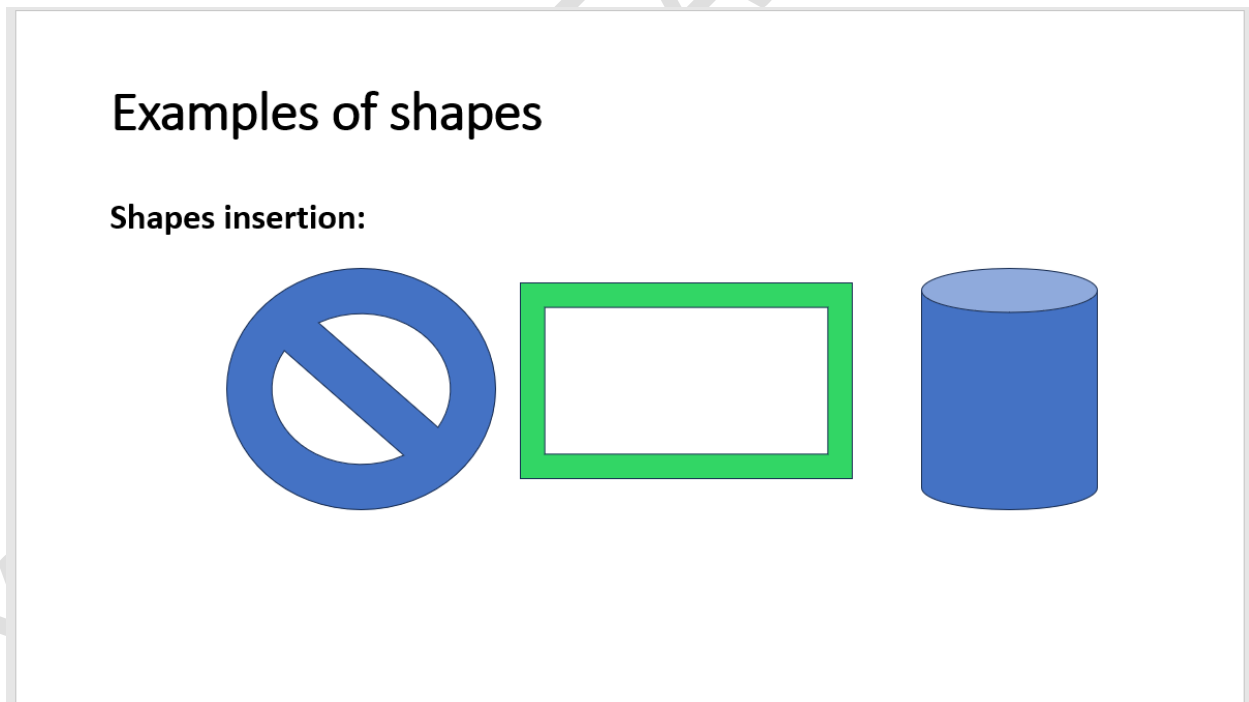
Step 1: Click on **Insert** tab.

Step 2: Click **Shapes** from the **Illustrations** group.



Picture 4. 27.Shapes from the Insert tab

Step 3: Select a shape such as a rectangle and then click and drag to draw the shape in the desired place. After drawing a shape, you can add text, bullets and numbering to them. You can also change their fill, outline and other effects on the Format tab.



Picture 4. 28. Different shapes in PowerPoint

4.3.7. Chart

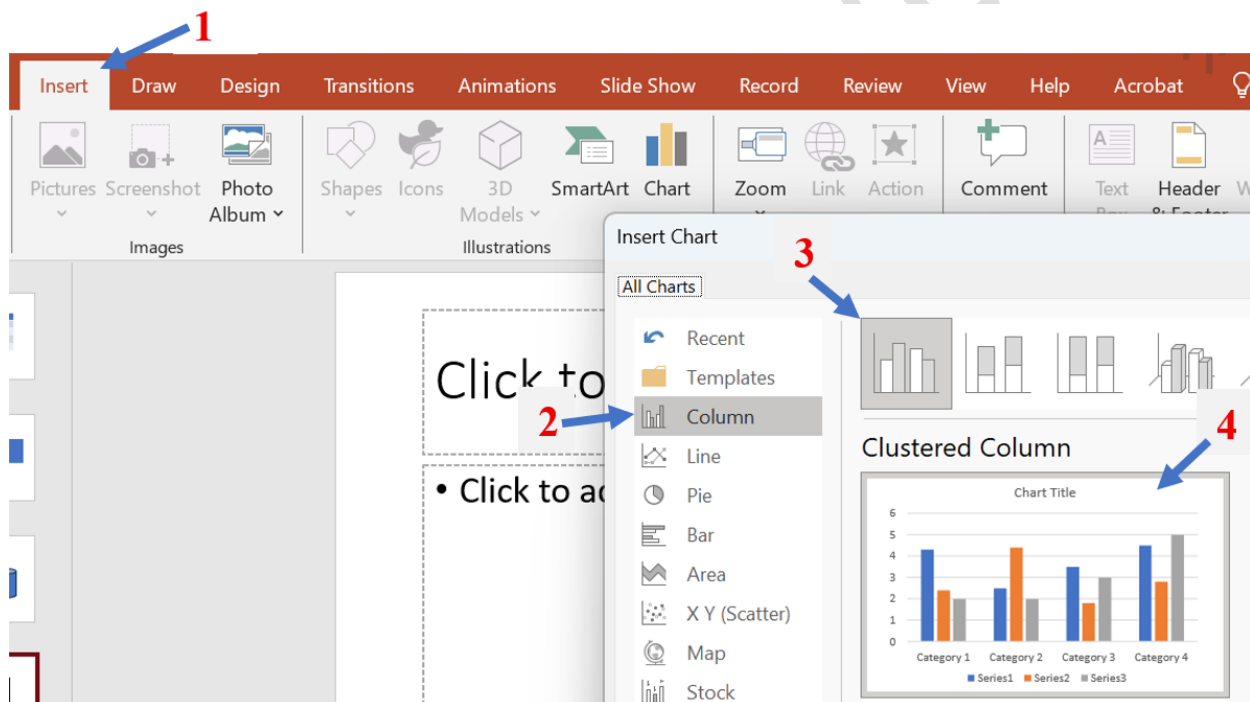
It is a pictorial representation of numerical data. There are many types of charts that can be created in a slide, but common charts frequently used are: column charts, line charts and pie charts.

Follow these steps to insert a chart:

Step 1: In the opened PowerPoint presentation, click where the chart will be then click on **Insert** tab

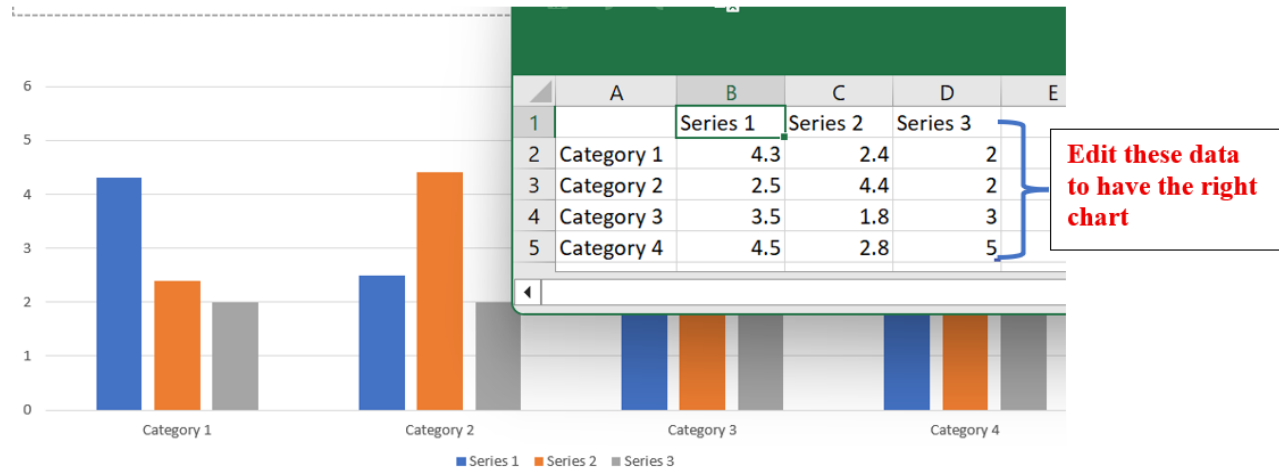
Step 2: Click on the **Chart** tool then choose the kind of chart to insert

Step 3: Choose any kind of chart such as **Column chart**. There are other charts namely Line, Pie, Bar, Area, X Y (Scatter), Map, Stock, etc. The steps to insert a chart are illustrated using labels **1, 2, 3 and 4**.



Picture 4. 29. Steps to insert a chart in PowerPoint

Step 4: Now the column chart appears together with its data sheet that you should edit to suit your chart. So edit that data to have a chart you want.



Picture 4. 30. A default column chart displayed together with data sheet to be edited

4.3.8. Media clips

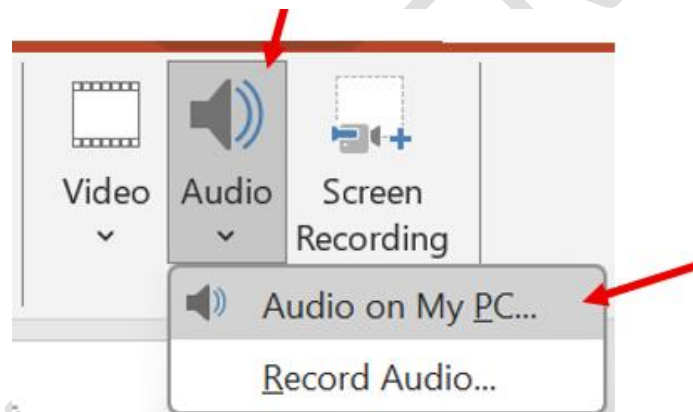
Media clip is a small segment of an electronic media either an audio clip or a video clip.

a. Inserting audio

To insert audio clip, follow the steps given below.

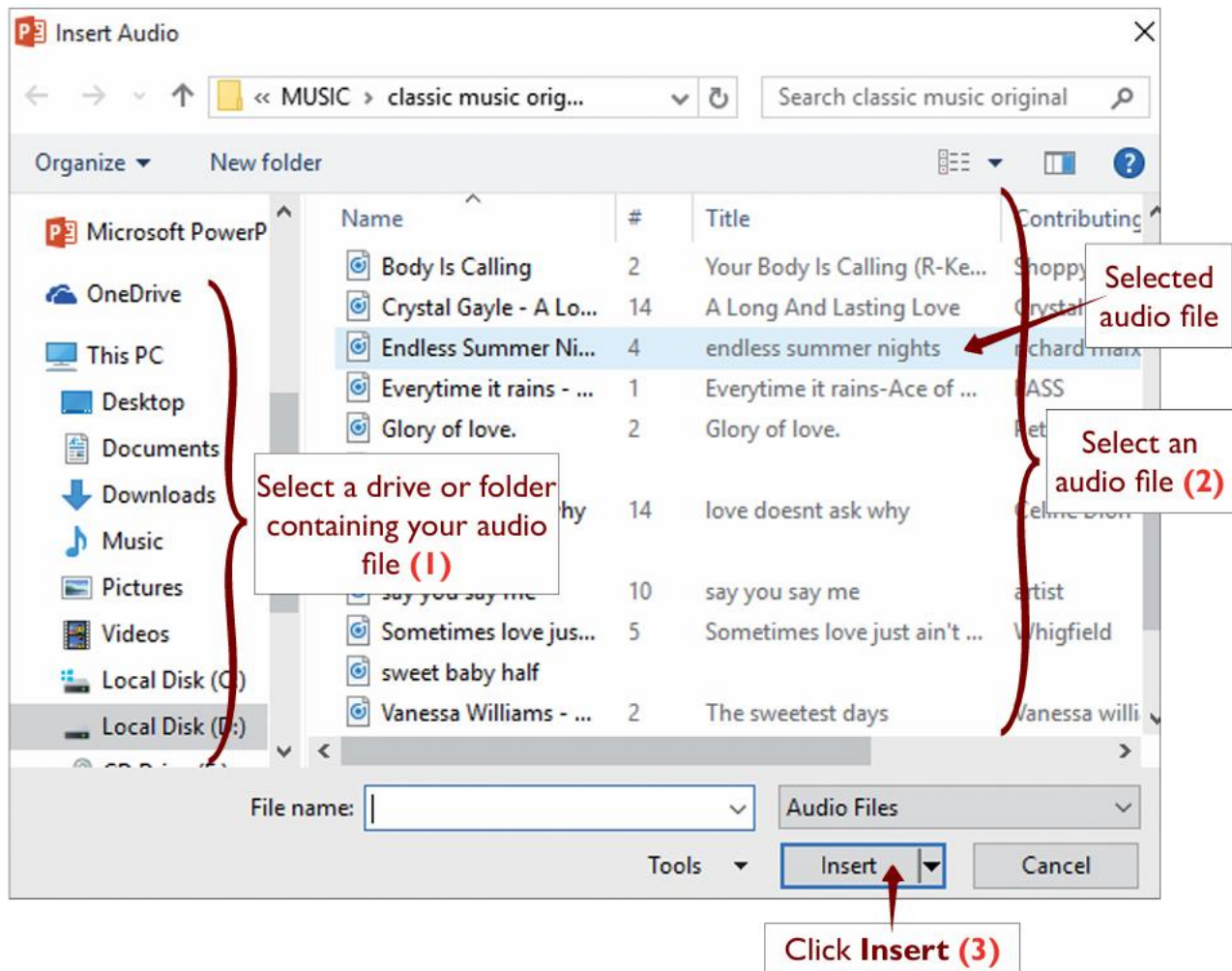
Step 1: Select the slide where you want to insert an audio clip.

Step 2: On the **Insert** tab, in the **Media** group, click the arrow under **Audio**, and then click **Audio on My PC**.



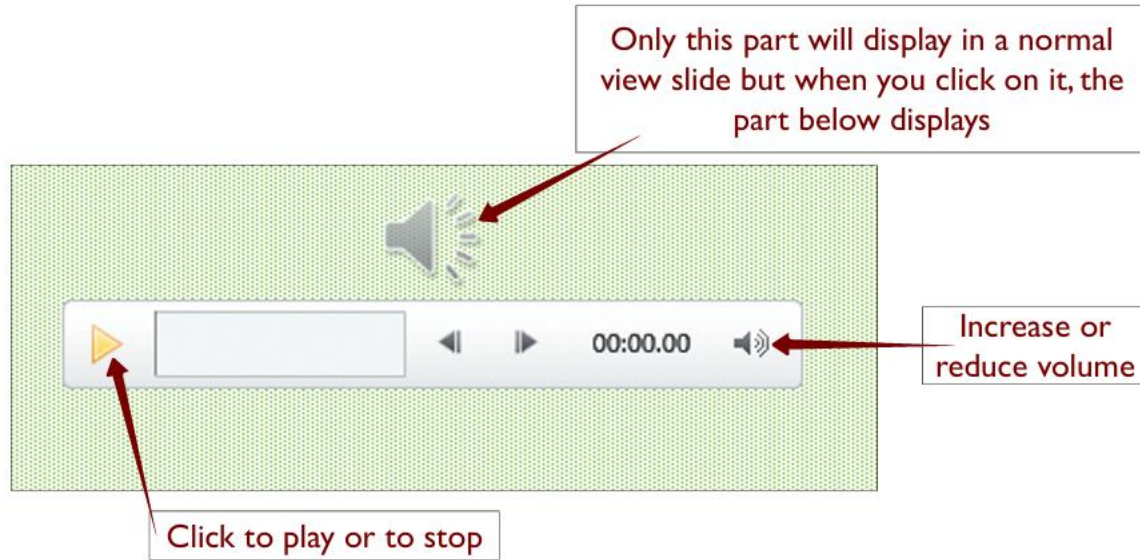
Picture 4. 31. Media group buttons displaying Audio menu.

Step 3: In the **Insert Audio** dialog box, select a folder containing the audio you want and then select the audio file and click on **Insert** button.



Picture 4. 32. Browsing the audio to insert

Step 4: The slide where audio has been inserted will have a graphic as shown in picture 6.45. Play using the media buttons displayed.



Picture 4. 33. Audio media clip

b. Inserting video

There are two ways to insert video: by linking and by embedding a video directly from your PC into your presentation.

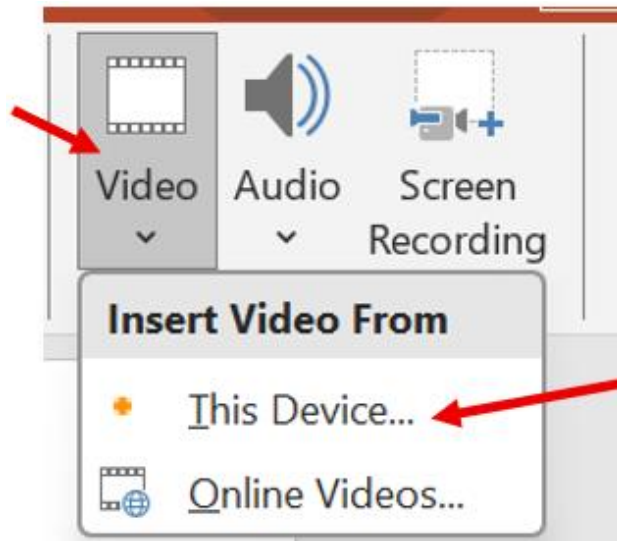
- **Embedded videos** are appropriate but tend to increase the size of your presentation.
- **Linked videos** keep your presentation smaller in size, because links can easily break, copy the video into the same folder as your presentation and create the link from there.

a) Embed a video clip

To insert a Video clip by embedding it, follow the steps given below.

Step 1: Select the slide where you want to insert video clip.

Step 2: On the **Insert tab**, in the **Media group**, click on **Video** then on **This Device**. Browse the video to insert in the presentation and click on **Insert**.



Picture 4. 34. Options for embedding a video in a presentation

Step 3: In the dialog box that appears, after browsing and selecting the video, click on **Insert**.

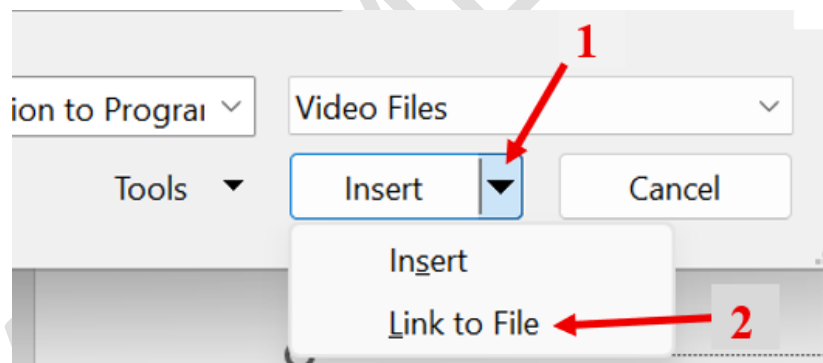
b) Link a video

To link to a video on your PC follow the steps below

Step 1: Select the slide where you want to insert the video clip.

Step 2: On the **Insert** tab, in the **Media** group, click on **Video**, and then click **This Device**.

Step 3: After browsing and selecting the video to link, click on the arrow (1) and click on **Link to File** (2).



Picture 4. 35. Options for linking a video file to a presentation

Application activity 4.3

- 1) Explain the difference between embedding a video and linking a video in PowerPoint.
- 2) Describe the steps you would follow to insert and resize an image in a slide.
- 3) Which command is used to insert a textbox?

- a) Insert → Shapes→ Textbox
- c) Insert → WordArt

- b) Insert → Textbox
- d) Insert → SmartArt

4.4. Formatting a slide

Activity 4.4

- 1) Give at least three examples of font styles?
- 2) Which layout to choose if you want to compare two sets of information side by side?
 - a. Title slide
 - b) Title only
 - c) Comparison
- 3) To change the font color of text, you should first:
 - a) Select the text
 - b) Click on the Design tab
 - c) Insert a new slide
 - d) Save the presentation
- 4) The Design tab in PowerPoint is mainly used to:
 - a) Add animations
 - b) Apply themes and variants
 - c) Insert pictures
 - d) Rearrange slides

Formatting a slide is adjusting the visual appearance and layout of the elements on a slide so that the information is clear, attractive, and easy for the audience to understand. This formatting can involve the font, alignment, themes and backgrounds, slide design, slide layout.

4.4.1. Changing the font

Follow these steps for changing the font in a slide:

Step 1: Select text on which to apply a font. The fonts that can be changed are font size, font colour, font style, shadow, etc.

Step 2: From the **Home tab** click on any one of the tools for changing the font. The tools to click on are in the table below:

Click on Home

**For changing
the font style**

**For changing
the font size**



Picture 4. 36. Tools for changing the font

In the slide below, the font of the text has been changed to make some text bold, italic and underlined. Also the color of some text has been changed to blue.

Rwanda Environmental Problems

- Land degradation
- Deforestation
- Wetland and biodiversity loss
- Natural causes
- Man-made disaster
- Reduction of water resources
- Human settlements



Picture 4. 37. The look of the slide after applying some font changes

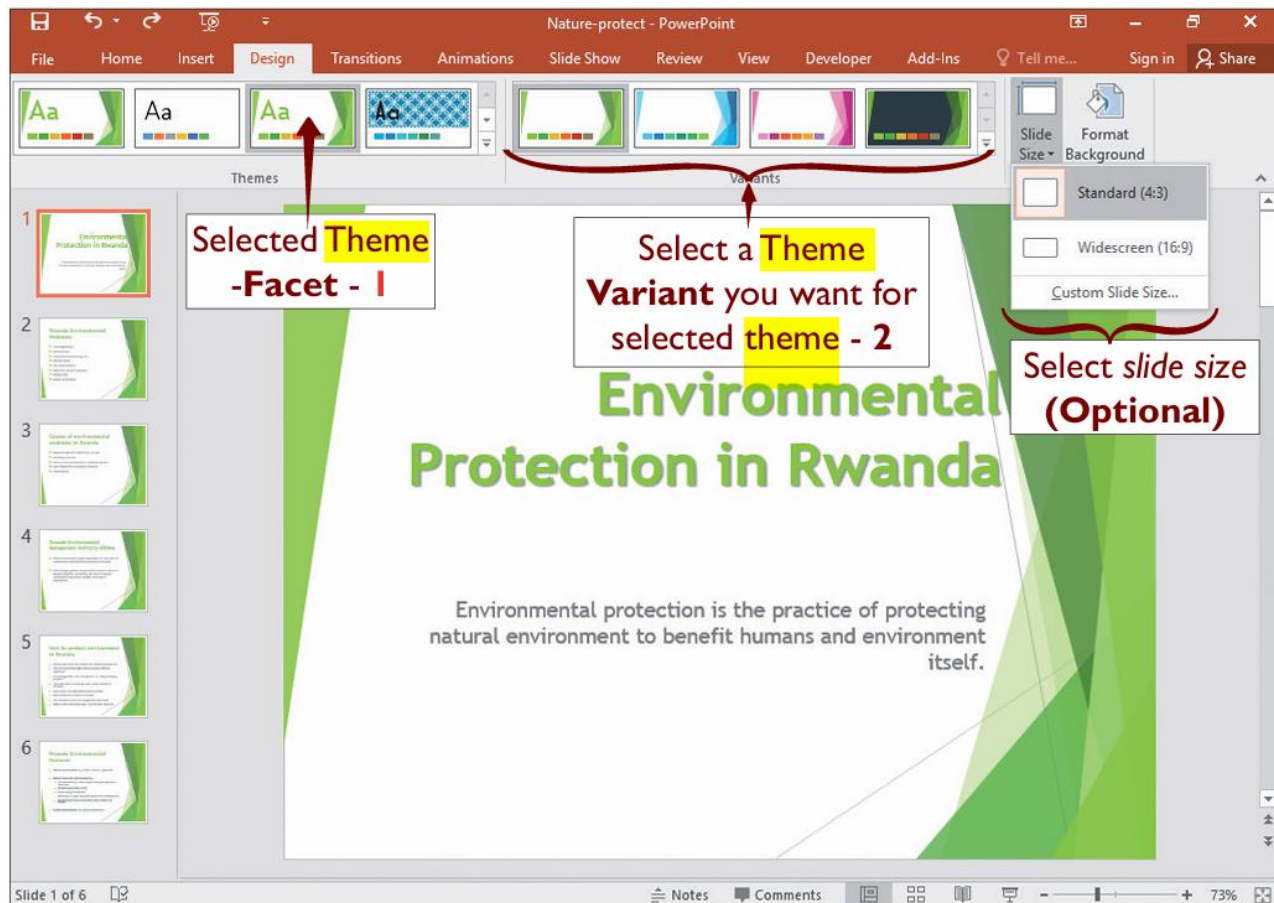
4.4.2. Slide design

Slide design is to the way a slide is arranged and styled to make information clear, attractive, and easy for the audience to understand. It focuses on how the slide looks and how the content is organized.

Follow these steps to change the design of a PowerPoint presentation:

Step 1: Click on the **Design tab**.

Step 2: From the **Themes group** select a theme you want.



Picture 4. 38. A presentation where a (Design) Theme has been applied

Step 3: Select a Theme Variant in case you need it and then save changes to your presentation. A theme variant is a set of colors that will appear in a theme.

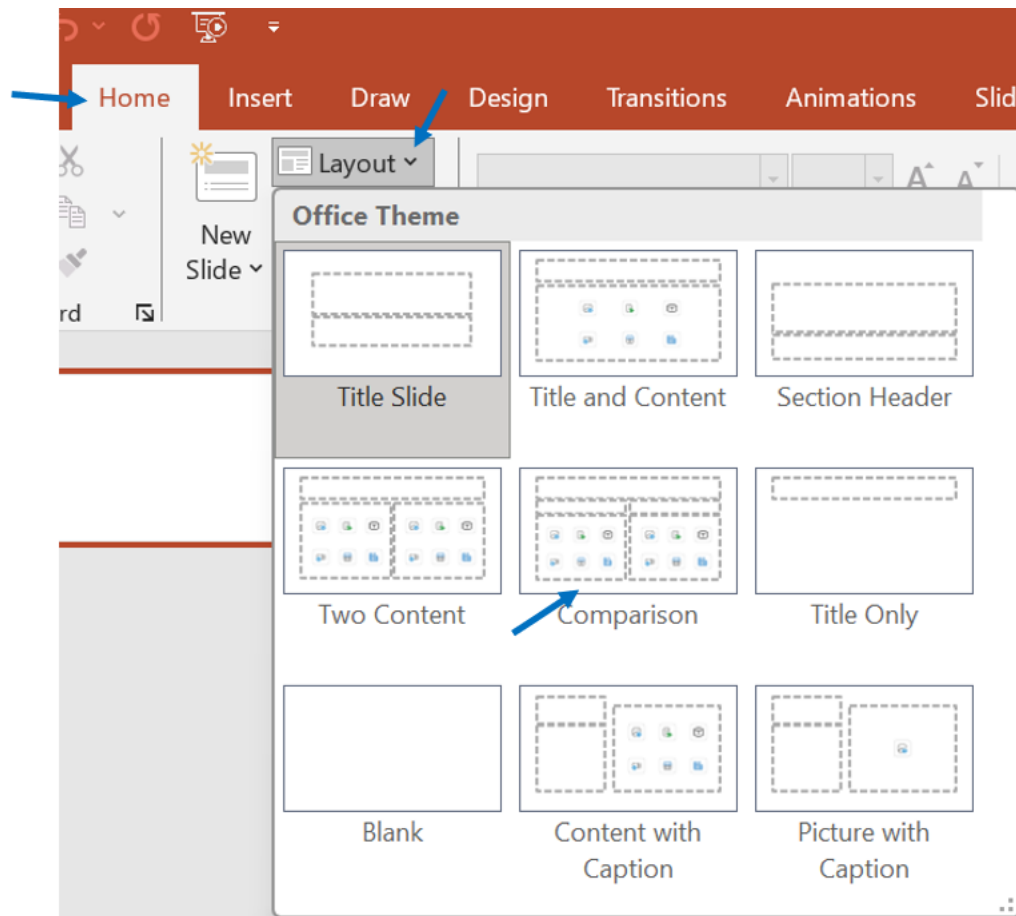
4.4.3. Slide layout

A slide layout in PowerPoint is a predesigned arrangement of placeholders that determines where content will appear on a slide. When you open a PowerPoint presentation, the default slide layout is **Title and Content**.

Follow these steps to choose a slide layout you want:

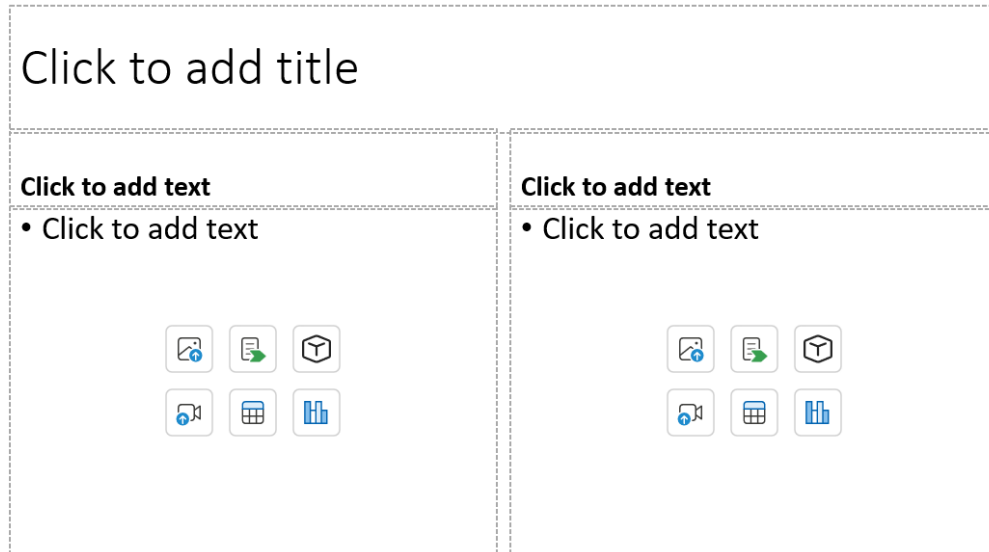
Step 1. Open PowerPoint and click on the **Home** tab

Step 2. Click on **Layout**. Immediately the different layouts appear. Choose one of them and write in it.



Picture 4. 39. Different slide layout

As per the image above, the different layouts are: Title Slide, Title and Content, Section Header, Two Content, Comparison, Title Only, Blank, Content with Caption and Picture with Caption.



Picture 4. 40. Comparison slide layout

Application activity 4.4

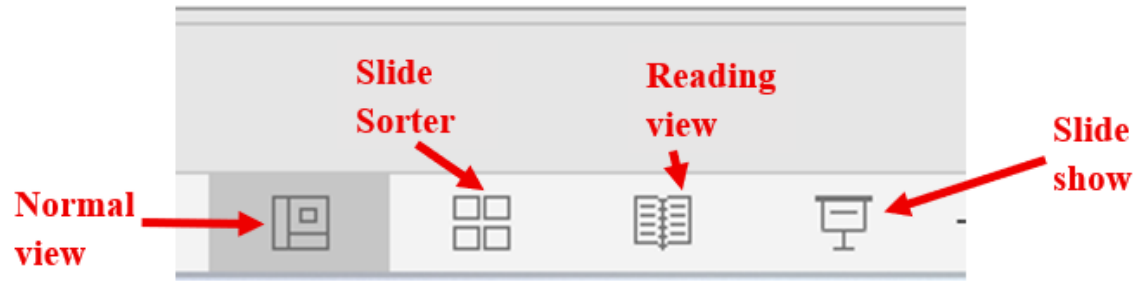
- 1) Which tab contains the tools for changing font size, color, and style?
 - a) Insert tab
 - b) Home tab
 - c) Design tab
 - d) View tab
- 2) Explain the difference between slide design and slide layout in PowerPoint.
- 3) Open a blank slide and:
 - a) Add text on any topic you choose
 - b) Change the font size to 28, make it bold, and change the color to blue.
 - c) Apply a theme of your choice to a presentation and then select a theme variant.
 - d) Create a new slide using the Comparison layout and insert text in both placeholders. Make sure the text in both placeholders is for comparing

4.5. Presentation views

Activity 4.5

- 1) A projector is connected to a computer using:
 - a) USB port only
 - b) VGA or HDMI port
 - c) Audio Jack
 - d) Printer cable
- 2) How do you move from one slide to another when presenting?
- 3) When you are presenting, how do you make the slides appear full screen?

Presentation views are different ways of displaying and working with slides. Each view serves a specific purpose such as editing, organizing, presenting, or designing the structure of a presentation. There are different views namely Normal view, Slide Sorter, Reading view and Slide show. Those views are found at the bottom right of a slide in PowerPoint.



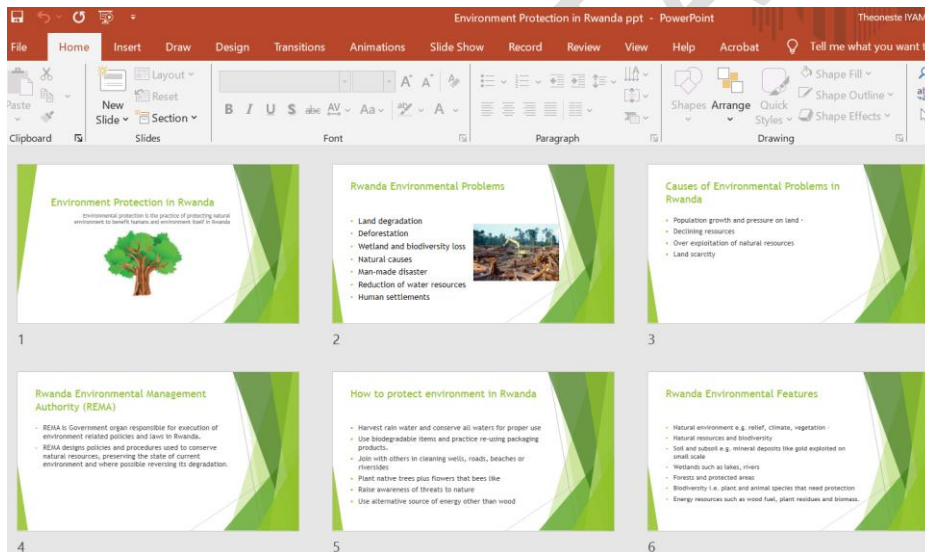
Picture 4. 41. The tabs for different presentation view

4.5.1. Normal view

Normal View is the default working view in PowerPoint. It is used for creating, editing, and formatting slides. It is characterized by a large slide workspace in the center, a slide thumbnail pane on the left, a notes pane at the bottom. It allows adding text, images, shapes, charts and other content.

4.5.2 Slide sorter

Slide sorter view displays all slides as small thumbnails on one screen. It shows the entire presentation at a glance, allows easy rearranging of slides and supports selecting multiple slides at once.



Picture 4. 42. Slides in a Slide Sorter view

4.5.3. Reading view

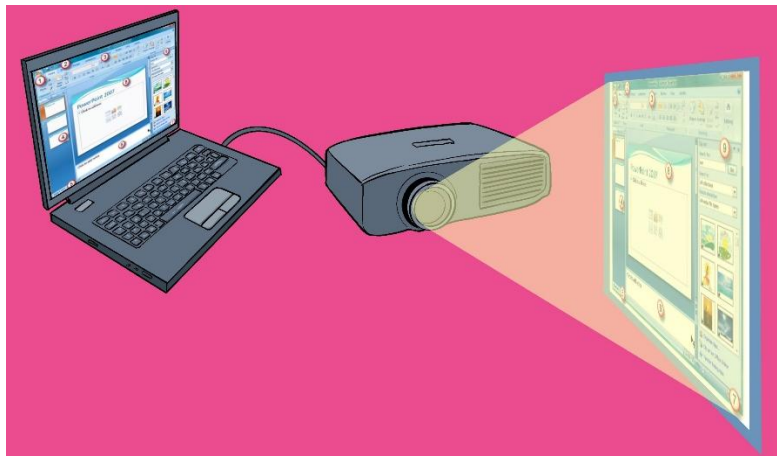
The reading view displays the presentation in a window mode, not full screen. It is similar to Slide Show but stays within the PowerPoint window. It is used when previewing the presentation.

4.5.4. Slide show

A Slide Show is a sequence of slides displayed one after another, usually on a projector or computer screen. A slide show helps you present information visually, combining text, images, charts, audio, and video to communicate ideas clearly and effectively. This view is the one used while presenting using a computer and projector.

a. Presenting using a projector

A projector is an output device that can take images generated by a computer and produce them by projection onto a screen, wall or another surface. A projector is connected to the computer through the VGA port but new projectors and computers are connected using the HDMI ports.



Picture 4. 43. Projector and computer connected

- To present using a projector, first connect the projector to the laptop, then open the presentation and click on **Slide Show**. Use the direction keys to move from one slide to another.
- **Conducting the presentation**

When everything is in order; the projector is properly connected and working, the handouts have been distributed and everyone is properly seated it is then time to start the presentation.

To be effective, a presentation must be short, capture only important points and provide short partial synthesis to make the audience easily remember what is so far presented. Also Rehearse the presentation and use scripts and notes to help you not forget the important points to mention. Be polite and use appropriate language.

Application activity 4.5

1) Which presentation view displays slides in thumbnail form for easy rearranging?

- | | |
|-----------------|----------------------|
| a) Normal View | b) Slide Sorter View |
| c) Reading View | d) Slide Show View |

- 2) You are preparing to present in class. Describe the steps you would take to connect a projector and start your presentation.
- 3) During a presentation, your slides are not appearing full screen. Which view should you select to fix this problem?

4.6. Master views

Activity 4.6

- 1) If you wanted all slides in your presentation to look the same, how would you do it?
- 2) Why might it be useful to edit all slides through one slide instead of editing each slide individually?
- 3) What does the Slide Master act as in a presentation?
 - a) A feature for adding transitions
 - b) A tool for printing slides
 - c) A blueprint or template for all slides
 - d) A single slide with animations
- 4) If you change the font size in the Slide Master, what happens?
 - a) Nothing changes until you print
 - b) Only handouts are affected
 - c) All slides based on that layout update
 - d) Only the current slide changes
- 5) How can you access the Handout Master in PowerPoint?
 - a) Click Design tab → Handout Master
 - b) Click Insert tab → Handout Master
 - c) Click Animations tab → Handout Master
 - d) Click View tab → Handout Master

In PowerPoint presentation or similar tools, a master view is a special view that let you control the overall design, layout, and formatting of slides, handouts, or notes. Instead of editing each slide one by one, a master view helps in making changes that apply across the entire presentation.

4.6.1. Slide master

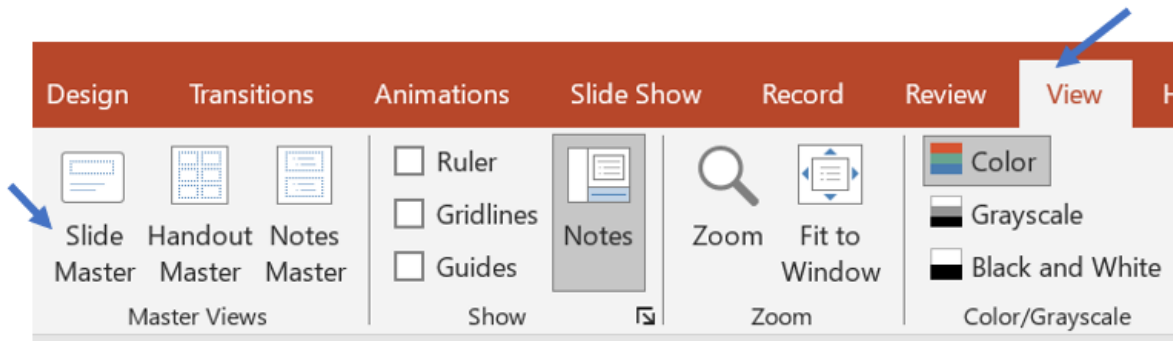
A **Slide Master** is a special feature in Microsoft PowerPoint that controls the **overall design and formatting** of your slides. Think of it as the "blueprint" or "template" behind your presentation.

When you make a change on the slide master, it replicates on all the slide layouts and slides. The slide layouts can receive all the formatting, position and common elements from slide master but you can also edit individual slide layouts to differ from the master slide. Any change you make to a slide layout affects only the slides based on that particular layout.

Follow these steps to apply slide master to your presentation:

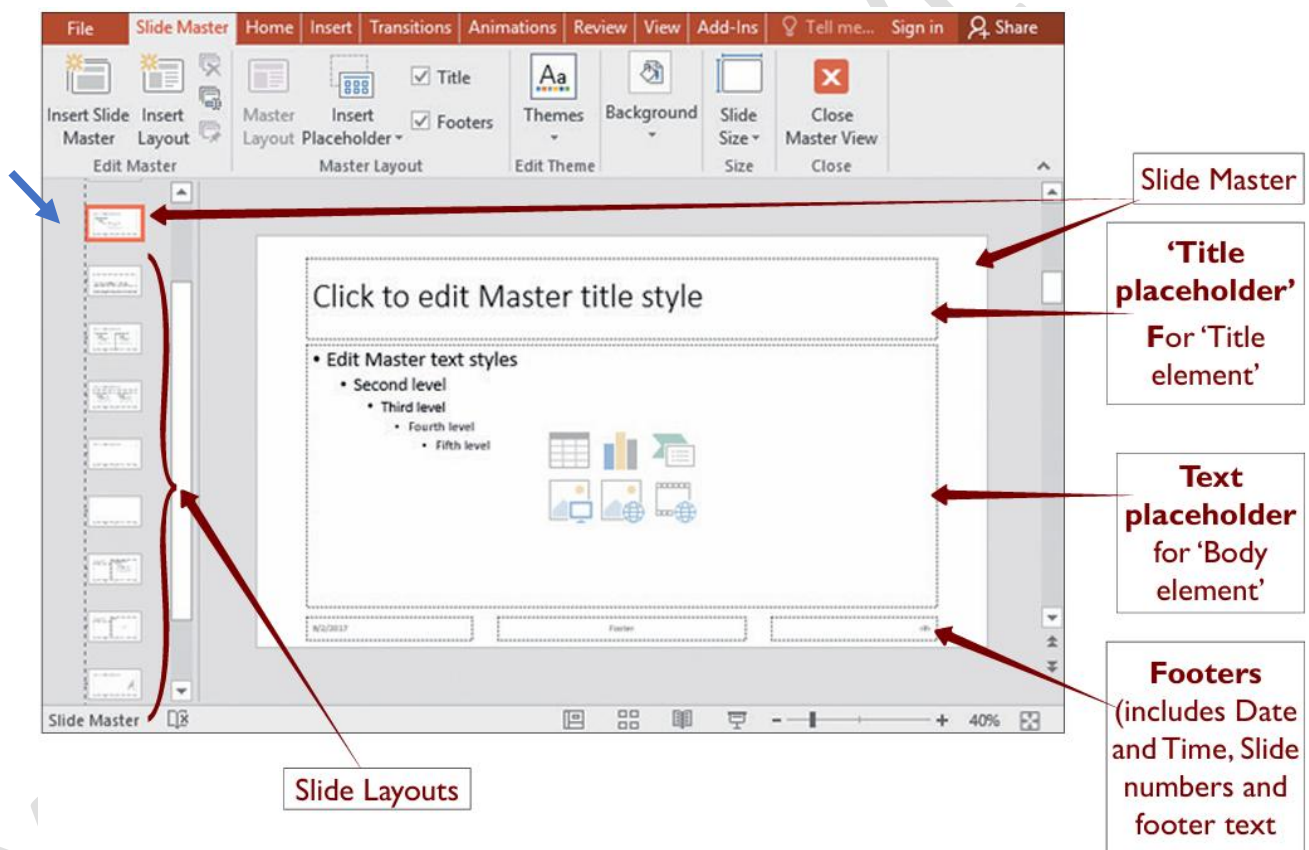
Step 1: Open Slide Master View by clicking on **View tab** and click on **Slide Master**.

Immediately a new panel opens showing the Slide Master at the top and all the layout masters below it.



Picture 4. 44. Menu and tools for starting a slide master

Step 2: Edit the Slide Master by selecting the top slide (slide master).then make your changes. You may set the background style or color, choose fonts and text sizes, add footers and headers. Add a logo, footer, or date.

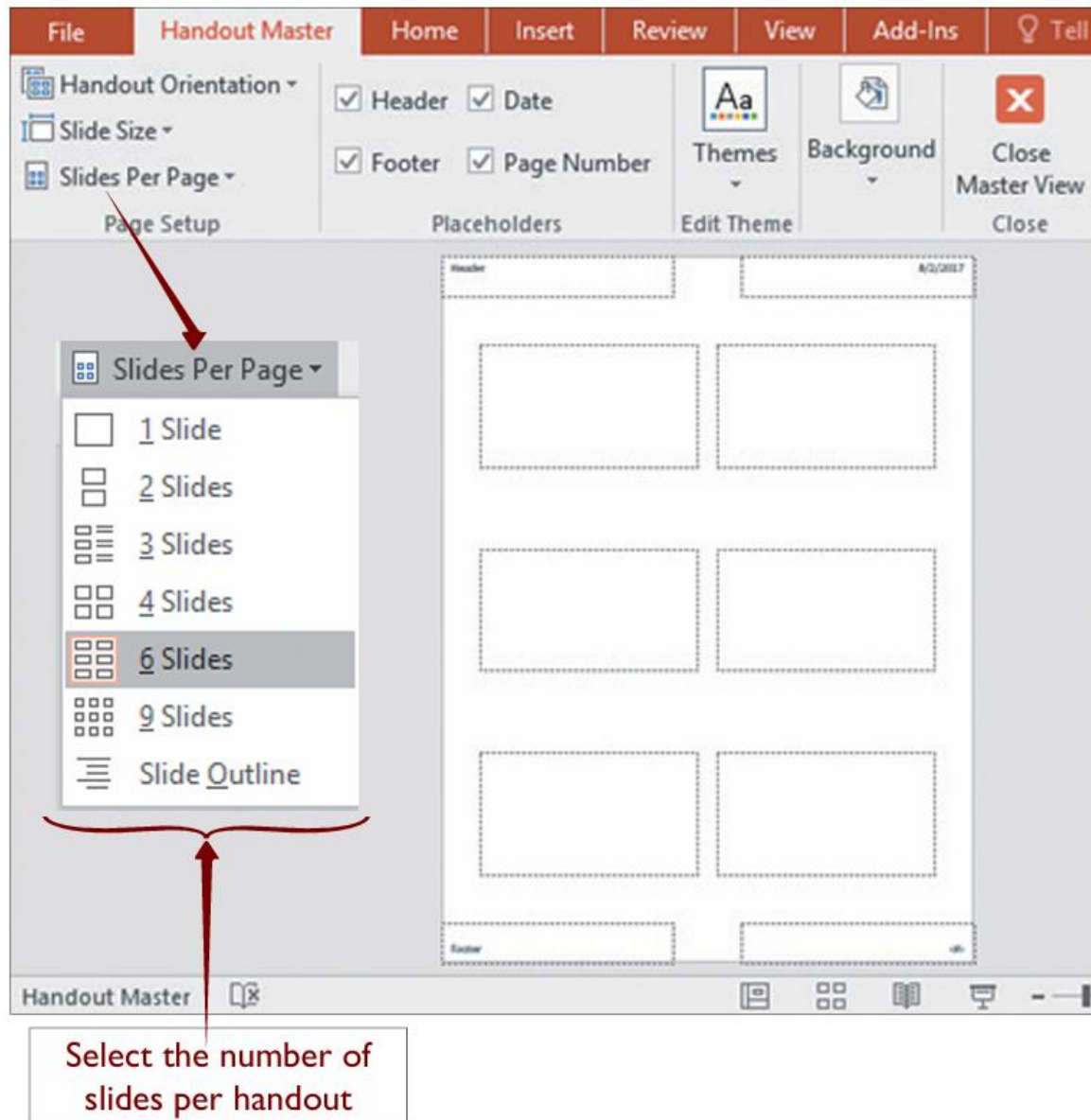


Picture 4. 45. Slide Master view

4.6.2. Handout master

In Microsoft PowerPoint, the Handout Master is a special view that lets you control the appearance of printed handouts. Unlike the Slide Master which affects slides as seen on the screen, the Handout Master focuses on how your slides look when distributed on paper. You can

have 1 up to 9 slides on a handout page. The appearance of a printed handout is determined by the handout master. To access handout master, click **View tab** on the Ribbon and select **Handout Master** from Master Views group.



Picture 4. 46. Handout master

Application activity 4.6

- 1) What is the difference between a Slide Master and a Handout Master?
- 2) How many slides can you place on a single handout page using the Handout Master?
- 3) Describe how changes made to a Slide Master affect slide layouts and slides.
- 4) Which tab on the Ribbon allows you to access the Slide Master view?
- 5) You are required to prepare a presentation on your school past. It must have a logo at each page and the date. Also text formatting must be uniform. Prepare that presentation.

4.7. Animation of a presentation

Activity 4.7

- 1) Which tab in PowerPoint is used to apply slide transitions?
 - a) Insert
 - b) Transitions
 - c) Design
 - d) View
- 2) What must you first select before applying a transition?
 - a) The background color
 - b) The text or image
 - c) The slide number
 - d) The file name
- 3) What is the purpose of a transition sound in a presentation?
 - a) To make the slide colorful
 - b) To signal that something is changing
 - c) To increase slide speed
 - d) To add more text
- 4) Which tab allows you to preview your presentation before showing it?
 - a) Home
 - b) Slide Show
 - c) Review
 - d) File
- 5) Why do you think adding sound might help the audience?
- 6) Why is it important to preview your work before presenting?

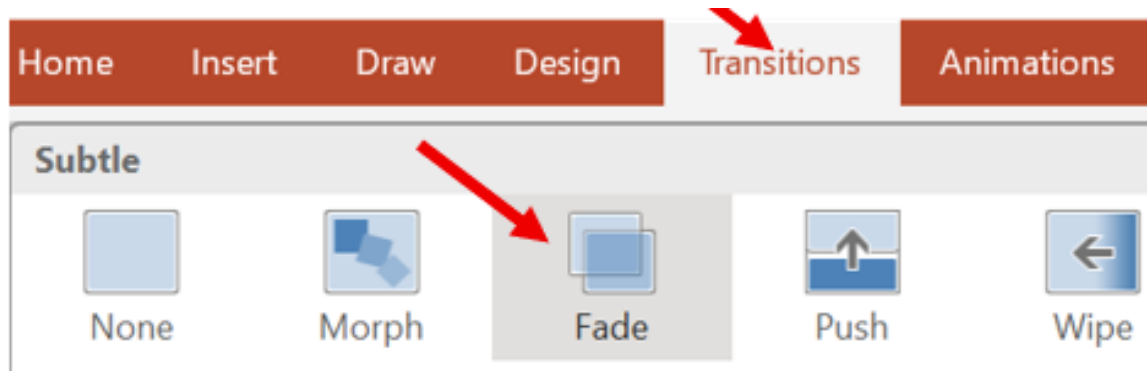
4.7.1. Slide transition

In PowerPoint there are three main slide transitions namely **Subtle**, **Exciting** and **Dynamic Content**.

In Subtle transition simple transitions are used to move from one slide to another, for Exciting additional visual effects are used to catch the eye of the audience while for Dynamic content will move only the placeholders, not the slides themselves.

To use the different transitions, do the following:

- Select the text or image on which to apply the transition
- Click on the **Transition tab** then choose one of the transitions. In the image below the chosen transition is “**Fade**”



Picture 4. 47. Applying transition

4.7.2. Transition sound and speed

A **transition sound** is a short audio used to signal that something is changing. It helps listeners understand that one part has ended and a new part is beginning. Transition speed is the amount of time a slide transition takes to complete in PowerPoint.

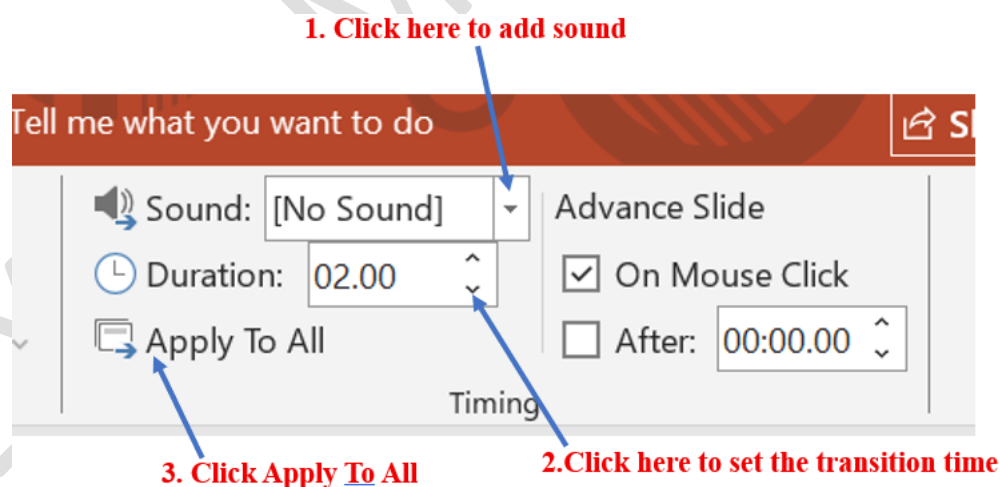
Follow these steps to add a transition sound and speed to a presentation:

Step 1. Open your PowerPoint presentation and select the slide where you want the sound

Step 2. Go to the **Transitions** tab. In the **Timing** group, click **Sound**.

Step 3. Click the dropdown and choose a built-in sound. If you want your own sound, choose **Other Sound...**

Step 4. Select an audio file and change the timing by clicking on **Duration** and apply to one slide or click **Apply to All** if needed.



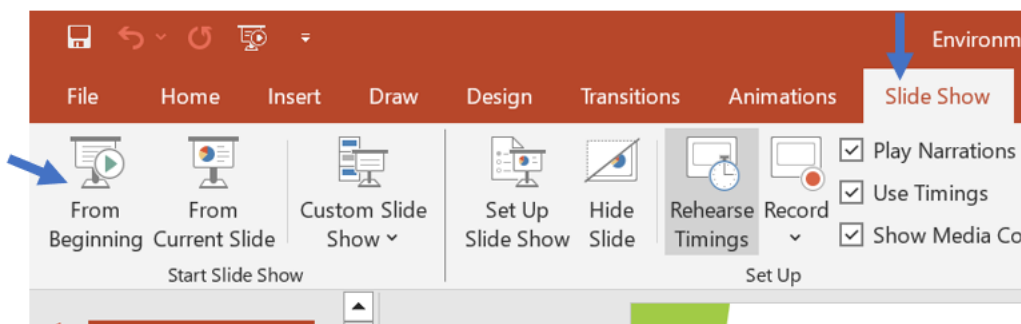
Picture 4. 48. Steps for adding slide transition

4.7.3. Preview presentation

A preview presentation is a feature in PowerPoint that lets you see how your slides will look when presented before you actually start the real slideshow.

Do the following to preview a presentation:

Open the PowerPoint presentation and click to the **Slide Show** tab. Click **From Beginning** or **From Current Slide** depending on where you want the slide show to start.



Picture 4. 49. Steps for slide show preview

Application activity 4.7

- 1) Create a PowerPoint presentation on a topic in Biology and do the following:
 - a) Insert three slides with different content (text, image, chart).
 - b) Apply a Subtle transition to the first slide, an Exciting transition to the second, and a Dynamic Content transition to the third.
 - c) Preview the slideshow and describe the differences you observed.
- 2) A classmate complains that their transitions are too fast and the audience cannot follow. What is the best way to improve their presentation?
 - a) Add pauses and slow down transitions
 - b) Use more complex animations
 - c) Skip transitions altogether
 - d) Increase the number of slides

4.8. End unit assessment

4.8.1. Project activity

Project Title: Presentation on Climate Change in Rwanda

Guidance: Climate change is a global challenge that affects Rwanda's environment and agriculture. Your task is to create a PowerPoint presentation that informs about climate change in general then how climate change happened in Rwanda and how it is affecting us Rwandans

- Do a research on climate change in Rwanda by using books, search engines and AI tools. Make sure you also get pictures illustrating climate change in Rwanda. The results of your research must be in a word document.

- From the document you have made, create a PowerPoint presentation. You can choose how slides follow one another but you can also make it like this:

Your presentation should have at least 5 slides:

- The first slide contains the slide title, the project title.
- The second slide contains the introduction to climate change in Rwanda in a bulleted format
- The third slide contains the impact on the environment. You should include some images to illustrate what you say
- The fourth slide contains data on rainfall or temperature changes. For this, you should include a chart or a table.
- The last slide is the proposal of what should be done to fight climate change in Rwanda. This will be followed by a slide with a thank you note.

Questions on the project:

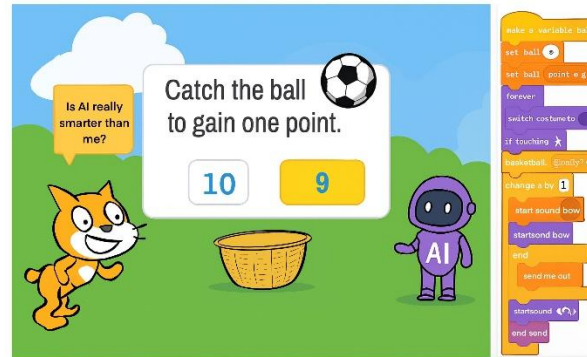
- 1) Why is it important to use a consistent theme and font size in a presentation
- 2) How does the use of Slide Master save time when you want to change the color of all slide titles at once?
- 3) Which view is most helpful if you want to see all your slides at once to rearrange them, and why?
- 4) Explain how adding a chart or a table on Slide 4 helps the audience understand climate data better than just using text.

4.8.2. Assessment questions

End of unit assessment 4

- 1) Differentiate between a presentation file and a slide.
- 2) Explain why PowerPoint titles should be short and clear.
- 3) State two reasons why large font sizes are recommended in presentations.
- 4) What happens to a slide when it is cut instead of copied?
- 5) State one reason why a chart is more suitable than text when presenting numerical data.
- 6) Explain why linked videos may stop working when moved to another computer.
- 7) Create a three-slide presentation on “Technology in Education” and do the following:
 - a) In the first slide insert an image from your computer and resize it to fit in the slide.
 - b) In the second slide insert a table with 4 rows and 3 columns, then enter sample data.
 - c) For all the slides apply the Office Theme.
 - d) Connect the projector and display the presentation in full screen mode
- 8) Why is it important to avoid overcrowding slides with information?

UNIT 5: PLATFORM GAME PROGRAMMING WITH AI



Key unit competence:

You will be able to develop an interactive game that integrates Artificial Intelligence.

Learning Objectives:

At the end of this unit, you will be able to:

Knowledge and understanding

- Identify platform game parts and the difference between programmed and AI behaviors.
- Define the roles of Motion, Control, and Variable blocks in game logic.
- Understand how algorithms and training data drive AI predictions and performance.

Skills

- Build game layouts and program player physics like jumping, gravity, and coordinates.
- Code enemies to follow players and make decisions using logic blocks.
- Create scoring systems and use Teachable Machine for gesture-controlled gameplay.

Attitudes & Values

- Develop a patient mindset for testing and debugging collision and sound errors.
- Value the importance of accurate data and fair AI behavior in game design.
- Appreciate using modern AI tools to create professional and interactive content.

5.0. Introduction

This is a unit on Platform Game programming with AI. In this unit you will transition from a player into a game developer, learning how to build games where characters sense and react to their environment. You will explore elements of a "Platformer" game, discovering how to design levels by creating ground, platforms, and hazards. You will learn to use the X and Y coordinate system to position objects and program physics like gravity and jumping so that the game character interacts with the environment.

Further in the unit Artificial Intelligence is introduced to move beyond basic movement and create adaptive behavior.

Introductory activity

A game in which, one student is the "Coder" and the other is the "Robot." The Coder gives a strict rule, like: "If I clap my hands, you must jump." The Robot only moves when that specific thing happens.

- 1) In this game, is the Robot thinking for itself, or is it just following an instruction?

- 2) Why is this different from true Artificial Intelligence?
- 3) If a robot has AI how will its movements change if it is playing a game?
- 4) By doing a research, give an example of a game that can be improved by integrating AI.

5.1. Platform games

Activity 5.1

Imagine a teacher and their students are designing a video game. Before they begin coding, they must create the "Game Architecture" by sketching the first level on paper. This layout must show the player's path from start to finish while highlighting potential dangers.

Working as co-designers, two students, Agahozo and Imena, draw a horizontal line for the Ground. They add a character called The Hero, a floating Platform, a spike labeled "the Hazard" and a gold coin representing the Reward. To test the mechanics, Agahozo moves their finger as the Hero while Imena acts as the Computer. Agahozo must jump over the spike and land on the platform to reach the coin. However, if they touch the hazard, Imena declares "Game Over," and they must restart.

- a) As a designer, what would happen to your Hero if you forgot to draw the Ground line?
- b) If you put the Gold Coin right next to the Hero with no spikes in between, would people buy your game? Why is the spike necessary?
- c) How can we translate your paper Spike into a rule for the computer to understand?

A Platform game (platformer) is a type of video game where the main character must jump between platforms and over obstacles to reach a goal. It different components and interactivity which makes the game possible.

The key components of a platformer are:

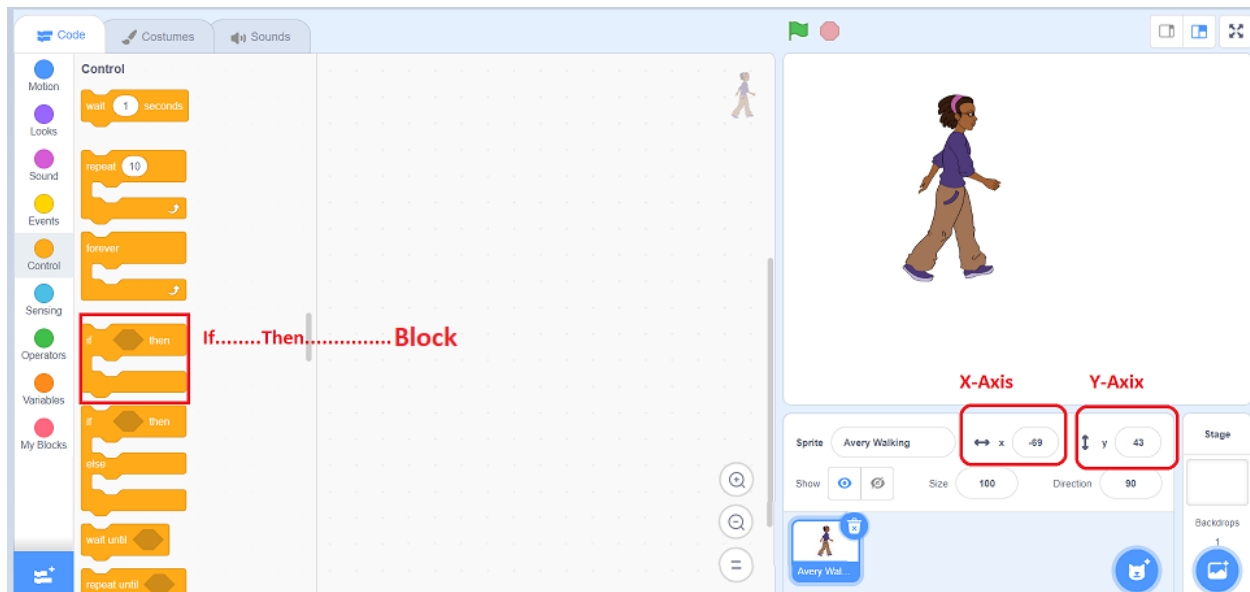
- **The player (Sprite):** The hero of your game that you control.
- **Platforms:** The surfaces such as ground, walls, floating blocks on which the hero walks.
- **Obstacles:** Things that make the game hard, like enemies, lava, or traps.
- **Collectibles (goals):** Items like coins or stars that give you points or finish the level.
- **Background (stage):** Shows the environment of the game, such as a forest, school.

Interactivity is the conversation between the player and the computer. It involves the following:

- **Input:** You press a key for performing particular task.
- **Action:** The computer runs code to make the Sprite jump after pressing a key.
- **Feedback:** You hear a clapping sound or see the score go up after action.

5.1.2 Scratch interface

In Scratch, coordinates are used to place the components on the stage. The X-axis moves the player left and right while the Y-axis moves the player up (jumping) and down (falling) while the If-Then blocks are used to create interactivity.



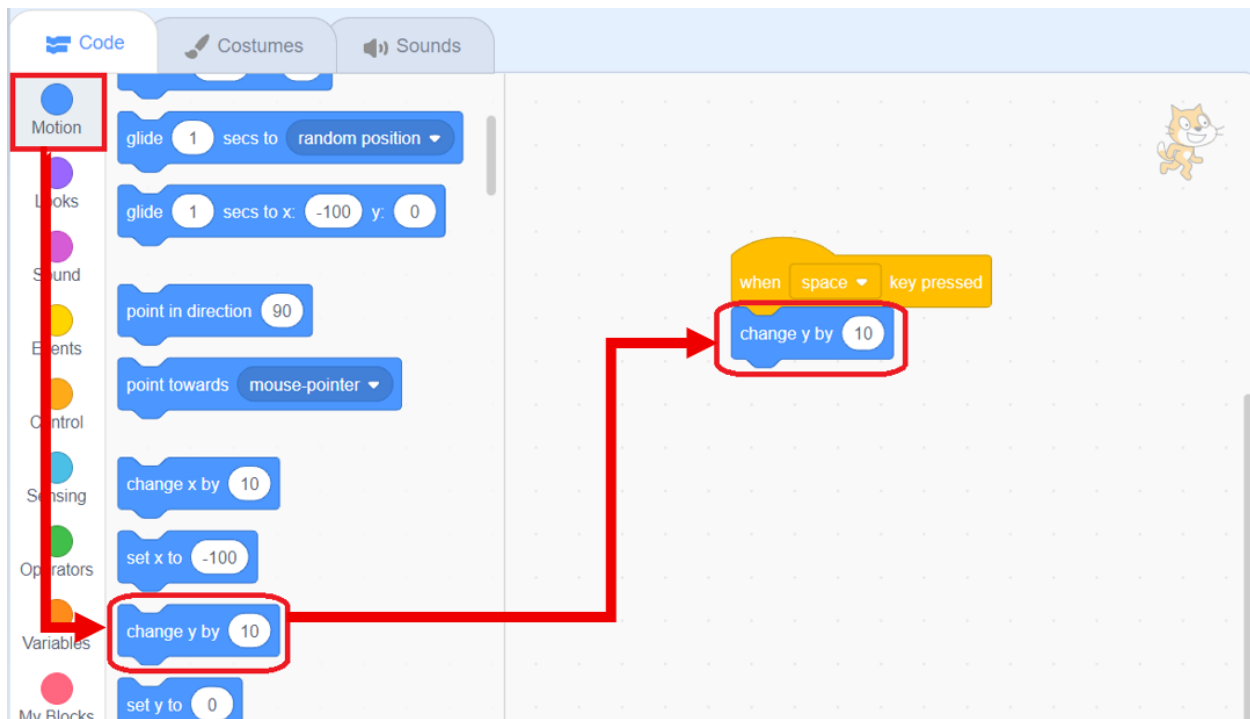
Picture 5. 1. The Scratch interface

5.1.3 Engaging games through interactivity

There are many example of interactivity in a platform game such as moving left and right using arrow keys, jumping using the space bar, collecting items and losing points when touching an obstacle

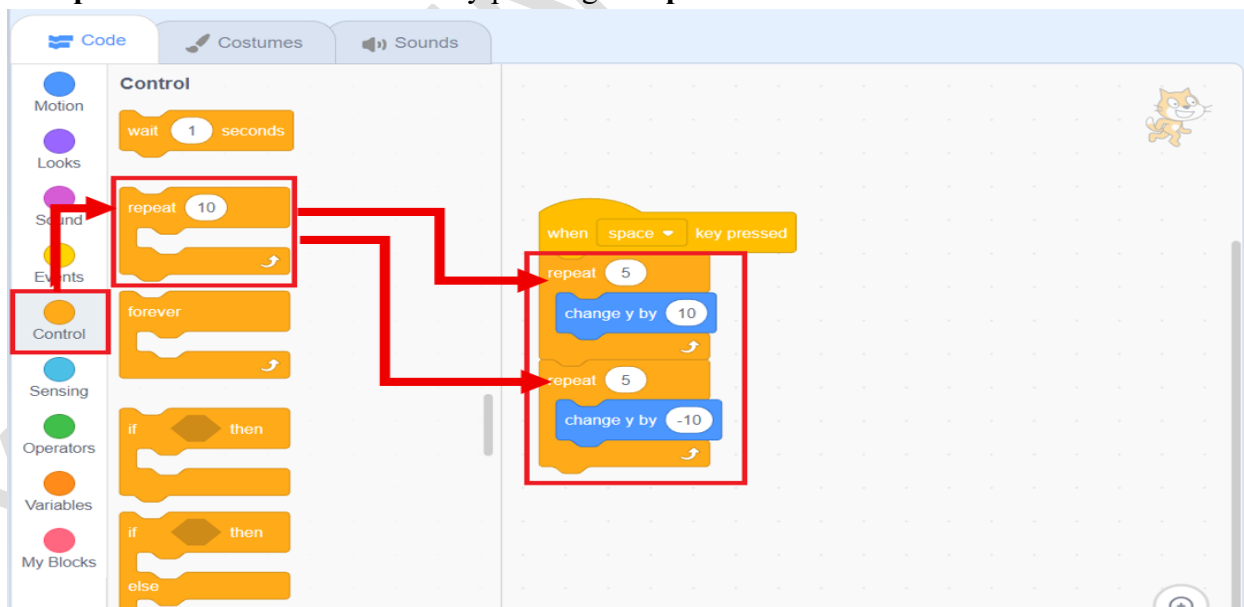
The procedure for creating a simple interaction in Scratch such as making a Sprite jump is outlined below:

- Step 1:** Open **Scratch** and choose a **sprite**.
- Step 2:** Click on **Events** and drag the block **when space key pressed**.
- Step 3:** Go to **Motion** and add “**change y by 10**”.



Picture 5. 2: Creating motion in Scratch

- Step 4:** Click on **Control**, Drag the **repeat** block, then add **repeat 5** to make the jump smooth.
- Step 5:** Add **change y by -10** to bring the sprite down.
- Step 6:** Test the interaction by pressing the **space bar**.



Picture 5. 3: Simple interaction in Scratch

Application activity 5.1

- 1) List the main components of platform games.

- 2) What Scratch block is used to control jumping?
- 3)
 - a) Create a simple platform scene in Scratch.
 - b) Add one sprite and one platform.
 - c) Make the sprite move left, right, and jump.

5.2: AI ethics in game programming

Activity 5.2

Consider a game in which an AI automated door decides who gets to enter a conference meeting. AI was trained only on photos of heroes wearing blue capes. Then two students, UMUHOZA and KWIZERA, try to enter the club. UMUHOZA's character has a blue cape, so AI lets her in. KWIZERA's character is a powerful hero but wears a red cape, so AI says "Access denied."

- a) Is this fair? Explain.
- b) Why did AI make a mistake?
- c) What AI ethics might be respected in the above scenario?

5.2.1 Fairness and non discrimination

In games, AI should treat every player equally, regardless of their gender, race, or background. In a game, fairness and non discrimination refers to making sure AI doesn't favor one group of people over another.

Example: If you use AI to generate enemy faces, make sure the AI doesn't only pick characters that look like people from one specific part of the world. It should create a diverse range of characters so everyone feels included.

5.2.2 Data privacy and security

When games use AI, they often collect information about how you play. Developers must protect this data to keep a player's personal information safe from hackers.

Example: If a game uses an AI voice-chat filter, the game should not record your private conversations and save them on a public server. Players should also be asked for permission before any data is collected.

5.2.3 Copyright and intellectual property

AI can create art, music, and code very fast. However, we must respect who owns the original ideas. By respecting the legal rights of creators you shouldn't use AI to steal someone else's work.

Example: If you use an AI tool to generate a Hero Sprite, you must ensure that AI wasn't trained on stolen artwork from a Rwandan artist without their permission. Always use AI tools that are Open Source or have a license.

5.2.4 Inclusive and ethical storytelling

AI can help write scripts and stories for games, but these stories must be respectful and kind. Using AI to tell stories that do not promote hate, bullying, or harmful stereotypes, and avoiding any kind of harassment in your created game. You must also avoid that the created game rely on bad ideologies.

Example: If you use an AI text generator in your game, check the text to make sure that AI isn't using offensive language or making fun of a specific culture.

Application Activity 5.2

- 1) What do you understand by the term ethic?
- 2) Why is data privacy important in online games?
- 3) How can we make sure our game characters are inclusive?
- 4) Does AI need to know my real-world home address to work? Explain.

5.3 Artificial Intelligence in games

Activity 5.3

Two students, Mugisha and Kezia, are going to demonstrate how a computer thinks using AI. The following are instructions to be followed to demonstrate the fact. Read them and answer the questions:

- The setup: Mugisha, the Player, stands at one side of the classroom. Kezia stands in the middle; she is the Enemy AI.
 - The rule: Kezia is given one specific instruction that "Every time Mugisha takes a step, you must take one step toward his current position".
 - The action: Mugisha moves left, right, or forward. Kezia follows him automatically based on the rule.
- a) Reproduce the same activity.
 - b) Did Kezia decide herself where to go?
 - c) How did it feel for the Player?

In a game, AI is the part of the computer code that makes characters or objects behave as if they have their own brain by sensing, thinking, deciding, and acting. With offline version of Scratch editor, AI sensors are found in extension to implement AI in Scratch projects. The following are some AI extensions in Scratch and what they do.

AI Extension in Scratch	What It Does (Simple Explanation)
Video sensing	Let's the computer see movement using the camera and react to it.
Text to speech	Makes the computer talk by reading text aloud.
Speech to text	Allows the computer to listen and turn spoken words into text.
Translate	Changes text from one language to another automatically.
Machine Learning (ML for Kids)	Helps the computer learn from examples (text, images, or numbers).
Pen	Allows the computer to draw and learn patterns on the screen.

Sensing	Helps the computer detect distance, touching, or mouse actions.
Operators	Helps the computer think and make decisions using logic and math.

5.3.1. AI behaviors

AI is used to create realistic actions in a game without a human controlling them, the following are some of behavior that can be used in game programming:

- **Pathfinding:** Enemies that track and chase the player (like Kezia did).
- **Auto-scoring:** The game logic that instantly recognizes when a player hits a target or crosses a line and adds points to the score.
- **Decision making:** If an enemy has low health, the AI might tell it to run away instead of attacking.

5.3.2. Algorithms in AI

An Algorithm is a list of steps or a recipe that the computer follows to solve a problem.

An example is an algorithm for hunting an enemy in game programming:

- Check where the player is.
- Is the player close? If Yes, move toward the player.
- If no, stay still.

a. Concept of algorithm in AI

In Scratch, AI works a bit differently than normal coding. Usually, you give a sprite a list of steps to follow, but with AI, instead of giving the sprite the answer, you give it the power to learn the answer.

A normal algorithm is like a cooking recipe while an **AI** Algorithm is more like a student. It looks at the examples, finds patterns, and makes its own decisions.

In Scratch, algorithms are used through **Extensions** (special extra blocks) or tools like Machine Learning for Kids.

b. Three main AI algorithms used

- **Classification (the sorter):** This is the most common AI algorithm used in Scratch. It helps the computer look at something and say that it knows what it is. You show the computer 20 pictures of apples and 20 pictures of bananas then AI algorithm looks for patterns.

Example: You can train an AI tool to recognize your hand gestures. If you show a Thumbs Up, the cat jumps. If you show a Thumbs Down, the cat sits.

- **Natural Language Processing (NLP):** This algorithm helps computers understand human language. It breaks down sentences to find keywords and the feeling (sentiment) behind them.

Example: Using the Text to Speech and Translate extensions, you can code a sprite to ask a question and, based on the answer, the AI decides if the player is happy or sad.

- **Computer vision:** This algorithm lets the computer **see** through your webcam. It identifies landmarks on your face like where your nose or eyes are, or tracks your movement.

Example: Using the Face Sensing extension, you can make a hat sprite follow your head as you move in front of the camera!

c. How AI learns

To use an AI algorithm in your Scratch game, you usually follow these three steps:

- **Train:** You give the computer training data such as lots of examples of photos, sounds, or text.
- **Learn:** The AI algorithm looks at the data and creates a model (its brain).
- **Use:** You bring the created model into Scratch. Now, instead of using the block When Space Key Pressed, you use a block that says When AI sees [Apple].

d. Role of AI in game

- **Personalization:** The game can respond to your voice or your face.
- **Smart enemies:** Instead of moving in a simple circle, an enemy can learn where you like to hide.
- **Accessibility:** You can make games that people can play just by moving their head or speaking, even if they can't use a keyboard.

Application Activity 5.3

- 1) What do we call the list of rules an AI follows?
- 2) Give one example of an AI behavior in a game.
- 3) AI characters are controlled by a second human player (True or False)
- 4) How do you control the character in the game?
- 5) How can an AI algorithm change the controls?
- 6) Could you move the player by tilting your head?
- 7) Could the background color change based on how loud you clap?

5.4. Building the game environment

Activity 5.4

Get a piece of paper and draw your dream game level. Include where the player starts, platforms to jump on, obstacles to avoid, and a goal to reach. Think about colors and themes such as space, jungle or city, etc.

- 1) What theme did you choose for your game?
- 2) Why did you place obstacles on the stage?
- 3) How does your layout help or challenge the player?

5.4.1 Platform and background creation

A background is the scenery behind everything (like wallpaper) while a platform is the surface where characters can stand on like floors and ledges.

In Scratch, a platform or a background can be created by:

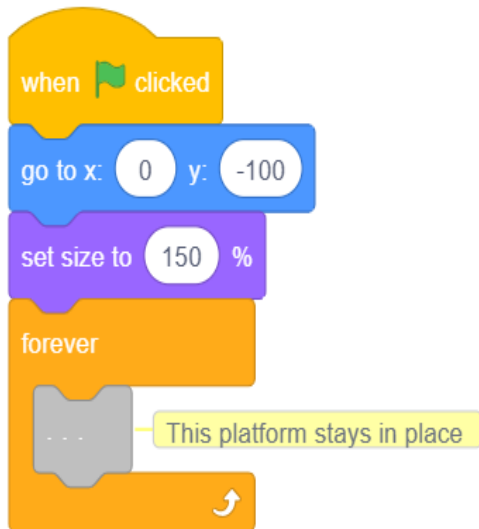
a. Choosing backgrounds:

- Step 1:** Click the Stage icon in the bottom-right
- Step 2:** Click Choose a Backdrop
- Step 3:** Pick from categories like Outdoor (Forest, Beach, Space), Indoor (Room, Castle, School) and Patterns (Stripes, Circles, Grid).

b. Creating platforms:

- Step 1:** Add a new sprite (like Block-A or Rocks)
- Step 2:** Make it wider using the Grow tool
- Step 3:** Change its color if needed
- Step 4:** Duplicate it to make more platforms

Example scratch blocks code for a simple platform:



Picture 5. 4: scratch blocks code for a simple platform

5.4.2 Sprites (Player, Obstacles, Goals)

Sprites are the characters and objects in your game. Every game need three main types:

a. Player Sprite: The character YOU control. Should be easy to see and control. Usually has movement scripts

- **Examples:** Cat, Dog, Giga, Wizard

b. Obstacle Sprites: Things to avoid

- **Examples:** Rocks, Fire, Ghosts, Spikes
- Should look dangerous or tricky
- Usually have movement or damage scripts

c. Goal Sprite: The target to reach

- **Examples:** Star, Gift, Door, Flag
- Should look rewarding or important
- Usually gives points or ends the level

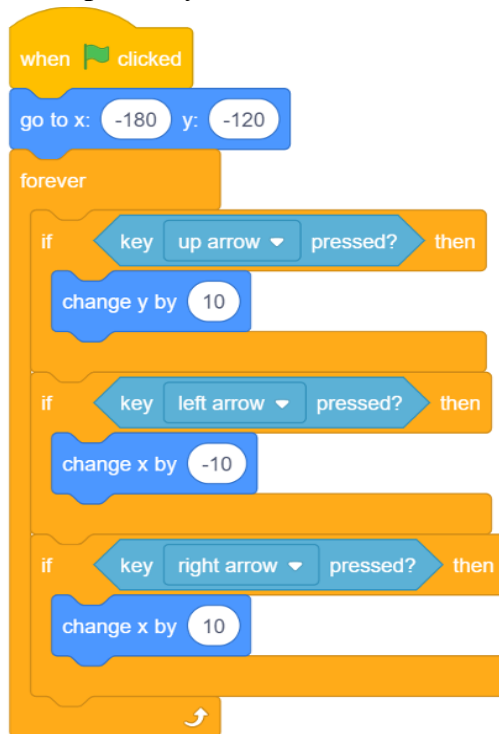
How to add and customize sprites:

- Step 1:** Click the **Choose a Sprite** button

Step 2: Browse categories or search

Step 3: Use the costume editor to change colors, add effects, draw your own!

Example: Player movement code



Picture 5. 5: Add and customize sprites

5.4.3 Plan basic game layout

Planning saves time and makes better games. It is like drawing a map before building a city. The game layout checklist includes several key elements that help in designing a clear and engaging game as follows:

- Start Point: Where the player begins
- Pathway: How they move through the level
- Challenges: Where obstacles are placed
- Rewards: Where goals or collectibles are placed
- End Point: How the level finishes

Layout planning steps:

Step 1: Draw on paper first by sketching your level design, marking important positions and planning the player's journey.

Step 2: Use Coordinates in Scratch by taking into account that Scratch stage is 480 pixels wide 360 pixels tall, the center is (0, 0), the left edge is x: -240, right edge is x: 240 and the bottom edge is y: -180, top edge is y: 180.

Step 3: Balance challenge and fun so that the game is not too easy, not too hard and frustrating.

Example layout plan:

- Start: (-200, -150) → Platform 1: (-150, -100) → Obstacle: (-50, -80)
- Platform 2: (0, -50) → Collectible: (50, -30)
- Platform 3: (100, 0) → Goal: (180, 20)

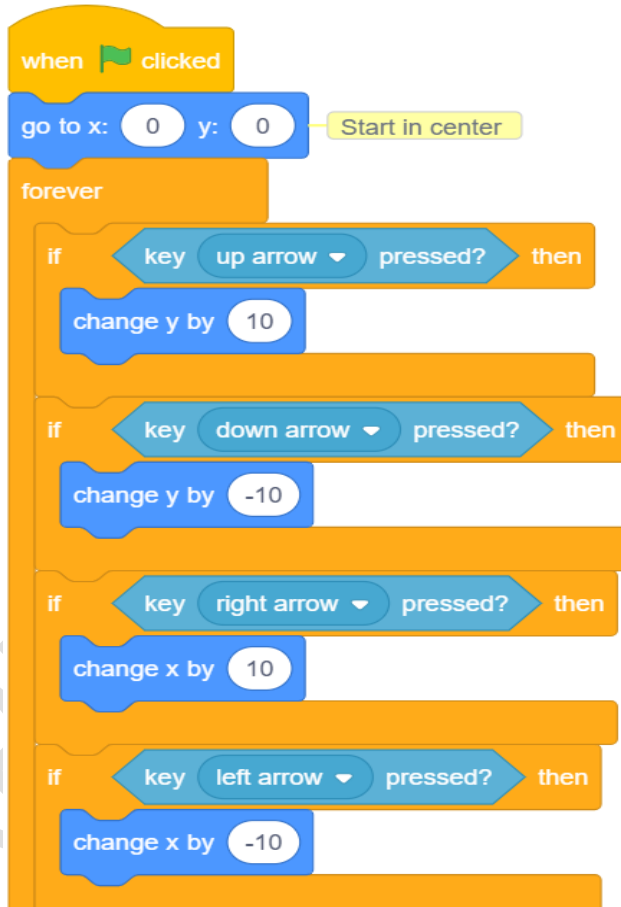
5.4.4 Example of AI chase challenge

The game to be created have the following elements:

- A player (Cat): Controlled by the player.
- An AI enemy (Net): Automatically follows the player using a simple algorithm.
- An AI scoring system: Automatically detects collisions, awards points, and resets the chase.

a. Step-by-step implementation plan

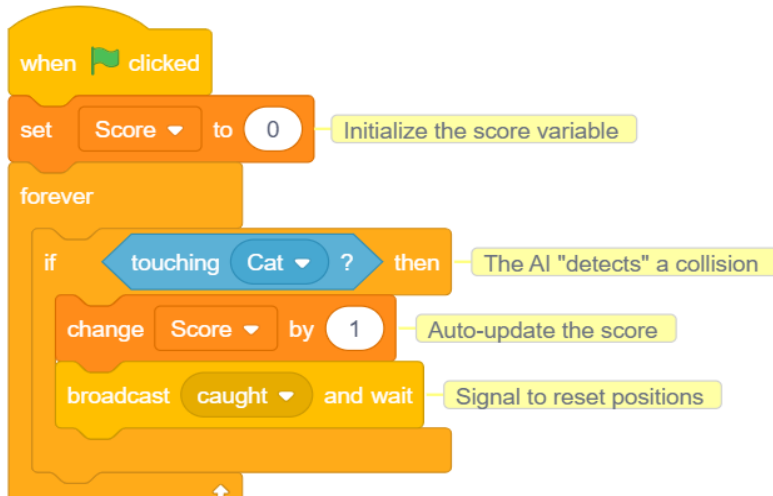
Step 1. Set the mechanisms to control the game by making it respond to keyboard input. It follows an algorithm When an arrow key is pressed, the character moves by changing its X or Y position, allowing the player to control movement using Scratch blocks.



Picture 5. 6: Player control

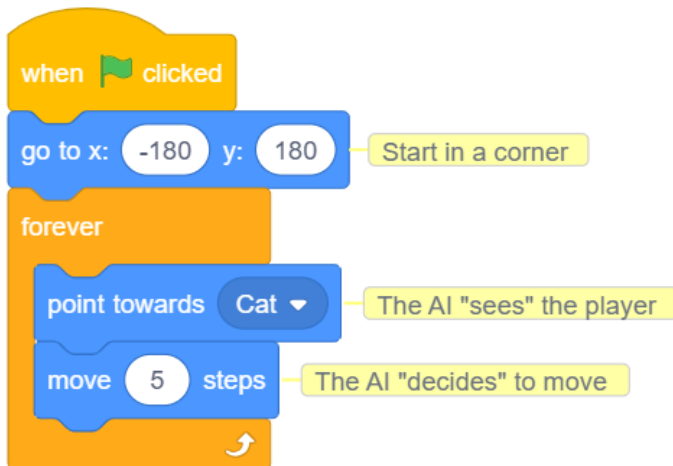
Step 1: Enemy AI will use the following behavior

- **Algorithm:** Forever → Point towards Player → Move a few steps.



Picture 5. 7: Enemy AI

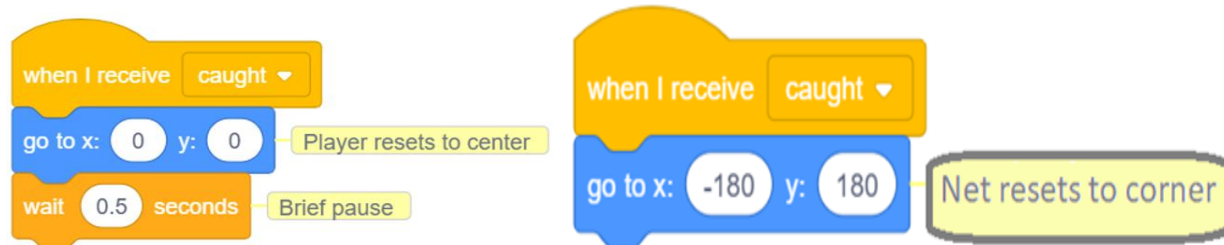
- **Scratch blocks Code:**



Picture 5. 8: Scratch algorithm

Step 2: Add an AI algorithm for auto-scoring and reset

- Algorithm: Forever → If touching Player → Change Score → Broadcast Reset.
- The broadcast message is used to Restart everything again (Reset). This is an important idea, because AI systems can send messages to each other and communicate.
- Scratch blocks Code (for Net):
- Scratch blocks Code (for Cat & Net - Reset Logic):



Picture 5. 9: Scratch block of codes

Application Activity 5.4

Create a simple platform game environment in Scratch called Jump & Run Adventure.

- Open a new Scratch project, name it, and select a suitable background.
- Design the game level by creating platforms at different heights using block sprites and colors.
- Add three main sprite types namely a player sprite with basic movement controls, an obstacle sprite that moves and a goal sprite placed at the end of the level.

5.5 Player movement and controls

Make a game in which one person is the programmer and another is a horse. The Programmer can only give three commands: Change X by 10 (Move right), Change X by -10 (Move left), Change Y by 50 (Jump up) while the horse must follow these steps to get from one side of the room to the other.

- What happens to the horse after a jump if there is no command to Change Y by -50?
- What will be the impact on the horse after using commands Up, Down, Left, and Right?

5.5.1. Player movement with arrows

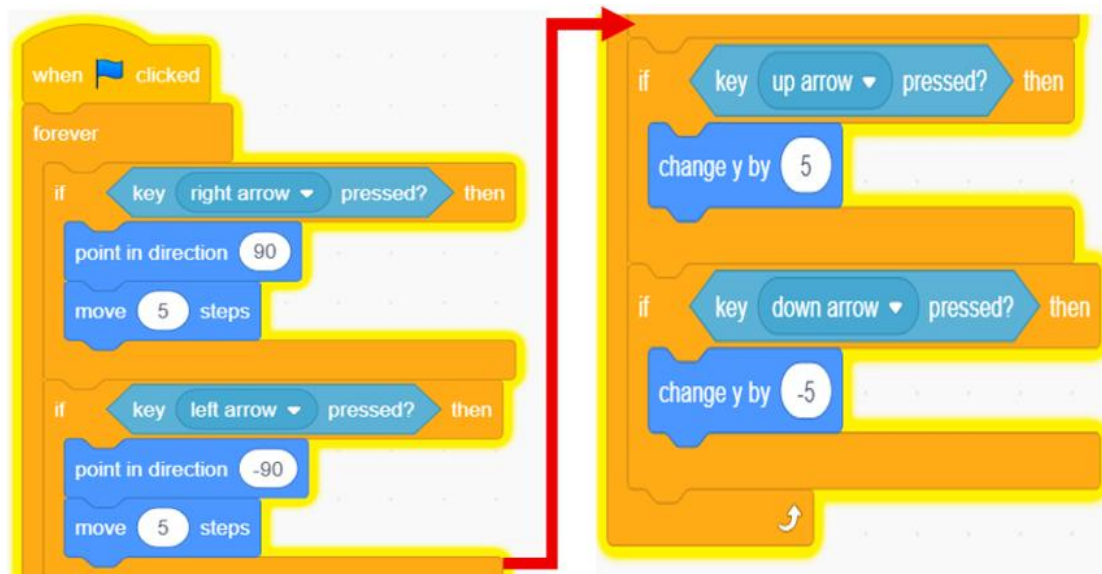
To make a character move, Scratch is programmed to listen to the keyboard by using the Event Blocks or Sensing Blocks. When you press the Right Arrow, the sprite should point right and walk and when you press the Left Arrow, the sprite should point left and walk.

Remember to set the sprite's Rotation Style to Left-Right so that they don't turn upside down when walking left.

Here are the steps for controlling the player movements

Step 1: Set up the player by adding a player sprite. Choose any character from the sprite library.

Step 2: Use appropriate blocks for arrow key controls. Here below is an example:



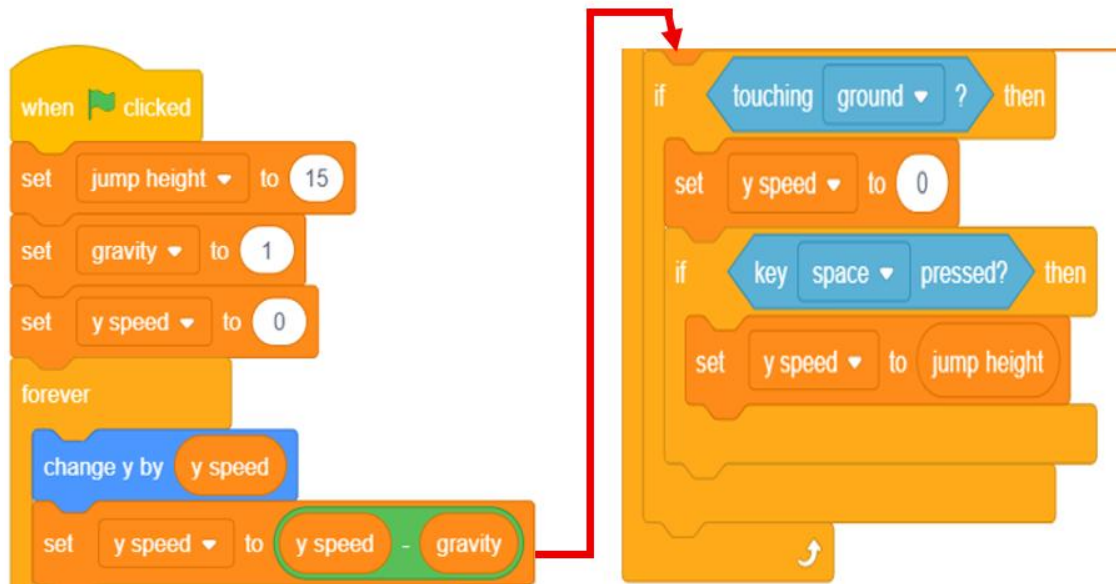
Picture 5. 10: Example of codes for player movements

5.5.2 Jump, fall, and collision control

- **Jumping:** This is when there is a change in the Y position (up).
- **Falling (gravity):** In the real world, what goes up must come down. For this a script that tells the sprite to fall until it touches a platform is needed.
- **Collision (touching):** This is how the sprite knows it is standing on something. Use a Touching Color or Touching Sprite block.
 - If the player is touching the **Ground** color, they stop falling.
 - If the player touches an **Enemy**, the game might end!

a. How to code them

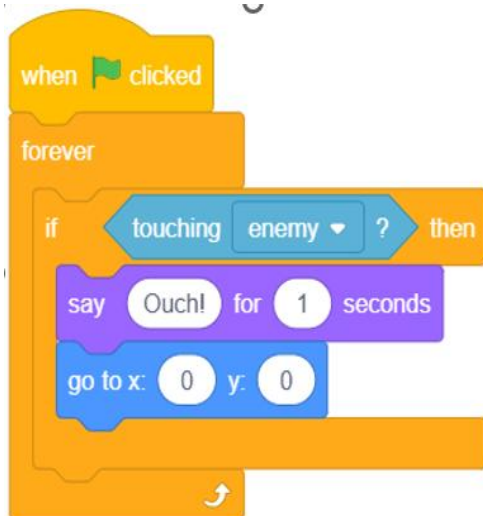
Step 1: Creating a jump by using codes like the ones below.



Picture 5. 11: Creating a Jump in scratch

Step 2: Adding a ground by creating a ground sprite such as a platform or floor and give it the name “ground” in the sprite info.

Step 3: Add collision detection features by using blocks such as the ones below:



Picture 5. 12: Collision detection

b. Coordinate movement in Scratch

To coordinate movements, use X and Y coordinates in order to position sprites accurately, and using coordinates for precise movement. The coordinates

Y: 180 (Top)



| (X: -240, Y: 180) (X: 240, Y: 180)



| (X: -240, Y: 0) (X: 240, Y: 0)



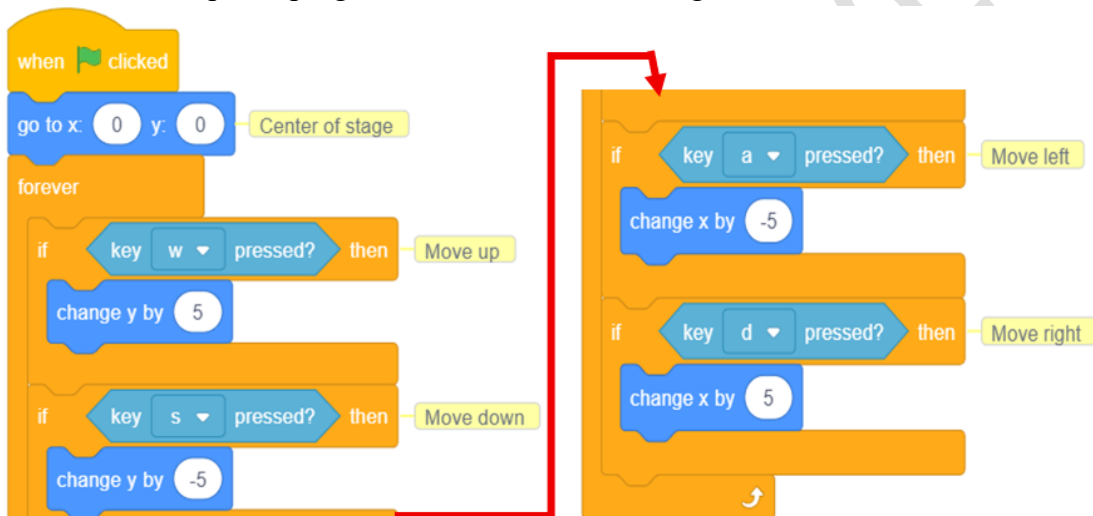
| (X: -240, Y: -180) (X: 240, Y: -180)



Y: -180 (Bottom)

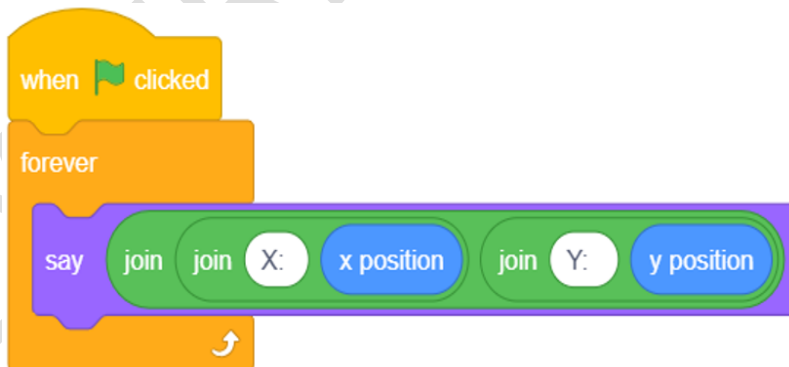
Picture 5. 13. The Scratch Stage Grid

Here is an example of program blocks for coordinating movements:



Picture 5. 14: Coordinate movement

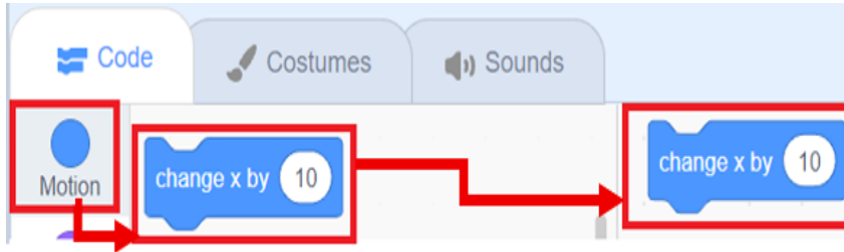
- Position checker tool: Add this code to see coordinates in action:



Picture 5. 15: Position checker tool

5.5.3 Coordinate movement in Scratch

To move sprites accurately, use the Coordinate System. A stage is like a big map where the x axis is used to move both right and left. To move right, increase X (e.g., Change X by 10) while for moving left, decrease X (e.g., Change X by -10).



Picture 5. 16: Movement on X-axis

Sprites can also move up and down by using the Y axis. To move up, increase Y (e.g., Change Y by 10) while for moving down, you decrease Y (e.g., Change Y by -10). The center in Scratch is the very middle of the screen which is X: 0 and Y: 0.

Application activity 5.5

- 1) Which axis (X or Y) do you change to move a sprite up and down?
- 2) If you want your sprite to move to the left, should you change X by a positive number (10) or a negative number (-10)?
- 3) What is the name of the yellow blocks used to start a script such as When [space] key pressed?
- 4) Why does a sprite need a falling or gravity script after it jumps?
- 5) What happens if your sprite's rotation style is not set to left right and you try to walk left?
- 6) Look at the Scratch stage. If a sprite is at X: 0, Y: 0, where on the screen is it located?

5.6 Adding obstacles and scoring

Activity 5.6

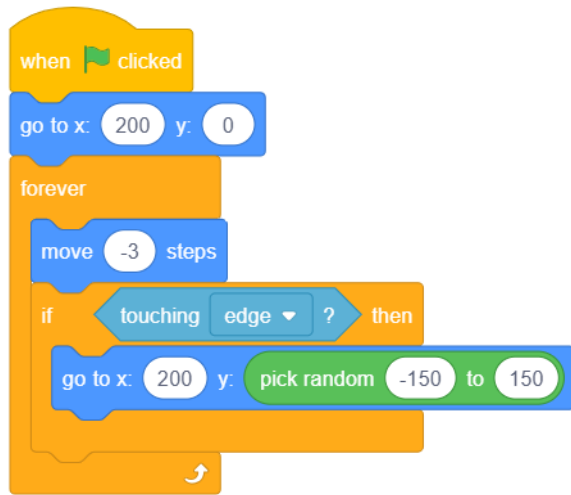
- 1) What is the difference between an enemy and a trap?
- 2) Which block category detects if two sprites hit each other?
- 3) If a player touches a reward, what should happen to the sprite?
- 4) What is a variable in your own words?
- 5) Why must we set Score to 0 when the green flag is clicked?
- 6) To make a score go down, do we use a positive or negative number?
- 7) Why do game developers test their games many times?
- 8) A variable can only store one number at a time? (True or False)

5.6.1 Interactive elements

Interactive elements are objects in a game that react when a player touches them. They make a game interesting. They are three types of interactive elements namely enemies, traps and coins (rewards to collect).

a. Enemy code

Here is a Scratch script to illustrate the logic for an **enemy sprite** that moves across the screen and reappears once it hits a boundary.



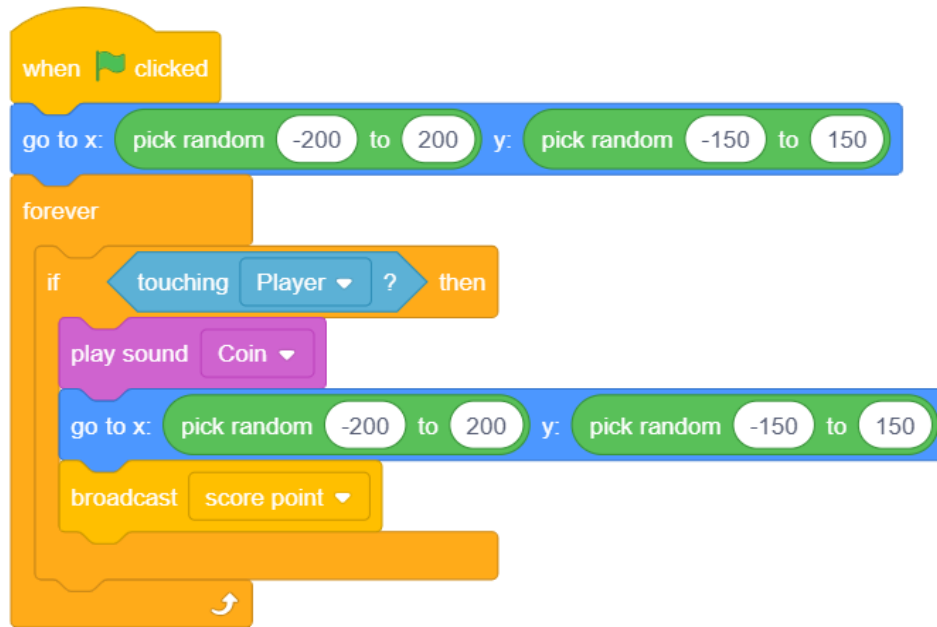
Picture 5. 17: Enemy code

Interpretation:

- When the game starts, the enemy is positioned on the far right side of the stage at the vertical center using the blocks When Green Flag Clicked and Go to x: 200 y: 0.
- The enemy continuously moves left using the blocks Forever and Move -3 steps, with the forever loop keeping the movement ongoing.
- Check whether the enemy has reached the edge of the stage by using the block If Touching Edge?
- When the enemy touches the left edge, it returns to the right side using go to x: 200 y: pick random -150 to 150, with a random y-position for unpredictable movement.

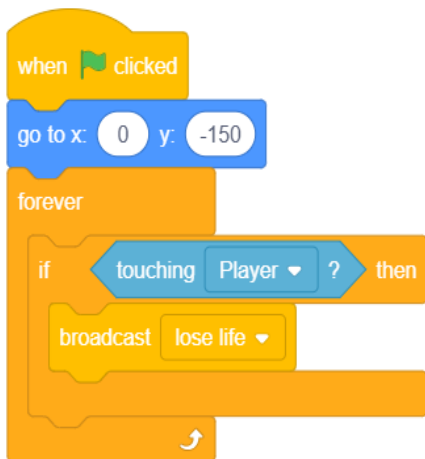
b. Adding a coin to collect

In many games, collecting items makes the gameplay more fun and rewarding. In Scratch, a coin can be added as an object that the player must touch or gather while playing. The coin can increase the score, create challenges, and give the player a goal to achieve during the game.



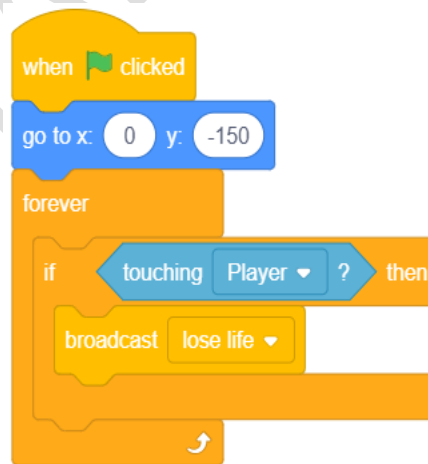
Picture 5. 18: Adding a coin to collect

c. Creating a simple trap



Picture 5. 20: Creating a simple trap

d. Visual guide: Game elements



Picture 5. 19: Visual guide: Game elements

5.6.2 Variable based scoring

A Variable is like a scoreboard that keeps track of numbers. It can go up, down, or reset to zero.

a. Creating Score variable

To create a variable in Scratch follow these steps:

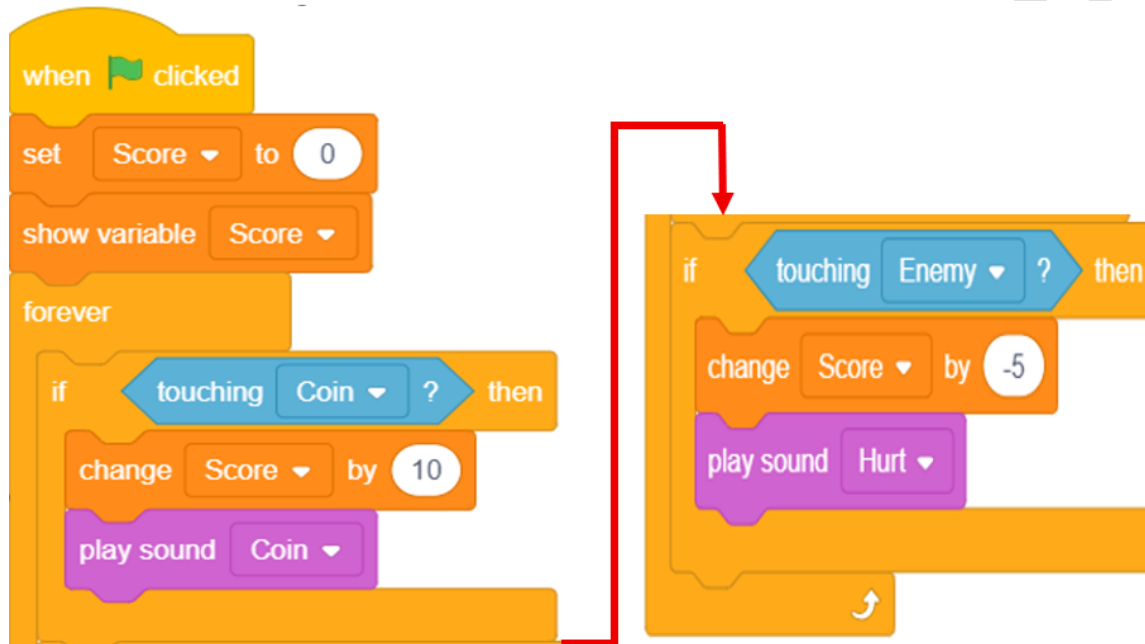
Step 1: Go to the **Variables** category and click **Make a Variable**

Step 2: Name it the variable as **“Score”** and choose **For all Sprite**

b. Basic scoring system

In many interactive games, a scoring system is used to track the player's progress and reward their achievements. By using variables and conditional logic, we can create a dynamic environment where the score changes based on the player's actions.

The following script demonstrates how to initialize a score at the start of a game and update it automatically whenever the player interacts with specific objects, such as collecting rewards or encountering obstacles.

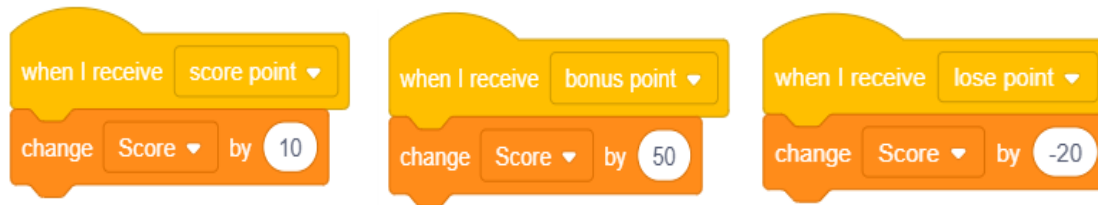


Picture 5. 21: Basic scoring system

c. Advanced scoring

In addition to basic interactions, a more complex game often requires different types of rewards and penalties. By using broadcast messages, we can trigger specific score updates from different parts of the program or different sprites. This allows the game to distinguish between standard achievements, special bonuses, and significant mistakes.

The blocks below illustrate how to use "When I receive" events to manage multiple scoring scenarios, ensuring the player's total score reflects the specific value of each item or event they encounter.



Picture 5. 22: Advanced scoring

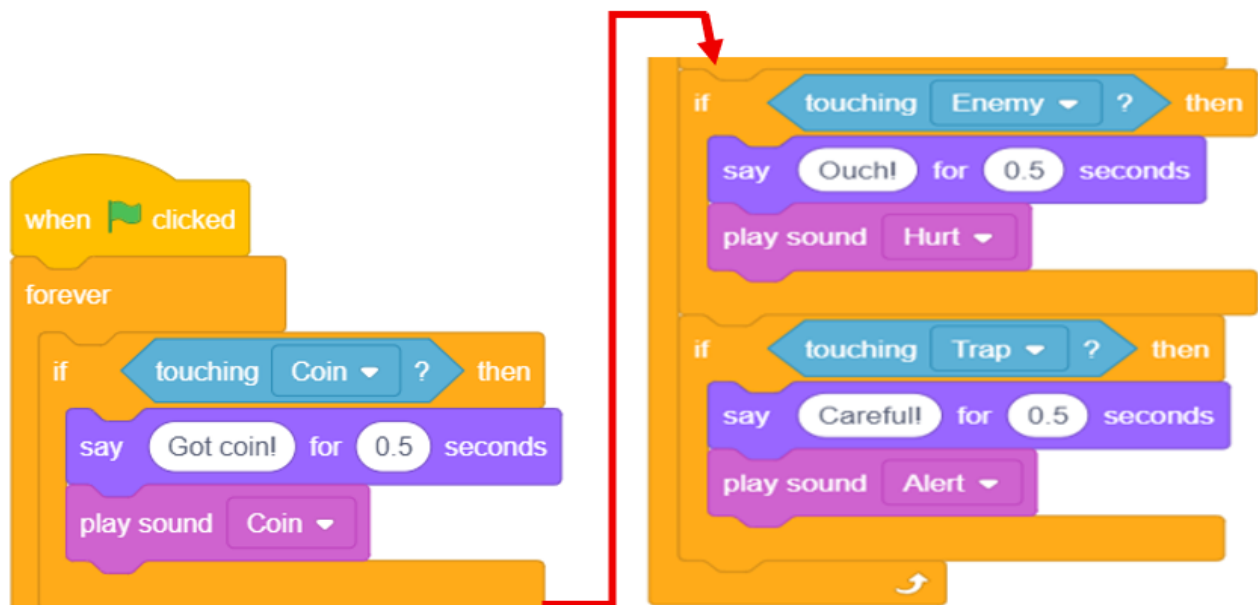
5.6.3 Testing and feedback

Testing helps us find problems and make the game more interesting. Every game played has been tested many times.

a. Collision detection testing

Before building a full game, it is important to check if the computer correctly identifies when two objects meet. Using simple visual and audio feedback, such as a sprite speaking or playing a sound, helps verify that the program is responding to interactions in real time.

The blocks below show how to test if a sprite is sensing various objects like coins, enemies, or traps. This process ensures that the detection logic works well before adding more features like scoring.



Picture 5. 23: Collision detection testing

b. Sound feedback library:

To make a game more engaging, adding sound effects provides immediate feedback to the player for different game events. For example, when a coin is collected, using a Coin or Pop sound signals a successful pickup. If the player is hurt, corresponding sounds such as Ouch can be

produced to highlight the mistake, while a bonus item interaction is best paired with celebratory sounds.

Application activity 5.6

- 1) What happens when you create a variable called “Score”?
 - a) It adds a new sprite to your game
 - b) It creates a place to store numbers that can change
 - c) It makes your game run faster
 - d) It changes the background color
- 2) Why do we use collision detection in games?
 - a) To make the game look pretty
 - b) To know when sprites touch each other
 - c) To increase the game speed
 - d) To add more colors
- 3) What is the purpose of sound feedback in games?
- 4) Which code block would you use to increase score by 10 points?
- 5) Create a scoring system
 - a) Make a variable called “Points”
 - b) Create a coin sprite that gives 5 points when collected
 - c) Create an enemy sprite that takes away 3 points when touched
 - d) Make sure the score shows on screen
- 6) Add sound feedback
 - a) Add a “Coin” sound when collecting coins
 - b) Add a “Hurt” sound when touching enemies
 - c) Add a “Tada” sound when score reaches 50 points
- 7) Design three different interactive elements:
 - a) A moving enemy
 - b) A stationary trap

5.7 AI behaviors in games

Activity 5.7

- 1) If a game character always moves in a straight line, is it easy or hard to beat? Why?
- 2) What happens if a game character hits a wall while chasing you? Should it stop, or should it try to find a way around?
- 3) What is the main reason developers use AI in games?
 - a) To let the player control all characters at once.
 - b) To make the game graphics look more realistic.
 - c) To create enemies that can think, react, and provide a challenge.
 - d) To make sure the player never loses a level.
- 4) Which term describes an enemy that actively pursues the player character?
 - a) Patrol behavior
 - b) Following behavior
 - c) Idle behavior
 - d) Static behavior

- 5) If an enemy always moves exactly 10 steps to the left and never changes, this is an example of:
- | | |
|-------------------------|----------------------|
| a) Programmed behavior | b) Adaptive behavior |
| c) Intelligent behavior | d) Random Behavior |

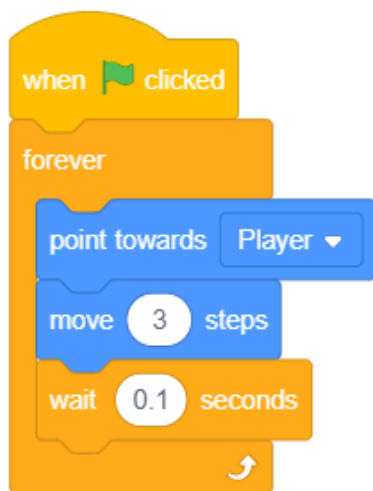
5.7.1 Program AI movement.

AI movement means programming characters to move on their own, without player control. The most common type is following behavior, where enemies chase the player

a. Basic following AI

To make a game more challenging, you can program enemies to actively track the player's movements. This is achieved by using motion blocks that constantly update the enemy's direction so it always faces and moves toward the player sprite.

The script below shows how to create a basic following behavior. By placing these commands inside a loop, the enemy will continuously adjust its path, creating a simple but effective chase mechanic.

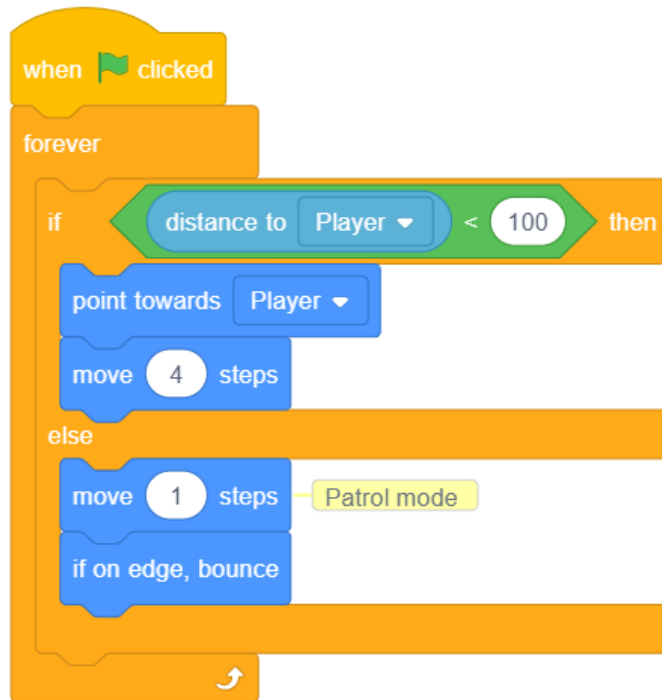


Picture 5. 24: Basic following AI

c. Smart following

To create a more realistic enemy, you can program it to sense how close the player is before it starts to chase. By using a distance check, the enemy can stay in a patrol mode, moving slowly or staying in one area until the player enters its detection range.

The following script uses an if-else block to decide between two behaviors. If the player is within a certain distance, the enemy switches to a faster chase. If not it continues its regular patrol movement.

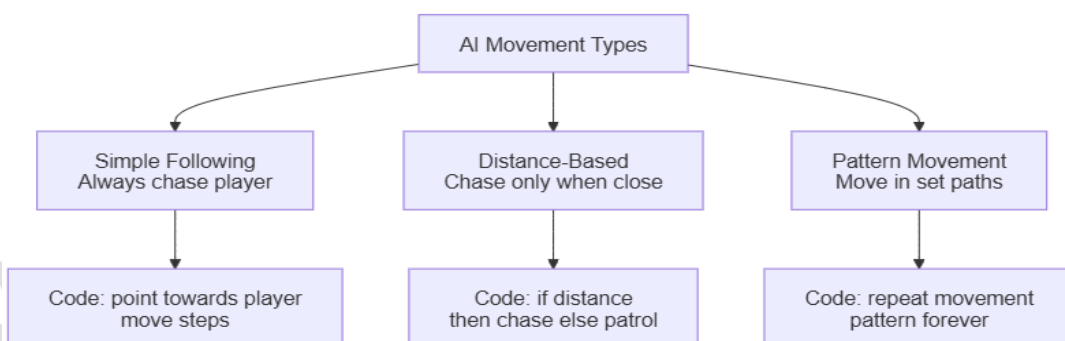


Picture 5. 25. Smart following (with distance check)

c. AI following patterns visual guide

To create a diverse and engaging game, different characters can be programmed with unique ways of moving. By selecting specific logic patterns, you can decide whether an enemy constantly pursues the player, waits until they are nearby to attack, or follows a predictable path.

The following flowchart outlines the three primary categories of AI movement. It connects each behavior type to the specific logic needed to implement it, providing a clear map for designing various game challenges.



Picture 5. 26: AI following patterns

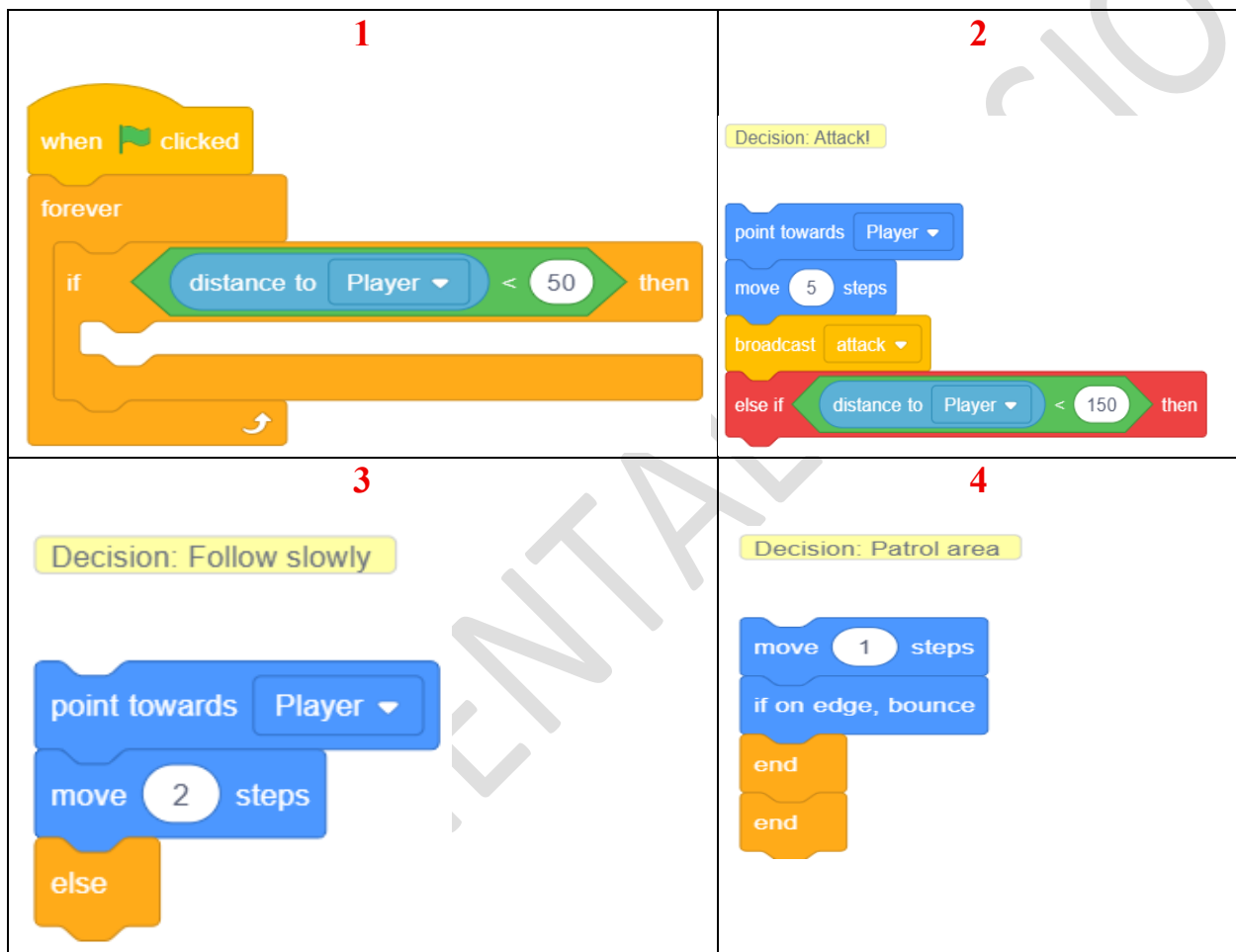
5.7.2 AI decision processes

Decision processes are the thinking rules to give to AI characters. They help AI decide what to do in different situations.

a. Simple decision making

By using nested conditional statements, you can create characters that make different choices based on the situation. This allows a sprite to evaluate multiple conditions in a specific order, responding with a unique action for each scenario.

The blocks below illustrate a decision making process where a sprite determines its behavior based on its proximity to the player. By defining ranges for attacking, following, or simply patrolling, you can build a more interactive and unpredictable experience for the user.



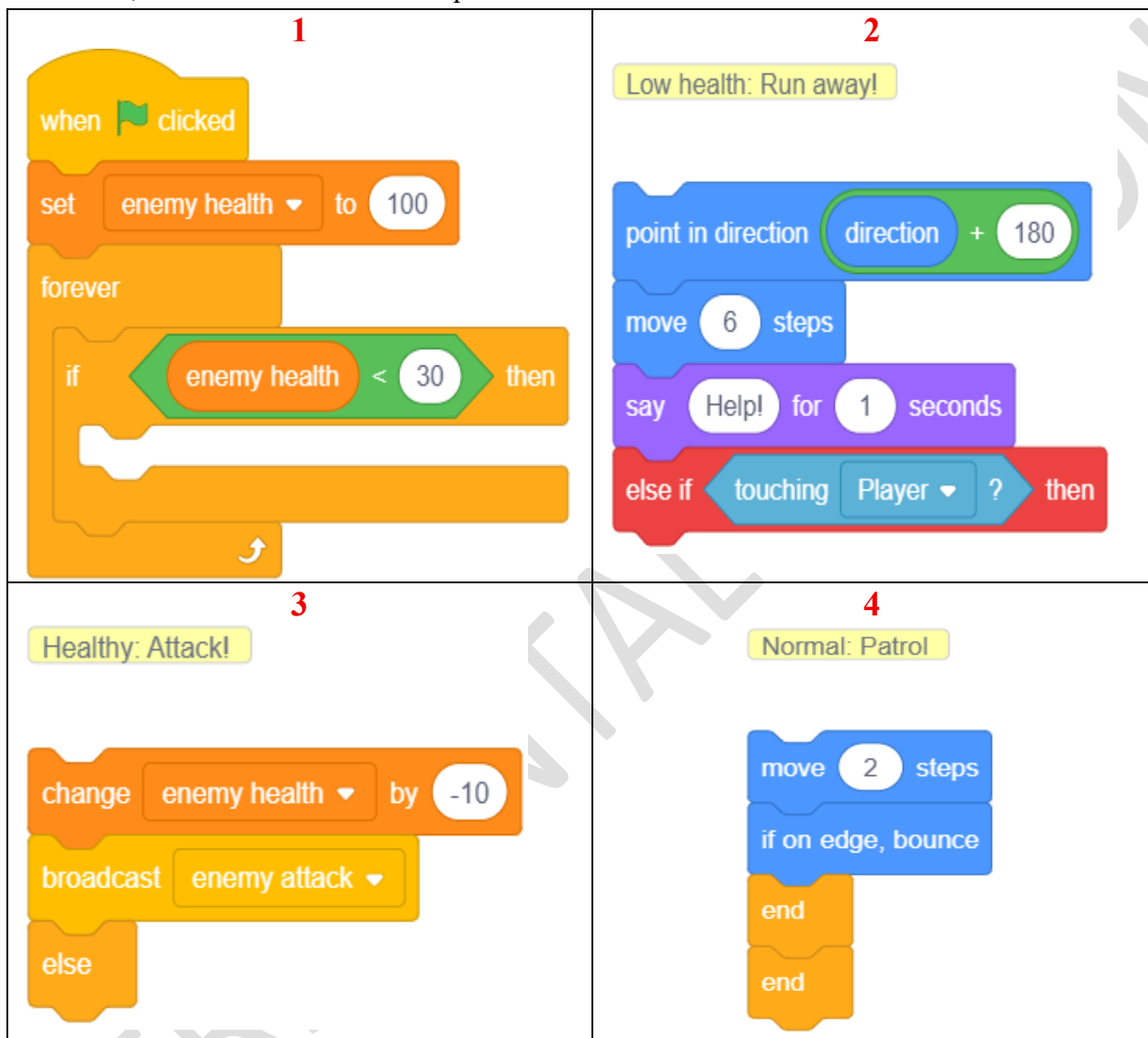
Picture 5. 27. Simple decision making

b. Health based decisions

Health based decisions refer to using a variable to determine how a sprite should behave or when the game state should change.

Beyond simple movement, Artificial Intelligence can be designed to change its behavior based on its current status. By monitoring a health variable, an enemy can switch from an aggressive state to a defensive one, making the game feel more reactive and lifelike.

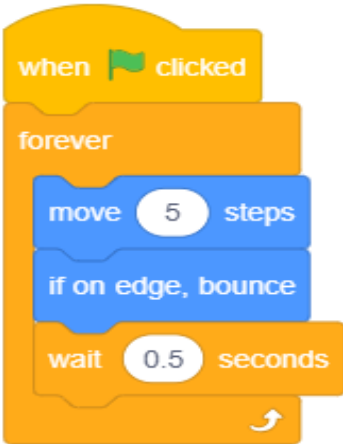
The blocks below demonstrate how a sprite can evaluate its own health to choose an appropriate action. This logic allows the enemy to attack when it is strong, patrol when no immediate threat is detected, or flee when its health drops below a certain level.



Picture 5. 28: Health-based decisions

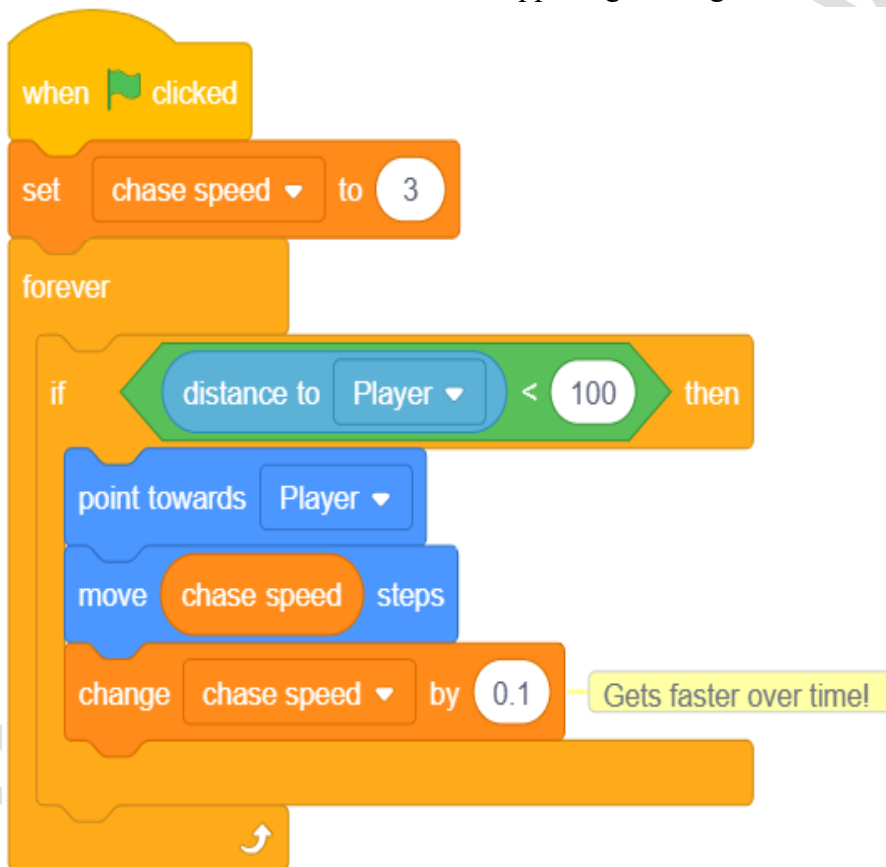
5.7.3 Programmed and adaptive behavior

Programmed behavior refers to a set of exact instructions that a sprite follows without any variation, ensuring that it always performs the same action in every situation. Because the movement is fixed, these behaviors are very easy for a player to predict, making them ideal for simple game elements.



Picture 5. 29. Example of programmed behavior

Adaptive behavior allows a sprite to change its actions based on the specific situation, making it more dynamic and harder for a player to predict. Unlike fixed instructions, this type of behavior enables a character to react to what is happening in the game environment in real time.



Picture 5. 30: Adaptive behavior example

Application activity 5.7

- 1) Why is it a good idea for an enemy to check how far away the player is before it starts chasing them? How does this help the game look more realistic?
- 2) If a bad guy is just walking back and forth on a set path and doesn't know that the player is there, in which AI state is it?
- 3) Why is a game more fun to play many times if the enemies change their behavior rather than doing the exact same thing every single time?
- 4) Using Scratch, build the following:
 - a) Make an enemy move left and right across the screen between X: -100 and X: 100 without stopping.
 - b) If the player gets close with a distance less than 150, the enemy should stop its walk, turn toward the player, and start chasing them.
 - c) Create a special rule where the enemy gets scared. If the player touches a Power up item, the enemy must stop attacking and run to the edge of the screen to hide.
- 5) AI is used in games but also for real things like self-driving cars. Why is it important to double-check the decisions we program into an AI to make sure it is safe for people?

5.8. Teachable Machine for AI

Activity 5.8

- 1)
 - a) Explain how children learn to differentiate a cat and a dog.
 - b) By analogy, explain how a computer can learn to differentiate a dog and a cat.
- 2)
 - a) If you saw a picture of a car once from the front, can you recognize it from the back or the side?
 - b) Explain why it is necessary for a computer to see many different photos of the same thing?
- 3) Here are the gestures and what they usually represent:
 - Open hand: Stop or Wait
 - Pointing up: Go up
 - Pointing down: Go down
 - Waving: Hello
 - a) How did you know what each gesture mean?
 - b) Could a computer learn these gestures too?

It is a free, web-based tool provided that allows you to create machine learning models quickly and without any coding. Teachable Machine has special tools that enable the training of AI to recognize gestures, sounds, or images. This is real AI that learns from examples, just like humans learn from practice.

5.8.1 AI prediction with Scratch

Prediction is when an AI looks at new information, like your hand in front of a camera, and makes a guess about what it is based on what it has learned before. It can be trained on images such as hand gesture, sounds such as clapping or speaking body positions.

a. Teachable Machine with Scratch

A Teachable Machine is like a brain creator which helps in building a custom AI without writing a single line of complex code, which you can then plug directly into Scratch projects. It involves these steps:

- Train the model in Teachable Machine
- Export it to Scratch
- Use the Machine Learning Model extension. The Scratch project can now recognize what the trained model has learned.

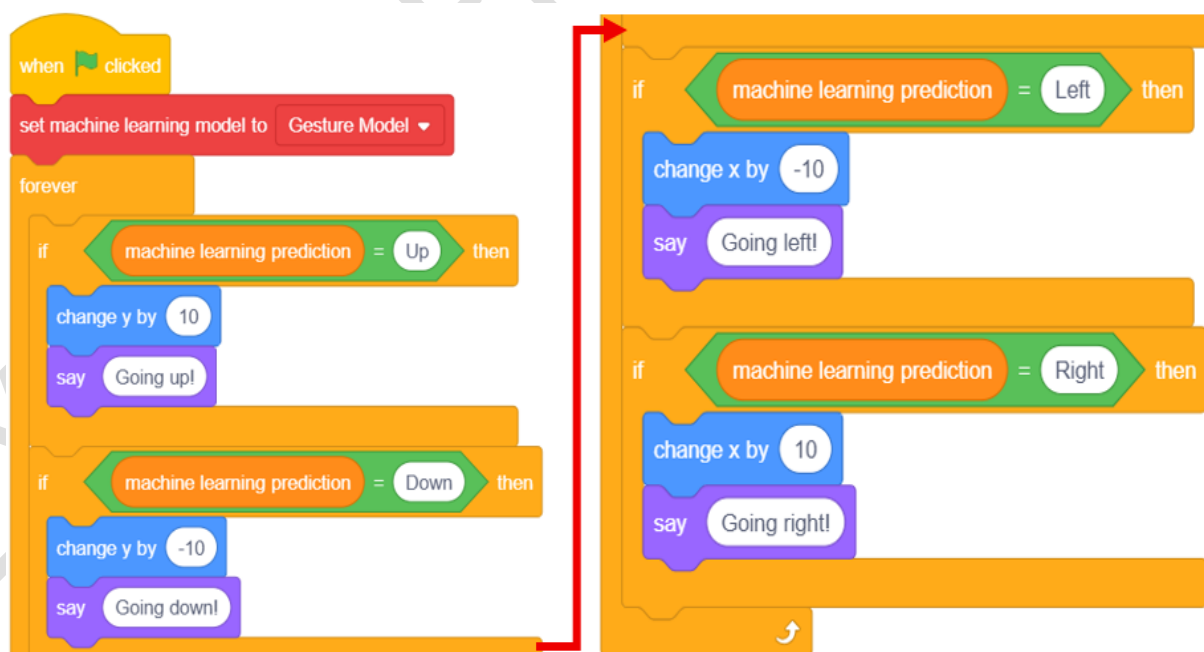
b. Simple gesture control example:

Step 1: Training in Teachable Machine

To create a simple gesture control follow this procedure:

- Go to teachablemachine.withgoogle.com and choose Image Project
- Create classes: Up, Down, Left, Right
- Take 50-100 photos for each gesture and train the model
- Export to Scratch

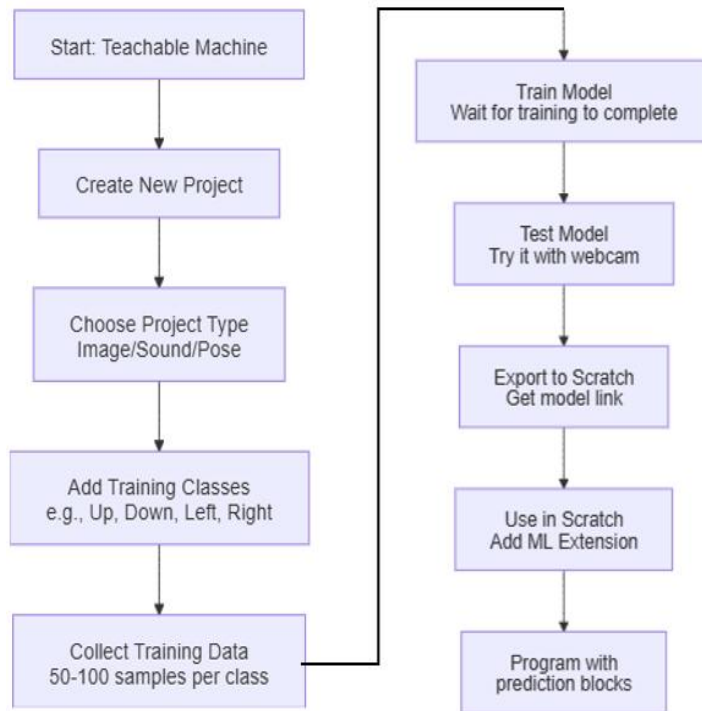
Step 2. Create the Scratch code for gesture control as shown below:



Picture 5. 31: Scratch Code for Gesture Control

c. Visual guide: Teachable Machine workflow

The diagram below illustrates the main steps involved in creating and using a machine learning model with Teachable Machine and Scratch. The process begins by creating a new project and selecting a project type such as image, sound, or pose recognition. Training classes are then added, and sample data is collected for each class. After the data is collected, the model is trained and tested to check its accuracy. The trained model can be exported and connected to Scratch, where prediction blocks can be used to create interactive AI-powered projects.



Picture 5. 32: Teachable Machine Workflow

d. Real world use of machine learning

Machine learning can be applied in many real-life situations including the following

- **Agriculture:** Recognizing healthy and diseased plants
- **Education:** Sign language recognition for inclusive learning
- **Healthcare:** Exercise movement tracking
- **Security:** Recognizing authorized personnel

5.8.2 Training data and AI performance

Training data is the examples we give to the AI to learn from. Like showing many pictures of cats so it learns what a cat looks like.

a. Collecting good training data

When collecting training data such as photos, it is important to take them in different lighting conditions and from various angles to help the model learn more accurately. Gestures or actions

should be performed consistently so that the model can recognize patterns correctly. A sufficient number of samples, usually between 50 and 100 for each class, should also be collected. Including some variety in backgrounds can improve the model's ability to recognize objects or movements in different environments.

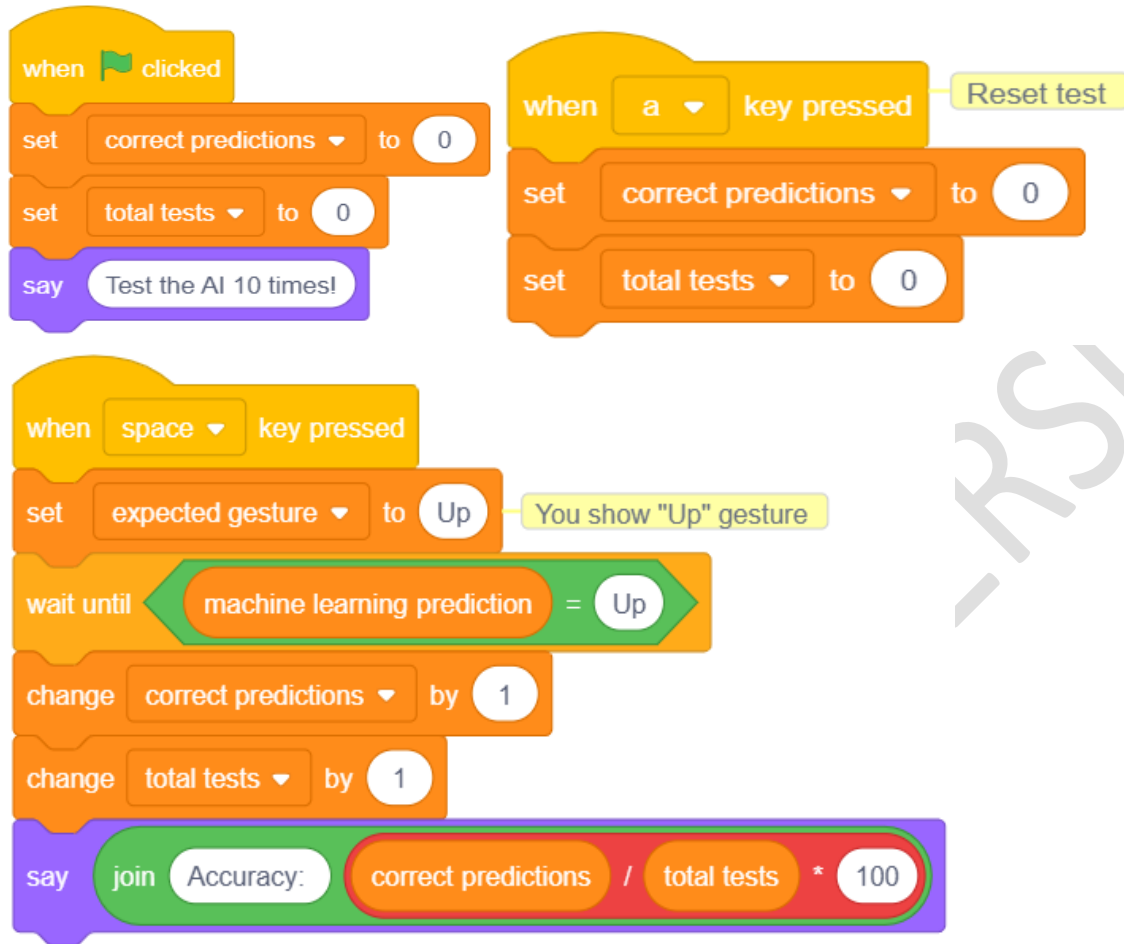
Avoid the following while collecting images:

- Use blurry photos
- Change gestures mid-training
- Use only one background
- Collect too few samples
- Include confusing examples

b. Code for testing AI accuracy:

Testing is an important step in machine learning because it helps determine how accurately a trained model can recognize and classify data. By comparing the expected result with the model's prediction, it is possible to measure the performance of the AI system and identify areas that may need improvement.

The blocks below demonstrate how to test the accuracy of an AI model in Scratch. The program begins by resetting the counters for correct predictions and total tests. Each time a specific gesture is shown, the AI checks whether its prediction matches the expected result. Correct predictions and total attempts are counted automatically, and the program calculates the accuracy percentage to show how well the trained model performs.



Picture 5. 33: Code for testing AI accuracy

c. Improving AI performance:

When AI confuses the Up and Down gestures, the model may respond incorrectly because the two gestures look too similar or were not trained with enough variety. To solve this problem, collect more clear and distinct training samples for each gesture. Ensure that the Up gesture is always performed differently from the Down gesture, and use different angles, lighting conditions, and backgrounds during training. Retraining the model with additional accurate examples helps AI better distinguish between the two gestures and improves its recognition accuracy.

- **Solution when AI doesn't recognize gestures quickly:**

- Step 1:** Reduce number of classes
- Step 2:** Simplify gestures
- Step 3:** Improve camera quality
- Step 4:** Add more varied training data

- **Ethical considerations:**

Privacy: The webcam should be used only after obtaining clear permission from the user, ensuring their consent is respected before any camera access is allowed.

Bias: This is to ensure fairness by training AI using a wide range of different examples so that it can perform well across various people, situations, and conditions.

Transparency: This is about being able to clearly understand how an AI system arrives at its decisions, so users can see the reasoning behind its outputs and trust the process.

Purpose: AI should be used in ways that bring positive benefits, such as supporting learning, solving problems, and improving everyday life, rather than for harmful or negative uses.

Application activity 5.8

- 1) What is the main purpose of Teachable Machine?
 - a) To play games online
 - b) To create machine learning models without coding
 - c) To edit photos and videos
 - d) To learn programming languages
- 2) Why is training data important for AI?
 - a) It makes the computer run faster
 - b) It provides examples for the AI to learn from
 - c) It increases internet speed
 - d) It changes the computer's color
- 3) What does "Garbage In, Garbage Out" mean in AI?
 - a) Computers need cleaning
 - b) Bad training data leads to bad AI performance
 - c) AI can sort trash automatically
 - d) Training data should be recycled
- 4) How many training samples should you collect per class?

a) 5–10 samples	b) 10–20 samples
c) 50–100 samples	d) 500–1000 samples
- 5) Create a Teachable Machine project with 3 classes namely Happy (smiling face), Sad (frowning face) and Neutral (normal face)
 - a) Collect 30 training images for each class
 - b) Train the model and test it
 - c) Export to Scratch

5.9. End unit assessment

5.9.1. Project activity

Project title: Jump & Run AI Adventure Game

❖ **Guidance:**

- This project should be done individually and use Scratch (offline or online).

- Design a simple game environment with: One player Sprite, at least one platform, One obstacle or trap, One goal or reward
- Program the player to: Move left and right using arrow keys, Jump using the space bar, Fall back down using gravity logic
- Add Artificial Intelligence by creating an enemy that follows or reacts to the player, using IF conditions or distance checking
- Use Teachable Machine to control movement using hand gestures.
- Test the game carefully to fix errors and ensure AI behavior is fair and safe.

Questions on the Project

- 1) What is the main purpose of the Jump & Run AI Adventure Game?
- 2) Which ICT tools or software were used to create and demonstrate the game?
- 3) Explain how the player controls the character and avoids obstacles in the game.

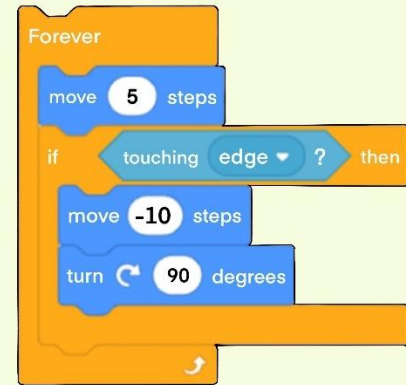
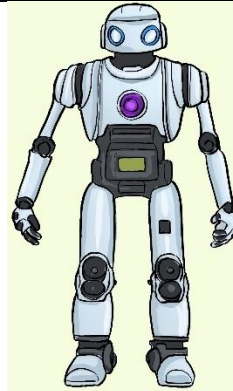
5.9.2. Assessment questions

- 1) In a game, Kamanzi is being pursued by an enemy.
 - a) Why is it more effective for the enemy to verify Kamanzi's distance before beginning the chase?
 - b) How does this behavior mimic real world logic?
- 2) If a bad guy in a game is just walking back and forth on a path and does not see Mutesi's character, what is the mood or state of that enemy?
- 3) Why is a game more fun to play many times if the enemies change what they do instead of doing the exact same thing every single time?
- 4) Using Scratch, build these parts of a game:
 - a) Make an enemy sprite move left and right across the screen forever
 - b) If the player gets close when the distance is less than 150, then the enemy must stop walking, turn toward the player, and start chasing them.
 - c) Create a rule where the enemy gets scared. If the player touches a Power-up, the enemy must stop attacking and run to the edge of the screen to hide.
- 5) AI is used in games, but also in real things like self-driving cars on the roads in Kigali. Why must we double-check the decisions we give to an AI to make sure people stay safe?
- 6) Create a game with the following characteristics
 - a) When the player touches a coin, add 5 points. If they touch an enemy, take away 3 points.

b) Add a pop sound for collecting coins, a crash sound for getting hurt, and a cheer sound if the player reaches 50 points.

EXPERIMENTAL VERSION

UNIT 6. ROBOT LOGIC IN SCRATCH



Key unit competence:

Simulate intelligent robot behavior using Scratch programming logic.

Learning Objectives:

Knowledge and understanding

- Explain robot mechanics, their real-world benefits, and Scratch coding blocks for control.
- Show Variables for robot memory and Sensors for environmental awareness and reactions.
- Issue multiple instructions to robots; use Lists for maze path memory.

Skills

- Link Scratch blocks to move robot sprites for real-life helpful tasks.
- Build programs with Sensors for surroundings detection and Variables for adaptive behavior.

Attitudes and values

- Recognize robots' life-improving potential; use coding ethically to solve safe, kind real-world problems.
- Embrace step-by-step thinking, persistence, and a "can-do" mindset for complex coding challenges.
- Value Sensors and Variables as a robot's "eyes and memory" for intelligent, safe decisions.

6.0. Introduction

This unit introduces learners to the foundational concepts of robotic logic using the Scratch programming environment. Scratch is a visual, block-based coding platform that is used to create interactive programs without needing advanced coding skills. This unit explore how robots make decisions, follow instructions, and respond to different conditions using logical commands.

Introductory activity

- 1) What is a robot?
- 2) You have a mechanical arm that can pick up a ball. Does it need a brain to know when to grab the ball, or does it just move on its own?

- 3) If you were a robot walking toward a wall, what should you do?
- a) Walk into the wall forever.
 - b) If you see a wall, turn around.
- 4) If a robot is counting how many precious stones it picks up in a game, it needs a special box in its brain to keep track of the number. In Scratch, what is the name of a box that holds a changing number of precious stones?

6.1 Introduction to robotics

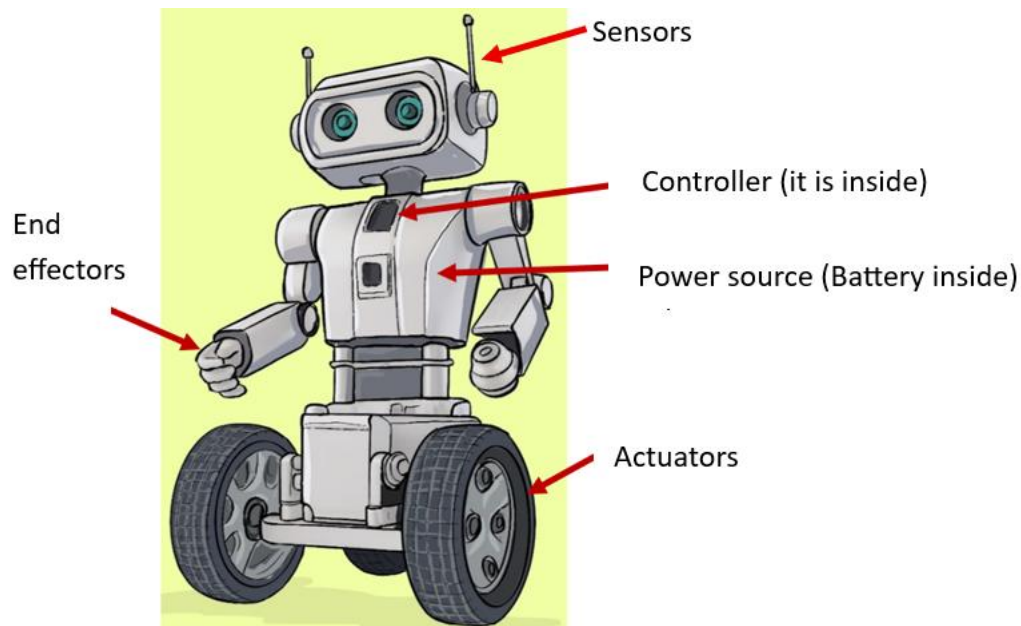
Activity 6.1

- 1) Answer True or False if each of the following is a robot or just an object.
 - a) A gas cooker
 - b) A self-driving car
 - c) An automatic vacuum cleaner
- 2) List areas where robots can be used.
- 3) What is a robot's job and why is it important?
- 4) What hardware do you think a robot needs?
- 5) We need to give instructions and thinking to a robot. What do we use for that?
- 6) If a robot takes over a person's job at a factory such as welding, is that a good thing or a bad thing?
- 7) How can we make sure a robot never hurts a human?

6.1.1. Robot parts

Robot are made of different parts that work together perform tasks. The common parts of a robot are in the table below:

Nº	Part Name	Role in a robot	Example
1	Controller	It is like the brain of a robot. It stores instructions and processes information	Arduino Uno, None MCU, etc.
2	Sensors	They detect changes in the environment and transmit it to the robotic system.	Light sensor, temperature sensor
3	Actuators	An actuator is a device that turns energy (like electricity or air) into physical movement.	Motors, hydraulic cylinders, LCD screen
4	Power source	They give electrical energy to the robot.	Batteries, electricity, solar power
5	End effectors	They handle, manipulate and sense objects.	Grippers, welding tools, spray nozzles



Picture 6. 1. Parts of a robot

6.1.2. Types of robots

There are different types of robots as they come in many forms depending on their design and purpose.

- **Industrial robots:** They are automated machines designed to perform specific tasks in manufacturing and production environments. They are programmed to carry out repetitive, precise, and often dangerous operations that would be difficult or unsafe for humans.

Examples: Welding robots

- **Service robots:** They are designed to help humans with everyday tasks in homes, hospitals, offices, and public spaces.

Examples: Cleaning robots.

- **Mobile robots:** These are robots designed to **move from one place to another** while performing tasks. They move using wheels, legs, or propellers.

Examples: Drones, delivery robots.

- **Humanoid robots:** They are robots designed to look and move like human beings. They often have a head, torso, arms, and legs, allowing them to perform tasks that are normally done by people.

- **Medical robots:** They are used in hospitals, clinics, and laboratories to support healthcare professionals. They are designed to work in environments where precision and reliability are very important. Examples include surgical robots which assist doctors in performing complex surgeries with high precision.

6.1.3. Importance of robots

Robots play a major role in modern life because they can do tasks rapidly and with accuracy. They help humans in many fields by doing difficult, repetitive, or dangerous work. Here is their importance:

- They increase speed and accuracy of work as they can perform tasks faster than humans and with high precision. They can also repeat the same task with consistent quality and never get tired.
- They can work in dangerous places such as chemical factories or do tasks that may cause injury or fatigue to humans.
- They improve productivity in factories because they can work all the time, speed assembly lines and ensure uniform product quality. Their manpower is less costly compared to humans.
- They Help in exploration of space, oceans, and hazardous Areas
- They provide services such as cleaning and delivery.

6.1.4. Robot ethics

As robots become more common in homes, schools, hospitals, and workplaces, it is important to ensure that they are used in ways that protect people and support society. Robot ethics provides guidelines that help designers, programmers, and users create and use robots responsibly.

- **Safety:** This is the most important principle in robot ethics. Robots must be designed, programmed, and used in ways that do not harm humans, animals, or the environment. This principle ensures that robots operate responsibly and support humans. It is therefore the responsibility of the robot maker to ensure that it is safe.
- **Privacy:** Robots that collect, store, or use any data must protect people's information. This means robots should not collect, share, or use people's private data such as names, phone numbers, photos, or locations without permission.
- **Responsibility:** Since robots are machines humans must remain responsible for robot actions. Therefore, robots cannot be blamed for what they do. The people who design, program, own, or use robots are responsible for their actions and any harm they can cause.
- **Fairness:** Robots should not discriminate people but treat them equally and not favor or harm anyone because of their gender, age, disability, race, language, background, etc. Those who make robots should make sure this criterion is met.
- **Transparency:** People should understand how robots make decisions and their actions must not be mysterious.

6.1.5 Hardware required for a robotics programming

Programming a robot requires more than just the robot itself. To write code, upload it, and test how the robot responds requires other hardware tools. These items help connect the robot to a computer, transfer instructions, test circuits, and troubleshoot any errors that may occur.

Here are the items required:

a. Controller

This is the most important part which is considered as the brain of a robot. It is the small computer that holds the code you write in Scratch.

- **Microcontrollers:** Small boards like the Arduino, micro:bit, or the mCore.
- **Single board computers:** More powerful brains like the Raspberry Pi, which can even handle cameras and AI. It has all functionalities of ordinary computer.

b. The senses

Sensors collect information from the world so the robot can make decisions.

- **Ultrasonic sensor:** It uses sound to measure distance. This helps robots avoid hitting walls.
- **Line follower:** Detects black and white colors on the ground (helps robots follow a path).
- **Light sensor:** Detects how bright or dark it is.
- **Touch bumper sensor:** Tells the robot if it has bumped into something.

c. Actuators

These parts are considered to be like the muscles and allow the robot to move and do work.

- **DC Motors:** Used for spinning wheels to make the robot drive.
- **Servo Motors:** Used for precise movements, like moving a robot arm or a head.
- **LEDs & buzzers:** Used for the robot to talk to us with lights or sounds.

d. The connection

You need tools to send your code from your screen to the robot's brain.

- **Computer or tablet:** To write your Scratch code.
- **USB cable:** For a wired connection to download the code.
- **Bluetooth or Wi-Fi module:** For a wireless connection so the robot can move freely.

e. Support hardware

- **Chassis:** The skeleton or frame that holds all the parts together.
- **Power supply:** These are batteries like Li-Po or AA batteries that give the robot energy.
- **Jumper wires:** The nervous system that connects the sensors and motors to the brain.
- **Memory card:** Used to store programs, sensor readings, and other data that the robot needs during operation.
- **Breadboard:** Used to build and test electronic circuits without soldering during programming and prototyping.

Application activity 6.1

- 1) Explain how robots help people do dangerous jobs.
- 2) Your robot uses a camera sensor to collect information while working. Which robot ethics principles is not respected? Explain your answer.
- 3) Choose the hardware devices needed to program the robot in Scratch.

a) Computer	e) Controller (Arduino, micro:bit, Raspberry Pi)
b) Sensors	f) Motors
c) Batteries	g) Breadboard
d) Jumper wires	
- 4) You want to design a robot helper for the school, fill the following information needed for that robot:

- a) Robot's name: _____
- b) Robot's job: Helps with _____
- c) One sensor: To sense _____
- d) Rule: My robot will always _____

6.2 Getting started with Scratch

Activity 6.2

- 1) How would a robot know where to move?
- 2) Can a robot understand English? Explain how it can get commands from humans.
- 3) Open Scratch Editor and do the following tasks:
 - a) Click on "Choose a Sprite" then search a robot
 - b) Use these blocks to program it: When Green Flag clicked, Move 10 steps, Forever

6.2.1 Categories of blocks for a robot

The main categories of Scratch blocks are categorized into Motion, Looks, Sound, Events, Control, sensing, operators, variables, My blocks. There are also other categories of blocks that are added as extensions depending on the functionalities you want to add to scratch. Scratch uses blocks to give instructions. Each block has a role, just like robot parts.

- a. Motion blocks** ● : These blocks help the robot move forward, reverse, turn, or follow a planned path.

Examples:

- Move 10 steps: Robot moves forward
- Turn 15 degrees: Robot turns left or right
- Go to x: ... y: ... : Robot moves to a position

- b. Control blocks** ● : They manage the program's logic by organizing loops, conditionals, and timing so the robot can repeat actions, make decisions, and sequence tasks. They include blocks such as repeat or forever, branching with if and if else, and pausing with wait.

Examples:

- Forever: Robot keeps moving
- Repeat: Robot repeats an action
- if ... then: Robot reacts to a situation

- c. Sensing blocks** : They read sensor input such as touch, light, distance, sound, or button presses and feed them to the robot so that it takes decisions basing on the data received.

Examples: Touching color?, distance to object, touching sprite?

- d. Operators blocks** ● : They perform calculations and comparisons by providing arithmetic, logical, and relational tools that process numbers and values. In robotics they let you compare sensor readings to thresholds to detect conditions, compute and adjust motor speeds using numeric expressions, and combine multiple tests with logical operators so decisions can be made.

Examples: <, >, =, and, or.

e. **Events blocks** ●: They start and coordinate actions by triggering scripts in response to inputs like button presses, broadcasts, or sensor signals.

Examples: when green flag clicked, when key pressed

f. **Looks:** It controls appearance and signals by changing visual and auditory outputs. In robotics, they are used show robot status (ON/OFF), display messages on screen and change LED colors in robot simulations.

Example: When green flag clicked, say "Robot ON" for 2 seconds.

g. **Sound:** It controls sound output by playing tones, melodies and recorded audio to signal states, provide feedback, alert users, or enrich interactions, allowing the robot to communicate status, warnings, or responses through sound. In robotics it plays beeps or alarms and give sound feedback.

h. **Variables** ●: A variable in robotics is a small storage space that holds important information the robot needs while it operates. It can store values such as speed, allowing the robot to adjust how fast it moves. Variables also help the robot count steps or keep track of a score during tasks or challenges.

Example: When green flag clicked, set mode to "ON"

They allow the robot to remember its current direction or mode, making it possible to change behaviors or continue a task.

i. **My blocks:** It is used to create your own command that you can reuse anywhere in the project. It allows to group several instructions into one block to make code clear and easier to understand. By creating My Block, you reduce repetition, improve readability, and make your robot or sprite perform complex tasks through a single, customized command.

Example: My block to avoid obstacle

Step 1: Create a My block

- Click My Blocks (pink category) then click Make a Block
- Give it a name such as avoid obstacle and click OK

Step 2: Define the My block

- This code tells the robot **what to do** when the block is used.
- define avoid obstacle
- turn 90 degrees
- move 10 steps

When the robot meets an obstacle, it turns and moves away.

Step 3: Use the My block in the Main Program

- when green flag clicked
- forever
- move 5 steps
- if touching color [black]
- avoid obstacle

j. **Extensions:** Extension block in Scratch is a special set of blocks that adds extra features to your project beyond the standard motion, looks, and control blocks. Extensions allow Scratch

to connect with external devices or provide additional programming capabilities. For example, you can use extensions to control robots like LEGO Spike or mBot.

6.2.2 Robot simulation

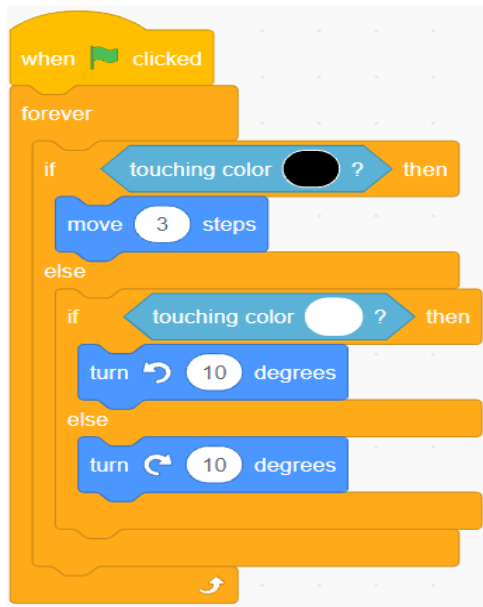
Robotics simulation is using software to model a robot and its environment so you can test how the robot behaves. Instead of using expensive materials to build a real robot with motors and sensors, you create a virtual robot that can do different activities such as moving, sensing obstacles, etc. To feel its surrounding environment, a robot must have sensors comparable to human's sensory organs. Here are the comm sensors used by robots:

No	Type of sensor	What It Does	Common Use	Example
1	Distance	Measures distance to objects	Obstacle avoidance	Ultrasonic, Infrared (IR), LiDAR
2	Touch	Detects physical contact	Detects collision	Bump sensor, Pressure sensor
3	Light	Detects light and color	Line-following robots	Light sensor, Color sensor
4	Sound	Detects sound	Voice or clap control	Microphone, Sound sensor
5	Motion	Detects movement and rotation	Balance and navigation	Gyroscope, Accelerometer, Encoder
6	Temperature	Measures heat or cold	Environment monitoring	Thermistor, Temperature sensor
7	Gas	Detects gases or smoke	Safety and rescue robots	Gas sensor, Smoke sensor
8	Vision	Allows the robot to see	Object recognition	Camera, Depth camera
9	Magnetic	Detects direction	Navigation	Compass, Magnetometer

Using simulations helps in experimenting freely, making mistakes safely and understanding the robot logic before testing the programs using real robots. Below are the common simulations of robots in Scratch:

a. Line following robot

This is a robot that moves by following a line drawn on the ground. It uses sensors to see the line and adjust its movement so it stays on the path. In real life, these robots are used in situations where a machine needs to move safely and accurately along a fixed path that must be followed with precision. Factories use line following robots to transport materials from one station to another.

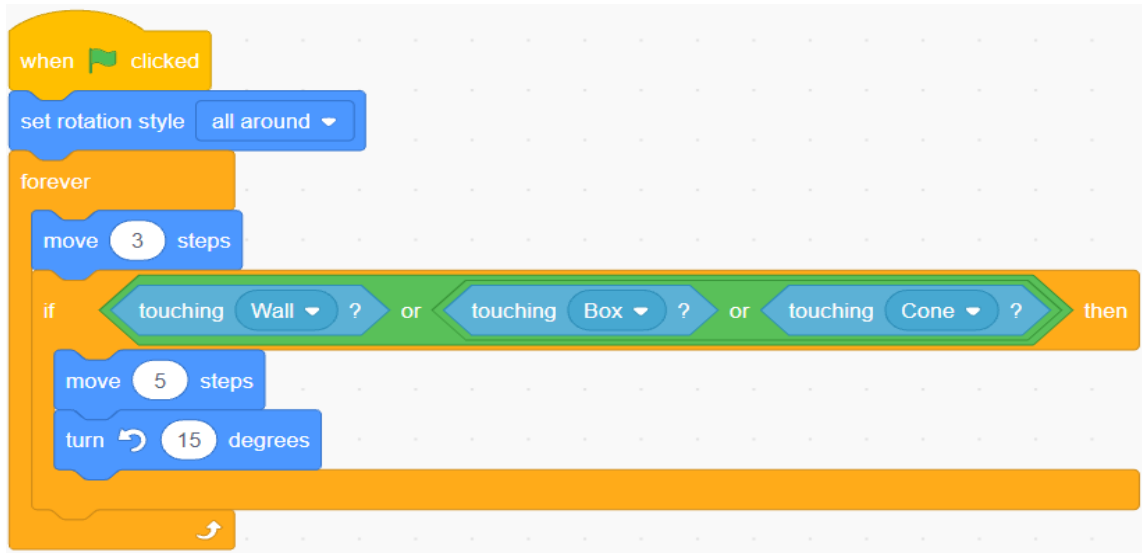


Picture 6. 2. Line following robot

b. Obstacle avoiding robot

An obstacle avoiding robot is a mobile robot designed to move around while automatically detecting and avoiding obstacles in its path. It uses sensors to see the environment and a control system to decide how to react. Instead of moving blindly, the robot uses sensors to detect objects in its path and automatically change direction to avoid collisions. It is like robots used in factories, warehouses, and even self-driving cars whereby those robots have to sense obstacles and avoid them.

In Scratch an obstacle avoidance robot is simulated by replacing a robot with a sprite while obstacles become other sprites (walls, boxes, cones) and sensors are replaced with the sensing blocks. Assume that robot sprite name is Robot, Obstacle sprites is Wall, Box, Cone, Sensing is done using (touching [sprite]? OR touching color? (if obstacles share the same color).

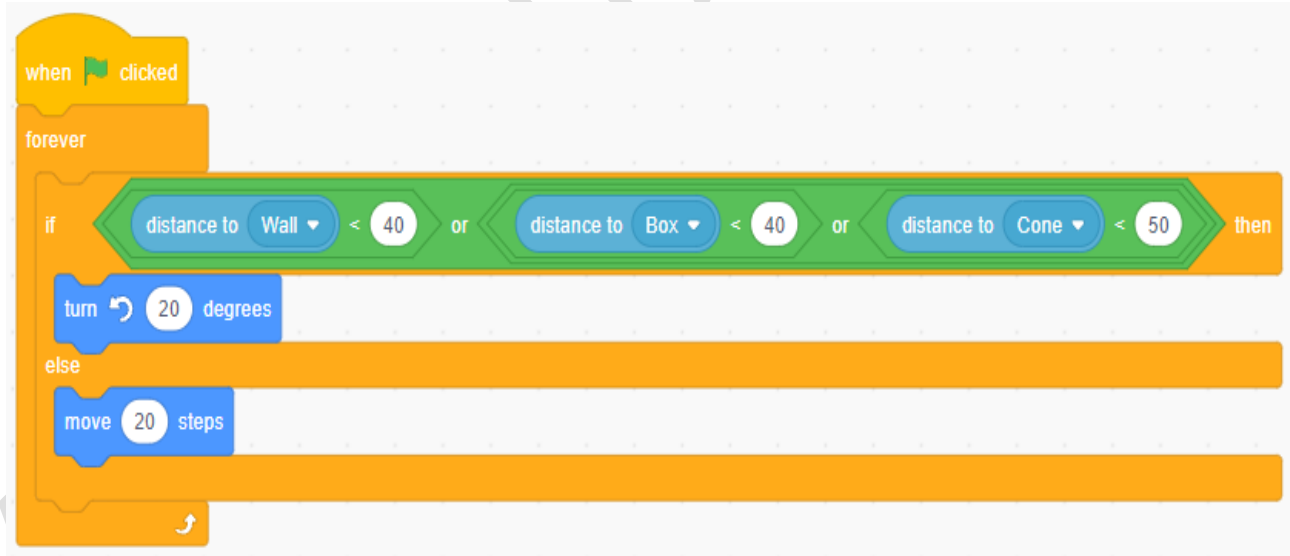


Picture 6. 3. Obstacle avoidance Scratch block code

The logic for the program above is that the robot sprite starts at the center of the stage and continuously moves forward. When it detects an obstacle, it moves backward slightly then turns 15 degrees clockwise and continues moving forward again.

- **Improved obstacle avoidance using distance sensor logic**

This version simulates a **distance sensor** using distance to block. The next is the full Scratch program, where the sprite compute the distance the decide which direction to taken:



Picture 6. 4. Obstacle avoidance using distance sensor logic

This program describes the behaviors of a robot while moving and become closer to the obstacle like Wall, Box, and Cone. If the distance is below 40 to these obstacles the Robot turn 20-degree backsword otherwise it moves 20 steps forward.

- **Real robots' components in Scratch simulation**

In Scratch, a robot sprite is fully equipped with all needed sensors. Making the sensor active is important to use Scratch sensor block for simulating a robot sprite. The above program mimics intelligent avoidance behavior of a robot.

The next table show most used sensors on a robot and its corresponding sensors' names in Scratch environment

No	Real Robot Component	Scratch Simulation
1	Ultrasonic sensor	distance to block
2	Infrared sensor	touching sprite?
3	Motors	move block
4	Turning wheels	turn block
5	Obstacle	Wall / Box sprite

Application Activity 6.2

- 1) Name three Scratch block categories used.
- 2) Which block helps the robot sense obstacles?
- 3) Which block helps the robot make decisions?
- 4) Why is simulation important before using a real robot?

6.3. Scratch variables

Activity 6.3

- 1) Explain how robots remember things like scores.
- 2) If you have a box labeled SCORE can the number inside it change, or must it stay the same forever?
- 3) How is a variable like a school bag or a cup that holds different amounts of water?
- 4) How can a robot change what it remembers such as a game score?
 - a) By deleting the whole program and starting over.
 - b) By using a variable that updates the value.
 - c) It is impossible for a robot to change a number.

6.3.1 Introduction to Scratch variables

a. Scratch variables

A variable in Scratch is a place where we store information that can change while the program is running. Examples of information stored in variables are Score, Speed, Number of steps, Game level, Lives in a game

b. Purpose of variables

Variables help programs and robots to **remember values, count things, make decisions, change behavior**. In robotics, variables help robots know how far they have moved, track how many times a sensor is triggered, control speed or direction.

c. Analogy

Variable is like a labeled box where you put something inside, a school bag that carries books, a cup that can hold different amounts of water. In this context the name of the variable is the label, and the value is what is inside.

6.3.2 Data types stored in Scratch variables

Scratch variables can store different types of data:

a. Numbers

These are values that count or measure.

Examples of variable that store number values in robotics:

- **Score** = 5, they are used in robotics to count movements
- **Speed** = 10, they are used in robotics to set robot speed
- **Steps** = 100, they are used in robotics to measure distance

b. Text

These are values composed with letters or words. The examples of variable that store text values in robotics are:

- **Status** = ON, used to show robot state.
- **Mode** = AUTO, used to display messages on screen

c. True or False

These are values used in decisions making. They can be either true (yes) or false (no).

The examples of variable that store yes or no values in robotics are:

- **LineDetected** = true. They are used in robotics to decide whether to stop or move
- **Obstacle** = False. They are used in robotics to check sensor conditions

6.3.3 Examples of variable use

Example 1: Score counter game robot

The variable name is score, then every time the robot touches a goal, score increases by 1. Then the robot shows the score on the screen, this helps the robot count success.

Example 2: Speed control

The variable name is speed, when speed is 5, the robot moves slowly, then When speed = 10, the robot moves fast. This helps the robot change movement.

Example 3: Obstacle detection (Robot Safety)

- Variable name: **obstacle**
- If obstacle = true, the robot stops
- If obstacle = false, the robot continues moving

This helps the robot **make safe decisions**.

Application activity 6.3

- 1) What is a variable in Scratch?
- 2) Give one example of information that can be stored in a variable.
- 3) Which data type is used to store scores or steps?
- 4) Which data type stores words or messages?
- 5) Why are variables important in robotics?
- 6) If a robot uses a variable called speed, what does this variable control?

6.4. Simulating one sensor

Activity 6.4

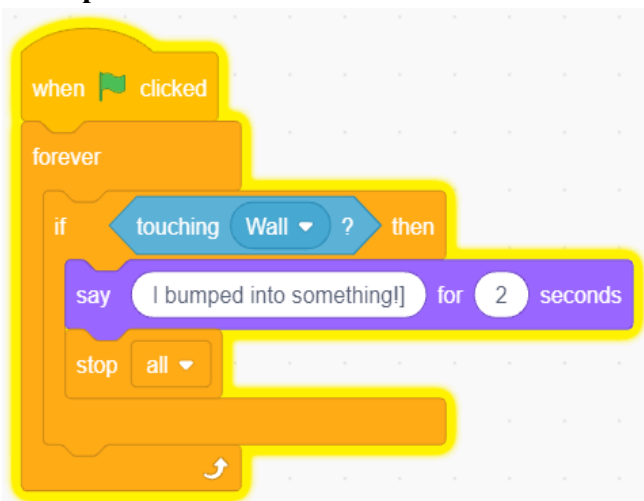
- 1) Robot don't have human eyes, what do they use to see a red line on the floor?
- 2) Bats fly in the dark by sending out sounds and listening for echoes? Explain how this works.
- 3) Some cars have a sensor on its front side. Which kind of information does this sensor send to the car?
- 4) If a robot doesn't have human eyes, what does it use to see a red line on the floor?
- 5) If a robot is programmed to stop when it hears a loud noise, which Scratch block can be used?
 - a) Move 10 steps
 - b) Loudness
 - c) Say Hello
- 6) Which sensor would a robot use to make sure it doesn't fall off the edge of a table?
 - a) A temperature sensor
 - b) A gas sensor
 - c) A touch or distance sensor

A Sensor is a device that let a robot see, hear, or feel. In Scratch, we use the light-blue Sensing Blocks to simulate these.

6.4.1 Touch sensor

A touch sensor tells a robot if it has hit a wall or a button. In Scratch, use <touching [mouse-pointer v]?> or <touching [edge v]?> block.

Example:

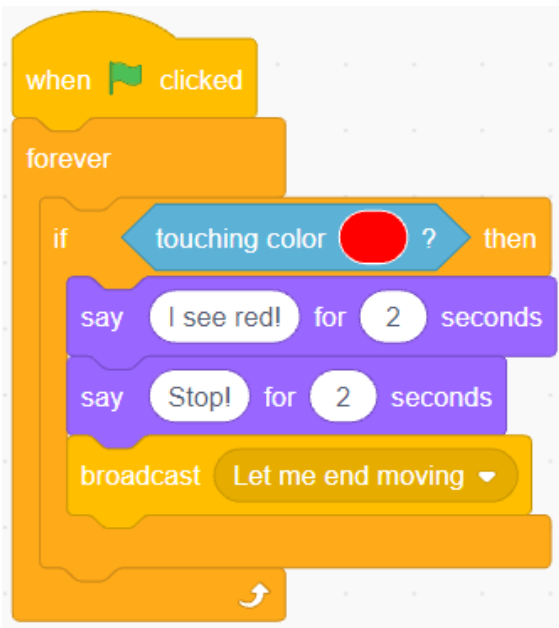


Picture 6. 5. Touching sensor

6.4.2 Color sensor

This sensor helps robots follow lines or sort objects by color. In Scratch, use <touching color [#ff0000]?> block.

Example:

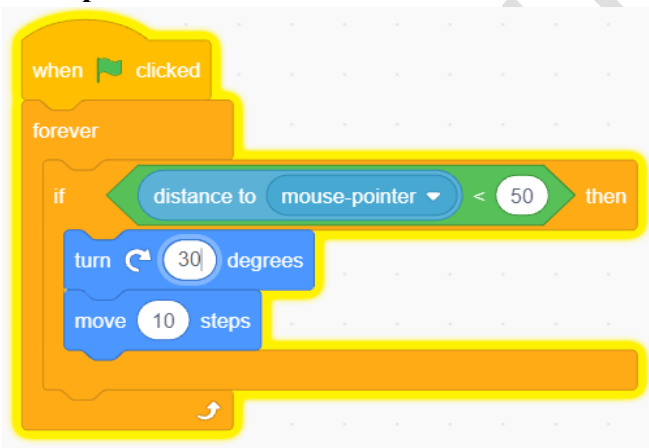


Picture 6. 6. Color sensor

6.4.3 Distance sensor

This sensor acts like a bat's ears. It sends out sound and listens for the echo to see how far away an object is. In Scratch use (distance to [mouse-pointer v]) block.

Example:

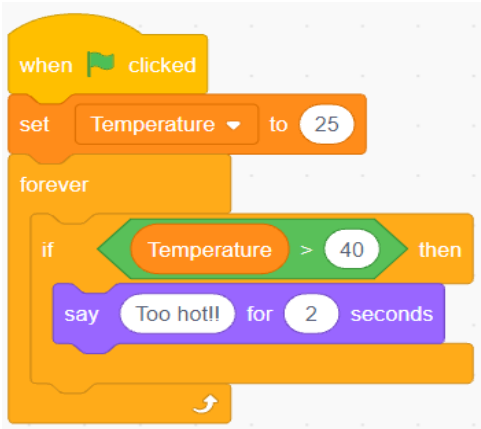


Picture 6. 7. Distance sensor

6.4.4 Temperature sensor

This sensor measures how hot or cold an environment is. Scratch doesn't have a thermometer block, so you use a variable called Temperature which acts as a sensor by storing the value of the temperature.

- **Example:**

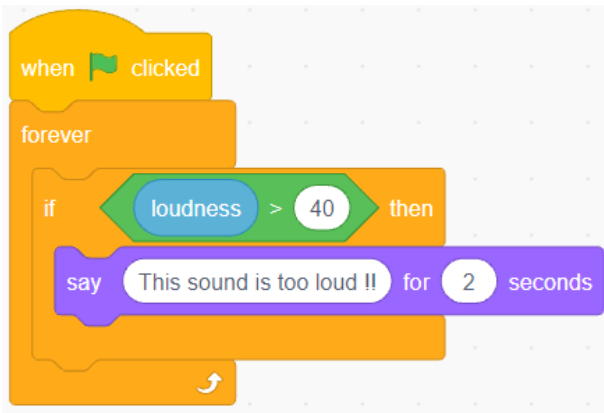


Picture 6. 8. Temperature sensor

6.4.5 Sound sensor

This sensor allows a robot to react to loud noises like a clap. In Scratch, use the (loudness) block.

Example:



Picture 6. 9. Sound sensor

6.4.6 Other sensors

Real world robots use sensors that Scratch doesn't have built-in, such as:

- **Gas sensors:** To smell smoke or dangerous gas.
- **Gyroscope or Tilt sensor:** To help a robot balance on two wheels. It tells the robot if it is leaning or falling over (used in drones and self-balancing robots).
- **Infrared (IR):** To receive signals from a remote control.
- **Pressure sensor:** Measures how hard something is being pressed (used in robotic grippers to hold an egg without cracking it).
- **Humidity sensor:** Measures how much water vapor is in the air (used in weather bots and smart plant systems).

Application activity 6.4

Open Scratch and answer these questions:

- 1) Which block would you use to stop a robot when it hits a black line?

- a) Distance to _____
- b) Touching color [Black]
- 2) An ultrasonic sensor uses light to see how far away a wall is. True or False?
- 3) If we want to simulate a temperature sensor in Scratch, we should create a _____
- 4) Complete the sentence: If a robot needs to know if it has bumped into a chair, it needs a _____ sensor.
- 5) Name one sensor that a real robot has but is not a standard block in Scratch.

6.5. Simulating many sensors

Activity 6.5

- 1) What if a robot meets a wall and hears a loud sound at the same time?
2. Can a robot make better decisions using more than one sensor? Explain.

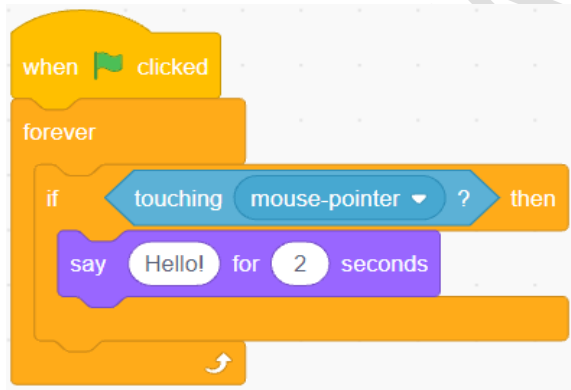
Simulating many sensors means, using more than one sensing condition in Scratch. This makes the sprite behave like a real robot that thinks carefully. An example is a robot that moves only if the path is clear and the sound is low.

6.5.1. Combined conditions

Combined conditions help the robot decide what to do based on more than one rule. In Scratch, we can make sprites react to more than one condition. This gets done using if blocks, if else blocks, and Nested if which is an if block inside another if block.

a. If condition

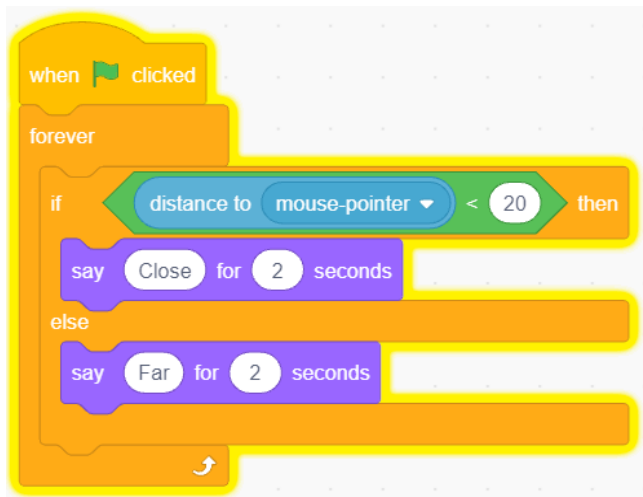
The robot checks one condition and acts as if it is true. The following is a Scratch block diagram example that makes a sprite say Hello! when it touches the mouse pointer.



Picture 6. 10. If condition in Scratch

b. IF ELSE condition

The robot chooses between two actions. The following is a Scratch block diagram example that makes a sprite say “Close” when near the mouse, and “Far” when it is far from the mouse.

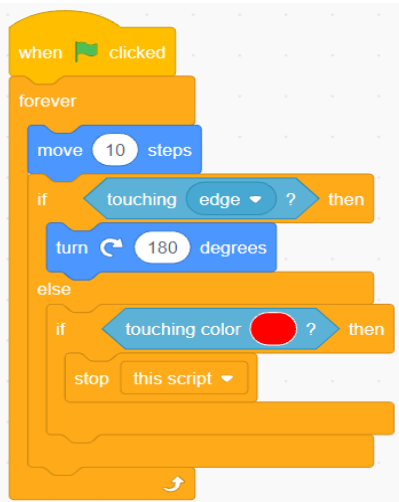


Picture 6. 11. If else condition in Scratch

c. Nested IF (IF inside IF)

This is a kind of an If condition inside another If condition. The robot checks one condition inside another condition before choosing which action to take.

The following is the Scratch block diagram example for Nested if that makes a sprite move forward but stop if it touches the edge OR if it touches red.

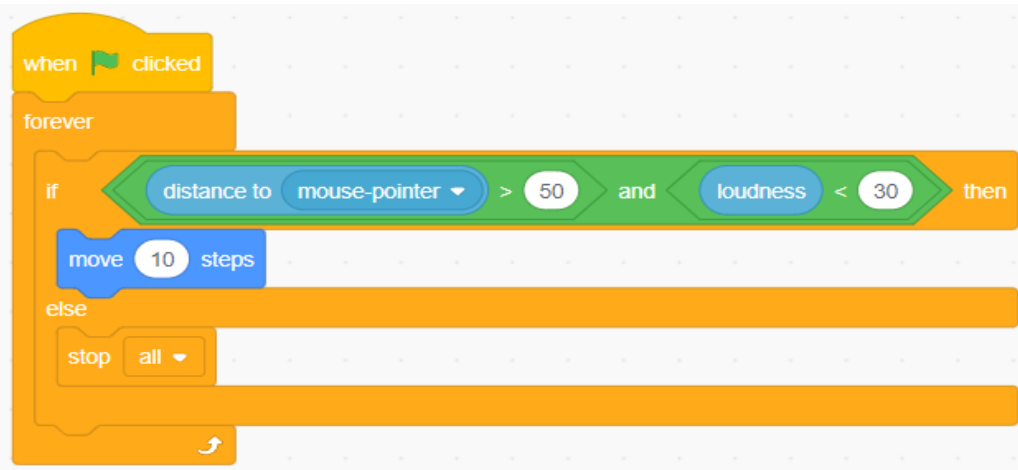


Picture 6. 12. Nested if in Scratch

6.5.2 Sensor fusion

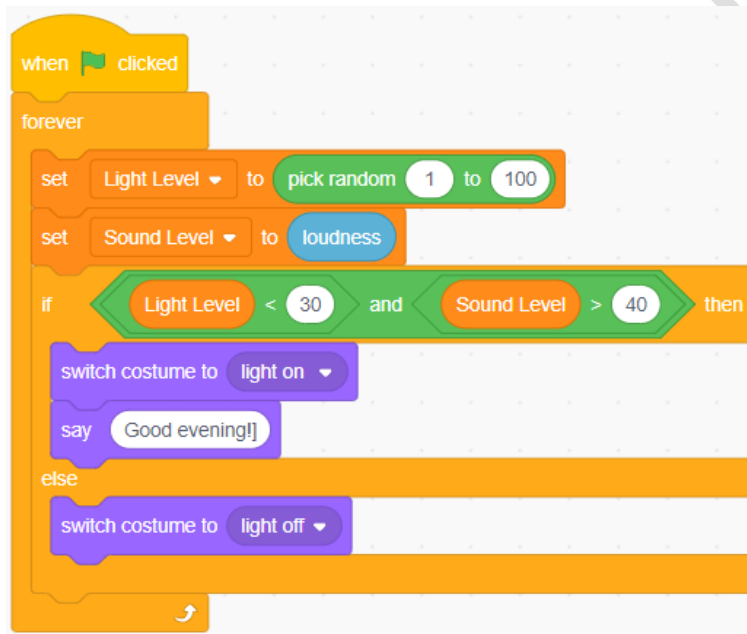
Sensor fusion is when a robot listens, looks, and feels at the same time. By using many sensors together and making one smart decision from all sensor data.

Example 1: A robot uses a distance sensor to check obstacles, color sensor to follow a line, and sound sensor to react to noise. Then, it moves only when all conditions are safe.



Picture 6.13. Sensor fusion example 1

Example 2: The car fuses the distance data with the camera data. The combined information leads to a much smarter decision: BRAKE NOW using the distance. Using the camera, it might see a stop sign that is far away. Below is a smart nightlight that turns on only when it's dark AND it detects movement.



Picture 6.14. Sensor fusion example 2

Sensor fusion is important because it helps robots make better decisions, avoid danger, and act like real intelligent machines making the robot more accurate and safer.

Application activity 6.5

- 1) What does it mean to simulate many sensors?
- 2) What does an IF ELSE block do?
- 3) What is a nested IF?

- 4) What is sensor fusion?
- 5) Why do robots use more than one sensor?
- 6) Give one example of two sensors that can work together.

6.6. End unit assessment

6.6.1. Project activity

Project Title: Robot Obstacle Avoidance Game

A. Project Guidance

- Do the project individually to demonstrate personal understanding of Scratch programming.
- Open Scratch (either online or offline version) and create a new Scratch project.
- Choose a robot sprite from the Scratch sprite library.
- Choose one Obstacle Sprite (for example: wall, box, rock).
- Position the robot on the stage where it has space to move.
- Place the obstacle somewhere in the robot's path.
- Select Robot sprite and start programming it.
- Add a When green flag clicked block to start the program.
- Use a forever control block so the robot moves continuously.
- Adjust the number of steps and degrees until movement looks realistic.
- Test the game by clicking the green flag.
- Observe whether the robot: Moves forward continuously, Detects the obstacle, Changes direction when it touches the obstacle
- Fix any errors if the robot does not behave correctly.
- Save the project with an appropriate name (e.g. *Robot Obstacle Avoidance Game*).

B. Questions on the Project

- 1) Which sprites did you choose for the robot and the obstacle, and why?
- 2) Which Scratch blocks did you use to make the robot move continuously?
- 3) How did you program the robot to detect the obstacle?
- 4) What happens when the robot touches the obstacle?

6.6.2. Assessment questions

- 1) Name two main parts of a robot and write one role for each.
- 2) What is a variable in Scratch? Give one example of a variable used in robotics.
- 3) Match the Scratch block to what it simulates in a robot:
 - a) Distance to [sprite]
 - b) Touching color [] ?
 - c) Loudness
 - i) Sound sensor
 - ii) Distance sensor
 - iii) Color sensor
- 4) A robot must move forward only when the path is clear and the sound is low. Which Scratch idea helps the robot make this decision?
 - a) One sensor only
 - b) Combined conditions
 - c) Motion blocks only
- 5) Name two Scratch block categories used to simulate an obstacle avoiding robot.
- 6) Why is safety important when designing and programming robots?
- 7) A robot uses a camera to collect information in a school. Which robot ethics rule must be respected? Explain your answer.